



TriCaster[®] TC Mini X

User Guide

NDI

August 25, 2023

Trademarks: TriCaster, 3Play, TalkShow, Video Toaster, LightWave 3D, and Broadcast Minds are registered trademarks of NewTek, Inc. MediaDS, Connect Spark, LightWave, and ProTek are trademarks and/or service marks of NewTek, Inc. All other products or brand names mentioned are trademarks or registered trademarks of their respective holders.

CONTENTS

CHAPTER 1 ABOUT THIS MANUAL.....	1
CHAPTER 2 INTRODUCTION.....	3
2.1.1 Overview	3
2.1.2 Launch Screen	3
2.1.3 Live Desktop	4
Section 2.2 Features	5
2.2.1 A/V Input and Output.....	5
2.2.2 TimeCode.....	6
2.2.3 Monitoring.....	6
2.2.4 Video Processing	7
2.2.5 The Switcher.....	7
2.2.6 M/Es	9
2.2.7 Virtual Sets	9
2.2.8 Record and Stream	10
2.2.9 Media Export	10
2.2.10 Automation	11
2.2.11 Import.....	11
2.2.12 Audio Mixer	11
2.2.13 Advanced Audio Connect	11
2.2.14 Integrated Media Players	12
2.2.15 Buffers	12
2.2.16 Titles and Graphics	13
CHAPTER 3 SETTING UP	14
Section 3.1 Command and Control	14
Section 3.2 Windows Login	15
Section 3.3 License and Registration	15
Section 3.4 Updates	15
Section 3.5 Enhanced Support (ProTek)	15
Section 3.6 Input Connections	16
3.6.1 Connect A/V Sources	16
3.6.2 Connecting a Control Surface	17
Section 3.7 Output Connections.....	17
3.7.1 A/V Output	17
3.7.2 Supplemental Video Outputs	17

Section 3.8 Starting a Session	18
Section 3.9 Configure A/V Output	21
3.9.1 Primary Outputs	21
3.9.2 Supplemental Outputs	23
Section 3.10 Configure Video Inputs	23
Section 3.11 Configure Audio	25
Section 3.12 Configure NDI Genlock	26
Section 3.13 Networking.....	27
 CHAPTER 4 WEB FEATURES	 29
Section 4.1 Password Protection	29
Section 4.2 Valuable Content	30
Section 4.3 Resources	30
Section 4.4 LivePanel™	30
Section 4.5 Video Training	31
Section 4.6 Operator Certification	32
 CHAPTER 5 THE LAUNCH SCREEN	 35
Section 5.1 Introduction to Sessions	35
Section 5.2 The Home Page	37
5.2.1 New (Session)	37
5.2.2 Titlebar Tools	38
5.2.3 NDI KVM	38
Open (Session).....	40
5.2.4 Shutdown	41
5.2.5 Help	45
5.2.6 Add-Ons	45
Section 5.3 The Session Page.....	46
5.3.1 Live	46
5.3.2 Graphics.....	46
5.3.3 Manage.....	47
 CHAPTER 6 LIVE DESKTOP.....	 51
Section 6.1 Display Requirements	51

Section 6.2 Overview	52
Section 6.3 Control Types	53
Section 6.4 Customizing the Desktop	53
6.4.1 Renaming Inputs.....	54
6.4.2 Workspaces	54
6.4.3 Express Switcher Mode	56
CHAPTER 7 THE DASHBOARD	57
Section 7.1 File Menu	58
Section 7.2 Options Menu	58
Section 7.3 Macros	59
Section 7.4 Workspaces	60
Section 7.5 Clock.....	60
7.5.1 LTC Timecode	60
7.5.2 Event Time.....	61
Section 7.6 Inline Help	61
Section 7.7 Notifications.....	62
CHAPTER 8 I/O CONFIGURATION	63
Section 8.1 Input Configuration	63
8.1.1 Input Tab	63
8.1.2 PTZ/Pan and Scan Presets	68
8.1.3 Image Tab	70
Section 8.2 Output Configuration	73
8.2.1 Output Tab	73
8.2.2 Record Tab.....	76
8.2.3 Genlock Tab	76
CHAPTER 9 SWITCHER, TRANSITIONS AND OVERLAY	79
Section 9.1 Switcher Modes	79
Section 9.2 Program/Preview Rows.....	80
Section 9.3 Background and DSK Layers	80
Section 9.4 Selecting Sources	81

Section 9.5 Linking Switcher Rows.....	81
Section 9.6 Transitions and Effects	81
9.6.1 Standard Mode.....	82
9.6.2 Background Controls	87
Section 9.7 Comps and MEMs	89
Section 9.8 Express Mode.....	89
9.8.1 Background Transition.....	89
9.8.2 Switching	90
9.8.3 DSKs.....	90
CHAPTER 10 MONITORING YOUR VIDEO	91
Section 10.1 Interface and Multiview	91
Section 10.2 Live Desktop Monitors	91
Section 10.3 Workspace Presets.....	92
Section 10.4 Scopes	93
10.4.1 LED Displays.....	93
Section 10.5 Viewport Options	94
10.5.1 Viewport Presets	95
10.5.2 LivePanel Preset Buttons.....	96
Section 10.6 Viewport Tools	96
Section 10.7 Program Monitor	97
Section 10.8 Look Ahead Preview.....	97
CHAPTER 11 MEDIA PLAYERS & BUFFERS.....	99
Section 11.1 Media Players.....	99
11.1.1 Playlists	100
11.1.2 File Operations	101
11.1.3 Speed.....	102
11.1.4 Macro Triggers.....	102
11.1.5 Transcode	102
11.1.6 Properties	103
11.1.7 Trimming Clips and the Scrub Bar	103
11.1.8 Media Browser	104
11.1.9 Player Controls	106
11.1.10 SHOW On (...).....	108
11.1.11 MEMs	109
11.1.12 Network Sharing.....	110

Section 11.2 Editing Title Pages	110
11.2.1 Header Tools	111
11.2.2 Footer Tools.....	112
11.2.3 DataLink.....	113
Section 11.3 Buffers.....	113
11.3.1 Buffer Types.....	114
11.3.2 Selecting Content	115
11.3.3 Buffers Menu.....	116
11.3.4 Animation Features	116
11.3.5 Live Link	116
CG Buffers.....	120
11.3.6 Buffer Watch Folders.....	121
11.3.7 Using Buffers	121
11.3.8 Keying, Proc Amps and More	122
11.3.9 Editing titles.....	122
11.3.10 Buffer Presets	122
 CHAPTER 12 PTZ CONTROL.....	 125
Section 12.1 Input Configuration, PTZ Tab	125
12.1.1 Connecting.....	125
12.1.2 PTZ Operations	126
12.1.3 Presets	127
12.1.4 PTZ and the Control panel	127
 CHAPTER 13 LIVEMATTE	 129
Section 13.1 Understanding Keying.....	129
Section 13.2 Matte	130
13.2.1 LiveMatte Modes	130
13.2.2 Color	130
13.2.3 Tolerance	131
13.2.4 Smoothness	131
13.2.5 Luma Limit	131
13.2.6 Strength	131
13.2.7 Offset.....	132
Section 13.3 Spill Suppression	132
Section 13.4 Compositing	132
Section 13.5 Fine Tuning.....	132
Section 13.6 Lighting for LiveMatte	133
Section 13.7 Crop Source	133

CHAPTER 14 MIX/EFFECT (M/E) TOOLS	135
Section 14.1 Overview	135
Section 14.2 M/E Modes.....	135
14.2.1 Mix Mode	136
14.2.2 Effect Mode	137
Section 14.3 The T-Bar	137
Section 14.4 Input Position Controls	137
Section 14.5 Default Effects.....	138
Section 14.6 Virtual Sets	139
14.6.1 Holographic LiveSets	140
Section 14.7 Key Channels	141
14.7.1 Key Layers and Autoplay	142
14.7.2 Augmented Reality	142
Section 14.8 Comps	143
14.8.1 Apply to Transitions.....	144
14.8.2 Managing Comps	144
14.8.3 Animation	145
CHAPTER 15 AUDIO.....	147
Section 15.1 Audio Specifications	148
Section 15.2 Headphones	148
Section 15.3 VU Meter Calibration	148
Section 15.4 External Sources.....	148
15.4.1 Connection Type.....	149
15.4.2 NDI KVM Audio.....	149
Section 15.5 Local Skype TX CALLERs and Mix Minus.....	150
Section 15.6 TalkBack	150
Section 15.7 Common Controls	150
15.7.1 Mute	151
15.7.2 Solo	151
Section 15.8 Internal Sources	152
15.8.1 Media Players	152
15.8.2 Effects (Transitions)	152

Section 15.9 Output and Primary Bus Controls	152
15.9.1 Headroom Notes	152
15.9.2 Stream	153
Section 15.10 Advanced Configuration	153
15.10.1 Input Tab	153
15.10.2 Processing Tab	155
15.10.3 Routing Tab	157
Section 15.11 Supplemental Audio Devices	160
Section 15.12 MEMs	160
CHAPTER 16 MACROS AND AUTOMATION	161
Section 16.1 Creating Macros	162
16.1.1 Speed and Snapshot Mode	162
16.1.2 Triggers	163
16.1.3 LivePanel Buttons	163
Section 16.2 Managing Macros	164
16.2.1 Session Macros	164
CHAPTER 17 STREAM/ENCODE	165
Section 17.1 Introduction	165
Section 17.2 Configuration	165
17.2.1 Source Setup	166
17.2.2 The Web Browser	166
17.2.3 Configuring Encoders	167
Section 17.3 Destination Presets	168
17.3.1 Custom Presets	169
Section 17.4 Initiating the Stream	169
Section 17.5 Capturing the Stream	169
Section 17.6 Streaming Strategies	169
17.6.1 On Demand or Live Streaming?	169
17.6.2 Streaming Media Providers	171
Section 17.7 Production and Capture Considerations	172
Section 17.8 Diagnostics and Troubleshooting	172
17.8.1 Testing your stream	172
17.8.2 Speed Tests	175
17.8.3 Where is the Problem?	175

CHAPTER 18 EXPORT.....	177
Section 18.1 Overview	177
Section 18.2 Export Menu	177
18.2.1 Preset List	177
18.2.2 New Preset	178
18.2.3 Social Media Sites.....	178
18.2.4 Transcode, SMTP, and FTP	178
18.2.5 Watermarking.....	178
Section 18.3 Export Media.....	178
18.3.1 Metadata	179
18.3.2 Presets	180
18.3.3 List Management.....	180
18.3.4 The Export Button	181
18.3.5 Other ‘Add to’ Methods	181
CHAPTER 19 RECORD, GRAB, AND REPLAY.....	183
Section 19.1 Record	183
19.1.1 Record Configuration	184
19.1.2 Capture Controls	184
Section 19.2 Replay.....	185
19.2.1 Instant and Deferred Replays	186
19.2.2 Switcher Source	186
19.2.3 Mixed Output	186
Section 19.3 Grab	186
CHAPTER 20 TITLE TEMPLATES.....	189
CHAPTER 21 CONTROL PANEL: TRICASTER FLEX	193
Section 21.1 Connection and Configuration	193
21.1.1 Safety.....	194
21.1.2 Cable Connections.....	194
21.1.3 Flex Configuration Webpage	194
21.1.4 Latest Versions	195
21.1.5 Choosing a Target.....	196
Section 21.2 TriCaster Flex Webpage	197
21.2.1 Administration Tab.....	197
21.2.2 Mapping Tab.....	199
Section 21.3 Control Layout.....	201
21.3.1 Display	202

Section 21.4 Switcher	202
21.4.1 Bank	202
21.4.2 DSK/Key	203
21.4.3 Stripe	203
21.4.4 MEM and COMP	203
Section 21.5 PTZ Controls	203
21.5.1 Control Button Group	204
Section 21.6 PAN/TILT	205
Section 21.7 Transitions	205
21.7.1 FTB	205
21.7.2 FADE & TRANS	206
21.7.3 RATE	206
Section 21.8 T-BAR	206
21.8.1 TAKE & AUTO	206
Section 21.9 Audio Features	207
21.9.1 Backplane Connections	207
21.9.2 Volume Knobs	207
21.9.3 TALK BACK	208
Section 21.10 Stream, Capture and Replay	208
21.10.1 MACRO	209
21.10.2 MEDIA PLAYER GROUP	209
 CONTROL SURFACE: TRICASTER MINI	 211
 APPENDIX A: ADVANCED FEATURES	 215
A.1 Introduction	215
A.2 Live Story Creator	215
Overview	215
Style-based Operations	218
Comment-based Commands	219
Teleprompter Output	226
A.3 Format Independent Production	227
Session Options	228
Input Rotation	229
Output Rotation	229
A.4 LiveGraphics	230
Use Layered PSD Files	231
Animate in After Effects	231
Using LiveGraphics	231
LiveGraphics and DataLink	232

A.5	LivePanel.....	233
A.6	Network Access to LivePanel	233
A.7	Home Page.....	234
A.8	Audio Mixer	234
A.9	DataLink	235
A.10	Media.....	235
A.11	Switcher	235
	Builder	236
A.12	Scoreboard.....	237
A.13	Advanced Audio I/O.....	238
A.14	Virtual PTZ	239
A.15	Virtual Set Library	239
A.16	Sound and Music	240
APPENDIX B: SKYPE AND SKYPE TX		241
B.1	Skype TX.....	241
B.2	TalkShow®	241
B.3	Skype TX Controller	242
B.4	Skype TX Caller.....	242
B.5	Skype for Content Creators	243
APPENDIX C: PERFORMANCE CONSIDERATIONS.....		245
C.1	Testing, One Two	245
C.2	IMAG and Latency.....	245
C.3	Relativity and the Speed of Light	245
C.4	Latency and Your Audience	246
C.5	Latency and Your NewTek System.....	246
C.6	Other Sources of Latency.....	247

APPENDIX D: VIDEO CALIBRATION	249
D.1 What (And Where) to Calibrate?	249
D.2 Calibrating Video Sources	250
D.3 Setting Black and White	250
D.4 Adjusting Color	251
D.5 Color Metrics	252
D.6 Calibrating Your Monitors	253
D.7 Computer Monitor	253
D.8 Program Output Monitor	254
D.9 Color Adjustments	254
APPENDIX E: KEYSTROKE SHORTCUTS	257
E.1 Switcher...	257
E.2 T-Bar	257
E.3 Record, Grab, and Stream	258
E.4 Tabs	258
E.5 Workspace	258
E.6 Media Players	258
E.7 General	259
E.8 Edit Title Pane	259
E.9 Selection and Navigation	259
E.10 Misc.	259
APPENDIX F: CUSTOMER SUPPORT	261
F.1 Product Support	261
F.2 The ProTek SM Advantage	261

APPENDIX G: RESTORE FACTORY DEFAULTS	262
G.1 Create a 'User Backup' Drive	262
G.2 Restore TriCaster Software.....	262
INDEX	264
CREDITS	267

PART I (GETTING STARTED)

Connections and registration, and a top-level overview of the primary features of your live production system.



This manual tells you everything you need to know to perform common operations with your NewTek live production system. It attempts to convey this essential information in a friendly yet concise way, while also providing a deeper reference section you can turn to when you really need more detail.

Even if you are the hands-on, 'I never-ask-directions!' type, please peruse this page. If questions about your live production system arise later, you may find that the information here allows you to jump directly to the details you need with a minimum of reading.

➤ *PART I – GETTING STARTED*

Introduction – connecting devices (cameras, monitors, etc.) and registration, ending with Chapter 4, Web Features, which among other things includes an overview of online resources to help familiarize you with common operations and features.

➤ *PART II – REFERENCE*

This section covers the fine details of using your system (for those who need it, or who simply like to know everything about everything).

➤ *PART III – CONTROL PANELS*

Your Live Production system can be taken to new levels of convenience and functionality with the addition of a supported external hardware control panel.

➤ *PART IV – APPENDICES*

Certain topics which benefit from in-depth coverage are in this section, along with cross-references to relevant information elsewhere in the manual and concluding with a keyword index. Appendix E: lists all shortcut keys.



Chapter 2 INTRODUCTION



Thank you for purchasing this NewTek product. As a company, NewTek is extremely proud of its record of innovation and its commitment to excellence in design, manufacture, and product support. This smaller version of our most advanced live production system software raises the bar again. You will find it exceptionally powerful and versatile. This chapter provides a tour of its components and features.

2.1.1 OVERVIEW

Time and again, NewTek's innovative live production systems have redefined broadcast workflows, provided new creative possibilities, and delivered significant cost benefits. Our products include the most complete, reliable, and efficient integrated systems available for live production and web streaming, with capabilities suited to almost any broadcast need. You can produce and distribute live video programs from diverse sources in ultra-high-definition formats (up to 2160/29.97p).

As well, support for NDI® (Network Device Interface) protocol places your new system squarely in the forefront of IP solutions for the video broadcast and production industries. And, thanks to NDI's versatile architecture, this does not isolate you in a proprietary 'walled garden'. NDI's widespread third-party adoption, support for interchange with other IP standards, and extensibility ensure your investment will *expand* your production horizons, rather than limit you to restrictive alternatives.

2.1.2 LAUNCH SCREEN

The *Launch Screen* appears shortly after you power up your live production system. This is the command center where your production projects are configured and launched.

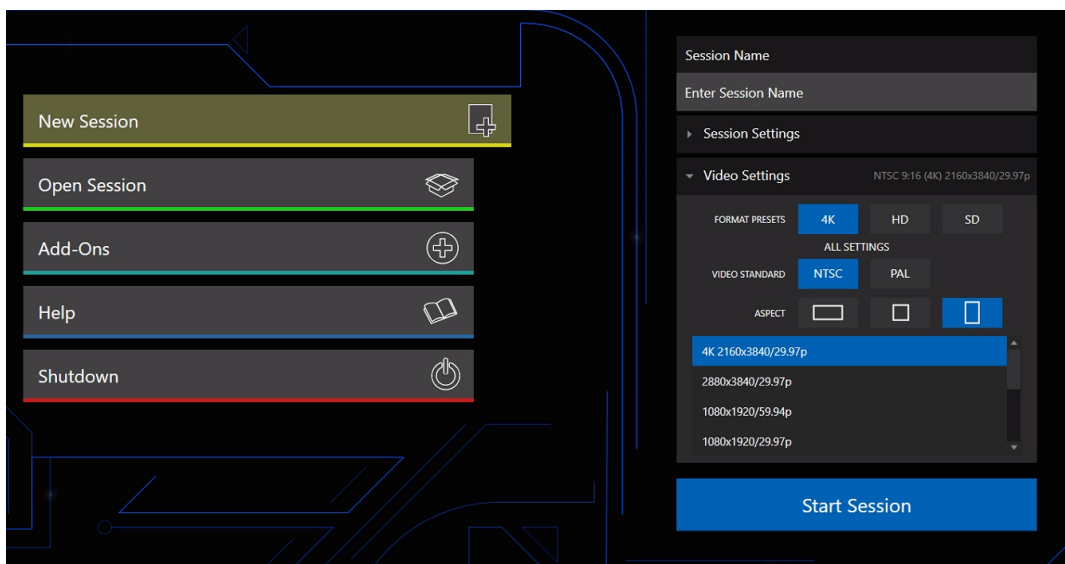


FIGURE 1

The *Home page* of the *Launch Screen* provides a number of important functions, notably allowing you to create (and re-open) *sessions*. Sessions are an important concept – essentially a custom preset prepared for an individual production or other purpose. Later, when you re-enter an existing session, all its assets, settings, and even control states are remembered.

Selecting a session takes you ‘into’ the session. TriCaster will show you the *Session Page* next, where you can prepare *Graphics* (title pages), *Manage* content, and more – or launch the *Live Desktop* to begin production.

2.1.3 LIVE DESKTOP

Your system’s live production features are all available from the *Live Desktop*, which in many ways mimics familiar video production equipment. However, the *Live Desktop* provides far more functionality in its integrated environment than similar single-purpose devices.

The various features, controls and modules comprising the *Live Desktop* are arranged in horizontal bands, as seen below.



FIGURE 2

- The top-most band comprises a convenient and powerful *Dashboard*.
- The area just below the *Dashboard* is *normally devoted to a multi-pane monitoring display, providing source and output views.
 - * This pane can be re-sized, even completely hidden; or the display can be customized to complement external *Multiview(s)*, or for numerous other purposes.

- The central *Live Control* section is home to the *Switcher*, *Transition*, *DSKs* (overlay channels) and *M/E* controls (*Mix Effect* panes can be minimized and are hidden from view in Figure 2).
- By default, the bottom third of the *Live Desktop* is home to tabbed control modules, including *Media Players*, *Buffers*, and the *Audio Mixer*.

SECTION 2.2 FEATURES

Let's take time for a brief overview of just *some* of the key features of your system.

2.2.1 A/V INPUT AND OUTPUT

It's worth taking a moment to consider that each NDI output potentially substitutes for multiple traditional outputs. For example – it is not unusual to need multiple physical outputs from a traditional video switcher for a single source – one for recording, another to broadcast, one for local monitoring and so on, with each destination consuming precious hardware connections. This is not the case with NewTek live production systems, which allows multiple downstream connections to a single NDI output.

Thus, it requires no hyperbole to state that your NewTek system provides virtually unlimited outputs, without expensive additional hardware.

- Connect and switch up to 8 external video sources, and even more internal sources. TriCaster Mini X supports HDMI video input and output, (varies by model) internal and network sources and both analog and digital audio input and output.
- Flexible, multi-destination delivery with multi-format video outputs, including 4 configurable mix outputs.
- Support for up to 8 simultaneous Pan-Tilt-Zoom (PTZ) robotic cameras via serial and network protocols, including RS232, RS422 and IP, with integrated controls and preset system.
- *LivePanel Buttons* add the convenience of presetting *and* editing macros, without ever leaving the user interface. (Varies by model)
- *Live Link* offers new opportunities by bringing the power of the internet into your TriCaster. Bring in live webpages, graphics or images into any of your 10 HTML *Buffers*.
- Importing a clip from *Media Player* now conveniently saves your imported/transcoded clip(s) in the same location.
- Unlimited creative potential for visual storytelling with re-entrant M/Es and easily accessible preset configurations for each M/E bus.
- Create non-standard aspect sessions like square or vertical, and discretely output specified video segments (with optional overlap).
- Multiple *Media Player* modules allow you to insert pre-recorded video, music, sounds and imagery into your live presentations.
- Select Microsoft Windows® or Apple Macintosh® computers on the same network as *Switcher* inputs for your live productions using NDI Screen Capture.
- Update *Buffers* in 'watch folders' from networked graphic workstations, even while these are displayed live.
- Transmit output to other NDI-enabled systems across the network.
- Integrated LiveSet™ technology with scores of virtual set and box effects included.
- Use switcher row color groups to automatically harmonize video sources for multiple *M/E* s, etc.

- Bring DSK's on and off screen independently or with background sources using combined layering and transition effects.

2.2.2 TIMECODE

- Default Quicktime® recording format includes full 'per frame' timecode.
- IsoCorder™ recorded clips from multiple sources are easily aligned for synchronized post show editing using embedded timecode.

2.2.3 MONITORING

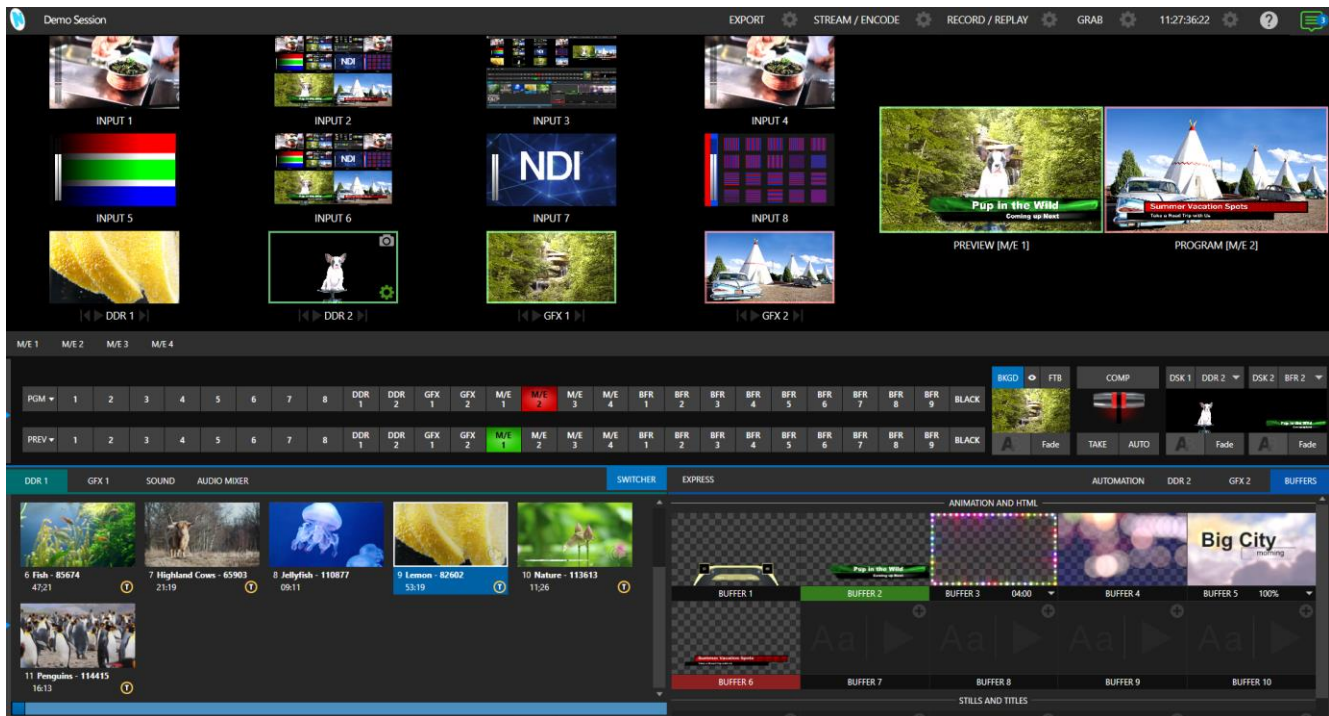


FIGURE 3

- The *Live Desktop* includes a large *monitoring pane* with multiple and easily customized *Workspaces*. By default, this area includes *Program* output and *Look Ahead Preview* monitors with associated controls, along with the most commonly used *Switcher* inputs; but many alternatives are available.
- *Quality Monitoring* provides resolution and frame rates of every video source coming into their TriCaster Mini to ensure quality of sources.
- TriCaster provides three more multiviewer outputs to drive local displays, with configurable workspace layouts and viewports to permit directors and operators to customize the control environment to their preferences.
- Beyond centralizing, organizing, and observing your choice of video signals with tally, you can calibrate monitor color characteristics and display visuals such as audio VU meters, production clocks, and *Waveform* and *Vectorscope*, (varies by model) for comprehensive management of your live operations.

- Monitor sources on both *Live Desktop* and external *Multiviews* are configurable, and include *Media Player* clip timecode counters, safe area overlays, custom logo display and more.
- *Waveform* and *Vectorscope* monitors help you calibrate your signal to ensure your output meets broadcast standards and looks great. (Varies by model)

2.2.4 VIDEO PROCESSING

- Full 4:4:4:4 32-bit floating point internal processing of all video sources.
- *Proc Amps* and *White Balance* controls for every source preserve pristine image and color fidelity.

2.2.5 THE SWITCHER

The powerful *Switcher* module, which provides native support for popular professional video routers and NDI® (Network Device Interface) protocol, puts countless sources at your fingertips in a familiar *Program/Preview* row control interface.

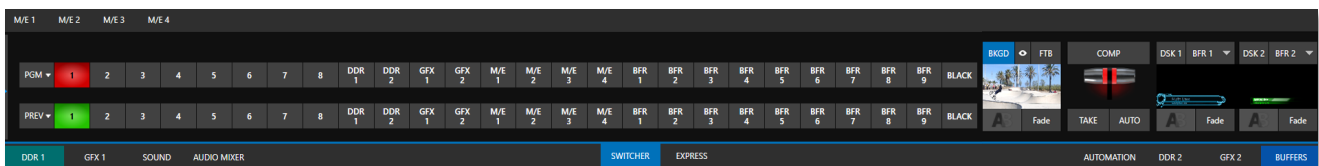


FIGURE 4

For less complex productions, the *Switcher's* convenient *Express mode* simplifies the process. This one-button operating mode will be especially welcome in environments where volunteers or less experienced operators are involved.

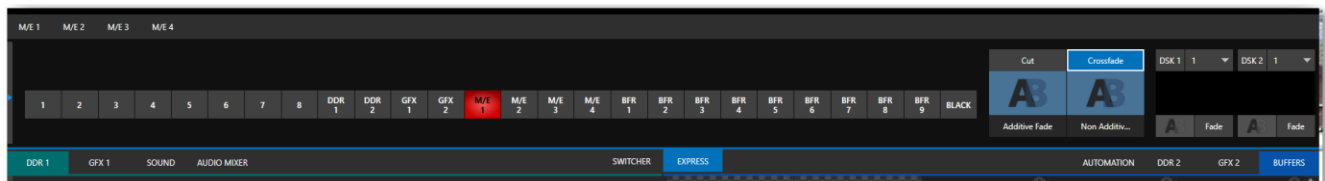


FIGURE 5

Additionally, support for switching using a touchscreen has been provided. See something interesting on a viewport in the monitoring section? Just *tap* the corresponding viewport to instantly send that source to *Program* output, using the current *background transition* (if any).

VIDEO LAYERS AND TRANSITIONS

The *Transition* section of the *Live Desktop* provides powerful tools for arranging and displaying the numerous video and graphic layers contributing to your ultimate program output.

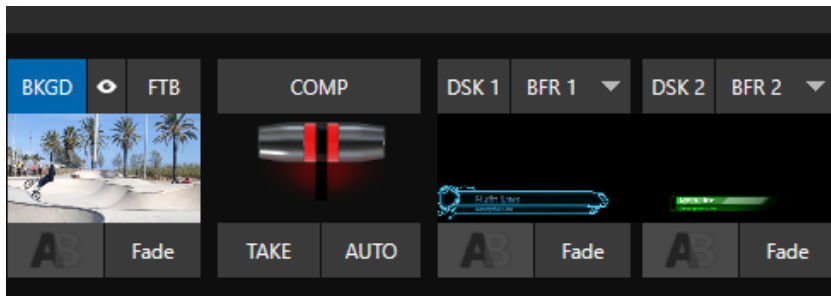


FIGURE 6

- Freely *hot-punch* or *transition* between external sources, internal *Media Players*, and *M/E effects*.
- Latch selected *Switcher* and *M/E* source rows in color groups to synchronize switching operations.
- Multiple *DSK* (Downstream Key) *Switcher* channels and independent *KEY* channels for each *M/E*, permit endless creative possibilities.
- *DSK* and *KEY* sources are always represented in small full motion ‘confidence monitors.’
- Use any of the hundreds of transitions supplied to reveal either *Background* or *DSK/Key* channels with animated wipes, trajectories, and dissolves, or select colorful and dynamic *Animation Store* transitions with overlays, sound and *TransWarp™* effects.
- Create an unlimited number of custom transition effects with embedded sounds, dynamic cloth effects, and full color overlays using the integrated *Animation Store Creator* application.
- Create eye-popping animated graphics using your favorite software (e.g., Adobe After Effects®), and convert them to animated *Buffer* effects that you can display without tying up a *Media Player*.
- Reveal video layers individually or in combination with one *T-bar*, *Take* or *Auto* operation.
- Select individual transition effects for any layer and adjust *Speed*, *Reverse* and *Ping Pong* options.

DSK CHANNELS



FIGURE 7

DSK layers support downstream overlays (*M/E Keys* are upstream overlays, i.e., ahead of the main *Switcher*).

- Display *DSK/Key* channels independently and swap them during a switch with custom transitions.
- Use *Media Players* to overlay title pages, including scrolls, crawls, and lower-thirds, or assign a *Camera* feed or *Network* source to an overlay channel.
- Alternatively, use the *Buffer* channels to assign a specific title, still image or animation effect to any overlay channel.

- Independent *Crop*, *Position*, *3D Rotation*, and *Scale* controls for each *DSK* channel permit you (for example) to configure multiple sources as 'Picture in Picture' elements, revealing them with individual custom effects.

2.2.6 M/ES



FIGURE 8

Color-coded *M/E* (Mix/Effect) banks provide stunning effects.

These include secondary video mixing capabilities, switching (with traditional wipes or *Animation Store* transitions with *TransWarp* capabilities), full alpha channel support and best in class *LiveMatte™* chroma and luma keying technology on all source channels and *M/E* buses.

Each *M/E* bank supports:

- Up to two primary video sources per effect.
- *KEY* (overlay) channels with individual transition capabilities.
- Sub-mix functionality, with individual transitions for *A/B Background* layers and *KEY* channels.
- Individual *Scale*, *Position*, *Crop* and *Rotation* controls for all channels.
- Integrated video composition engine on the switcher and each *M/E* bus to create, store, and apply layer configurations and DVE-style motion sequences
- Powerful *Comp* (composition) preset features let you animate layer and overlay position, rotation, and other attributes in full motion.
- Convenient *MEM* system allows instant access to pre-configured *M/E* setups.

2.2.7 VIRTUAL SETS



FIGURE 9

- Achieve the look of large, sophisticated studio sets in a very small space with NewTek's *LiveSet*™ virtual set technology.
- Virtual sets support up to two keyable video sources.
- *Zoom* and *pan* the virtual 'camera' during live production (without touching the physical camera).
- Or use a *Holographic LiveSet* to freely rotate and zoom the camera view.
- Set up custom virtual set animation targets even while live.

2.2.8 RECORD AND STREAM

- Record your live-switched production at full resolution to a high-quality QuickTime® format.
- Recorded files include embedded timecode.
- IsoCorder™ technology supports up to multiple internal a/v recordings at once, chosen from among local hardware camera inputs and all four outputs.
- Easily grab still images from *Switcher* input sources, or batch capture selected outputs.
- Simultaneous outputs for broadcast, projectors, and live web streams.
- Streaming: RTMP, RTSP, HTTP, SRT and more.
- Stream live to multiple Internet destinations for redistribution and archive your stream locally.
- Perform instant replays with custom transitions and playback speed with just one click.
- Or use NewTek's TimeWarp™ instant replay control surface to mark, replay and manage highlights.

2.2.9 MEDIA EXPORT

Export support makes it easy to distribute content to multiple targets in one pass. This module stores credentials for social media and upload sites and provides a queue for uploading and file conversion.

Clips and stills from live sessions can easily be added to the *Export* queue without interrupting production, for distribution to multiple hosting services and social media sites (including Twitter®, Facebook®, YouTube®, Vimeo, and more) all with a single click – even when live production recording is still underway.

This is great for concerts, galas, sport events, game shows and more. The *Export* feature directly addresses FTP sites and network locations too; transfer media to multiple locations at the same time to meet post-production needs or other collaborative and archival requirements.

2.2.10 AUTOMATION

Record, edit, and play back macros, whether a simple operation or more involved series of events at the touch of a button. Use powerful integrated automation systems to trigger these, simplifying complex tasks and ensuring consistent operations.

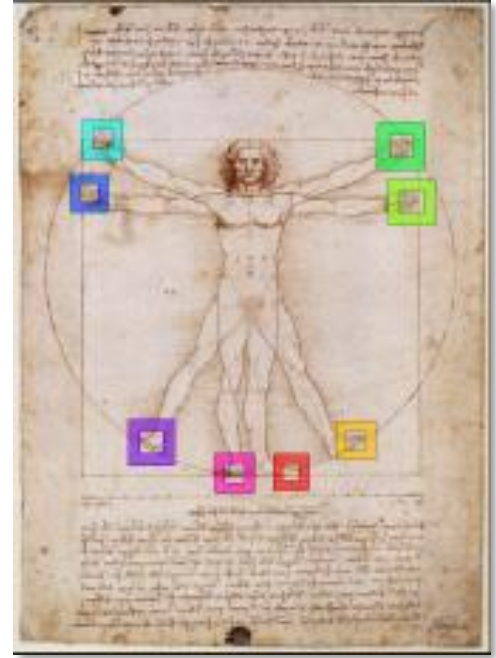
Hint: See also Live Story Creator in the Appendix on Advanced Features.

2.2.11 IMPORT

IMPORT MEDIA

The integrated *Import Media* feature provides a convenient one-stop approach to gathering productions assets.

- Batch import media files for use in *Media Players*.
- Imported files are automatically categorized as stills or clips, and can be pre-assigned to the specific players, making them easy to find and use during live productions.
- When necessary, imported files are transcoded to formats that ensure smooth playback.
- Importing a clip from *Media Player* conveniently saves your transcoded clip along with the original file in a 'Transcoded' folder.



2.2.12 AUDIO MIXER

The integrated *Audio Mixer* provides extensive multi-channel control and management.

- *Neural Voice Isolation* (AI Audio) is a powerful tool that cleans audio at the touch of a button. Using AI, TriCaster can cancel or reduce background noise and automatically detect voices, maintaining all-important production quality.
- Adjust numerous internal and external audio sources and introduce them into your program manually or automatically.
- Control output levels for program, headphones, auxiliary outputs, recorders, and stream separately.
- Convenient *Audio Mixer MEM* system allows single-click access to prepared setups.
- Automation features include complex '*Follow Program Video*' options.
- Exclusive or latched *Solo* controls.
- Independent 7-band *EQ*, *Compressor/Limiter* and *Noise Gate* for each source and output.
- Complete multi-bus matrix routing options.
- Configurable *VU meters*: Show calibration on digital (dBFS) or analog (dBU or dBu) scales.

2.2.13 ADVANCED AUDIO CONNECT

- VST plugin with NDI offers unlimited support for mix minus, intercom, and other advanced capabilities without any restrictions on scale, all with floating point audio, and without any channel restrictions.

- Supports comprehensive integration of system audio with external mixers.
- Provides independent routing of any audio input or audio output channel(s) for external routing.
- Enables extraction of embedded audio from NDI® sources for external routing.

2.2.14 INTEGRATED MEDIA PLAYERS

Media Players and title system serve up graphics, titles, video clips and audio files quickly and easily during your live productions.

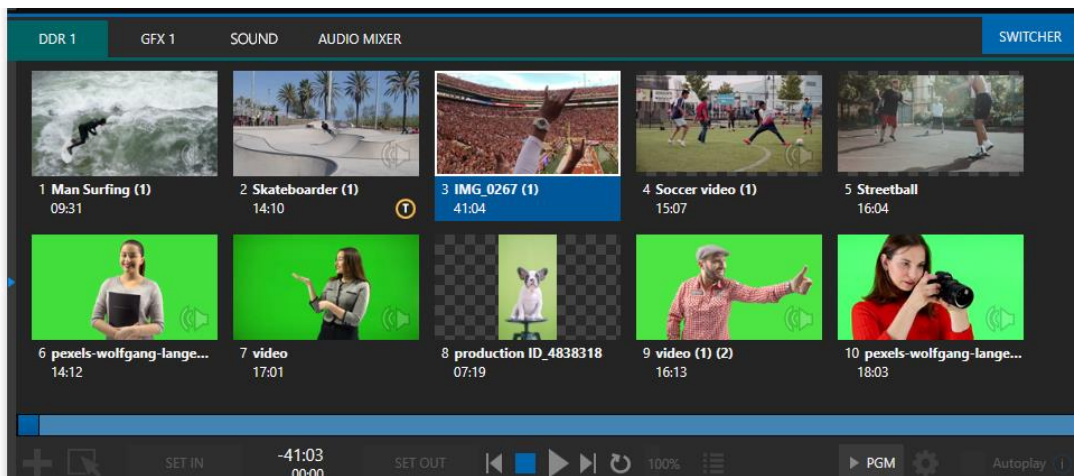


FIGURE 10

- Add a/v clips, sound or still images to *Media Player* playlists.
- Trim and re-order playlist entries.
- Add colorful transitions between clips.
- Flexible playback controls include variable *Speed* and *Loop* options.
- Each *Media Player* has its own volume and other advanced audio controls in the *Audio Mixer*.
- Convenient 'per clip' audio level and playback *Speed* control.
- *Autoplay* feature starts and stops playback automatically based on *Switcher* activity.
- Control *Media Players* with *macros* triggered by other events.
- Assign titles, graphics, clips, or buffers to multiple *DSK* or *M/E Key* channels.
- Edit titles live, even while 'on air'.
- Transition a clip to the *Main Switcher* or an *M/E* with a custom effect, play it, and then restore the target's original source with a single click using the *Show On* feature.
- Store playlists and player configurations in *MEM slots* for easy recall.

2.2.15 BUFFERS

The *Buffers* module provides quick review and management of the internal media buffer system, providing combined playback and asset management tools, and is fully supported by the *Macro* system. (Figure 11).

Render web pages directly into your *Buffers* with *Live Link* and pull live web page elements into your production. From graphics or images on a web page to videos from your organization, *Live Link* provides agility and new opportunities to your production. (See Appendix: 11.3.5 Live Link)

Buffers can be applied in any video channel, including all main *Switcher* and *M/E* rows, or as overlays in the countless *DSK/KEY* channels.

Buffers can contain animated effects (compiled from your own clips using the supplied *Animation Store Creator* application), still graphics or title pages, or dynamic LiveGraphics™ – all with full alpha channel support.

Buffer effects are resident in reserved *Switcher Memory*, making it possible to use them all at once without tying up *Media Players*. Thus, *Buffers* effectively substitute for *many* additional *DDR*s.

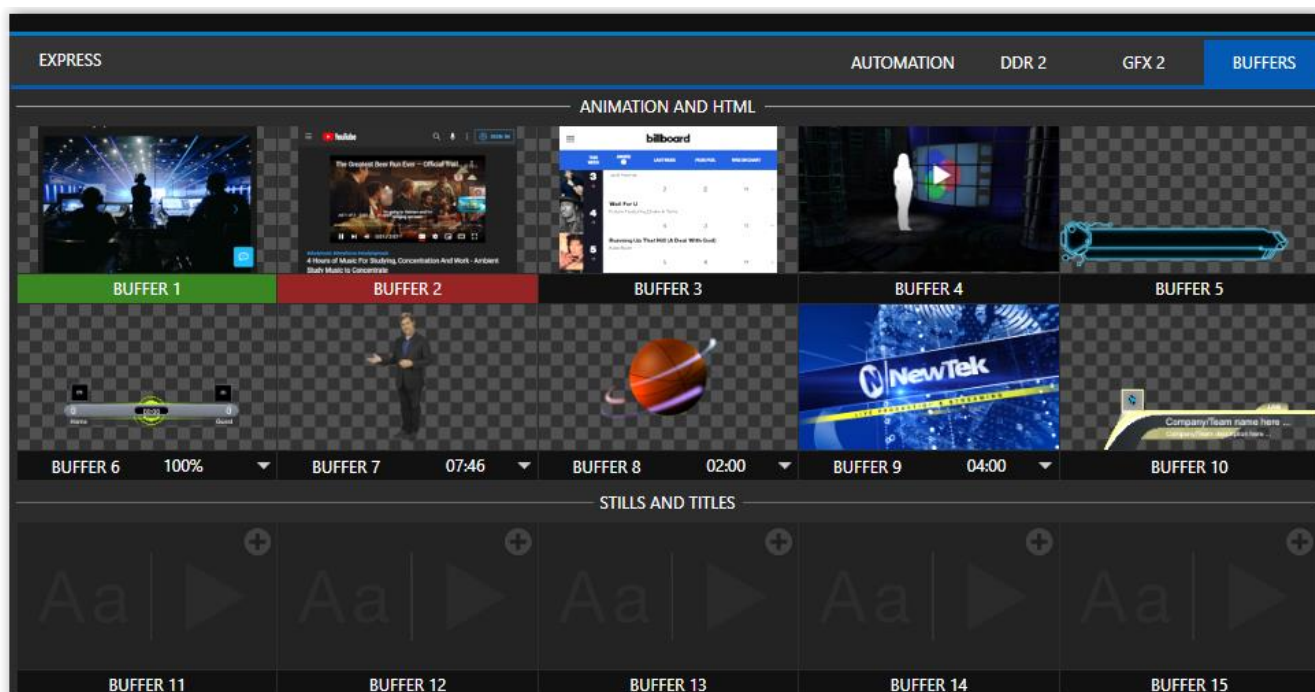


FIGURE 11

2.2.16 TITLES AND GRAPHICS

An extensive set of live-editable title pages in various theme packs are supplied, all ready for immediate use in your productions.

Hint: See also LiveGraphics in Appendix A: Advanced Features



This chapter explains how to connect power, monitors and audio visual sources, and external control devices to your NewTek live production system. After completing this short section, you'll be all set to begin using your new unit.

To begin, let's review 'what came in the box':

- Your NewTek live production system
- A/C power cabling
- NewTek mouse and keyboard
- Four Mini DisplayPort adapters

SECTION 3.1 COMMAND AND CONTROL

Hint: The user interface requires a minimum monitor resolution of 1920x1080.

1. Connect an external computer monitor to a video output port on TC Mini's backplane (Figure 12).
2. Connect the *mouse* and *keyboard* to any of the USB ports.
3. Connect the *power cord* to an A/C power receptacle.
4. Connect a network or ethernet cable to either of the gigabit ethernet network connections, connect other end of network cable to your computer network. (This connection is not necessary for basic operation but is required for initial set up)
5. Turn on the computer monitor.
6. Press the *Power* switch, located on the faceplate of the TriCaster Mini X.

At this point, the blue *Power LED* will illuminate, and the device will boot up. (If this does not happen, check your connections, and retry).

Though not a requirement, we do strongly recommend that you use an uninterruptable power supply (UPS), as for any 'mission critical' system.

Likewise, consider A/C "power conditioning", especially in situations where local power is unreliable or 'noisy'. Surge protection is especially important in some locales.

A word about UPS devices:

'Modified sine wave' UPS devices are popular due to low manufacturing costs. However, such units should generally be viewed as being of low quality and possibly inadequate to fully protect the system from abnormal power events.

For a modest added cost, consider a "pure sine wave" UPS. These units can be relied on to supply very clean power, eliminating potential problems, and are recommended for applications demanding high reliability.

SECTION 3.2 WINDOWS LOGIN

Although the system can be operated without a password, it is recommended to use one to secure your system from unwanted use or intrusion. Use of a password is required if you intend to use the system's support for Skype TX™ a/v sources. Thus, a password is required by default.

Important note: The default password for new (or 'newly restored') systems is simply "newtek", entered without the quotation marks, of course.

SECTION 3.3 LICENSE AND REGISTRATION

On first launch, your system will present an *End User License Agreement* dialog. After you accept this, one or more *Registration* dialogs are presented. If requested, enter the unique *Serial Number* for your system.

Hint: If the Serial number doesn't appear automatically and you can't find it on your unit, contact NewTek's Support center – see <https://www.newtek.com/support/get-support/>

You can register and obtain your registration code directly on the local system, or on another system connected to the Internet. If the system is connected to the Internet, the registration process guides you through the necessary steps.

Hint: Information on connecting to a network can be found in Section 3.13.

After registering, enter the resulting *registration code* into the field provided in the *Registration dialog*.

Hint: It's a good idea to record the login name and password you choose when creating your website profile and keep them in a safe place. Jot down the registration code too; it could come in handy if you ever need to restore the system's software to as-shipped state when you don't have access to the Internet.

SECTION 3.4 UPDATES

This would be a good time to check for recent free software updates for your system – visit <https://www.newtek.com/downloads/> to do so.

SECTION 3.5 ENHANCED SUPPORT (PROTEK)

NewTek's optional ProTekSM service programs offer renewable (and transferable) coverage and enhanced support service features extending well beyond the standard warranty period. Please see <https://www.newtek.com/support/protek/> or your local authorized NewTek reseller for more details regarding ProTek plan options.

SECTION 3.6 INPUT CONNECTIONS

3.6.1 CONNECT A/V SOURCES

Note: Hardware details as depicted are subject to change without prior notice.

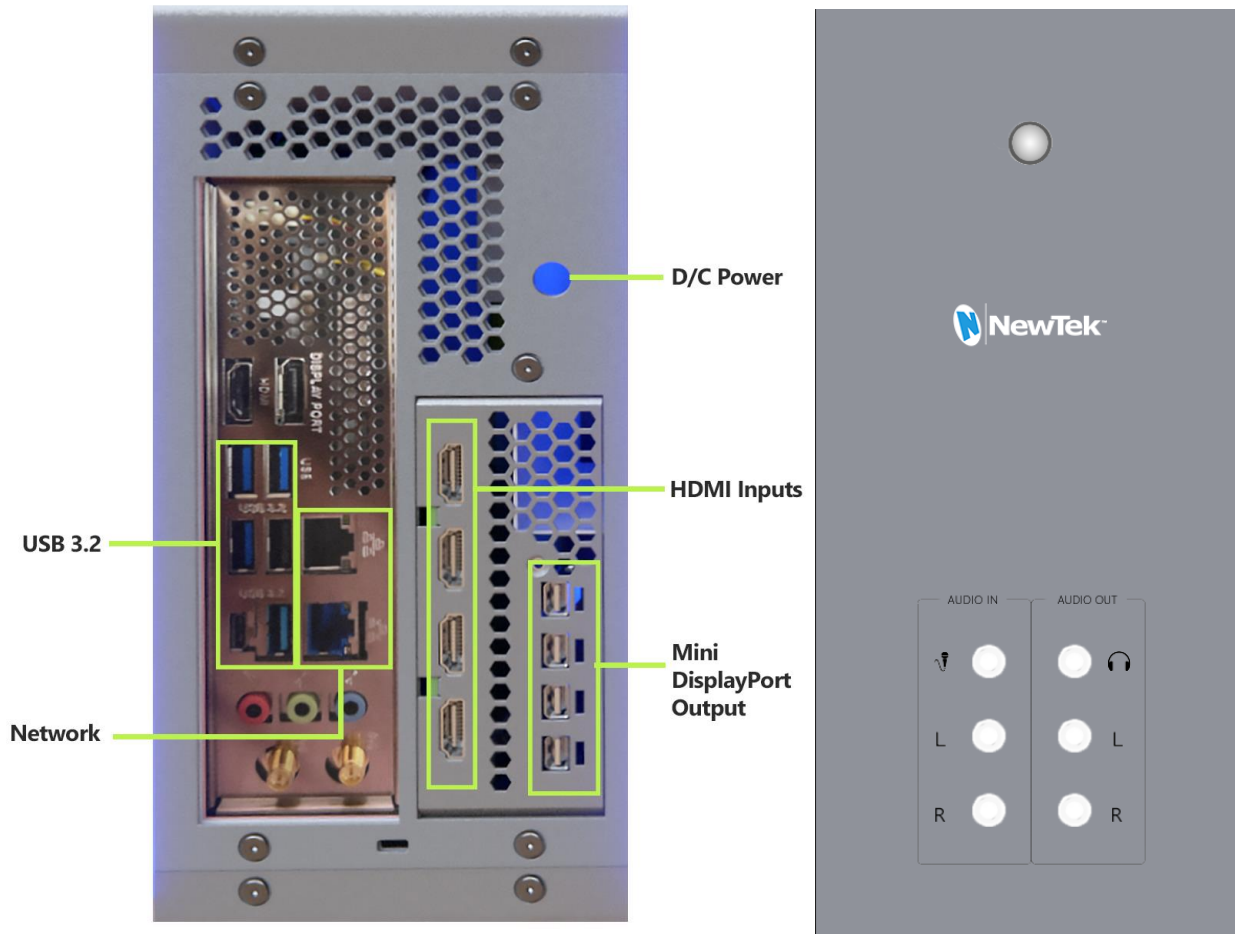


FIGURE 12

Once you have connected your mouse, keyboard, computer monitor and power cord (as described in Chapter 3 Setting Up) you are ready to connect your A/V sources.

1. Included in your TriCaster Mini X are four Mini DisplayPort Outputs to HDMI adapters. Connect the correct adapter for your monitor or display to the TC Mini.
2. Connect the HDMI cable to the adapter, and the other end of the HDMI adapter to the monitor or display.
3. Connect a network or ethernet cable to either of the gigabit ethernet network connections, connect other end of network cable to your computer network. (This connection is not necessary for basic operation but is required for initial set up)
4. Connect video sources to the HDMI inputs.

3.6.2 CONNECTING A CONTROL SURFACE

Please refer to the Control Surface chapters later in this user guide for information on connecting and configuring control surfaces.

SECTION 3.7 OUTPUT CONNECTIONS

Next, we'll discuss the audio and video output connectors along with relevant settings. We'll start with video output, not only so you can view your results but because certain items are best considered before beginning a live production session.

3.7.1 A/V OUTPUT

Your NewTek system provides multiple video and audio output connector groups, and very flexible display options. Here are some of the possibilities:

- TriCaster Mini can output a number of different sources – including as many as four different mixed compositions, NDI and streaming outputs, two streaming encoders, multiple internal recorders, and three additional local multiview monitors.
- All hardware inputs and all outputs are always presented to your local network as NDI sources, ready for varied utilization at any convenient place where your network extends.
- Send program output, secondary mixes, or custom 'monitor bridge' layouts to as many as three supplemental monitors.
- Stream two different outputs to multiple Content Delivery Networks via a suitable network connection.

Each of the four primary outputs can be configured independently. Video format settings for these outputs are provided in the Live Desktop's *Output Configuration* panel.

1. Connect downstream video production devices to the connectors in the *HDMI OUT* section as desired.
2. NDI and Internet Streaming outputs obviously require a network connection, and in the latter case, an Internet connection.
3. Connect audio outputs.
 - a. *Analog audio* – Connect external audio devices to the three quarter-inch audio jacks on the faceplate.
 - b. *Digital audio* – TC Mini X supports digital audio over HDMI outputs.

3.7.2 SUPPLEMENTAL VIDEO OUTPUTS

As mentioned back in Section 3.1, there are multiple monitor ports on the TC Mini backplane. The operating system will have set one of these as the "main display", and this is where the *Live Desktop* (the primary user interface and Multiview on the same screen) appears.

Hint: You can choose which monitor is the “main display” by choosing Administrator Mode, then clicking Display under the Settings heading.

Four mini display ports are available, three of the outputs support multiviewers (using standard display resolutions) to connect more monitors or projection devices (etc.) for various broadcast or IMAG (Image Magnification) installations.

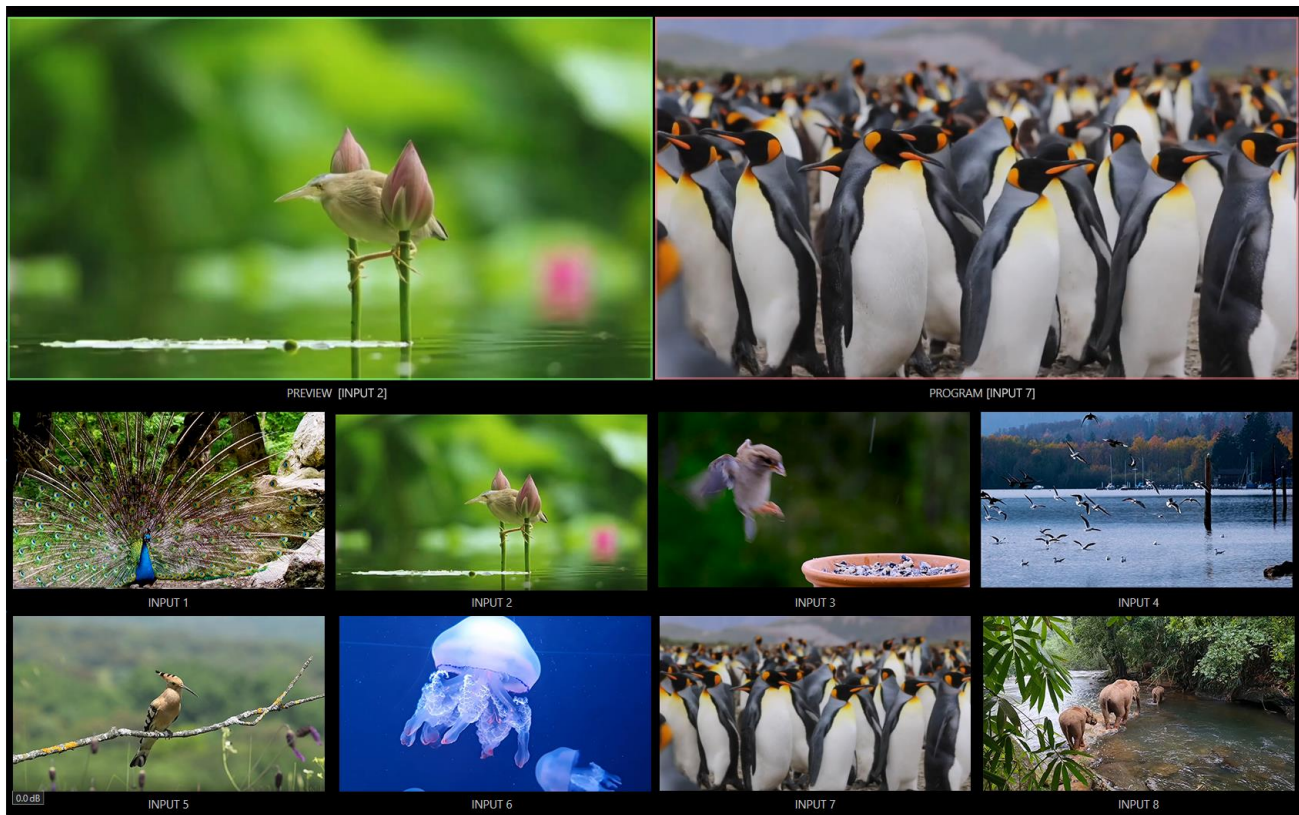


FIGURE 13

These supplemental monitors are referred to as *Multiview* outputs and can display a variety of optional layouts that can be selected and configured at any time in the *Live Desktop*.

Note: A mismatch of output format and connection type is possible at times. For example, imagery in a Graphics player may not match the current output format or the selected connection type. Display mismatches can often be handled, but in rare cases the source may simply not be shown.

SECTION 3.8 STARTING A SESSION

If your system is not already running, power it up now to reveal the *Launch Screen*. (If it is already running, but is in the *Live Desktop*, you will need to exit, then click the large triangular *Back* button at left on the *Session Page* to return to the *Home Page*.)

The *Launch Screen* consists of two similar pages labeled *Home* and *Session*. Among other things, the *Home Page* is where you choose basic session settings (see Section 5.1 for a discussion of sessions).

THE HOME PAGE

When no previously created sessions exist, the panel dominating the *Home Page* defaults to *New*, inviting you to create a new session (Figure 14).

A link containing the text “*Enter Session Name*” is shown at the top of the right-hand pane when the *New Session* link is selected at left. Click to type in this text box to replace the default name if you like (otherwise, the session will use the current date as its name).

Note: A variety of optional session configurations are provided for your live production needs. You can choose either 4K (UHD), HD (High Definition), or SD (Standard Definition) operating modes. SD options include both 4:3 and 16:9 (widescreen) image aspects. You can also select between different video Standards according to your locale, choosing NSTC or PAL.

Continue with session creation by designating the video *Standard* used in your locale. For the moment, let's click the HD button in the *Preset* list, which establishes *1080/59.94p*, then click the *Start Session* link below. By default, new sessions are created on the D: (Media) drive (see Section 5.2.1 for a discussion of session *Volume* options).

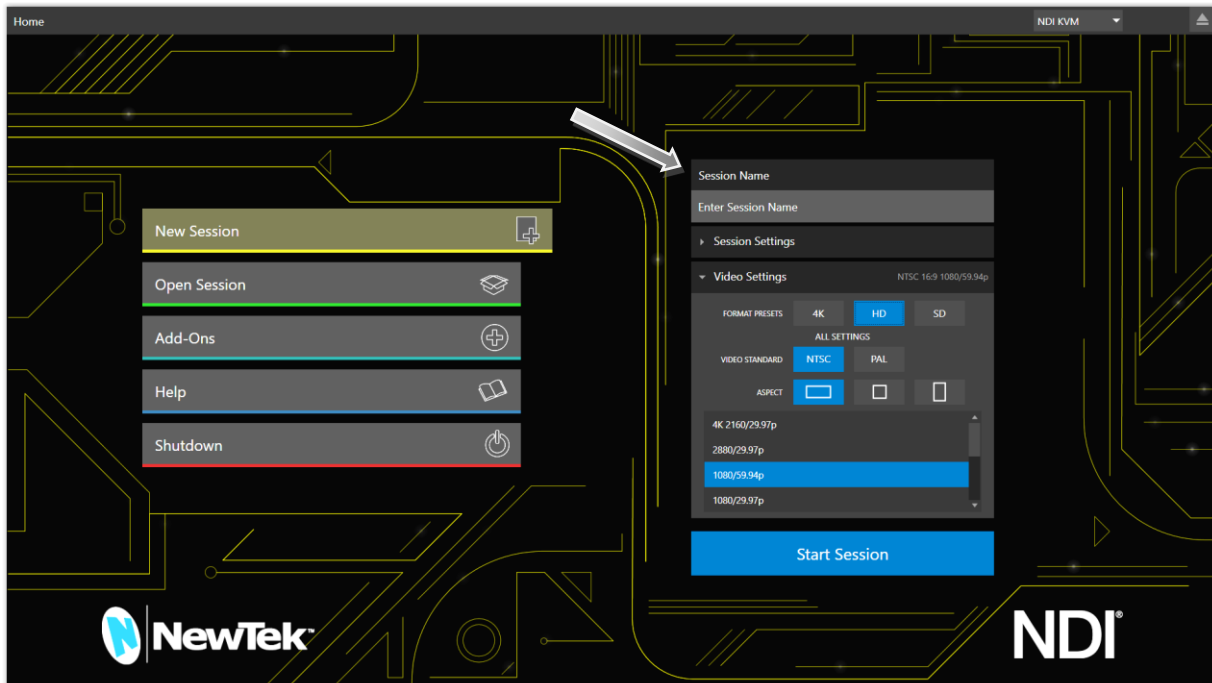


FIGURE 14

THE SESSION PAGE

Clicking *Start Session* opens the *Session Page*, providing access to a number of other possibilities. For now, as our intention is to configure our connections in the *Live Desktop*, please click the *Live* link at left.

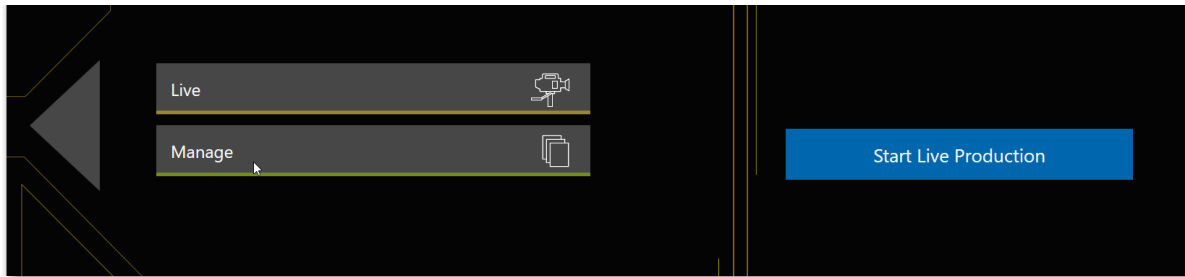


FIGURE 15

Simply click the link labeled *Start Live Production* to launch the *Live Desktop*, which is where you will spend all your time during live production.

THE LIVE DESKTOP

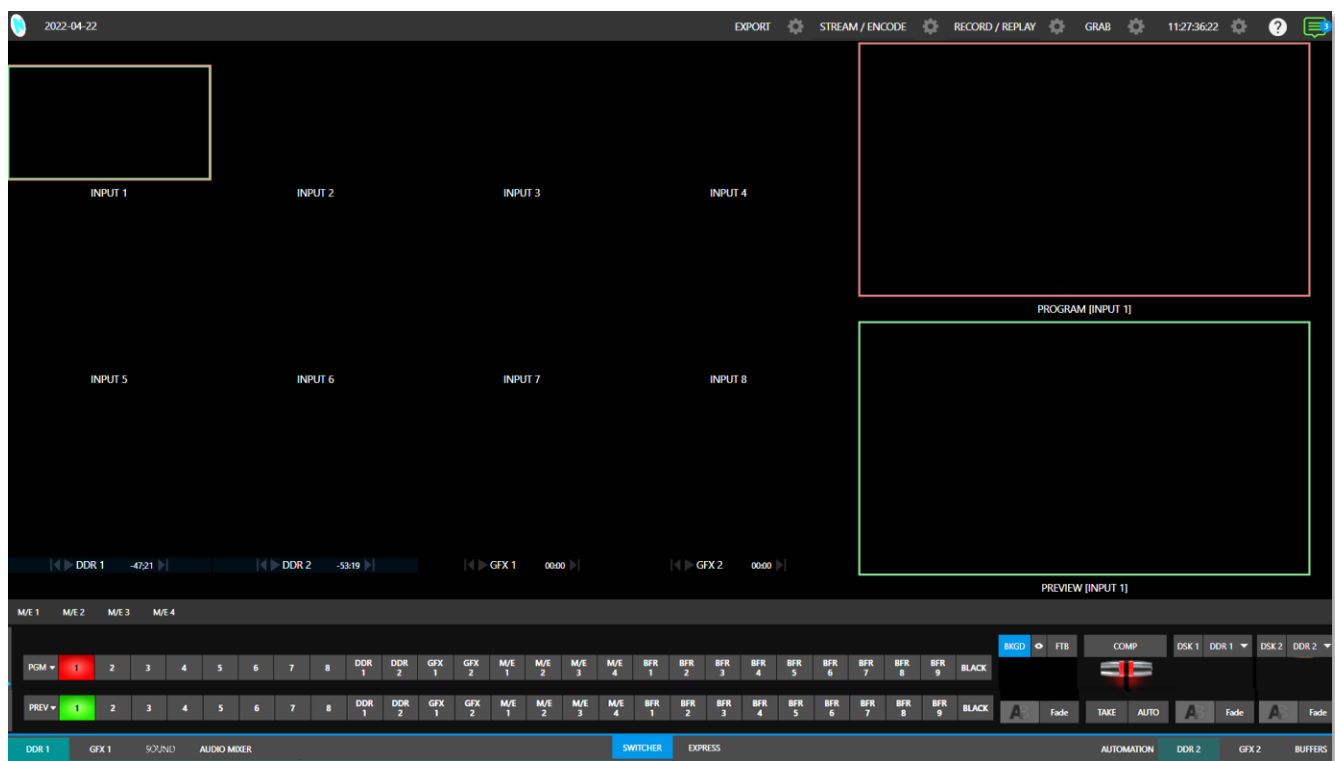


FIGURE 16

Initially, as you have yet to configure input devices or add content, the *Live Desktop* will look a bit barren. Take a quick look around, but then let's continue to configure your devices. (We had a brief glimpse at the *Live Desktop* back in 2.1.3, but we'll examine it further in Chapter 4, Web Features, coming up soon.)

3.9.1 PRIMARY OUTPUTS

Complete video monitoring is provided right on the *Live Desktop* in a multiview occupying the upper section of the screen. This lets you operate without even connecting downstream video monitors or devices. (For facilities using NDI-enabled broadcast devices and systems, there would be no necessity to bother with *any* traditional connections.)

Often, though, you will want to connect external monitors or other downstream devices to the *HDMI OUT* connectors of the unit. (See Section 3.7). In either case, you will be faced with decisions about the output formats, and audio and video sources provided to downstream systems.

Hint: There can be many additional outputs apart from these primary ones and the Stream outputs configured in the same pane. We'll discuss supplemental outputs in Section 3.9.2.

Let's take a closer look at output configuration options. Move your mouse pointer to the large *Program Output* monitor on the *Live Desktop*.

1. Click the *Configure* button (gear) that appears at lower right over the viewport to open the *Output Configuration* panel (Figure 17).

This panel hosts tabbed panes, and the first one is labeled *Output*. This is where the signals sent to each of the video output busses designated *MIX 1-4* are configured. There are various decisions to make for each output:

- What video source will you assign to the output?
- What audio source will accompany it on digital connections that carry both signals?
- Which video format will be transmitted?

And so on ...

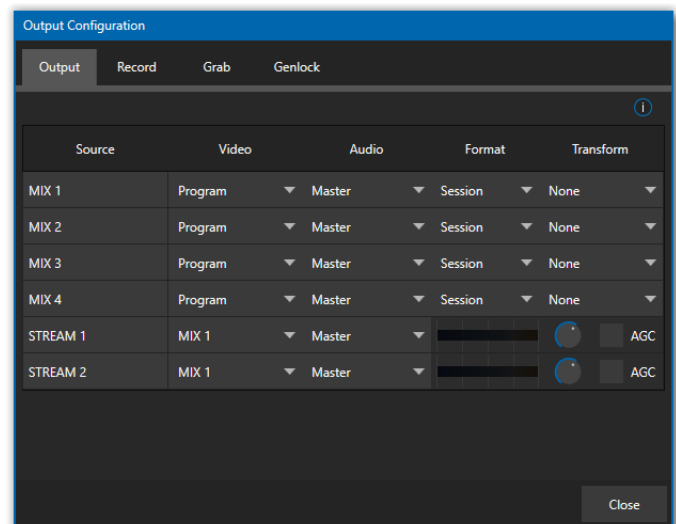


FIGURE 17

Hint: Format and color adjustments for devices connected to the various 'monitor ports' on the backplane are located in the Display Settings panel, accessible from the Workspaces menu in the main Dashboard.

OUTPUT CHANNEL OPTIONS

Let's review video *Source* selection first. A drop-down menu by that name permits you to select sources for the primary video outputs (Figure 18).

The primary output connectors are identified by number in the HDMI OUT group on the system backplane; these sources also appear on your network as NDI sources.

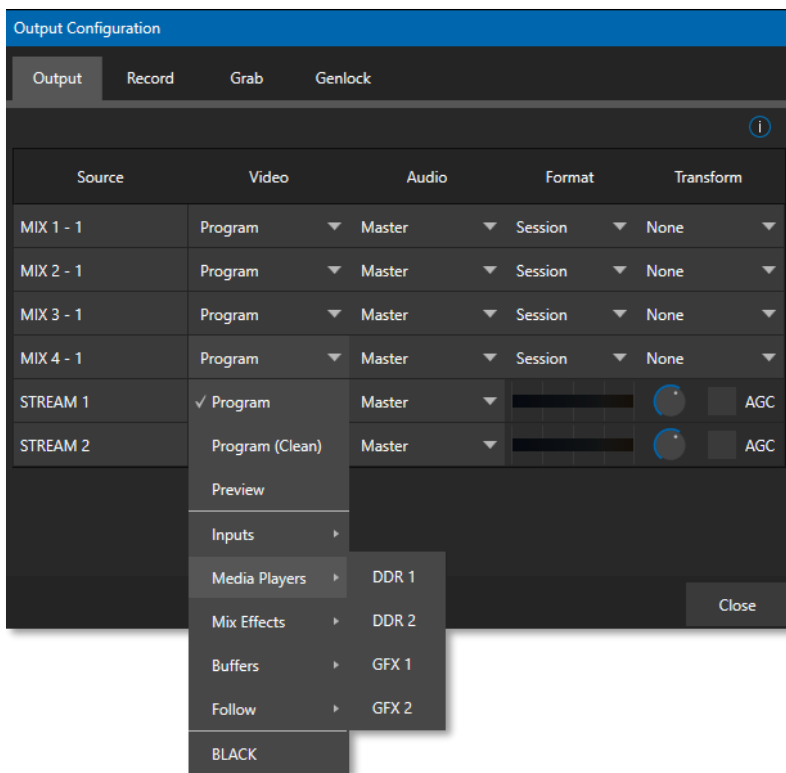


FIGURE 18

Hint: NDI sources are listed by supporting systems as machine_name (channel_name). So, if your system is named MyMini, the 2nd output appears as “MyMini(MIX 2)”.

Optional *Source* selections for *MIX* output channels include any of the following:

- *Program* output
 - A clean *Program* feed
- The ‘look ahead’ *Preview*
- The feed from any input
- Graphic or video output from a *Media Player*
- *Mix Effects*
 - Direct output from any *M/E*
 - Clean output from any *M/E*
- Output from any of the *Buffers*
- *Follow* selected *M/E Program* or *Preview*, or a *Switcher* color group
- *Black*

Audio menu options include any of the *Master* or the sound from a selected audio input. The *Format* menu lets you independently configure the format for each output, while *Transform* permits rotation and selection of segments of the output image.

Note: See Section 8.2 for detailed information on all output options.

3.9.2 SUPPLEMENTAL OUTPUTS

The mixed outputs mentioned above are just the tip of the iceberg. Direct support for internet streaming, multiviews, and recorders drastically reduce the need for ‘outputs’ as traditionally defined in this space. But this is just the beginning: Extensive NDI (Network Device Interface) options provide NewTek systems with more powerful and plentiful output capabilities than almost any other video mixer on the market.

FULLTIME NDI OUTPUTS

Among NDI sources provided (automatically, without any configuration steps required) to outboard systems over the network are the following:

- **Local HDMI sources** – all eight HDMI inputs, if in use.
- **Mix 1-4** –all mixed *Switcher* outputs.
- **Multiviews** – with the NDI KVM feature enabled each monitor screen is available over NDI, with or without remote mouse and keyboard connection capabilities.
- **Teleprompter** – The *Automation* tab at right in the lower third of the *Live Desktop* provides a great teleprompter feature over NDI at no additional expense.

SECTION 3.10 CONFIGURE VIDEO INPUTS

Source selection settings for video inputs can be accessed in the *Live Desktop*’s monitoring section, so let’s spend a few moments there before continuing.

The individual monitor viewports on the *Live Desktop* can be flexibly assigned to different *Switcher* sources and outputs. Complete monitor layouts can be stored and recalled using tools located in the *Workspace* menu, located in the *Dashboard* across the top of the screen.

We’ll look into this in depth in the *Reference Section* of this manual, but for now let’s configure the video sources you connected earlier (Section 3.6.1).

1. Click *Workspace* in the *Dashboard* (at the top of the *Live Desktop*). Monitoring *layout presets* are listed at the top of the menu as *Interface A* through *D*. Select the layout labeled *A* under *Multiview 1* (if it isn’t already check-marked).
2. By default, this default monitoring pane shows individual viewports for the various *Switcher* sources.
3. To continue to assign sources, you could select *Configure* from an input monitor’s (right-click) context menu, but let’s take a different approach.

Move your mouse over the viewport for *INPUT 1* and notice that a *Configure button* (gear icon) appears at lower right. Click this to open a tabbed settings panel (Figure 19).

4. Click the *Source* drop-down menu to see a list of sources you can select. HDMI inputs and *Black*, along with *Skype TX Caller* connections are listed first under the *Local* heading. Additional NDI sources discovered on your network are grouped below under source device headings.

For NDI sources, simply selecting completes the connection. If you choose an HDMI source, further selections may be required.

5. The *Format* menu defaults to *Auto-Detect*. Generally, this option will suffice. Otherwise, for HDMI sources you can manually select the correct *Format* (Figure 20).
6. Close the *Configuration* panel for now (click the *Close* button, or simply ‘click outside’ the panel to close it).
7. Continue to configure additional video sources in the same manner.

We’ll look at the other options and settings later, but at this point you should be able to view the video inputs you have configured on the *Live Desktop* multiview monitors.

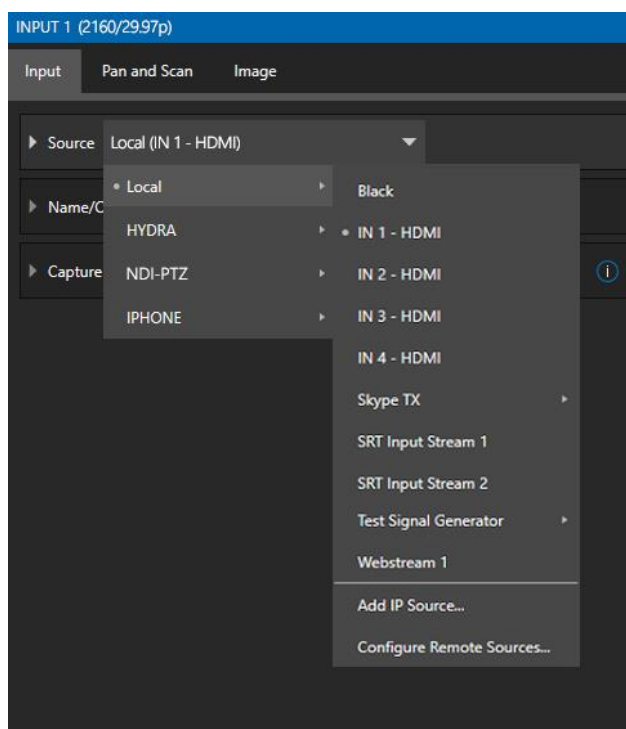


FIGURE 19

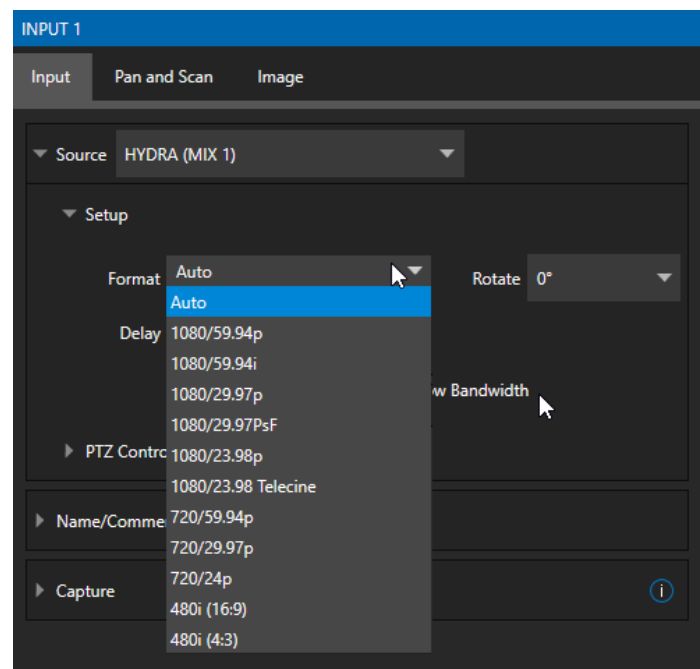


FIGURE 20

SECTION 3.11 CONFIGURE AUDIO

Click the *Audio Mixer* tab (centered in the lower third of the *Live Desktop*) to reveal audio features, including configuration controls for all internal and external audio sources and outputs, including streaming.

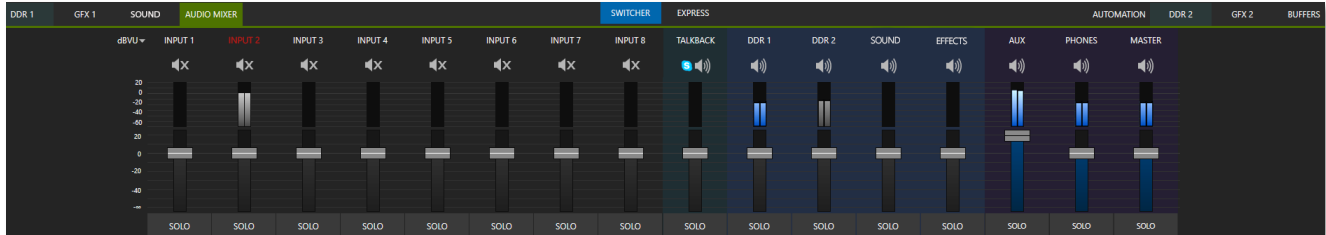


FIGURE 21

Each input and output has its own control column with *Volume* slider(s), VU meter(s), and other convenient features. An identifying label sits at the top of each control panel. Roll the mouse pointer over the label to reveal a *Configuration* button (gear) at right which, when clicked, opens the *Configuration* panel for the input as shown in Figure 22.

In this latter panel, click the *Connection* menu to display options for an input. You will see the local hardware inputs listed in the *Local* group as “IN 1”, “IN 2”, etc.

Local connections can be assigned to ‘listen’ to either an HDMI embedded audio source connected to the corresponding video input, or analog audio inputs provided on the unit.

Note: Analog audio levels conform to SMPTE RP-155. The maximum input/output level is +24 dBu and the sample rate is 48 kHz.

Beyond this, you have the option of assigning the audio delivered over the network from any NDI or other supported network audio source (such as Audinate’s Dante™ sources) available on the system.

Returning to the audio *Input Configuration* panel, note that it holds both basic and advanced audio features.

In the former category, volume sliders are provided below *VU* meters for each audio source and output. Source sliders default to their 0dB gain setting on first launch. After adding audio sources, adjust these sliders as required.

Hint: Most numeric controls in the interface can be reset to their default values by double-clicking on the slider or control knob. The default value for Gain sliders is 0dBVU.

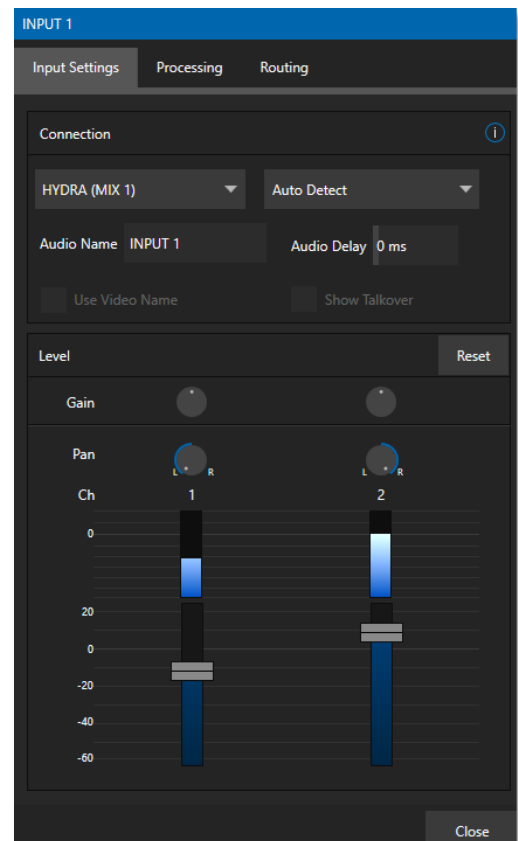


FIGURE 22

In digital audio systems, levels exceeding 'legal' values are 'clipped' (uniformly assigned the maximum value). This results in audible issues that cannot be easily corrected later. For this reason, it's customary to configure normal operating level (also referred to as the 'alignment level', and sometimes, 'nominal level') well below the clipping limit – sufficiently so that occasional excessively loud sounds (say, loud laughter or applause) can be accommodated without risk.

This range above between nominal level and the highest *possible* level is commonly referred to as 'audio headroom'. What is considered a suitable headroom allowance can vary from one locale to another, in different industry applications, and even in individual studios. NewTek systems follow well-established audio conventions, providing 20dB of headroom above nominal level (+4dBu at 0dB on the VU scale).

Hint: Confusion can sometimes arise because different calibration scales are common in various audio realms, and even for different device types and software.

For example, analog mixers commonly show levels on VU scales indexed as just described. In contrast, digital devices and editing software usually display levels in dB FS (Decibels Full Scale) with 0dBFS – the absolute maximum signal level that can be recorded – at the top. NewTek VU meters have selectable indexing, allowing you to view a traditional dB VU scale or dBFS as you please (see Section 15.9.1).

Whatever scale you choose, use *Volume* controls (and, for *Mic* connections, the *Gain* controls in the *Configuration panel*) to avoid over-modulation. The *Compressor/Limiter* feature (also located in the *Audio Configuration panel*) is another powerful tool to help you prevent clipping – see Section 15.10.

SECTION 3.12 CONFIGURE NDI GENLOCK

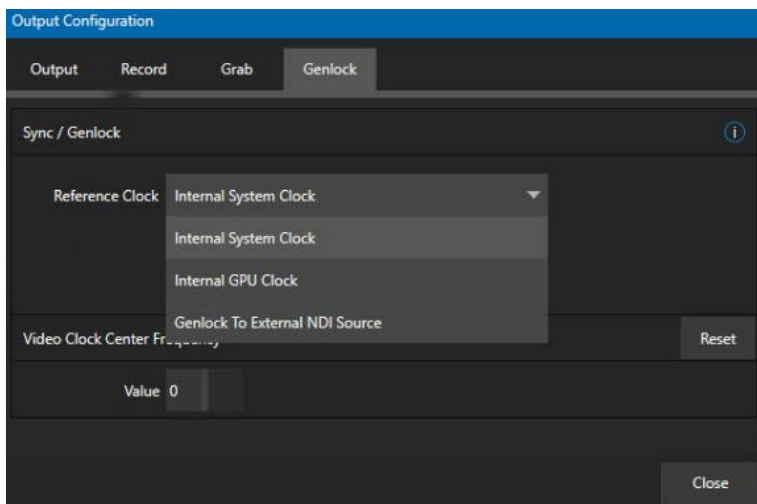


FIGURE 23

NDI Genlock synchronization allows video sync to reference a network-supplied external clock signal over NDI. This type of synchronization will be key to future 'cloud-based' (and hybrid) production environments.

The Genlock feature allows TriCaster to 'lock' its video output or NDI signal, to timing derived from an external reference signal (house sync, such as 'black burst') supplied to its genlock input connector.

This allows TriCaster output to be synchronized to other external equipment that is locked to the same reference. TriCaster comes with additional options for Synchronization, (Figure 23) the pull-down menu conveniently centralizes all sync options and allows them to be changed on the fly.

Ideally both the cameras and your NewTek system's output should be genlocked. If they are not genlocked to the same reference signal, by default the *Switcher* will automatically apply a TBC (Time Base Correction) operation. (Time base correction may drop or insert frames as necessary to maintain sync, hence is a less desirable approach.)

Genlocking is not an absolute *requirement* in most cases, but it is recommended whenever you have the capability.

Tip: "Internal Video Clock" means clocking to the SDI output (best quality when connecting a projector to an SDI output).

"Internal GPU Clock" means following the graphics card output (best quality when connecting a projector to a Multiview output).

When genlocking is active and properly configured, the timecode in the *titlebar* will display in green.

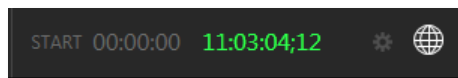


FIGURE 24

Note: For a deeper discussion of genlocking, please see Section 8.2.2.

SECTION 3.13 NETWORKING

Of course, the system provides extensive support for NDI sources and outputs across a network. Few will be surprised to find that this next generation feature requires your system to be connected to the network. 😊

TriCaster Mini X provides two - one Gigabit ports to network connections. Generally, simply connecting a suitable cable from an Ethernet port on the backplane to your local network is all that is required to connect to a *local area network* (LAN). In some settings, additional steps may be required.

You can access the system *Network and Sharing* control panel to accomplish more extensive configuration tasks – see *Settings: Configure Network* under the heading *Administrator Mode* in Section 5.2.4. If further help connecting is required, please consult your system administrator.

The next section (Chapter 4, Web Features) will introduce you to online tutorials, training materials and other assets to guide you through your learning experience.

Chapter 4 WEB FEATURES



This chapter provides information meant to guide you to other resources that will introduce the major components and functions of your NewTek system. This includes extensive online training resources, friendly and helpful online communities, web-based external control, and more. More detailed reference material on all aspects follow in Part II (Reference).

Having made the essential external connections in the previous section, you'll want to start your first practical live production session, and skim through some important fundamentals. This would be a good time for us to introduce you to a special feature of your system: Your TriCaster has its very own local webpage. TriCaster Mini X Advanced Features also includes a browser-based control system, referred to as LivePanel™.

SECTION 4.1 PASSWORD PROTECTION

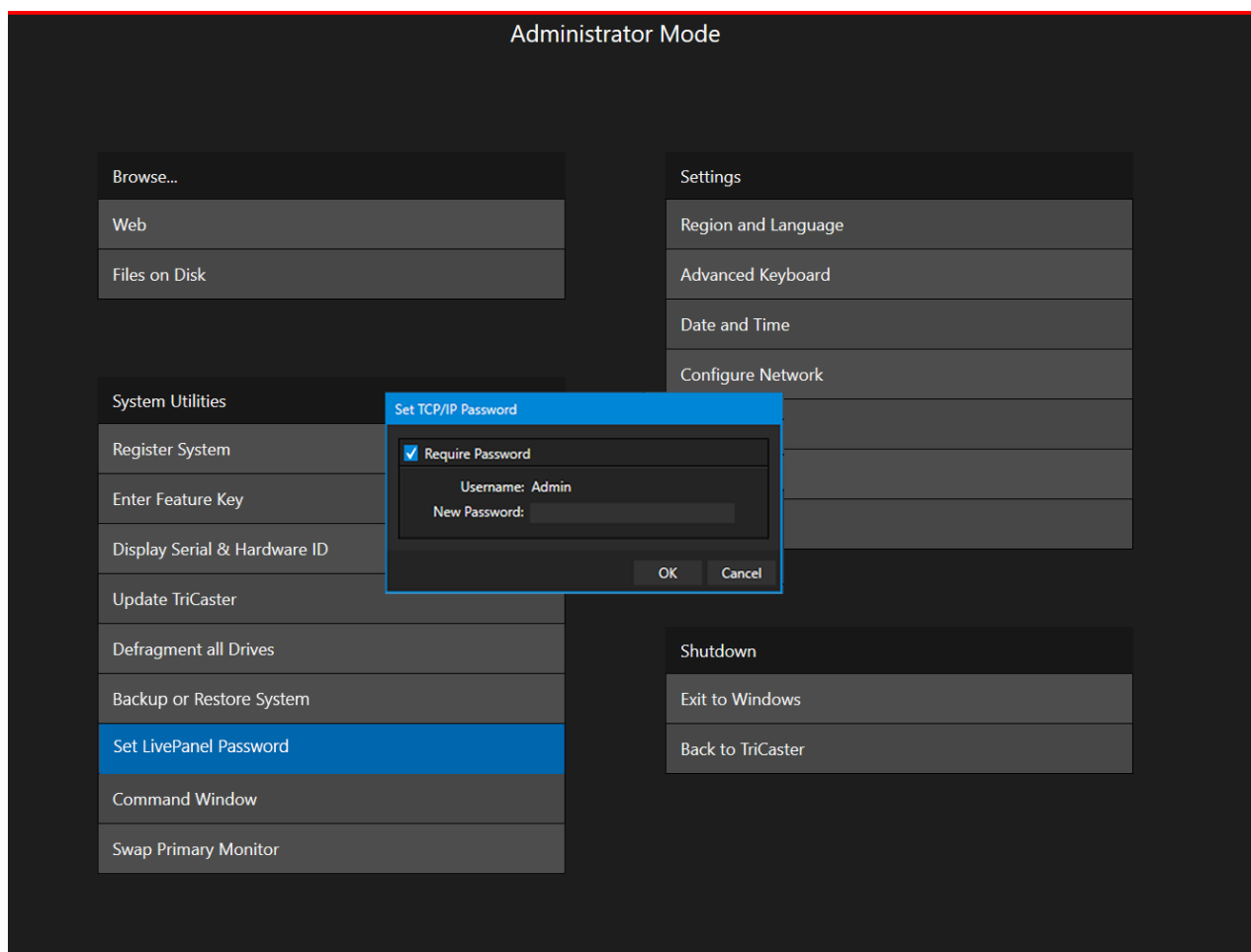


FIGURE 25

First though, let's note that (for security reasons) features that can control your production over the network are under password protection by default. Initially, the username and password are both set to "admin."

To enable or disable this security feature, or to modify the password, select *Administrator Mode* from the *Shutdown* menu on the *Home Page* main menu in the *Launch Screen*.

SECTION 4.2 VALUABLE CONTENT

You'll notice different types of content including the following linked on the *Resources* page:

- **The NewTek DataLink™ web browser extension.**
- **Documentation**
 - *User Guide* – the document you are reading
 - Macro and keyboard shortcut listings, and more
- **NewTek University** – online video classes and *Operator Certification* programs.
- **Extras** – download updates, codecs, and utilities
- **Communities** – forums, blogs, YouTube™ videos, Facebook®, and more ...
- **Support** – Customer Service and product protection programs.

The 'hamburger' menu at left in the titlebar of TriCaster's web page provides access to additional web assets, including:

- **Preview Presets** – which among other things provides a very convenient extension to TriCaster's PTZ control features.
- **Shortcut Commands** – a categorized listing of shortcut commands for use in macros or custom network control systems.

SECTION 4.3 RESOURCES

The basic webpage system provided by all TriCasters primarily features a *Resources* page, discussed shortly in the context of the more sophisticated *LivePanel* implementation. In addition to the *Resources* page, you will find pages listing system keystroke shortcuts, as well as a reference to your 3Play system.

SECTION 4.4 LIVEPANEL™

Having launched a session (refer to Section 3.8 if you need a refresher), click the *Notification* button at extreme upper-right in the *Dashboard* at the top of the *Live Desktop*.

Click this to open the *Notification Pane*, and then click the *Web Browser* button in the footer of that panel to view the webpage in a local browser.



FIGURE 26

IMPORTANT NOTE: This view is provided so you can preview the NewTek LivePanel features. It is not intended that you use the NewTek LivePanel web apps in a browser running on the local system.

To view the *LivePanel* webpages externally instead, simply connect any suitable device (be it a laptop, tablet, or desktop computer) to the same network, open this external device's web browser and navigate to the URL shown beside the *Web Browser* button.

Hint: Access to *LivePanel* does not require an external Internet connection, though some of the links provided on the *Resources* page will fail without one.

The *Resources* page is always available, and – by virtue of links to online documentation and assets – and thus is also always up to date and relevant. You can, for example, access the *Resources* page from a tablet or other mobile device to follow along with the steps outlined in an online video tutorial, or view details of a feature in the *User Guide* while operating live.

SECTION 4.5 VIDEO TRAINING

NewTek is in the video production business, and it couldn't be more appropriate for us to provide online classes to help you get the most out of your system. NewTek University <https://www.newtek.com/newtek-u/> provides an opportunity to gain the skills and knowledge you or your team need to succeed. Classes range from 3Play Operation to Live Production with TriCaster to NDI and Performance Media Networking. Also included are a sample of NewTek University online classes with several free courses.

Visit our webpage to find more or see our YouTube channel <https://www.youtube.com/user/NewTekInc> for a long list of tutorials for NewTek products designed for all users, novice, or adept. From initial product registration to live production, streaming and exporting to social media services, you'll find it illustrated and explained here.

It's very likely that you will be able to easily learn the basics of live production with NewTek systems by viewing these videos. When you need deeper knowledge, you will find Part II of this guide, the Reference section, helpful.

SECTION 4.6 OPERATOR CERTIFICATION

NewTek's official Operator Certification program (see <https://newtek.learnupon.com/store>) might interest you too, both as a way to obtain relevant skills and knowledge as well as to assure potential clients that your attainments have been validated and recognized.

PART II (REFERENCE)

A thorough examination of the various aspects of your TriCaster. Every button, menu item, feature and control are considered in this section, so you can take full advantage of your system.

Chapter 5 THE LAUNCH SCREEN



The Launch Screen is the gateway to a suite of applications as well as maintenance and management features for both sessions and the system. We'll discuss each of these in the pages that follow, beginning with a review of the concept of sessions.

The Launch Screen's *Home Page* appears whenever you launch your TriCaster Mini X. From this screen you will create and launch *sessions*, then choose what sort of operations you wish to perform within it by selecting a link on the (similar) *Session Page*.

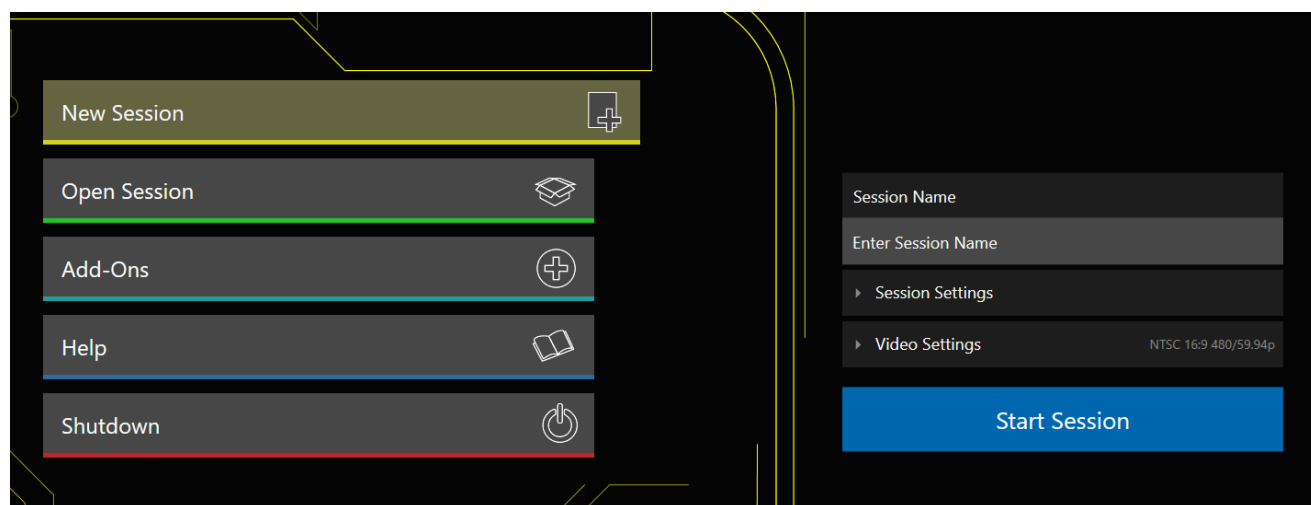


FIGURE 27

Your intention may be to begin a new live production, or to produce another episode of a live series. Perhaps you wish to prepare title pages for an upcoming event or perform system maintenance. We'll look into each of these in turn, but first let's consider a fundamental production concept, the *session*. What is a session, and why are sessions both important and valuable to you?

SECTION 5.1 INTRODUCTION TO SESSIONS

Any production involves a specific operating environment. The *session* is where TriCaster stores the details of that environment. Obviously, then, configuring session settings properly is important:

- What broadcast standard is used in your locale? Is it *PAL*, common in Europe among other places, or perhaps *NTSC*, standard throughout North American?
- Are cameras connected using hardware inputs, NDI, or a mixture of both connection types?

As you continue, you may make other adjustments relevant to your current production requirements:

- You might adjust cameras using the *Proc Amps* provided in *Input Configuration* panels.
- If your production plans include the use of greenscreen staging, you will doubtless adjust the *LiveMatte* settings for one or more cameras to provide optimal keying.
- Perhaps you will use *Workspace Display Settings* to adjust the preview monitor color characteristics.
- What are your output device connection preferences?
 - What video mixes and other internal sources do you want to route to which outputs?
 - Do you intend to configure a connection for a projection system? What external audio connections and adjustments are required?
- You might create a playlist of custom title pages in a *Media Player*, along with additional graphic elements for the *Buffers*.

During the actual live production, you may perform further fine-tuning, and add to the media content used in the presentation:

- You might grab a series of still images from *Program Output*.
- Or record video clips from the production for replay.
- And capture the network *Stream* output as a file.

The list of adjustments, activities and assets involved in a specific production goes on, but the main point to grasp is that the *session* is comprised of all of the above collectively. If you do not deliberately delete the session, all your session media and all your session settings are ready for immediate recall and re-use.

When you re-open an existing session, it's just as if you were continuing an earlier event. Thus, if you return to the same venue another day under similar conditions, simply re-open your prior session and you are virtually ready to go. (Of course, it's the professional course of wisdom to test *everything* before beginning any event.)

Naturally, you can store multiple sessions and re-open them freely. This greatly simplifies management of files associated with producing unique programs, and easily accommodates different users with their own individual requirements and preferences.

Finding your files – automatically.

Typical sessions include many media clips, along with graphics, titles, and buffers. Naturally, users must ensure make sure that all of media required is on hand for the session.

To help with this task, the system will automatically locate the necessary media, even when drive settings have been changed on the system (as, for example, when a session backup has been restored onto different drives compared to the original setup).

Having discussed sessions, let's go on to consider how sessions are created, and how you choose which one to work on. These items, along with a few other top-level functions, are found in the *Home Page*.

This primary screen is the first thing you encounter after launch. It is dominated by the main menu shown in Figure 28.

Choosing a link from the menu updates the right-hand pane of the *Home Page* to provide options and controls related to your choice.

For example, the first thing you will likely do on launching a brand new unit is create a session.

In anticipation of this, the *New* icon is pre-selected automatically whenever there are no existing sessions. This results in the relevant session options being displayed at right, as shown in Figure 29.

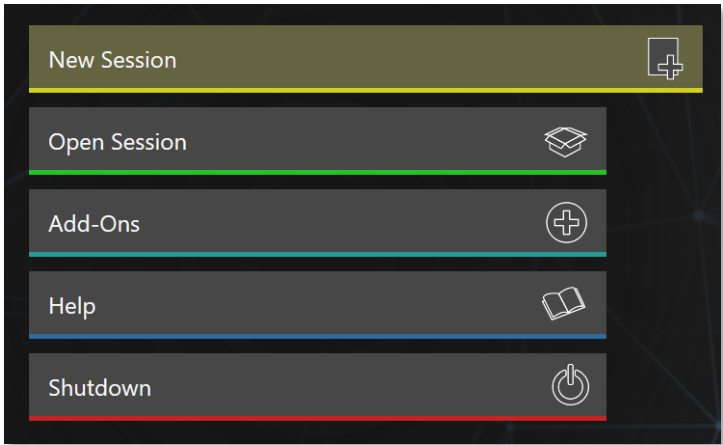


FIGURE 28

5.2.1 NEW (SESSION)

As just mentioned, clicking *New* on the link populates the pane at right with related options.

Principally, for each session you must designate (by selecting switches) your local video standard.

- Select NTSC or PAL
- Select a session format, whether by clicking one of the convenient Preset buttons, or making a manual selection at right (such as 1080/59.94i)

SESSION NAME

You will see a name field for your session listed above the options pane on the right. You can click in this field to modify the default name (which will be the current date) using the keyboard.

SESSION SETTINGS

The *Volume* drop-down menu designates a primary hard drive for the session (and its associated content) to be created and stored on.

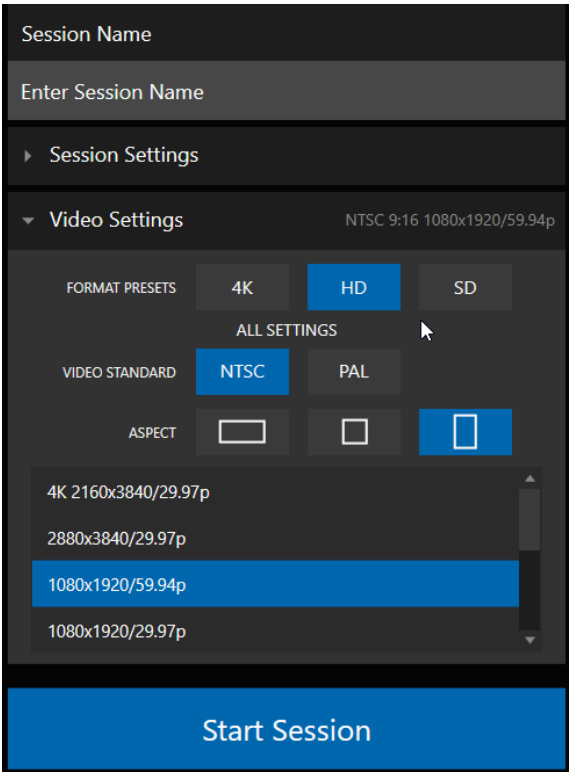


FIGURE 29

The *Template* drop-down menu lists all sessions currently available on the system. If you select an entry, the session you create subsequently retains all settings of the source session.

FORMAT AND OUTPUT CONSIDERATIONS

Remember that for live production, you can simultaneously supply both SD and HD output for *either* SD or HD sessions. The session setting does have some significant ramifications, though. For example, if your session format is SD 4:3, this is the file format captured by the *Record* function when capturing *Program* output – even when the video cameras supplying your inputs are HD; and the opposite is also true.

Note: See Section 8.2 for more information on Video Output configurations.

Having made your selections, you would normally click the *Start Session* button at lower right to launch the *Live Desktop* – but first, let's explore a few other features of the *Home Page*.

5.2.2 TITLEBAR TOOLS

EJECT



FIGURE 30

An *Eject* button at the top-right corner of the page allows you to safely disconnect external drives. (Drives can also be ejected from *the Live Desktop*.)

Note: Media configured as Read Only is not shown for the New or Open (session) features, since they cannot be used for sessions. However, Read Only media can be used for import purposes.

5.2.3 NDI KVM

KVM is an abbreviation for "keyboard, video and mouse". You will notice an NDI KVM menu just left of the Eject button. This feature leverages NDI® protocol to provide superb KVM functionality, giving you complete control of your system from anywhere on the network where you can run the Windows version of NDI Studio Monitor.

To enable this feature, go to the Launch screen's Home page, and use the titlebar *NDI KVM* menu (Figure 31) to select which operating mode you wish to apply, choosing between *Monitor Only* or *Full Control* (which passes mouse and keyboard operations to the remote system).

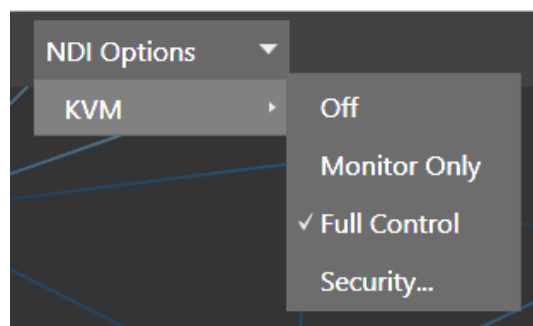


FIGURE 31

Use the *Security* option in this menu to apply *NDI Group* control to limit who on your network can view the *NDI KVM* output from the host system. To view the output from the remote system and control it, select the machine's NDI output in the *Studio Monitor* application supplied with the free NDI Tool pack, and enable the KVM button overlaid at upper-left when you move the mouse pointer over the screen.

Hint: Note that Studio Monitor's KVM toggle button can be relocated to a more convenient spot by dragging.

You can select to view or control the User Interface or another *Multiview*, or both with multiple instances of *Studio Monitor* (Figure 32) open on the receiving unit.

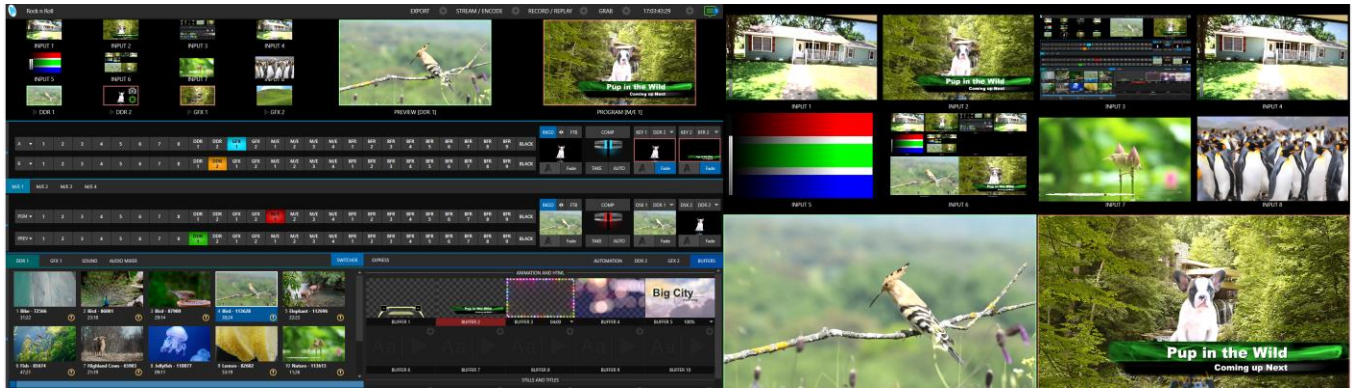


FIGURE 32

Note, too, that this feature gives you a great way to multiple your *Multiviews* around your studio or campus. You can even use these displays as video sources for another NDI-enabled system. System audio will also pass to the receiver, and you can even copy and paste text between the two systems.

With the User Interface running full screen in *Studio Monitor* on a receiving system, it's hard to remember that you're controlling a remote system. Even touch is supported, meaning you can run the User Interface output on a Microsoft Surface™ system for portable touch control over your entire live production system.

Hint: To learn about using NDI KVM with audio, see Section 15.4.2 NDI KVM Audio.

OPEN (SESSION)

If there are existing sessions on the system, the main menu at left on the *Home Page* will default to *Open* (Figure 33). Clicking *Open* causes the *Sessions List* to appear at right. This pane lists all sessions stored on currently mounted storage media.

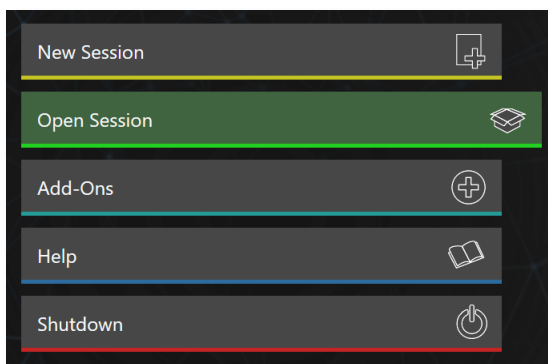


FIGURE 33

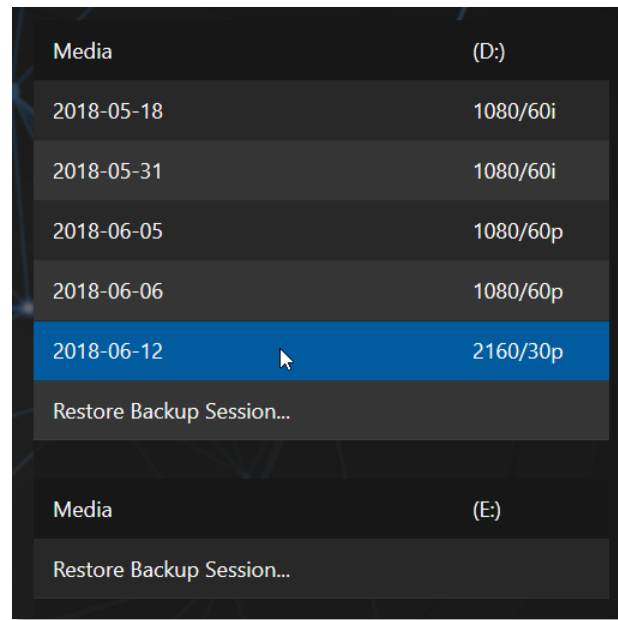


FIGURE 34

Available sessions are grouped under the names of the storage volume they are located on (Figure 34). The listing shows the *Session Name* and *Format* for each session, on each drive. Let's look at the *Sessions List* briefly before we open a session.

AUTO-LAUNCH SESSION

Normally, a few seconds after launch, TriCaster automatically reloads the last session you were in, allowing you to basically power up into the last session unattended. You can, of course, interrupt this process by selecting another session or pressing any key. Advanced users can modify this behavior, either by designating a specific session to auto-launch irrespective of the most recent manual selection, or by disabling the feature entirely.

Important Note: Modifying this feature requires changes to the Windows® registry. We strongly recommend that edits to the registry be performed only by experienced persons, as the system can be rendered inoperable if this is done improperly.

To do this, add a new value named *auto_launch_session* to:

HKEY_LOCAL_MACHINE\SOFTWARE\NewTek\XD\Admin Screen

Set its value data to the name of an existing session, and that session will auto-open on launch (instead of the 'last open' session). If the specified session does not exist, the auto-launch feature will not proceed, by

setting *auto_launch_session* to “_Disabled” (or some other suitable string) rather than a real session name you can effectively disable auto-launch.

CONTEXT MENU

Right-click a session name to open a context menu with two items listed in it: *Rename* and *Delete*. Be aware that the *Delete* function should be used thoughtfully, as it cannot be undone.

All content stored in the named session’s folders will be deleted, including imported clips, images and titles, any clips captured to the default session folders.

Note: Content that is not local to the session, but which was used in it is not deleted. For example, you may have added still images to your playlist directly from a USB thumb-drive, doing so without actually importing the files into the session. Those external files will not be deleted.

RESTORE SESSION BACKUP

In another section of the *Launch Screen* (*Session Page>Manage*) there is a feature that allows you to back up the current session to another storage volume.

The *Restore Session Backup* link at the bottom of the *Session List* (for each drive) is provided to complement this feature. To restore a session you have previously archived, click *Restore Backup Session* (Figure 34), and use the file explorer provided to select a *Session Backup* file.

Click *Open*, and a progress gauge will track the restoration process. In due course, the newly restored session will be added to the *Session List* for the specified drive.

Hint: Restoring a session that has a lot of content can take considerable time. Be judicious, then, if considering a restoration shortly before a scheduled live production.

OPENING A SESSION

Click or tap a name in the *Session List* to launch that session and display the *Sessions Screen*.

Administrator Mode – the DMZ...

It’s worth considering why this mode is provided. Your NewTek system plays a central role in your video productions. Though technically ‘a computer’, it is un-wise to expect it to serve ‘dual-duty’ as both a live production system and a general purpose computer.

Administrator Mode serves as a demilitarized zone between the live production system and host operating system.

It is important not to burden your system with software or accessories that compromise its performance or reliability. The Administrator Mode screen provides access to key settings and options, while limiting exposure to system options that could have that sort of negative impact.

5.2.4 SHUTDOWN

This selection provides *Restart*, *Shutdown*, and *Administrator Mode* links. The first two options are presumably self-explanatory, but the latter deserves a bit more discussion.



Warning: Risk of Electric Shock. Disconnect all power sources before servicing.

ADMINISTRATOR MODE

Selecting *Administrator Mode* from the *Shutdown* menu on the *Home Page* icon ring exits the live production environment, and takes you to a new screen (Figure 35) that offers various functions and settings having to do with system maintenance and management.

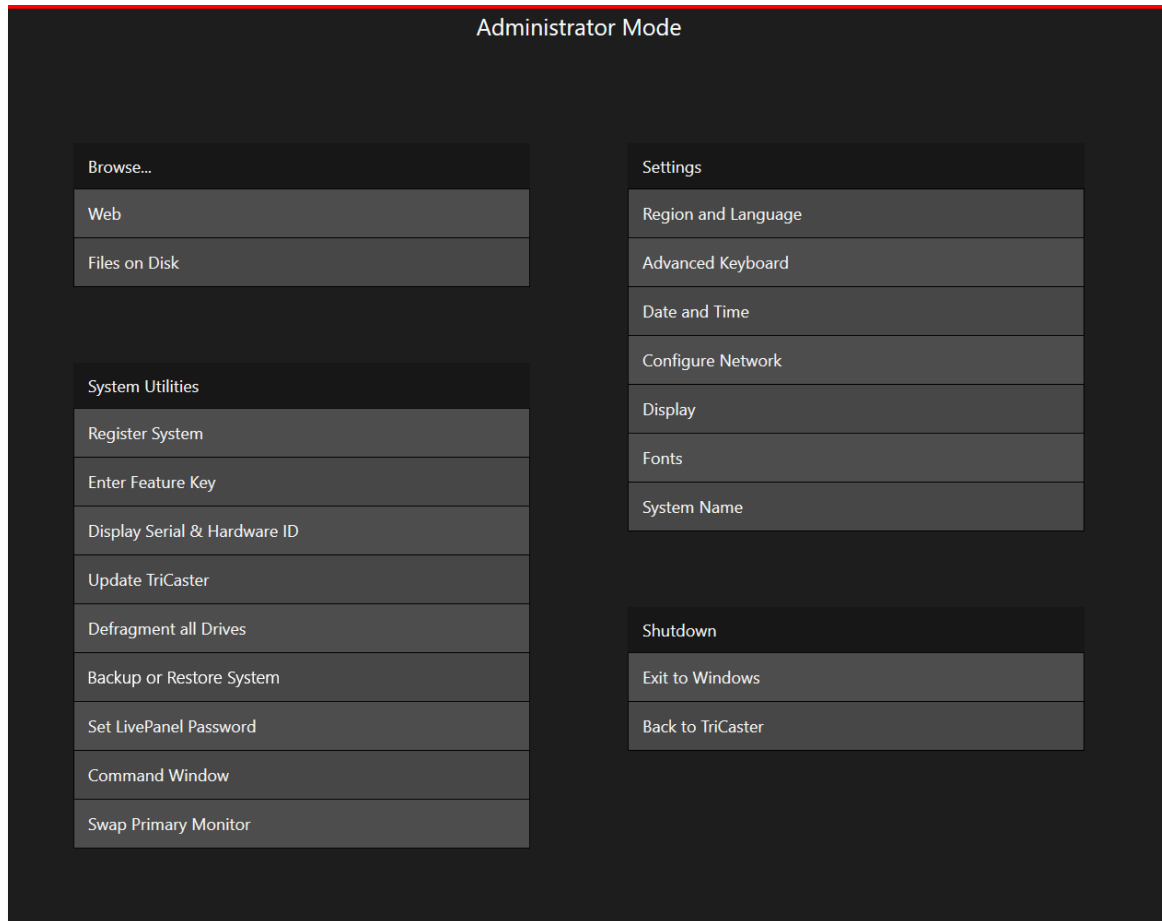


FIGURE 35

Administrator Mode options are grouped under several headings, as discussed next.

BROWSE

- *Web* – Opens the default system Internet browser.
- *Files on Disk* – Opens a system file window.

SYSTEM UTILITIES

- *Register System* – The registration process was outlined back in Section 3.2.
- *Enter Feature Key* – This link allows you to enable optional features.
- *Display Serial and Hardware ID*

- *Update* – NewTek may periodically provide software updates. Updates can enhance performance, security or even add useful new features. If the unit is connected to the Internet, clicking *Update* takes you to the *Updates and Downloads webpage* where you can locate the most recent software version.

Having done so, you can download and run the *Auto Updater* for the new version, or use the *Download Tool* option, which allows you to store the related files on another system for later transfer to your NewTek live production system.

- *Defragment all Drives*

Over the course of lengthy use, data on hard drives devoted to audio and video storage tends to become fragmented, which degrades playback performance. Defragmenting will correct this condition.

You could think of your hard drive as being like a shelf in your library. As time goes along, the shelf fills with books. Some are quite large, others smaller – just as media files may be larger or smaller.

As time goes along, you decide to remove books you have finished using from the shelf, in order to make room for new additions. You remove one book here, another there, opening gaps between the remaining books.

This makes some shelf space available by creating gaps between the remaining books. Sadly, another large new book may be too big to fit in any one of the gaps.



FIGURE 36

A foolish librarian might tear the new book into smaller sections, just big enough to fit into the open spaces on the shelf. This would obviously be unwise. When you wish to refer to the book later, you will waste a lot of time locating and assembling its sections before you can access it.

How much better it would be to slide the remaining books closer together, combining all the free space first. Unfortunately, computers are ‘foolish librarians’. They tend to want to fill in all the gaps in their storage areas, even if this requires literally shredding your ‘book’ (in reality large video files from your sessions) into tiny fragments.

Defragmenting the storage volume has the same effect as sliding the books together, resulting in one or more larger gaps. The end result is that the system doesn’t have to frantically search in many different places to assemble the video streams from your recorded session.

Hint: Defragmentation can sometimes take considerable time, so it’s not something you want to begin just before an important event.

- *Backup or Restore System* – A complete set of *Backup* and *Restore* features are available to help in the event of an unforeseen problem. We strongly encourage you to protect against any unforeseen

occurrence by using the *Backup* feature soon after registering. See Appendix G: Restore Factory Defaults.

- *Set Live Panel Password* – Configure the password for access to the web pages and network control features provided by the unit.
- *Command Window* – Open a system command line window, allowing you (among other things) to use advanced commands to manage your system and network settings.
- *Swap Primary Monitor* – The *Live Desktop* will appear on the monitor designated as primary.

SETTINGS

- *Regions and Language* – Open the system language control panel.
- *Advanced Keyboard* – Open the system keyboard controls panel.
- *Date and time* – Open the system keyboard controls panel.
- *Configure Network* – Open the system network controls panel.
- *Display* – Open the system display controls panel (useful for configuring multiple monitors, among other things).
- *Fonts* – Open the system font management console, allowing you to add that new font your client wants you to use for their title pages.
- *System Name* – It can be very useful to modify the default name by which your system is identified on the network. This is especially true when more than one unit might be on the same network. The *System Name* feature opens the System Properties console's Computer Name tab. Click the *Change* button to modify the current name.

SHUTDOWN

- *Exit to Windows* – Leave the *Administration Mode* screen and display the standard system desktop.
- *Back to (TriCaster)* – Exit *Administration Mode* and return to the *Launch Screen*.

5.2.5 HELP

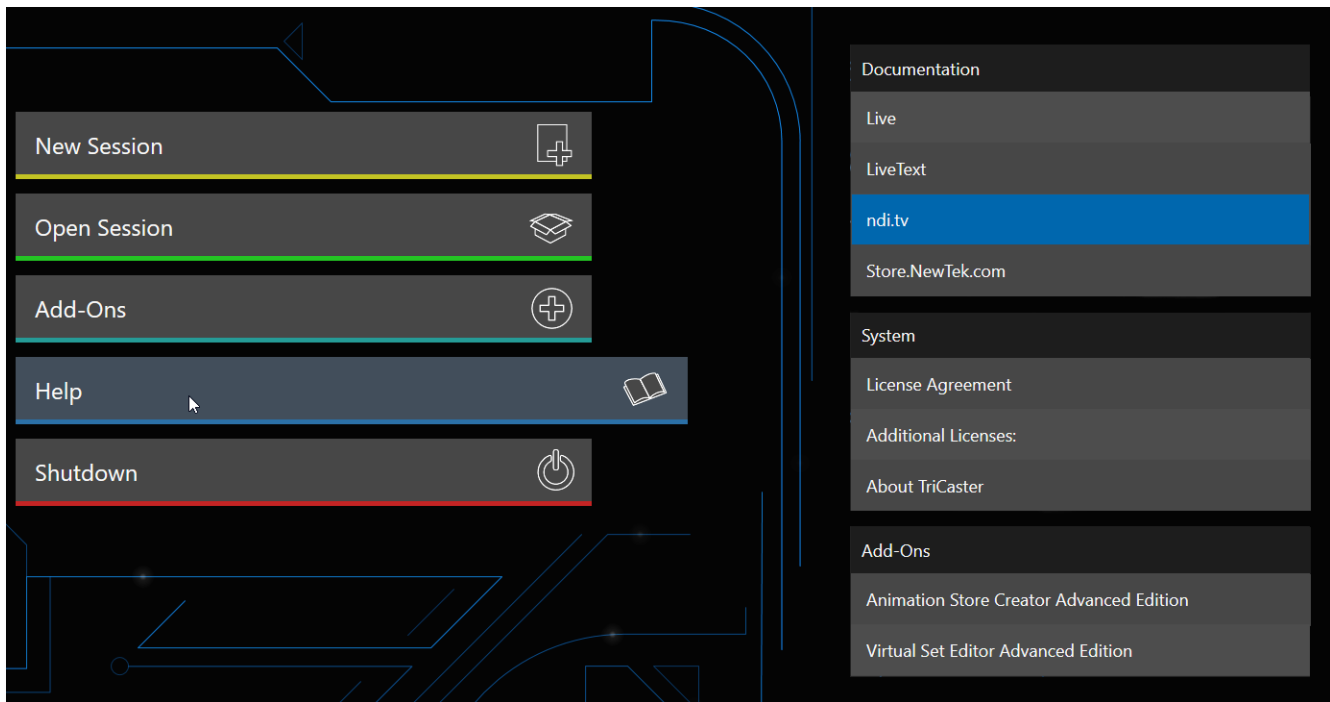


FIGURE 37

Select the *Help* icon to reveal links to open manuals and other information.

The primary User Guide is found here. Click the *License Agreement* link to review the end user license requirements, or the *About* link to list software version information, credits, and acknowledgments.

5.2.6 ADD-ONS

NewTek offers additional software tools to expand the power of your NewTek system. The icon labeled *Add-Ons* on the *Home Page* provides access to some of these tools.

When you select the *Add-Ons* icon, links are displayed at right for installed software applications, allowing you to launch them.

For example, you will see links to the included *Animation Store Creator* utility as well as a demo version of the optional *Virtual Set Editor™* program, the very useful *DataLink Configuration* applet (see also Section 5.3.3; please refer to the accompanying *Automation and Integration Guide* for full details on DataLink's many powerful source options), and much more.

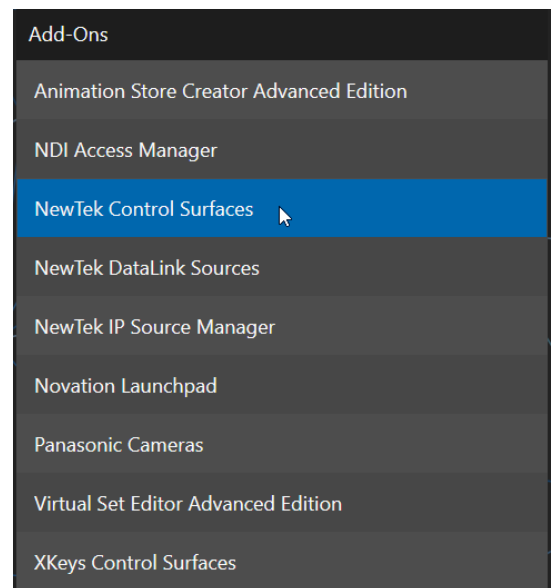


FIGURE 38



FIGURE 39

Opening a session from the *Home Page* (whether using *Open* or *New*) takes you to the *Session Page*, providing a new set of selections on the main menu.

The name of the current session is shown in the top-left corner of the screen. The current session is displayed in the form “Home/session name”. If you wish to leave the current session and return to the *Home Page*, you can click the large *Back Arrow* displayed at left, or the word “Home” in the session name label.

The main menu on the *Session Page* offers three selections. These will be discussed next.

5.3.1 LIVE



FIGURE 40

The icon labeled *Live* represents the *Live Desktop*, your live production center. Selecting it (by clicking or tapping the link), presents a *Start Live Production* link button in the *Session Page*’s right-hand pane; click this link to launch the *Live Desktop*.

5.3.2 GRAPHICS



FIGURE 41

TriCaster includes an integrated variant of LiveText™, NewTek’s standalone titling and CG application. This version cannot be used while live, but gives you the ability to author title pages with text entries and images that *can* be updated while live. These pages are displayed from Media Players or Buffers during live production.

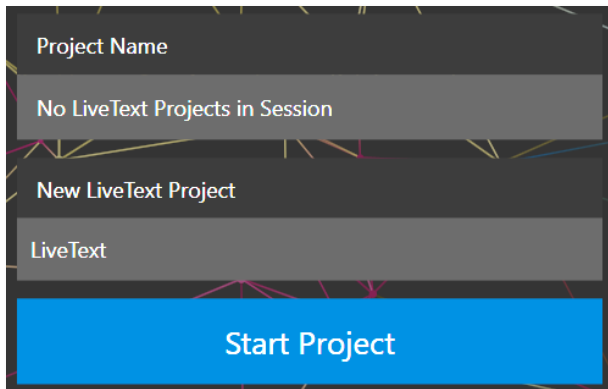


FIGURE 42

When you first enter a new session, no LiveText projects are listed in the area under *Project Name*. After you create one or more projects, the name(s) will appear here and, as you would expect, you can re-open the project by clicking its name.

Note: New LiveText projects, by default, are stored inside a folder named for the session, for example, you might find a project at D:\LIVETEXT PROJECTS\sessionname\projectname.cg.

LiveText's internal File menu and Save As functions default to the same location. However, it is possible to use File>Save As to store a project in another location outside the session structure.

Be aware of a new LiveText project, click in the default project name field (just above the Start New Project link) and modify it, then click Start New Project. When you finish working in LiveText, click the (x) button in its upper-right corner (or select Exit in the File menu) to return to the Session Page.

Note: The integrated version of LiveText cannot be accessed during live production operation. It is provided as a title page and graphics authoring tool only, not a CG server. A number of excellent third-party offerings, as well as the standalone version of LiveText (available from NewTek as a separate purchase) are listed and described in the accompanying Automation and Integration Guide.

5.3.3 MANAGE



FIGURE 43

Various internal production modules provide file bins to permit you to manually administer files related to their projects, playlists, and so on. The *Manage* icon in the *Session Page* provides an alternative approach to file management.

At times you may find it useful to be able to quickly access the various files associated with specific sessions. Selecting *Manage* refreshes the options pane on the right-hand side of the *Session Page* with a number of convenient items

BROWSE

Under the label *Browse*, you will see direct links to the *Clips*, *Still*, and other content associated with the current session. Clicking one of these links (or selecting it with the up/down arrow keys and pressing Enter) opens a system file explorer.

You can use the familiar features and shortcut keystrokes in these windows (Cut, Copy, Paste, Rename, Delete and so-on) to manage the session content.

Hint: Of course, you can open several of these folders simultaneously, and navigate these file windows to other locations as well. For example, you might copy the title page (.cxml) files from the Titles folder of one session into the Titles folder of a different one before deleting the first session.

Several other important features are contained in the *Manage* group, discussed next.

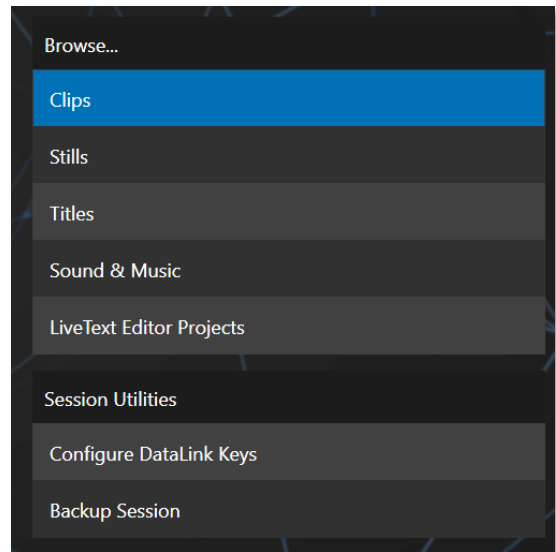


FIGURE 44

CONFIGURE DATA LINK KEYS

DataLink™ is an astonishingly useful NewTek exclusive, allowing the handling and updating of text and image data from both internal and external source for use in title pages in real-time, as well as other purposes. Briefly, *DataLink* tracks variables (called *DataLink keys*) and their values and updates these values on-the-fly when used as entries in title pages.

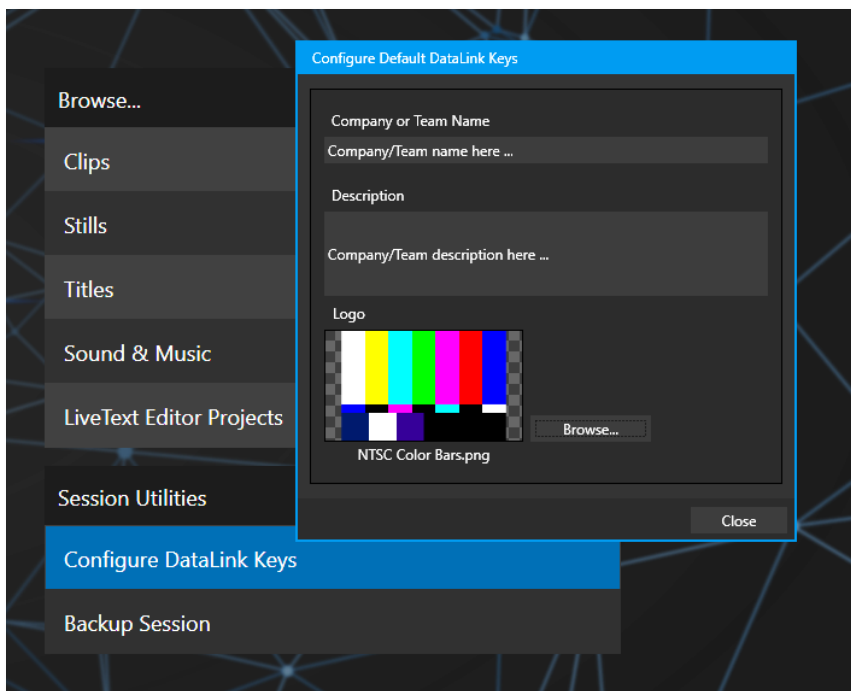


FIGURE 45

Hint: DataLink keys can serve in other ways, too, for example to automatically insert information such as time or scores into labels and comments of social media uploads, or as instant replay clip metadata.

Since *DataLink* falls into the general realm of automation and has access to external sources, complete coverage of its features and capabilities is found in the companion *Automation and Integration Guide* included with this product; but we'll mention a few highlights as we go along as well.

Hint: If not today, we urge you to make definite plans to read the DataLink Chapter in the Automation and Integration Guide. Without exaggeration, DataLink is a game-changer. It can save you a great deal of time, prevent embarrassing and potentially costly mistakes, and lift your productions to new heights.

The *Manage* group link labeled *Configure Datalink Keys* opens a dialog that allows you to pre-assign values to three special *DataLink* keys.

These *session keys* serve as the default first and second line text inserts and image used by many of the supplied template title pages. (By taking a moment to populate these keys with, for example, a company name, motto, and logo, you will find much of the included graphic content ready to use for a given production without ever having to type another line.)

BACKUP SESSION

Clicking the *Backup Session* button opens a system file explorer that you can use to assign a storage location for the backup files. A progress gauge is displayed during processing, and if necessary, you may *Cancel* the operation. Of course, the session to be backed up is the current session (to back up a different session, return to the *Home Page* and *Open* a different session.

Note: The backup operation feature does not 'gather' media files in the backup, but it does store Media Player playlists. Provided the media files are still available, they will appear as expected when the session you backed up is restored.

Chapter 6 LIVE DESKTOP



The Live Desktop is the control center for all your live production work. It Provides control over switching, transitions and overlays, titles and graphics, audio mixing, playback of digitized content, and much more. In addition, streaming and recording features are located here, along with keying and virtual set tools.

The *Live Desktop* is launched by creating (or opening) a session in the Launch Screen's *Home Page*, then selecting the *LIVE* icon and clicking *Start Live Production*.

SECTION 6.1 DISPLAY REQUIREMENTS

The *Live Desktop* requires a *minimum* screen resolution of 1920x1080



FIGURE 46

SECTION 6.2 OVERVIEW

The *Live Desktop* provides visual feedback for operations, including monitoring, live switching, and so on. While it is seldom necessary for it to appear as heavily populated as seen in, we display it this way to illustrate that it can be logically divided into five horizontal bands, described from top to bottom in the following table.

1 - Dashboard

- Quick access to interface options and important tools, including *Publish* and *Macros* along with *Record*, *Stream*, *Workspace* and *Timecode* options.

2 - Monitors

- User configurable layouts – monitor live inputs along with internal sources (such as *DDR*s, *M/E*s, and *Buffers*) plus *Look Ahead Preview* and *Program* output
- *Waveform* and *Vectorscope* monitors

3 - M/Es & Matrix Router

- *Effect* mode – control up to four primary video layers plus 4 overlay channels
- *Mix* mode – secondary switcher controls plus 4 overlay channels
- Each *M/E* includes dedicated an extensive complement of *Keyers*, transition generators, scaler/positioners, and more

4 - Switching

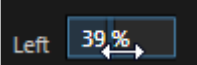
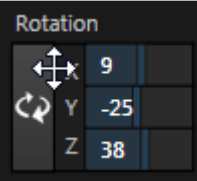
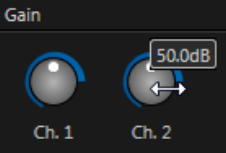
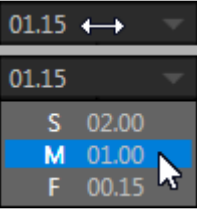
- *Switcher* rows – *Program* and *Preview* rows
- *Transition* controls – *T-bar* and *Delegate* buttons
- *DSK* video layer configuration and controls

5 - Tabbed Control Modules

- Internal *Media Players* and *Buffers*
- *PTZ* (robotic) camera controls
- *Audio* (mixer) – input configuration, level control, *EQ* and *Compressor/Limiters*

SECTION 6.3 CONTROL TYPES

You will find various control types used repeatedly in the *Live Desktop*. Some examples are shown in the table below.

Control Type	Type	Examples	Usage
	Numeric Slider	Gain, Pan	Drag in line with the control's orientation to raise or lower the current value.
	Rotary slider	Position, Size, Rotate	Drag up-down to adjust one value, left-right to adjust the other value, diagonally to adjust two different values simultaneously. Hold down Alt while dragging vertically to modify the third value (Z) when provided.
	Rotary Knob (single-value)	Audio Gain	Drag left-right to adjust the current value.
	Combo-Slider	Transition, Zoom Duration	• Drag left/right to raise or lower the current value. • Click the digits to type in a new value. • Or click the triangle to open a drop-down menu.

To make very fine adjustments to slider values, hold down the Ctrl key while dragging the mouse (this increases accuracy by 10x). Hold down the *Shift* key and double-click most controls to reset them to their default values.

SECTION 6.4 CUSTOMIZING THE DESKTOP

The *Live Desktop* layout can be adapted to many different scenarios and provides a number of customization options that can be of benefit in your workflow. In this section, we'll touch on several of these.

6.4.1 RENAMING INPUTS

As charming as descriptive names like *Camera 1*, *DDR 2* and *M/E 5* are, you may wish to change these labels to something more appropriate for your local needs.

To do so, you will need to open the *Input Configuration* pane for the *Switcher* source in question. There are several ways to open this pane. You can use any of the methods described below to do so.

- Right-click a viewport in the *Live Desktop* or external *Multiview*, or a Switcher button, and select *Configure* from the context menu.
- Roll the mouse pointer over the viewport for the source and click the *configuration* (gear) icon shown at lower right.
- ‘Two finger tap’ the viewport.

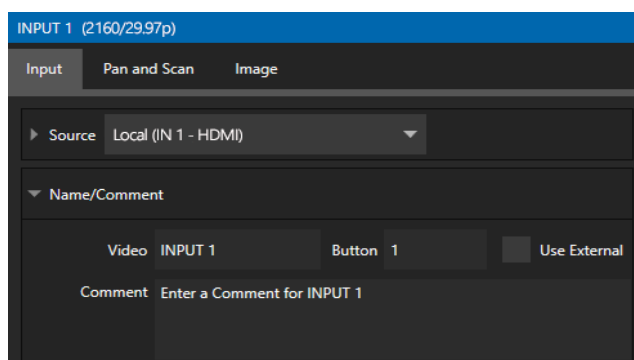


FIGURE 47

Expand the *Name/Comment* group in the *Input* tab of the *Input Configuration* pane, by clicking the twirl-down triangle at left. This group contains several editable text fields, including *Video* (the label shown under monitors), a shorter Switcher *Button* label, and the longer *Comment* field.

Hint: The Comment entry is more than just memory aid. NewTek's DataLink feature can automatically update title pages as inputs are displayed by drawing on the information you enter here.

6.4.2 WORKSPACES

Various layouts and custom monitoring options for the *Live Desktop* and *Multiviews* are provided in the *Workspace* menu.

When one or more *Multiviews* monitor is in use, *Workspace* preset options can be combined with great flexibility, letting you see *what* you want to see *where* you want to see it.

It's easy to set up complementary displays. Each connected monitor, including the *Live Desktop* screen (listed in the *Workspaces* menu as *Multiview 1*) has four unique presets, labeled A-D. You can load a different viewport layout into each of these presets, and recall them easily using this menu, or perhaps using Macros assigned to keystrokes.

In addition, each individual viewport can be assigned to any *Switcher* input source or output using its own context menu. So, for example, even if you initially use *Load Default* to assign identical viewport layouts to preset A and preset B for a given screen, you can configure each viewport differently in the two presets.

Switching between presets, then, will allow you to view different sources. Even the overlays (such as *Safe Area* display) for each viewport are fully independent.

SCREEN DIVIDERS

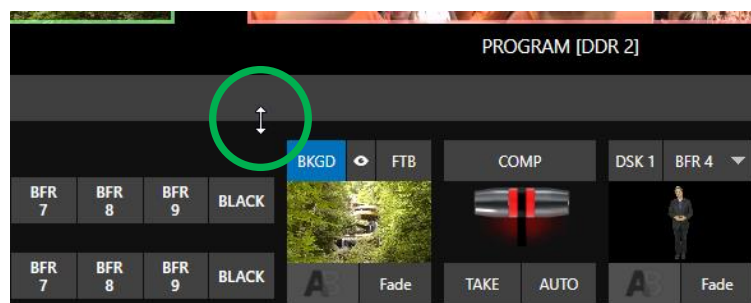


FIGURE 49

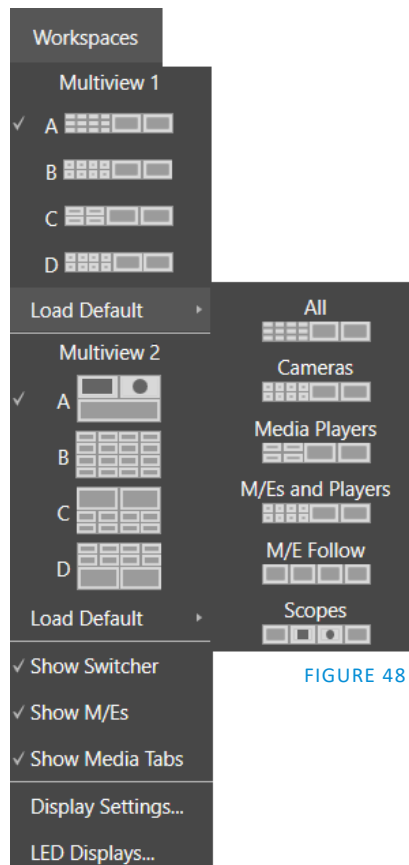


FIGURE 48

Notice too that, when the lower tabbed modules are visible, you can also drag the horizontal divider between the *Live Desktop* monitor pane and the *Switcher* up or down to modify its position.

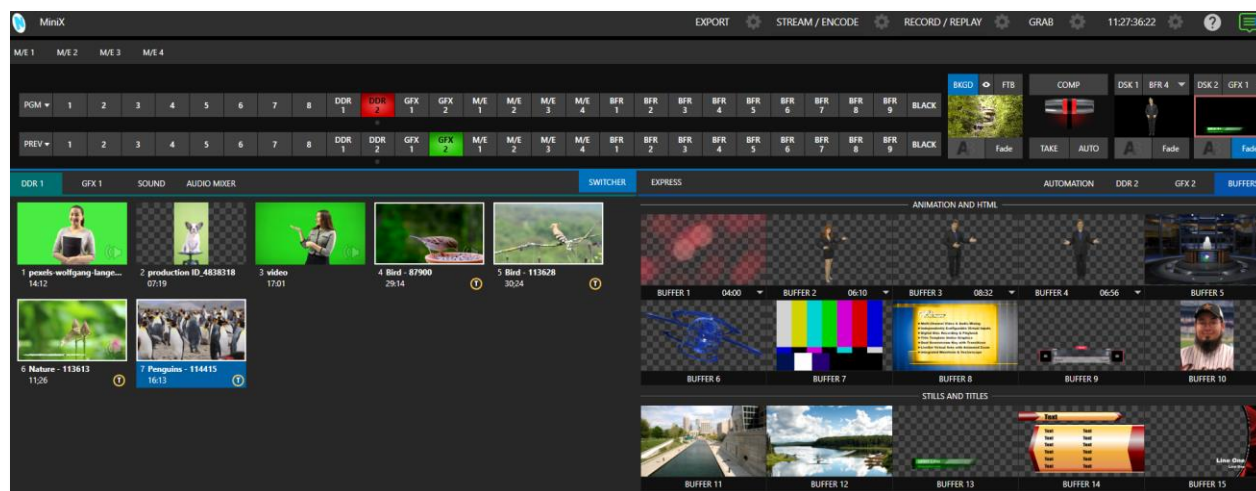


FIGURE 50

You can even move the horizontal divider all the way up to the bottom of the *Dashboard*, thus hiding the *Desktop* monitor pane entirely, providing much more room for other modules in the process.

Hint: Reset the horizontal divider to its default locations by double-clicking it. Also, note that a vertical drag bar is provided just left of the Program and Preview viewports in some Workspaces.

Neither is it necessary to display modules you don't access frequently (say, for a simpler production) at all times. For example, the M/E pane is hidden by default; click the *Mix Effects* label or an *M/E tab* above the *Switcher* to toggle display of the *M/E* pane – or press the keyboard shortcut “m”.

6.4.3 EXPRESS SWITCHER MODE

TriCaster Mini's alternative *Switcher* interface provides new opportunities to customize both the interface and workflow. The *Switcher's Express* mode is very compact, freeing valuable *Live Desktop* space. Viewports in the multiview area above are significantly larger than in the more familiar two row (Program/Preview) *Switcher* mode.

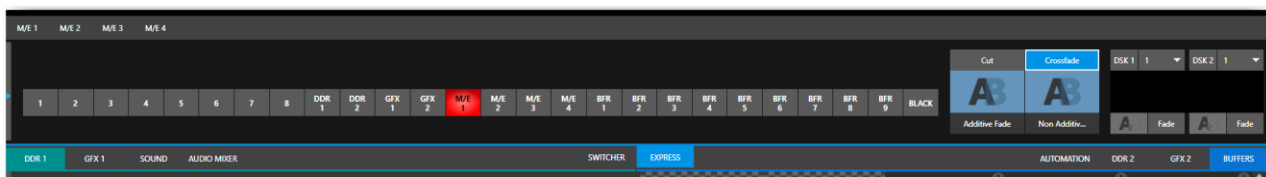


FIGURE 51

Too, *Express* mode (Figure 52) provides a simple 'single-click' workflow, ideal for less complex productions, when a control panel is in use, or in environments involving student or volunteer operators who may struggle with traditional switcher workflows.

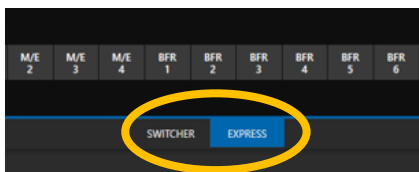


FIGURE 52

You can easily switch between the standard *Switcher* mode and *Express* operating modes by clicking their eponymous tabs at right in the horizontal bar just above the *DSK* controls.

For more information on these two modes of operation, see Chapter 9, *Switcher, Transitions and Overlay*. Several additional workflow and interface options are hosted in the *Options* menu, which we will discuss shortly (in Section 7.2).



The Dashboard groups important production features along with configuration and display controls in one convenient place for quick access and review. As well, the Dashboard serves to provide information on the current session, status updates and storage usage, along with timecode and the Macros and Options menus.

The *Dashboard* is home to several important displays, tools, and controls. Prominently located at the very top of the *Live Desktop*, it occupies the full width of the screen.

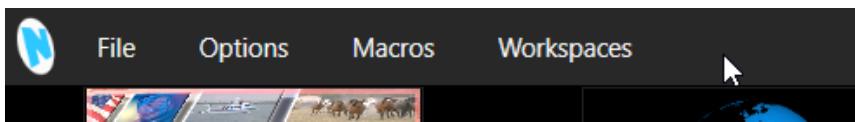


FIGURE 53

Initially, to avoid overwhelming us, the left end of the *Dashboard* simply shows the name of the current session. Moving the mouse pointer to the top of the screen reveals the set of menus shown in Figure 53.

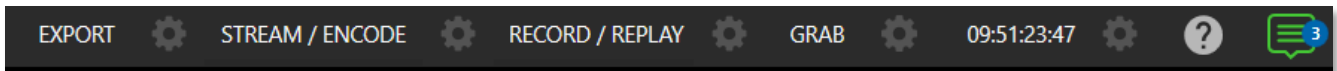


FIGURE 54

The right half of the *Dashboard* (Figure 54) contains additional features and tools that are more frequently accessed, for which reason they are displayed full-time. The various elements comprising the entire *Dashboard* are listed below (starting from the left):

1. *File menu* – see Chapter 7
2. *Options menu* - see Chapter 7
3. *Macros menu* – see Chapter 16
4. *Workspaces menu* – see Chapter 10
5. *EXPORT button* and *Configuration (gear)* – see Chapter 18
6. *STREAM/ENCODE button* and *Configuration (gear)* – see Chapter 17
7. *RECORD/REPLAY buttons* and *Configuration (gear)* – see Chapter 19
8. *GRAB button* and *Configuration (gear)* – see Chapter 19
9. *Clock(s)* and *Configuration (gear)* – see Chapter 7
10. *In-line Help (question mark)* – see Chapter 7
11. *Notification Panel* – see Chapter 7

Some of these items are so important that they rate their own chapters. Others are detailed in various sections of this guide (cross references to the relevant sections of the manual are provided above). In this chapter, we'll focus on those *Dashboard* features not discussed in greater depth elsewhere.

SECTION 7.1 FILE MENU

This menu drops down to reveal the following items:

- *Eject* – opens a sub-menu that lets you safely disconnect selected storage volumes
- *Import Media* – easily add media files from external sources to the appropriate session folders, and automatically transcode them to friendly formats for optimal playback if necessary.
- *Share Media Folders and Buffers* (see Section 11.3)
- *Exit* – close your live production session, and return to the *Session page* of the *Launch Screen* (all session settings are stored on exiting)

SECTION 7.2 OPTIONS MENU

Several very useful interface and workflow options are presented in the *Options* menu.

- *Tabs Follow*
 - Enable *Follow Preview Row for All Inputs* to display the tabbed pane for a *Preview* row source automatically on selection changes (for *M/E* selections, the *M/E* pane must be maximized to see the *M/E* tab update).
 - When *Tabs Follow Preview for M/Es* is checked, only the *M/E* pane selection updates (when you select an *M/E* as source on *Preview*).
 - *Tabs Follow All Delegates* updates the *Desktop* in sync with control panel delegate changes.

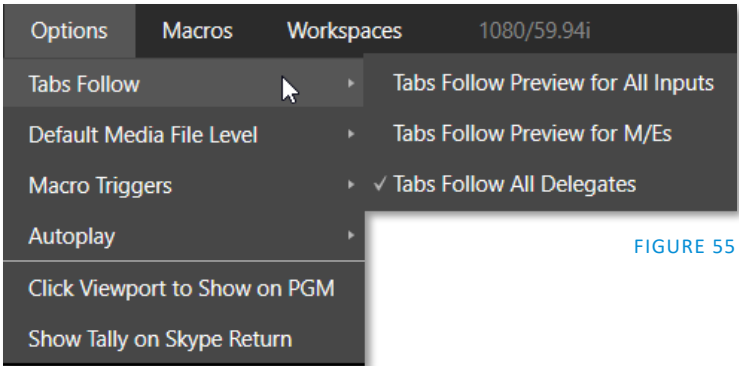


FIGURE 55

Hint: Control panel delegate operations can sometimes result in a tabbed module that you want to view being obscured when certain options are enabled. When this happens, press the control panel Delegate button again to re-display the tabbed pane you are interested in.

- *Default Media File Level* – the default ‘per-clip’ audio level applied to imported media files is controlled by this option.

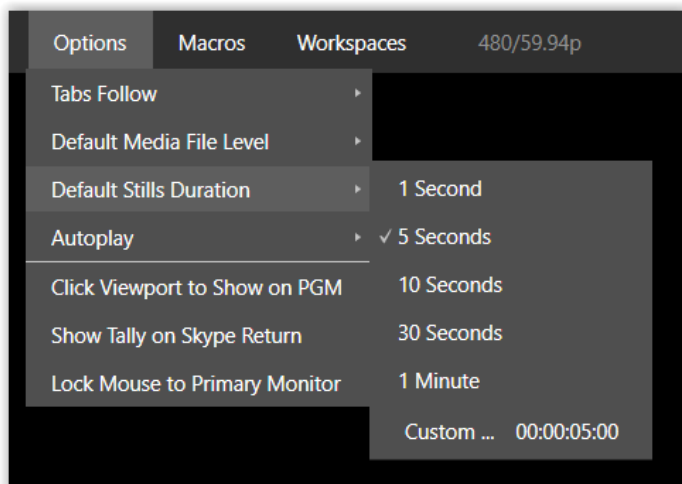


FIGURE 56

- *Default Stills Duration* – the default duration of a still can be applied, along with creating a custom default.
- *AutoPlay*
 - *Enable Autoplay Out on M/Es* – Enables *Autoplay*'s transition-out behavior for *M/E*'s (*Mix* mode) displayed on *Program* out (off by default).
 - *Extend Play* – When enabled along with *Autoplay* in the *DDR ...*
 - Playback will extend past any marked out point, up to the last frame of the file – even if the operator manually removes the *DDR* from output.
 - After the last frame is displayed, the playhead advances to the next playlist item.
- *Click Viewport to Show on PGM* – When this feature is enabled, clicking a viewport (or, on a touchscreen, tapping a viewport) selects the corresponding video source on the main Switcher's *Preview* row and performs the current *Background transition* to display it on *Program* output.
- *Show Tally on Skype Return* – The return video sent to remote Skype TX callers shows a tally overlay.
- *Lock Mouse to Primary Monitor* – Stops the mouse from freely moving across monitors and locks to one primary monitor.

SECTION 7.3 MACROS

Click *Macros* to show a menu containing a *Configure Macros* item. This opens a large panel that allows to you create, organize, and even edit macros.

Macros provide extremely important production benefits – sufficiently so that we have not only have given them a place of their own in the User Guide (Chapter 16), but they are also discussed in greater detail in the accompanying *Automation and Integration Guide*. Also, note that the *Shortcut Commands* used in macros are listed on your TriCaster's locally served Resources web page.

SECTION 7.4 WORKSPACES

The *Workspaces* menu opens when clicked to reveal control groups for each detected screen. These control the monitor viewport layouts shown on the *Live Desktop* and external *Multiview* displays. *Workspace* features are discussed in Section 10.3.

SECTION 7.5 CLOCK

The Dashboard also hosts a clock which, in addition to showing the current time, can display countdowns to event start and end times when enabled.

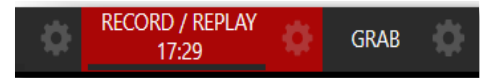


FIGURE 57

To access time features, click the small gear icon next to the timecode display at right in Live Desktop's *Dashboard*.

Hint: You can set the system clock, as for any computer; simply exit to the Windows® Desktop to do so.

7.5.1 LTC TIMECODE

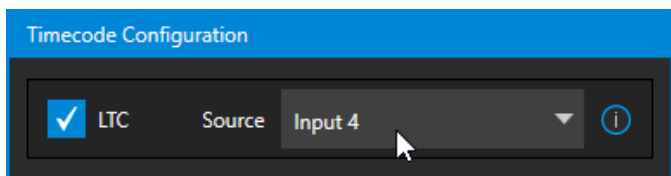


FIGURE 58

Linear timecode has long served as a method of sharing an external timecode reference in video production. Output from an external *timecode generator* is supplied to devices in the video pipeline using a standard audio connection.

Choose an LTC source in the *Timecode Configuration* panel, and the feature will decode time stamps from the audible signal, using the values for clock displays as well as embedded timecode when recording video (if the file format supports this). This is a great asset for post-production purposes.

Hint: The timecode display in the Live Desktop titlebar is tinted blue when external timecode is in use. If the external connection is lost for any reason, the display changes to white. TriCaster will attempt to maintain continuous timecode from the interruption on, until a valid signal is restored (in which case the display turns blue once more).

7.5.2 EVENT TIME

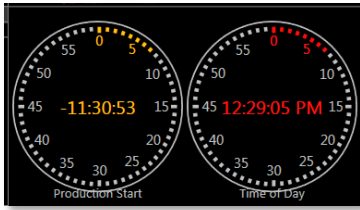


FIGURE 59

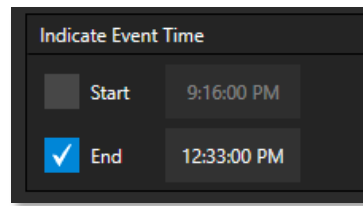


FIGURE 60

Two switches under the label *Indicate Event Time* allow you to configure *Start* and *End* times for your upcoming show. Beside each of these is an editable time field. Enable the *Start/End* switches, then click inside the field to modify the time value by direct entry or drag left right to raise or lower the current value.

During live production, the *Dashboard* and as well as monitor panes and default *Multiview* layouts offer *Broadcast Clock* displays based on the current timecode. Secondary clock displays show a useful countdown to the (production) *Start* and *End* times when the corresponding switches are enabled in the *Timecode Configuration* panel.

SECTION 7.6 INLINE HELP

Are you new to TriCaster or need a refresher on a certain feature? Click on the *Open Help* icon in the dashboard, next to the *Notifications* icon (See Figure 61).

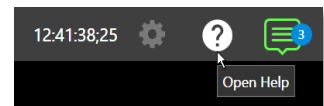


FIGURE 61

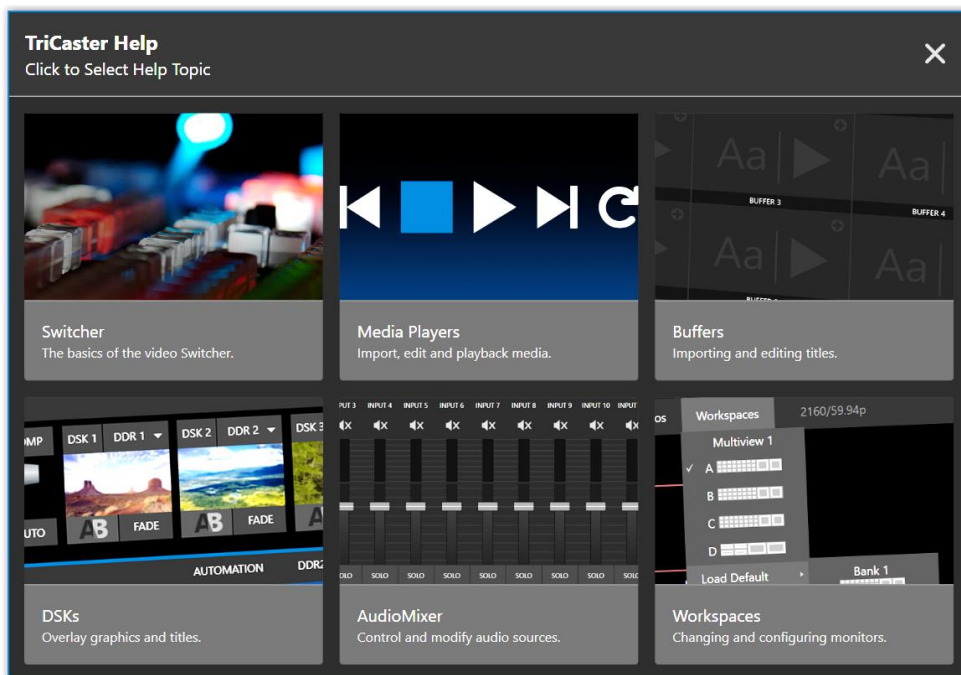


FIGURE 62

A menu will open to offer help with an introduction to TriCaster Mini X, tutorials, videos and easily accessed contextual help. We will continue to add more topics over time.

SECTION 7.7 NOTIFICATIONS

The last item at right on the *Dashboard* is the *Notification* icon (see Figure 61). A 'number bubble' indicates how many un-viewed entries have accumulated. Clicking the icon displays the *Notification Pane*, which provides both helpful information and some very useful tools.

The *Notification* icon can be colored white, green, amber or red. The color indicates the type of messages available for review:

- A green icon denotes an informative message, as well as the availability of new media.
- An amber icon indicates a warning message has been received.
- Higher priority alert messages are denoted by a red icon.

The highest priority (un-viewed) item in the list at the moment determines the *Notification* icon color. After you open the panel to review the notifications in the list, the icon turns white.

Individual items can be cleared from the list using the context menu opened with the triangle gadget shown at right when you roll the mouse over an entry, or you can empty the list with one click using the *Clear All* button in the footer of the panel.

Information appearing in the *Notification Pane* can include the following:

- Session name, format, and software build number (press *Alt + b* on the keyboard to update this item)
- Status messages pertaining to operations or system conditions; these may be benign notifications or cautionary. For example, a message indicating that the connection to a control panel has been lost is given an icon with an amber color. A higher priority warning triggers a message with a red icon.

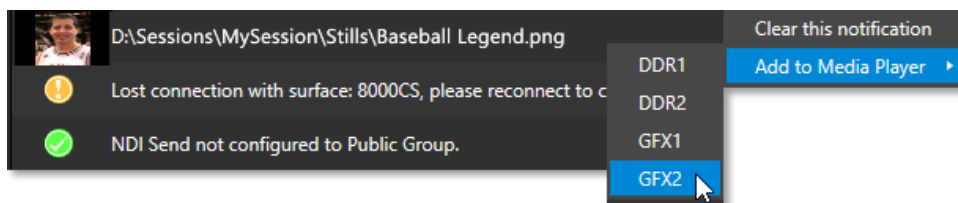


FIGURE 63

- A special message is added if a clip or still image is added to one of the *Session's* media file locations (such as the session *Clips* and *Stills* folders). These entries show a thumbnail icon at left, along with the filename and path. In this case, the item's context menu includes an *Add to Media Player* option (Figure 63), which lets you immediately append the new file to a selected *Media Player* playlist.

Hint: This last feature is particularly handy when adding files across the network, as perhaps when using the DataLink™ for TriCaster plugin to import images from the popular Chrome web browser.



Your NewTek system provides extensive control over video sources, along with endless creative features. Each source has Proc Amp, keyer (LiveMatte™) and Crop (a.k.a., ‘garbage matte’) features. The number and flexibility of outputs supplied is impressive, too, and we’ll discuss these fully in this chapter.

The addition of NDI® (Network Device Interface) support provides you with virtually unlimited input and output possibilities. Generally, an NDI source needs little if any configuration; other source types may require you to choose between optional connection methods and settings.

SECTION 8.1 INPUT CONFIGURATION

8.1.1 INPUT TAB

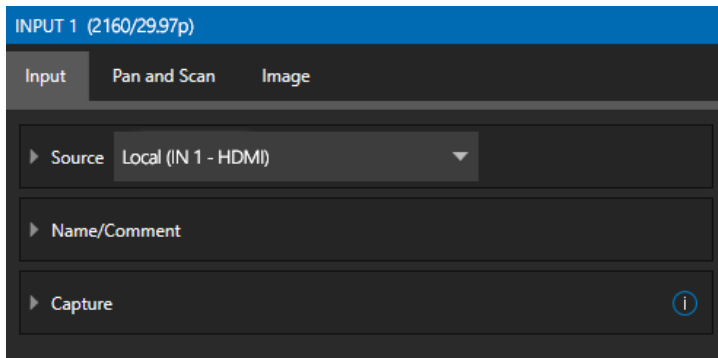


FIGURE 64

- Any external NDI source, Skype TX Caller, or a local hardware source connected to one of the system’s hardware input connectors, can be flexibly assigned to any *Switcher* input.
- This also means that sources can be easily re-ordered on the *Switcher*.

(Likewise, default audio sources for *Switcher* inputs can be flexibly re-assigned in the *Audio Mixer* module. For example, audio Input 1 and video Input 1 are not inextricably linked.)

The assignment of one of the various types of sources to a *Switcher* button (e.g., “Input 1” on the *Switcher*) is made in the *Input Configuration* panel (Figure 64) introduced to us back in Section 3.9.2.

Open *Input Configuration* by any of the following methods:

- Double-click the monitor viewport for a *Switcher* input.
- Click the *Configuration* (gear) icon shown at lower right when the mouse pointer is rolled over above the viewport.
- If you have a touchscreen, you can two-finger tap the viewport.
- Or right-click a *Switcher* input button and select the *Configure* menu item.

SOURCE

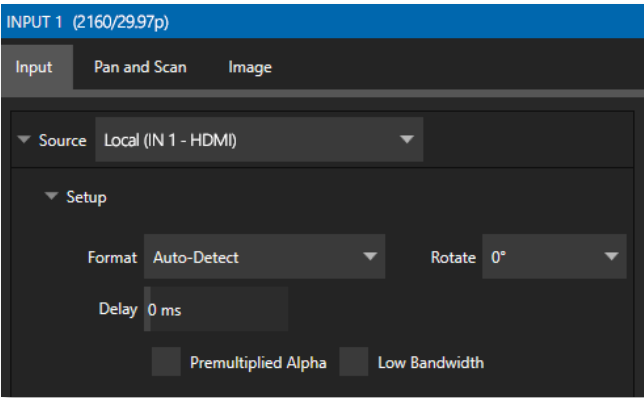


FIGURE 65

Use the *Source* menu in this panel to assign one of the many sources available to the corresponding *Switcher* input. Available video sources are grouped under labels named for the device supplying them.

The *Local* group in the *Source* menu includes those sources connected locally to the system's hardware inputs, any other local hardware sources detected (such as a webcam), video conference application, along with *Skype TX Caller* connections hosted by the local system (in contrast to externally hosted Skype TX sources, such as those supplied by a NewTek Talkshow® VS-4000 system), and *Black*.

Note: Quality Monitoring displays Resolution and frame rate of every video source coming into your TriCaster Mini at the top of the Input Configuration panel, to ensure the quality of your sources. See Figure 65.

* Please refer to Appendix B: Skype and Skype TX, for more on this useful source option.

IP SOURCE

The *Add IP Source* option is near the bottom of the local source list. Clicking the *Add IP Source* entry (Figure 67) opens the *IP Source Manager*. Adding entries to the list of sources shown in this panel causes corresponding entries for new sources to appear in the Local group shown in the source menu of the Configure panel (Figure 66).

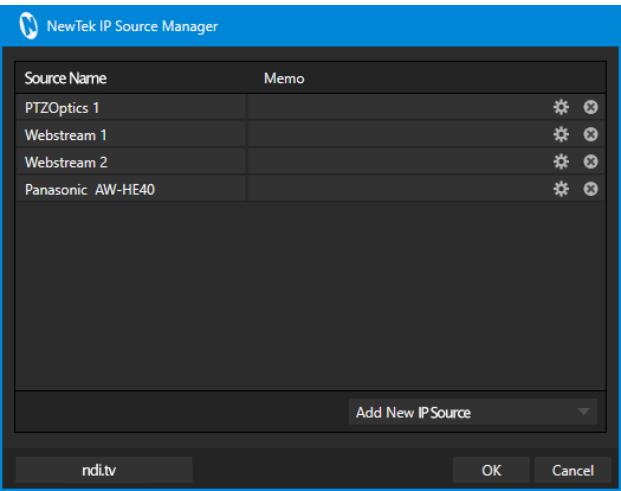


FIGURE 66

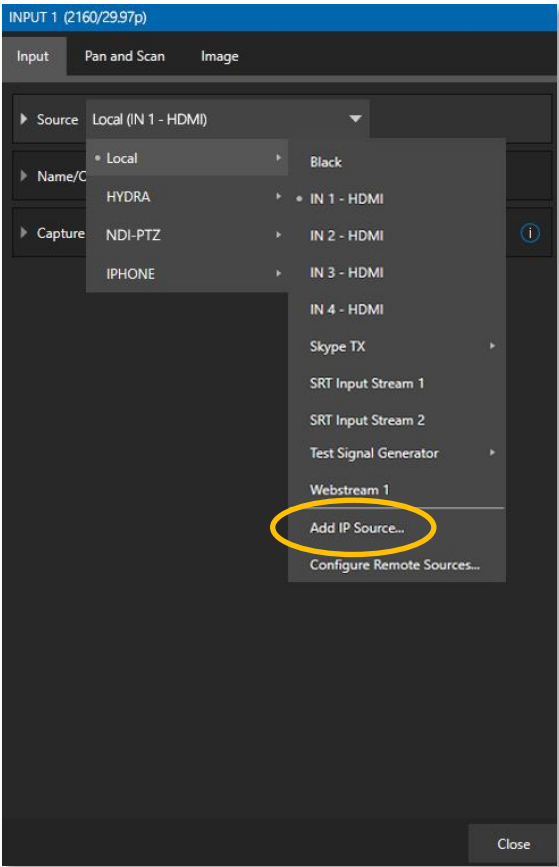


FIGURE 67

To *Add IP Source* select a source type from the dropdown list provided (Figure 68). This opens a dialog suited to the particular source device you wish to add, such as one of the numerous supported PTZ camera brands and models.

Additional protocols have been added to provide more options for video sources. RTMP (Real Time Message Protocol), a standard for delivering streams to your online video platform. RTSP (Real Time Streaming Protocol), used for establishing and controlling media sessions between end points. SRT Source (Secure Reliable Transport) is an open-source protocol that is managed by the SRT Alliance. SRT can be used to send media over unpredictable networks, like the internet.

To open an SRT source, you will need to fill out the following information to configure the *SRT Input Connection* (Figure 69).

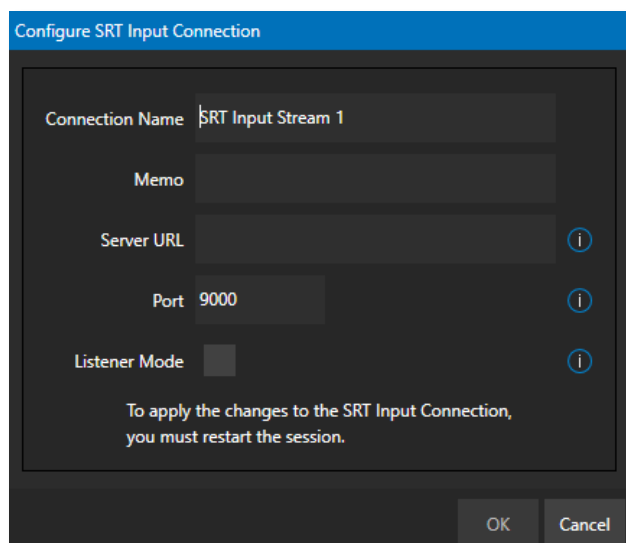


FIGURE 69

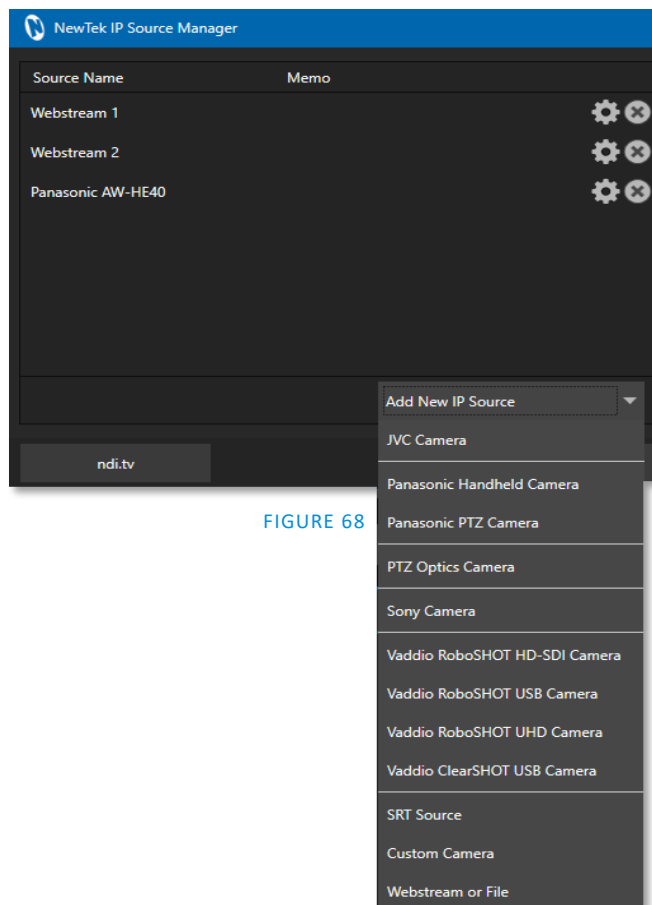


FIGURE 68

Note: It is strongly recommended to get the stream working in VLC first, then copy the URL into TriCaster.

Memo – enter a brief description for later reference.

Server URL – the server URL must be the public IP address of the remote source, either as a numerical address, or named such as "entrypoint.cloud.website.com".

Port – each SRT stream must have a unique port number. This can be any valid port, but ports in the 9000 or 10000 range are common.

Listener Mode – the stream connects to the Caller machine, then waits for it to initiate streaming. Otherwise, this machine is the Caller, and the other side must be the Listener.

The NewTek IP Source manager panel displays the selected source, here you can edit by clicking the gear to the right of the source name or click the x to remove the source (Figure 68).

Note: After adding an IP source, you must exit and restart the software for the new settings to be applied.

SETUP

FORMAT

If you use the twirl-down triangle gadget at left to expand the Source>Setup control group, you will see that the Format for these diverse source types of defaults to Auto. In the case of an NDI source (and often, for hardware sources too) no further settings are required.

For hardware video sources, additional *Format* options are provided in this menu, allowing you to choose a setting manually if the *Auto* option is unable to correctly identify the format. Let's explore other features located in the *Input Configuration* panel.

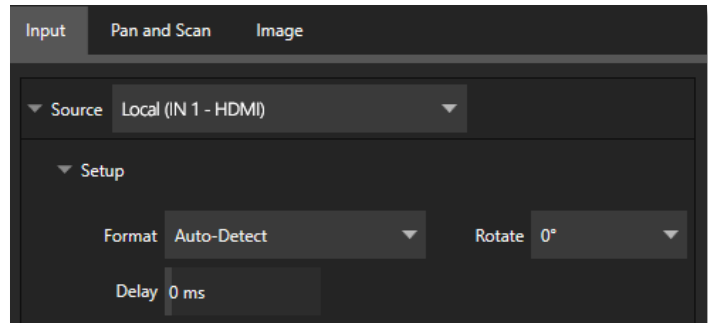


FIGURE 70

DEVICE WEBPAGE

For network connected sources (such as NDI sources), a *Device Webpage* button may appear just right of the *Source* menu. Click this button to access the remote device's own configuration webpage.

VIDEO DELAY

At times, typically due to upstream processing and architecture, video may arrive at the system's inputs ahead of the corresponding audio. The *Video Delay* feature allows you to compensate for these issues to establish a/v sync.

PREMULTIPLIED ALPHA

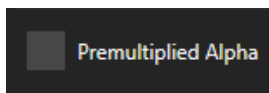


FIGURE 71

If you are supplying imagery (video sources, video clips, or still images) that support transparency by means of an embedded alpha channel, your choice here will be important. There are two 'flavors' alpha channel pixel encoding. The first is often called 'straight' or may be referred to as 'non-premultiplied'. Unsurprisingly, the alternative is 'premultiplied'.

The *Premultiplied Alpha* switch is off by default. Making the correct selection is necessary for correct compositing over other imagery.

LOW BANDWIDTH

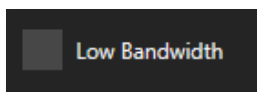


FIGURE 72

For NDI sources, a *Low Bandwidth* option is shown. This allows you to force the sending device to a lower quality video stream that may nevertheless be quite useable if the source is not intended to be displayed full screen. This option may be preferable in network settings with limited capacity (such as WiFi).

PTZ CONTROL

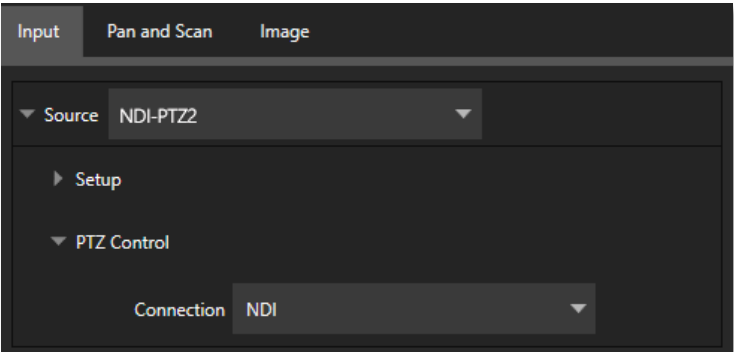


FIGURE 73

For many source types, a *PTZ Control* menu is shown in the *Source* control group. The default control *Connection* type is *NDI* which, assuming you are configuring an NDI source, makes life a lot easier because there is nothing else to configure.

Otherwise, if you *must select a ‘legacy’ control connection type like RS422, etc., or perhaps a non-NDI network connection, additional controls may be shown in this group to let you configure things like *Baud Rate*, *Com Port*, *IP Address*, and the like.

* We apologize that mankind has not yet completed its planet-wide upgrade to NDI (shouldn’t be much longer now, we hope). 😊

NAME/COMMENT

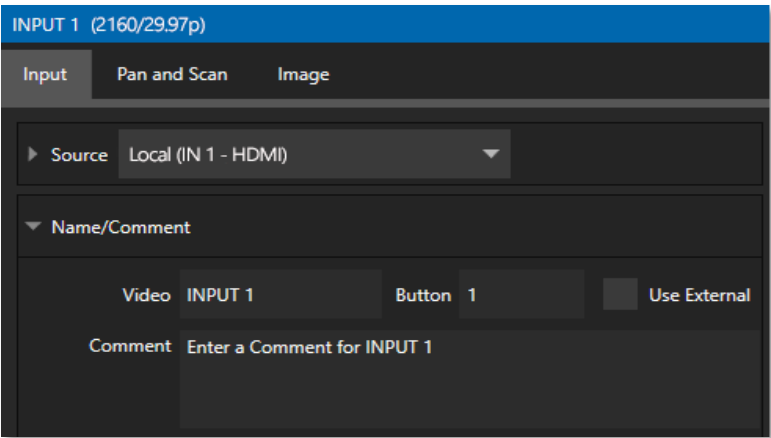


FIGURE 74

Expanding the *Name/Comment* control group reveals text entry boxes that allow you to provide labels for your video sources – specifically, the *Video* entry will appear beneath viewports in multiviews and some menus (where space permits), and the short *Button* text will be used to label *Switcher* buttons.

Enable the *Use External* switch to automatically pass the channel name for a remote source such as an HDMI router or NDI channel to the *Video* label field.

The *Comment* entry lets you enter memo text that can serve as memory aids, but or another very valuable purpose, too. Specifically, the values entered in these two fields supply the values for special *DataLink* keys. Among other things, the values from these *DataLink* keys can be used to update text values in title pages, or to add information to the filename of recordings.

Hint: For example, you could use a macro to automatically display a title page briefly any time you switch cameras. The Name and Comment entries for inputs update the values assigned to DataLink Keys named %PGM Source Name% and %PGM Source Comment% base on Program row selections.

You might enter “Bill Jones, CEO” as the Video name for a camera, and “Megadyne Computronics, Inc.” as the Comment. Continue to give unique values to inputs in similar fashion. Then enter %PGM Source Name% on the first line of a title page, and %PGM Source Comment% on the second line. When you change cameras, your macro will display the page, correctly identifying the talent based on the input Name and Comment.

CAPTURE

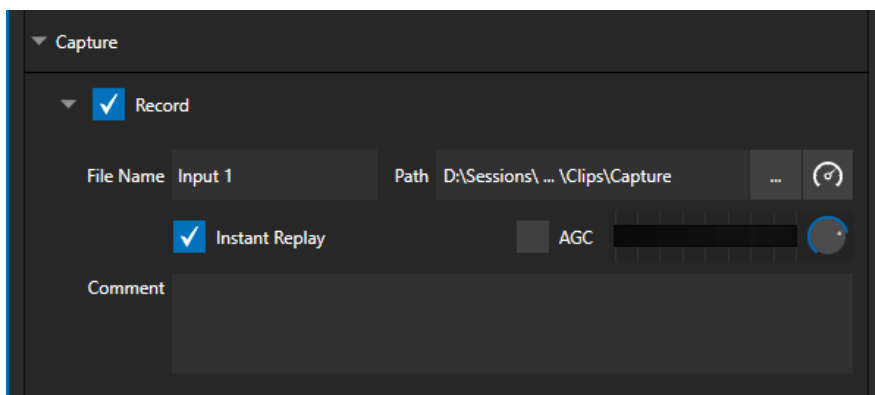


FIGURE 75

Each Switcher source has a *Capture* group in the *Input* tab. This control group shows settings and options for grabbing still images and, for appropriate sources, recording. These important capabilities are discussed in full in Chapter 19, Record, Grab, and Replay. The controls shown in Figure 75 are detailed in Section 19.1.2

Note: ISO Recording of UHD NDI HX is not supported. If you have selected such a source, please route through a MIX to record. Go to ndi.tv/formats for additional information.

8.1.2 PTZ/PAN AND SCAN PRESETS

Another control group appears on the lower part of the *Input* tab when appropriate. This group may be labeled either *PTZ Presets* or *Pan and Scan Presets*, depending on the source type.

In either case, the features and options presented in this group are similar. At the top, you will see 16 numbered preset slots. Rolling over these slots reveals two gadgets: Click the snapshot (camera) gadget to store or update a preset. Click the configuration (gear) gadget to show a *Preset Properties* panel (Figure 76) with two text boxes labeled *Alias* and *Comment*.

The entries in these two text boxes, like the *Name* and *Comment* values discussed earlier, provide the values for special *DataLink* keys that store the values from the last PTZ preset applied, as follows:

- PTZ PGM Alias
- PTZ PREV Alias
- PTZ PGM Comment
- PTZ PREV Comment

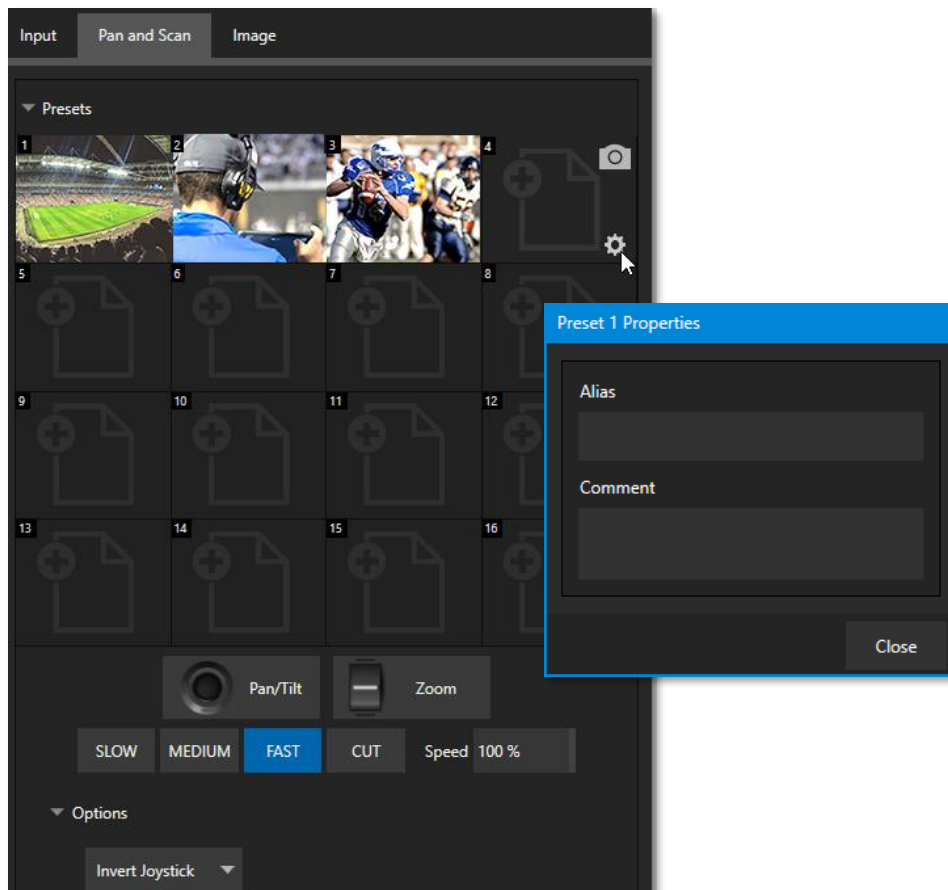


FIGURE 76

A set of controls located below the *Presets* bin allows you to control connected PTZ cameras, or to affect sources supporting *Pan and Scan* features in similar fashion (you might think of these sources as providing a sort of ‘virtual PTZ’ functionality).

Note: NewTek NC1 Spark units connected by NDI will show Virtual PTZ (Pan and Scan) controls. With Advanced Features, most Switcher sources feature Virtual PTZ. – see Appendix A: Advanced Features .

SLOW, *MEDIUM*, and *FAST* preset buttons complement the numeric *Speed* control slider at right. These affect the speed of the transition from the current position to that stored in a newly selected preset.

The *Options* group, when expanded, reveals *White Balance* options along with a menu that allows you to invert the operation of the *Joystick* on individual axes (both in the interface and on connected hardware control panels).

Note: Focus, Iris, and White Balance features are only shown when a PTZ cameras is connected to the input. However, features in the Input Configuration panel’s Image tab, discussed next, can provide similar functionality to White Balance.

8.1.3 IMAGE TAB

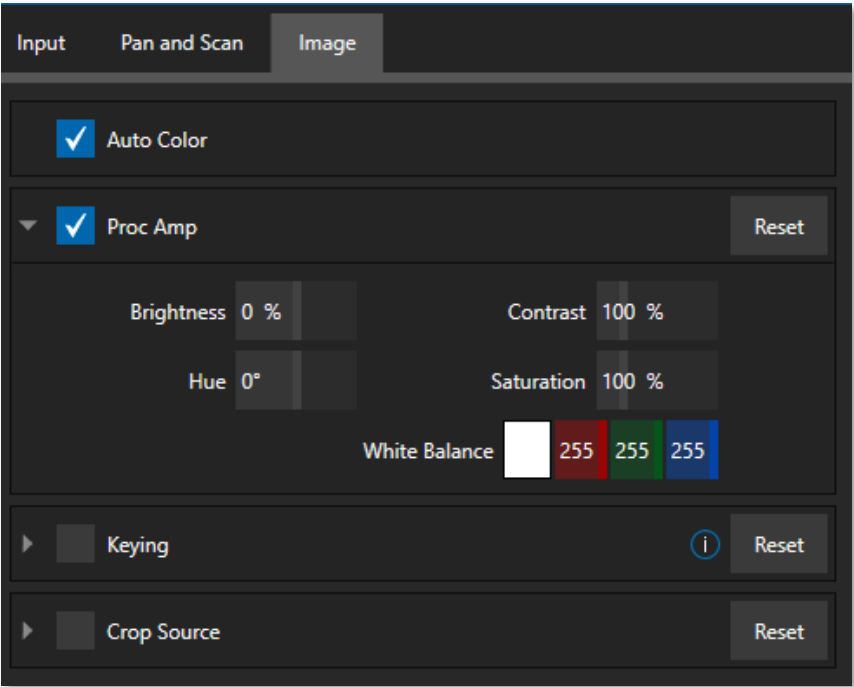


FIGURE 77

The *Image* tab in the *Input Configuration* panel hosts a set of features that provide extensive color control processing, chromakeying, and cropping options for every video source.

AUTO COLOR

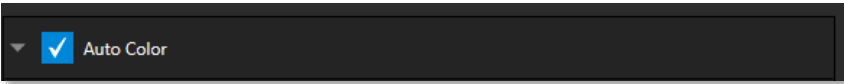


FIGURE 78

Lighting conditions can change dramatically during many live events, especially those held outdoors. Adding to this problem, production usually involves multiple cameras and, all too often, these may not have uniform color characteristics. Ensuring consistent color when switching from one angle to another, avoiding unwelcome brightness or color shifts as evening falls or when a cloud obscures the sun briefly can be troublesome, and expensive.

Auto Color, a unique feature capable of dynamically adapting the color characteristics of your video sources as lighting conditions vary, can minimize these problems. For many productions, simply enabling *Auto Color* is all it takes to produce a show that looks amazingly consistent.

PROC AMP

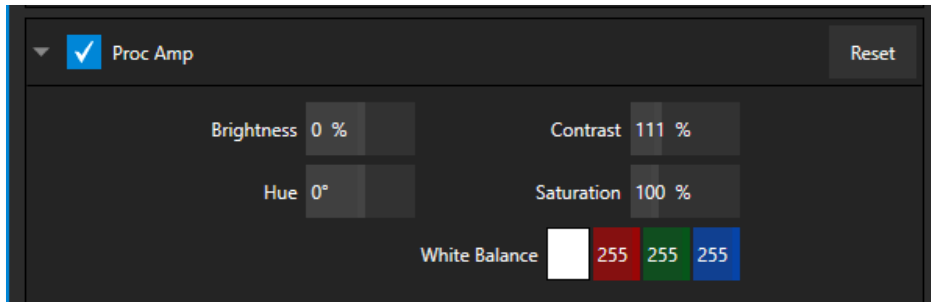


FIGURE 79

A switch at the top of the *Proc Amp* control group toggles the feature on/off. Other controls operate as follows:

- *Brightness*: Adjustment range from -50 to +50 IRE (the default being 0).
As reference, the full luminance range of the visible portion of a video signal can be thought of as '100 IRE units' (named for the Institute of Radio Engineers) – ignoring minor regional variations.
- *Contrast* – Adjustment ranges from 25 - 400% (default 100%).
- *Hue* – Adjustment range between -180° and +180°. Adjusts the master color of the video signal from the attached source, swinging the entire image through the color wheel's spectrum.
- *Saturation* – Adjustment ranges from 0-500%. Zero saturation results in a 'black and white' picture; increased saturation results in richer colors. High saturation values can exaggerate the color portion of the signal.

(Note that over-saturated colors are considered illegal for broadcast transmission and may result in display problems on some devices.)

Hint: Proc Amp adjustments are applied downstream of LiveMatte, which can help when composing greenscreen shots to match a background or LiveSet.

White Balance – to automatically white balance, click and hold the mouse button on the *Color* well, and then slide the 'eyedropper' pointer onto the monitor for the corresponding source. Release the mouse button over a part of the image that should appear as white after processing.

KEYING

The Keying control group in the *Image tab* hosts *LiveMatte*, a powerful real-time keying system for live production.

Keying is a popular and powerful method of compositing multiple images, whether photos, video clips or live camera streams.

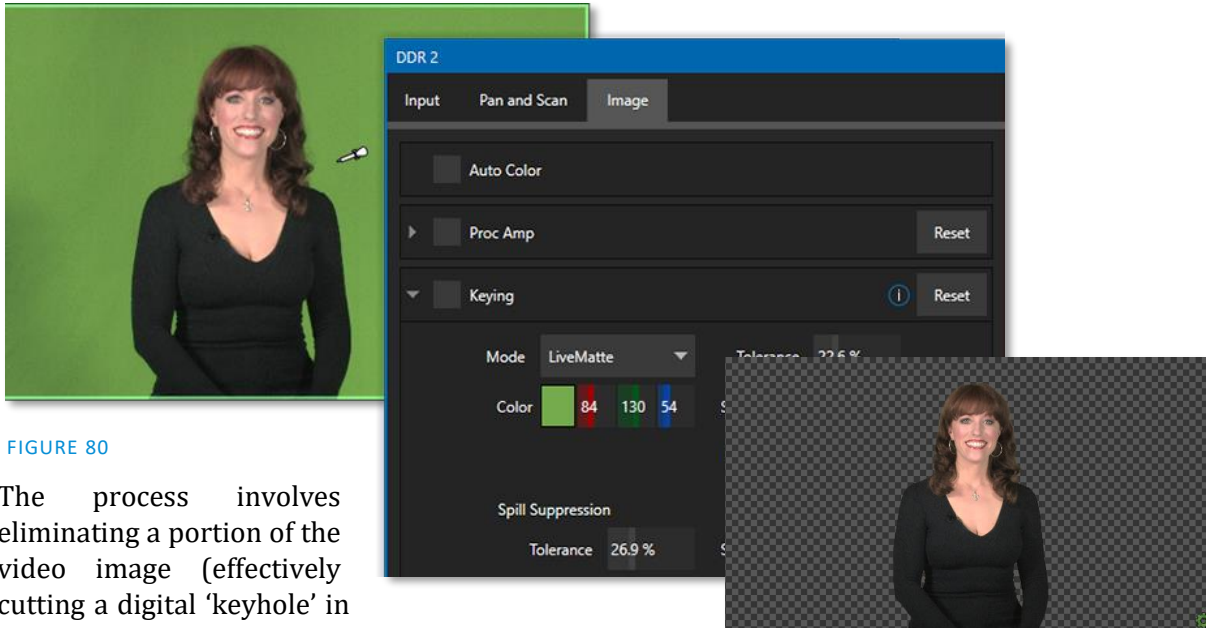


FIGURE 80

The process involves eliminating a portion of the video image (effectively cutting a digital 'keyhole' in it) to reveal a user-defined background scene.

This feature also plays an important role in the workflow of *LiveSet*, NewTek's powerful virtual set technology.

LiveMatte's controls are deceptively simple, making a great deal of complex digital manipulations easy to use. Even so, much can be said about getting the best results. For that reason, we've devoted a whole chapter in this manual to discussing it – please see Chapter 13, *LiveMatte* .

Hint: When LiveMatte, Proc Amp, or Crop settings are active for a source, bright green, blue and yellow indicators are lit under its monitor.

CROP SOURCE

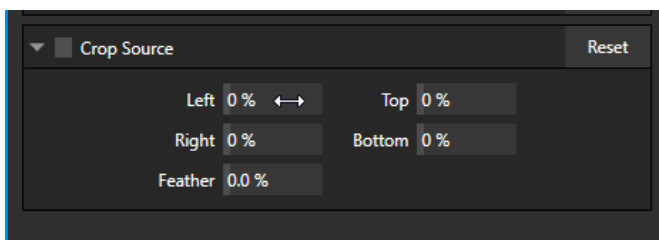


FIGURE 81

It is very common for a source to be supplied with unintentional inclusions; these are often items that remain after chromakeying is applied, but which need to be removed along with the background. (Common examples include microphones or lighting fixtures dangling from above, or perhaps a harsh crease, blemish, or tear in the background screen.) Or, as is frequently the case, the source video itself may have a few pixels of black or video 'noise' along one or more of its edges.

The settings in the *Crop Source* control group can be used to remove such unwanted 'garbage' from the scene, and for other purposes, too – such as to isolate a portion of the screen for use as a 'Picture in Picture' overlay.

Numeric controls in this group let you define margins for each side of the frame. Drag left or right on the number fields to adjust the values interactively or click a field to enter an exact value using the keyboard. The region defined by these controls is completely removed. Use the *Feather* setting to soften the edges.

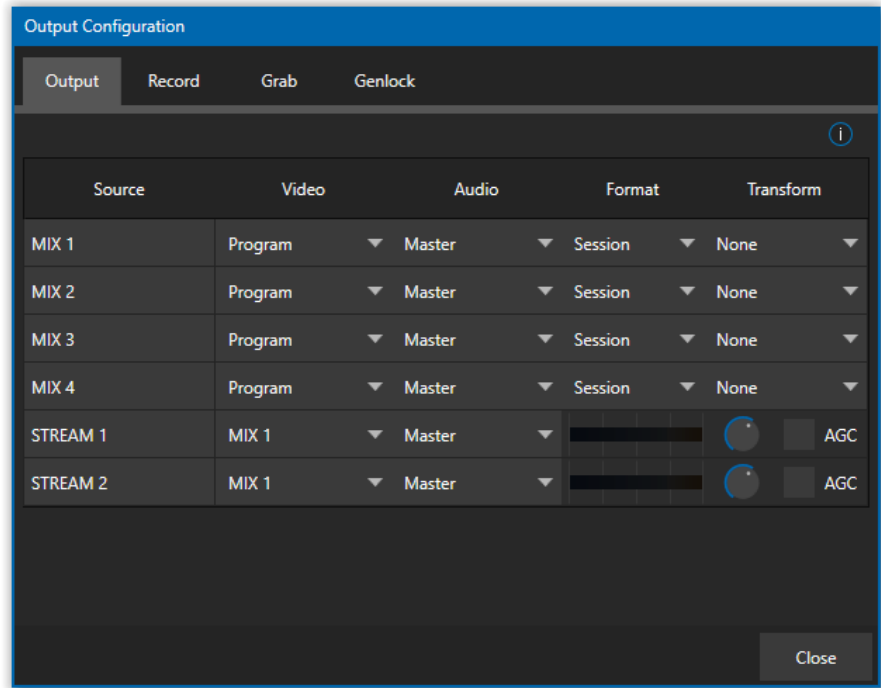
Hint: For added convenience, similar cropping tools are available separately in the Position panels of DSKs, along with the Key and, for LiveSet Effects, each layer's settings for MEs.

SECTION 8.2 OUTPUT CONFIGURATION



FIGURE 82

Roll the mouse pointer over the *Program* monitor to reveal a *Configure* button (Figure 82) at right in the titlebar below the display. Click it to open the *Output Configuration* panel.



Hint: As for other viewports, If Click Viewports to Show on PGM is disabled in the Options menu, double-click the mouse on the viewport to open this panel.

FIGURE 83

The *Output* tab in the panel above contains controls governing the system's primary outputs.

8.2.1 OUTPUT TAB

We discussed *Primary* and *Secondary* outputs back in Section 3.9. The first four entries in this panel are primary outputs; typically, these are also 'mixed' outputs, hence their default labels – MIX 1, 2, etc. These

video sources are sent to the corresponding SDI output connectors (when provided), and as NDI® (Network Device Interface) outputs.

VIDEO

The *primary* outputs support the largest number of optional video sources and, uniquely, can follow a delegated *M/E*, or a *Switcher* color group. All other outputs can be assigned to follow a primary output or show another designated *Switcher* source (excluding M/Es).

AUDIO

In similar fashion, you can choose which audio source accompanies any of the primary outputs. Choose any individual audio mixer input, or any of the mixed audio outputs, Master, or Aux (audio mix options vary by model).

FORMAT

The *Format* menu allows you to choose the video format for each output. Select the video format for downstream devices you intend to connect to the corresponding output here. The formats available are drawn from the list below (modes available vary according to session mode):

- 2160p
- 1080i and/or 1080p
- 720p
- 480p – progressive standard definition NTSC sessions only
- 480i (4:3) –interlaced standard definition NTSC sessions only
- 480i (16:9) –interlaced standard definition interlaced NTSC sessions only
- 576p – progressive standard definition PAL sessions
- 576i (4:3) –interlaced standard definition PAL sessions
- 576i (16:9) – interlaced standard definition PAL sessions

Generally, source formats that are inconsistent with the current output format setting are automatically confirmed when possible. In some cases, such as non-standard format sources, the output format may be modified to provide a suitable display. That said, it's best to avoid non-standard sources if possible.

TRANSFORM

Video Mix also support independent transform and flip controls, as well as Section options as seen below.

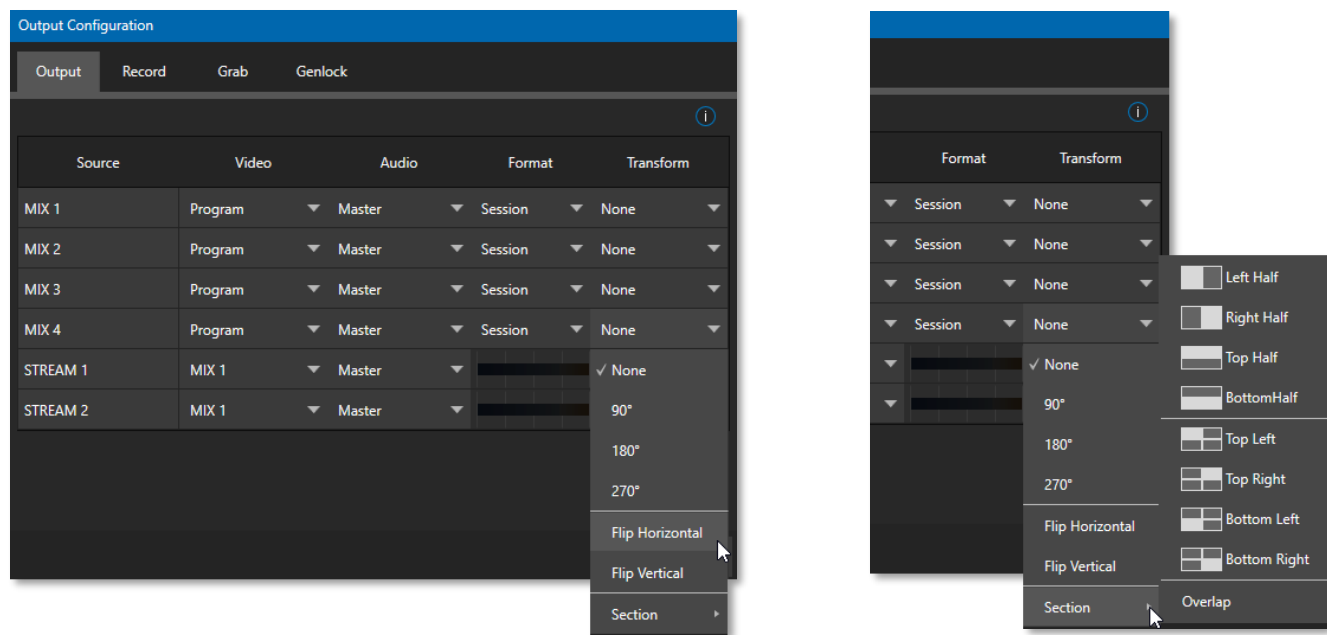


FIGURE 84

STREAM

The *Stream* controls in the *Output* tab let you independently assign any of the primary mixes to the two streaming encoders. Likewise, for models with multiple audio busses, you can send audio from the *Master* audio mix or any of the *Aux* busses to one of the three streaming encoders.

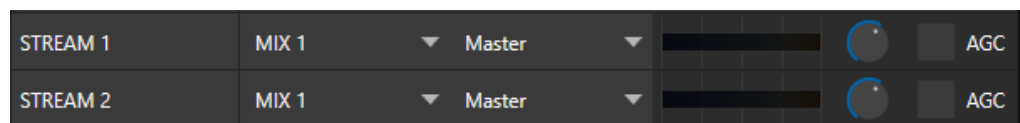


FIGURE 85

The audio controls also include individual *VU* meters, *Gain* knobs, and an *AGC* (Automatic Gain Control) option. These allow you to modulate audio for the streams separately from your primary audio outputs.

Hint: Streaming output is always de-interlaced.

Streaming Output involves more options, too, since there are so many ways to stream. In this panel, you simply configure the audio and video sources sent to the streaming output. All other options and settings relevant to streaming are discussed in Chapter 17.

8.2.2 RECORD TAB

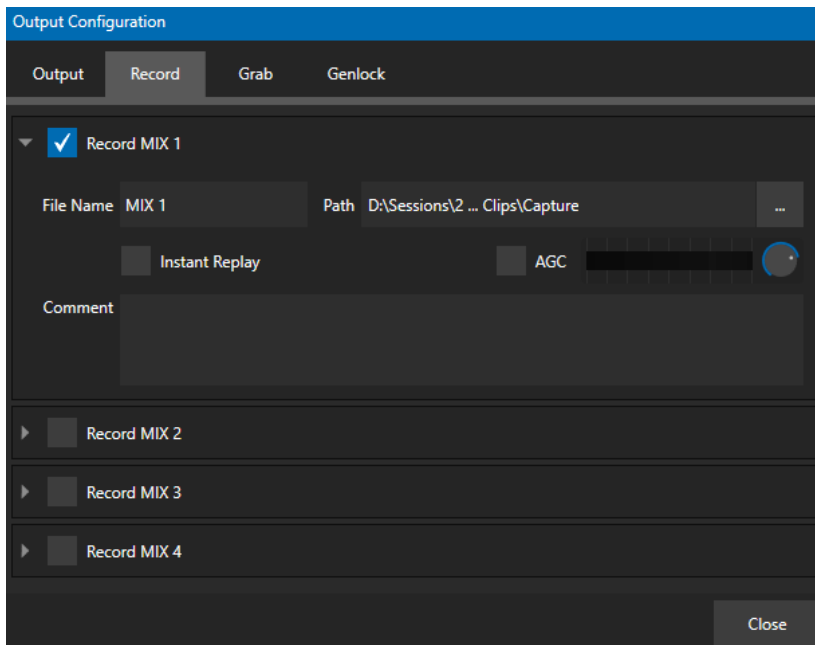


FIGURE 86

Each MIX output source has a corresponding *Record* control group to provide settings and options for capture. These important capabilities are discussed in full in Chapter 19, Record, Grab, and Replay. The controls shown in Figure 86, are detailed in Section 19.1.2.

8.2.3 GENLOCK TAB

The *Genlock* feature allows your NewTek system to ‘lock’ its video output to a reference video signal supplied to its *Genlock* input connector.

This synchronizes system video output to external equipment locked to the same reference. Genlocking is not a *requirement*, but it is very beneficial, and you should use it if you have the capability.

Miniscule local timing differences between these may force tiny delays during switching operations, which can contribute to throughput latency. Thus, serving i) the *Genlock* input and ii) other video devices in the chain with a single reference is the best approach.

You could think of it this way:

- ❖ Genlocking your *cameras* has the effect of locking their output together, ensuring optimal synchronization for live switching. This may result in throughput latency benefits.
- ❖ Supplying the same sync source to the *Genlock* input ensures a match between the system’s video output and any downstream video devices required to handle both it *and* other (genlocked) sources.

Genlock settings are hosted in a tab by the same name in the *Output Configuration* panel (Figure 86) Video Clock Center Frequency

This setting is applied when a genlock reference signal is *not* in use. To adjust the setting, supply color bars to an input and pass video output to a downstream *Vectorscopes*. The *Vectorscopes* display is completely stable when *Center Frequency* is properly adjusted.

Hint: The term “genlock” refers to “generator locking”.

Professional video devices often provide a “genlock input”, which allows an external reference signal (often referred to as ‘house sync’) to control its video timing.

The output of video devices connected in this manner is synchronized to the reference signal, and they are referred to as ‘genlocked’.

Chapter 9 SWITCHER, TRANSITIONS AND OVERLAY



Many Live Desktop features replicate traditional video switcher controls in an easy to comprehend and use fashion. The Live Desktop features powerful transition controls, downstream overlay channels, interactive layer monitors, and powerful automation features.

The central part of the *Live Desktop* (between the monitoring section and the tabbed modules) is taken up by the *Switcher* and related controls and features, including *Layer Controls*, which include main and *DSK* (Downstream Keyer) *Transition* controls and configuration features.

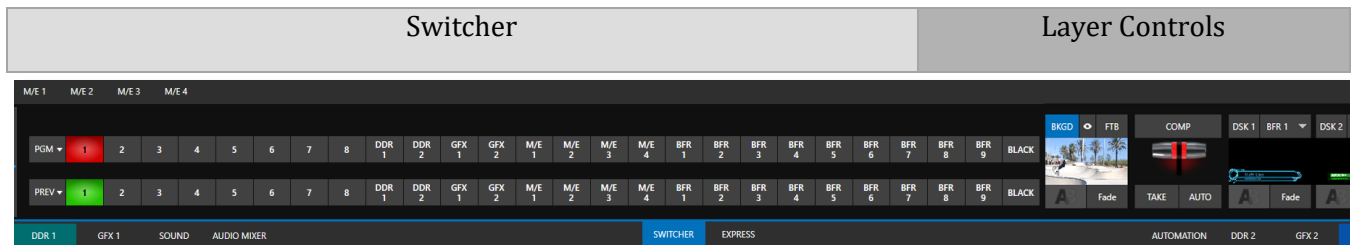


FIGURE 87

SECTION 9.1 SWITCHER MODES

TriCaster Mini X supports two different *Switcher* modes, allowing you to choose which one is best suited to your need for a given program and environment.

The standard *Switcher* interface provides control over your main *Program* video output using the familiar *Program/Preview* row method.

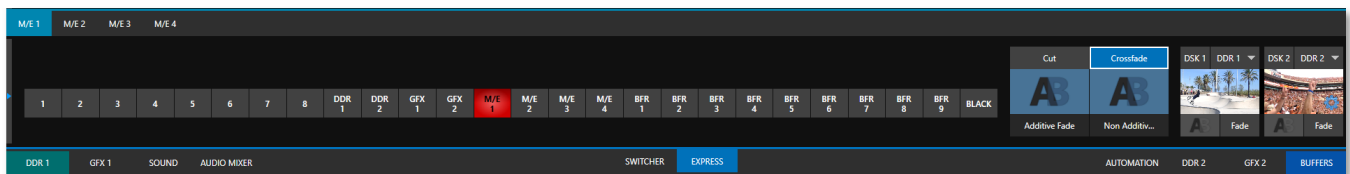


FIGURE 88

For less complex productions, the *Switcher's* convenient *Express mode* simplifies the process (see Section 9.8). This one-button operating mode will be especially welcome in environments where volunteers or less experienced operators are involved.

To select the current operating mode, simply click either the *SWITCHER* or *EXPRESS* tab provided at right in the horizontal bar immediately above the *Switcher* pane (Figure 89).

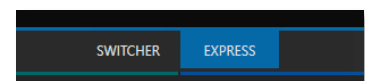
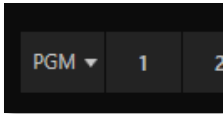


FIGURE 89

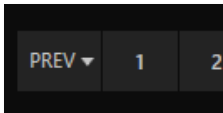
SECTION 9.2 PROGRAM/PREVIEW ROWS

The video source selections in the *Switcher* rows include all external inputs, including video router sources, internal sources (*Media Players* and *Buffers*), and the output from all M/Es.



The *PGM* (Program) row selection determines the dominant video stream of the *Background* (BKGD) *video layer* – that is the ‘rear-most’ layer of the composition sent to *Program* output.

Other sources may be mixed above the *Background* layer at times as you apply *LiveMatte*, or as portions of an incoming *Preview* row video source appear during a transition.



The *Preview* row appears in standard *Switcher* mode. Selections determine which source is queued up for display in the *Background* layer by a subsequent (BKGD) *Take* or *Transition* operation.

SECTION 9.3 BACKGROUND AND DSK LAYERS

The concept of *video layers* is central to understanding how the *Switcher*, *M/E* and *Transition* controls relate to one another, and how they combine to form the video seen on *Program* output.

- The *Background* layer (often shortened to simply ‘BKGD’) is always the base for the video composition displayed on *Program* output.
- *DSK* (Down Stream Keyer, or ‘overlay’) layers may appear above (in front of) the *Background*.

DSK layers are typically used for overlaying graphics, titles, etc., though they may serve other purposes as well. In addition to BKGD, up to five additional ‘primary layers’ (varies by model) can contribute to the final *Program* output at any given moment:

- Overlay layers (*DSKs*) are composed above the *BKGD* layer on output. *DSK 2* appears ‘in front of’ *DSK 1* on *Program Output* – that is, closest to the viewer – and so on in order.
- *FTB* (Fade to Black) constitutes a final overlay layer – one that obscures all other layers when applied.



Recall, too, that the *BKGD* layer itself is often a composite of sub-layers:

- It may include mixed video from both the *Program* or *Preview* rows.
- Selecting an *M/E* as source on *Program* or *Preview* can add many sub-layers to the *BKGD* composition, including the *M/E*'s primary *Inputs* and dedicated *KEY* layers (which are similar to *DSK* layers but being upstream of the main *Switcher*, appear composed in the background layer).

Note: Since M/Es are reentrant, the BKGD layer can at times reach very high numbers of sub-layers.

SECTION 9.4 SELECTING SOURCES

In standard *Switcher* mode, video sources for *PGM* (Program) and *PVW* (Preview) rows are selected individually by pressing buttons on those rows. By contrast, selecting a button on the single row in *Express* mode first places the designated source on the (unseen) *Preview* bus, then immediately switches it to *Program* output.

For *DSK/Key* layers, source selection is made using a drop-down menu above the integrated viewport located in the *DSK/Key* control group.

SECTION 9.5 LINKING SWITCHER ROWS

It can be useful to link two (or more) *Switcher* or *M/E* source rows together, to cause them to operate synchronously. *Program* and *Preview* rows and *M/E* source rows all show a triangle beside the row label at left. Click it to open a menu that lets you set up linking.

As you'd expect, rows assigned to the same color groups are linked. A selection made in any linked row updates the selection of all other rows in the same color group to match. Thus Figure 90 shows the *Input A* row for an *M/E* linked to the *PGM* row of the main *Switcher*. The *Ungroup* menu item removes the current row from a group, while *Clear this group* removes all rows from the current group.



FIGURE 90

SECTION 9.6 TRANSITIONS AND EFFECTS

We discussed video layers in Section 9.3. With this in mind, it's easy to comprehend the layout and use of the *Transition* controls. Let's consider the *Transition* controls in the standard *Switcher* layout first.

9.6.1 STANDARD MODE

On the left of this group are the main *Transition* controls, including the *T-bar*. The control groups right of the T-Bar provide configuration and control options for the individual *DSK* layers.

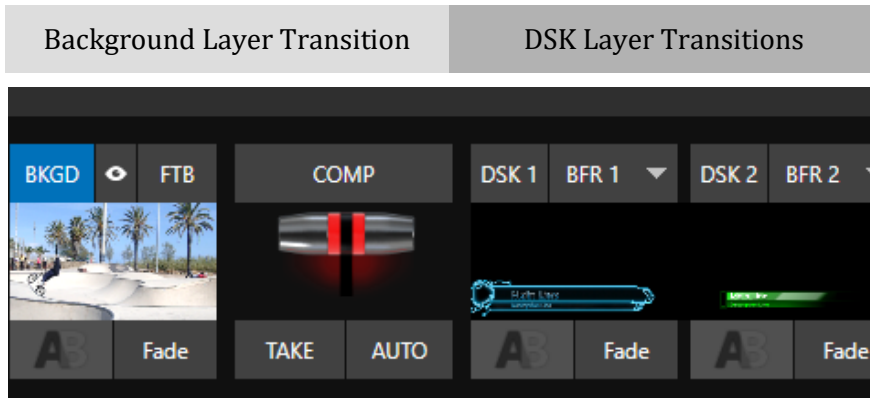
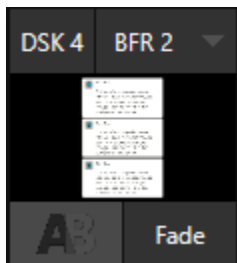


FIGURE 91

DSK CONTROLS



Each *DSK* layer has a live video viewport showing the current source assigned to it (using the menu right above the viewport) and its own transition effect.

Click the transition icon at lower left below the viewport (Figure 92) to reveal a palette of different transition presets provided for quick selection (Figure 94).

Click an entry in the palette to select it or move the mouse pointer to the “+” sign that pops up for each icon and click to open the *Custom Media Browser*.

FIGURE 92

Hint: The frequently used Cut and Fade effects are always available in the transition palette. As these cannot be replaced, no + sign appears for these icons.

In the *Media Browser*, you can choose from the hundreds of transition effects, or even *Animation Store* effects that you prepared yourself with the supplied *Animation Store Creator* application. The selected effect will replace the current one in the palette. To display or hide the *DSK* video layer over the *BKGD* layer on *Program* output using the currently selected effect, click (or tap) the viewport or the effect name label just below.

Hint: You can halt an unfinished effect in progress by clicking again during the transition. Then click it once more to continue performing the effect.

TRANSITION DELEGATES

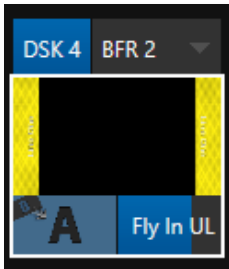


FIGURE 93

You can also control transition progress using the Switcher's *T-Bar* control, at left. The *T-Bar* operates on all *delegated* video layers. To delegate a *DSK*, click its label at upper left (Figure 93) to turn it blue. Clicking it a second time will un-delegate the layer.

The *T-Bar* acts on all delegated layers, including the *Background* video layer. For example, if *DSK 1* is visible, but *DSK 2* is not, when both *DSKs* and *BKGD* are delegated performing a T-Bar (or BKGD AUTO) operation reverses the visibility of the two *DSKs* on *Program* output when the *BKGD* transition occurs.

TRANSITION TIMING

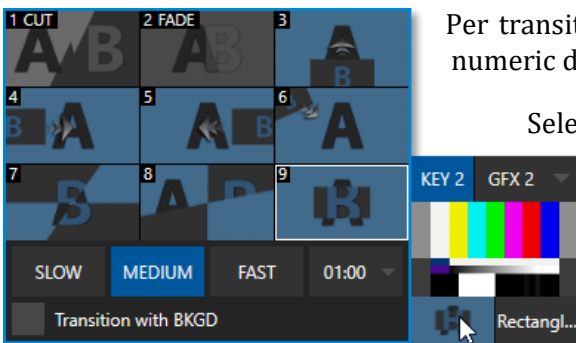


FIGURE 94

Per transition timing is set and stored in the effect palette, using the numeric duration control beside the effect icon.

Select transition speeds using the menu below the *Transition Palette*. You can also drag the mouse pointer over the numeric display to set a custom time or click it to enable keyboard entry of the effect duration.

Hint: The direction of Transitions applied as DSK (and M/E KEY) layer effects automatically alternates. If the first click displays the layer using an effect, the next click removes it using the reverse effect. This 'Ping Pong' behavior is optional for BKGD (Background) layer transitions.

ANIMATION STORE TRANSITIONS

You can also choose special transition effects called *Animation Stores*. These powerful effects normally include an embedded full color animated overlay, along with sounds for transitioning in and out (the audio level for *Effects* is controlled in the *Audio Mixer* tab below the *Switcher*).

These special *Animation Store transitions* are loaded into the *Transition Palette* in the same way as their less colorful cousins, using the *Browse* feature. Several *Animation Store transitions* are supplied, but you can generate your own using the supplied *Animation Store Creator* application and custom animation content you have access to or create using art software.

Note: The Animation Store Creator application is covered in its own manual, which can be opened from the Help menu on the Home page in TriCaster's Launch Screen.



FIGURE 95

Many more configuration options are available for *DSKs* (and their siblings, *M/E* key layers, too).

To access these settings and features, roll the mouse pointer over the *DSK* viewport, and click the configuration (gear) gadget that appears at lower right (Figure 95).

Doing so will open the *Input Configuration* panel for the source assigned to the *DSK*, but with the addition of a supplemental tab labeled *DSK (1-2)*.

The control groups in this new tab expand to reveal *Position*, *Crop*, *Apply with COMP*, *Borders, Edges and Shadows* settings.

POSITION (AND CROP)

The *Position* control group includes *Position*, *Zoom*, *Rotation*, and *Priority*. *Position* settings can be toggled on and off together using the switch provided in the group header. (Figure 96).

Click and drag on the *Position* button (diamond) to relocate the *DSK* layer vertically or horizontally within the frame.

Drag left or right on either of the two nearby numeric controls to adjust a single axis only.

Dragging the cursor on the *Zoom* button (magnifying glass) affects the apparent size of the overlay. Again, if you drag just one of the associated numeric gadgets you can adjust just one dimension of the corresponding *DSK* layer – width or height.

In similar fashion, drag the pointer over the *Rotation* button with the left mouse button depressed to turn the overlay source on three axes as follows:

- Drag left/right to rotate the source about the Y (vertical) axis.
- Drag up/down to rotate about the X (horizontal) axis.
- Drag while holding *Alt* down to rotate about the Z axis.
- Drag on a single numeric slider or hold down *Ctrl* to constrain rotation to one axis.

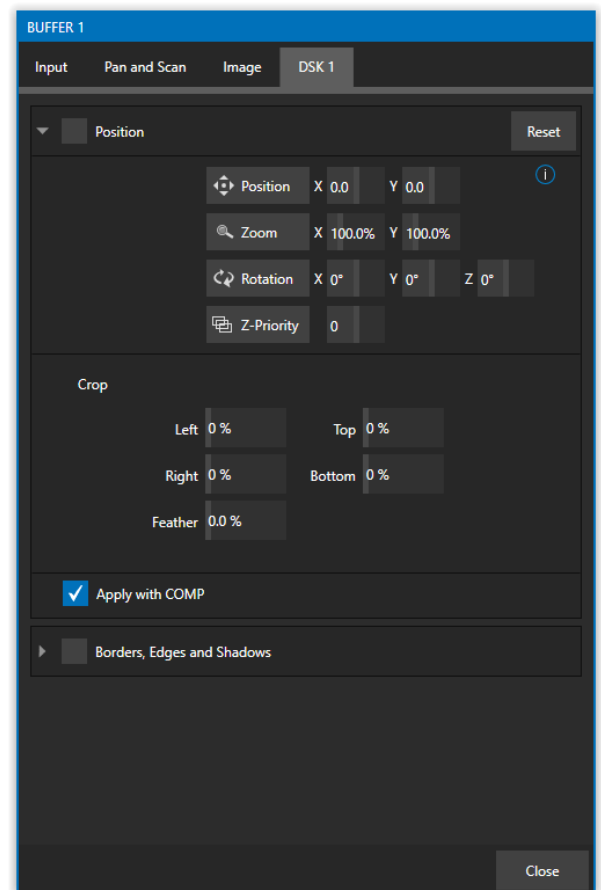


FIGURE 96

Hint: If you click a numeric field (or right-click it), you can type a value into the gadget using the keyboard; press Enter to complete the editing action, or Esc to cancel it).

The *Crop* *DSK/KEY* controls in this group are similar to those found in the *Input* tab, as discussed back in Section 8.1.1. However, these settings are applied to the *DSK/KEY* layer, without any impact on the source itself as it may be displayed elsewhere in the *Switcher*.

Z-PRIORITY

Normally, *KEY* and *DSK* layers appear in numeric order from ‘back’ (furthest from the viewer) to ‘front’. This is if *DSK 1* and *DSK 2* are both displayed and occupy the same position in the frame, the content in *DSK 2* will occlude *DSK 1*.

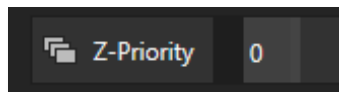


FIGURE 97

The *Priority* setting in *DSK* and *KEY* layer *Positioning* panels allows you to revise the default layer order on a selective basis. This feature was specially implemented to provide additional flexibility for use with the *Comps* feature.

For example, imagine an *M/E* set up with 4 *KEY*s supplying a quad-box setup for four remote interviewees over a background supplied by the *M/E*. You might want to use *Comps* to zoom the top-left input up to fill the screen while the moderator chats with that person. Normally, *KEY*s 1-3 would always appear *behind* *KEY* 4 – not what you want at all. The *Priority* feature lets you move any *KEY* to the front (and the setting is stored in your *Comps*).

The range of *Priority* settings runs from -10 to +10; the default is 0. A layer with a higher index is shown in front of those with lower indices. When two layers have the same layer priority, they are rendered in their natural (*DSK/KEY* layer) order.

APPLY WITH COMP

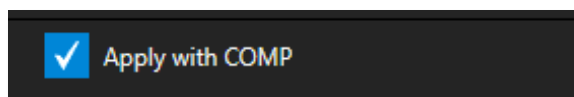


FIGURE 98

We will discuss the powerful *Comp* system a bit later (Section 9.7), but we’ll mention it in passing here to highlight the *Apply with Comp* switch provided in the *Position* control group. At the lowest level, *Comps* can be thought of as presets that store complete *Switcher* or *M/E* setups.

By default, the settings stored in a *Comp* include the *Position*, *Crop*, and visibility state for each *DSK* or *KEY* layer. Disable the *Apply with Comp* feature if you want to exclude a given *DSK/KEY* channel from *Comp* control, handling it manually instead.

Hint: You might find this useful, for example, to ensure that a station ID ‘bug’ shown over output is not accidentally removed by application of a Comp.

DSK and *KEY* layers automatically appear on the *Preview* and (*M/E Preview*) monitors when the *Position* panel is open (regardless of layer display options).

This allows you to modify a layer’s position without the result being inadvertently shown on output.

BORDERS, EDGES AND SHADOWS



FIGURE 99

The *Borders, Edges and Shadows* group also provides each *DSK*, *KEY* and *M/E* layer with three quick access *Border* preset slots (Figure 99).

Hint: Since these are per-layer Position effects and can be controlled – even animated – by Comps, you can use the Borders feature to create custom multi-box compositions in M/Es.

These powerful effects can include full color overlays, backgrounds, matte layers for ‘keyhole’ effects, and shadows.

You can freely scale, position and rotate various *Switcher* sources, add custom borders, overlays, shadows, and so-on, over custom backgrounds or even live or animated sources – all without special skills or resorting to *Virtual Set Editor*.

In addition to hundreds of supplied borders, you can easily create elaborate custom effects using Photoshop. You need merely define a Photoshop format file with three (rasterized) layers. The uppermost layer contains foreground elements (such as a bezel). The next layer is treated as a mask based on opacity and defines the part of the source image that will appear in the result. The ‘bottom’ layer supplies a background to appear behind transparent parts of the source (as, for example, when *LiveMatte* is applied to a source).

A template PSD file is supplied to assist you to do this. You will find the multi-layer Photoshop file in the Borders folder at C:\ProgramData\NewTek\(*TriCaster*)\Effects\Borders.



FIGURE 100

Hint: Since the opacity of the mask layer can vary between fully opaque and fully transparent, you can easily prepare soft-edged effects such as vignettes. Also, as foreground and background layers can optionally be empty, a simple opaque shape in the mask layer can serve a variety of imaginative purposes.

TRANSPARENCY

Sources assigned to *DSKs* are often partially transparent. This might be because they are drawn from a *Media Player (DDR)* file that includes an embedded *alpha channel*, or because *LiveMatte* or *Crop* options are enabled for the source, or perhaps because a *Network* source includes an alpha channel, or even all these factors operating together.

In all these cases, *DSK* layers automatically respect transparency when supplied by the source. The *BKGD* layer and all visible content in lower-numbered *DSKs* will appear through or around sources with transparency as appropriate.

Important Note: It's best to use files with straight (a.k.a. "non-premultiplied") alpha channels in TriCaster's Media Players. Premultiplied files will generally not yield correct results when overlaid on other imagery.



FIGURE 101

DSK layers offer a lot of creative possibilities. You might use *DSK* channels to display a permanent station ID 'bug', superimpose a company logo onto a title page, perhaps to add a 'spinning globe' animation playing in the *DDR* to a lower-third, 'frame' a keyed source composed over a title (Figure 101), or set up many other elaborate effects in this manner.

9.6.2 BACKGROUND CONTROLS



FIGURE 102

BACKGROUND TRANSITION GROUP

Transition controls in this section apply to the *Background* video layer only. In most respects, these tools are identical to the *DSK* transition controls discussed earlier, but there is one difference worth mentioning.



FIGURE 103

The *Duration* menu for the *Background* transition offers two items not included in the similar *DSK/KEY* controls:

- *Reverse* – configures the current transition to run in reverse direction the next time it is applied.
- *Ping Pong* – when enabled, this option causes the direction of the transition to be automatically swapped after each time it is applied.

FTB

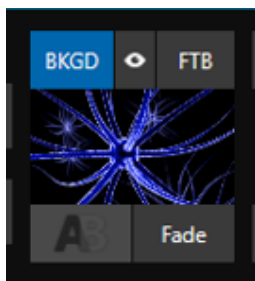


FIGURE 104

Let's discuss the *FTB* feature next. The acronym stands for *Fade to Black*. The result of clicking the *FTB* button will not likely surprise you very much. ☺

FTB offers a convenient method of doing what its name implies – fading *Program* output completely to black. It might help to think of *FTB* as a final video layer added above all others before *Program* output, completely obscuring everything below it.

As a memory aid, the *FTB* button pulses during operation. *FTB*'s fade duration is drawn from the *BKGD* transition setting.

Note: Hold down Shift while pressing the FTB button on a control panel to initiate an FTB operation. Hiding or displaying FTB triggers both Autoplay and Audio Follow Video when enabled for Media Players. It also fades Master Audio to mute when displayed, and back up again when hidden.

TAKE AND AUTO

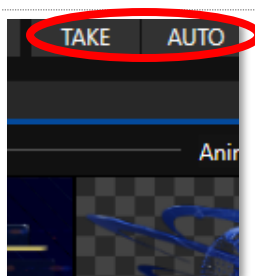


FIGURE 105

Clicking the *Background* layer's *Take* button (keyboard shortcut *Enter*) performs a straight cut for all video layers that are currently delegated.

Likewise, if you press *Auto* (or the keyboard *Spacebar*), the transitions assigned to all delegated video layers are performed.

Hint: You can halt an Auto operation partway by clicking the button a second time during the transition. The operation will be completed the next time you click the button.

T-BAR

The *T-bar* mimics the similar control on a traditional video switcher and allows you to *manually* transition between selected video layers. To use the *T-bar*, pull it downward by dragging it with the mouse pointer. Drag it all the way to the bottom and release to complete a transition; the *T-bar* then pops back to the top. Naturally, when the *T-bar* is dragged part way, a partial transition occurs. With certain transitions this can be useful for split screen effects.

Hint: The QuickSelect button (marked with an 'eye' icon to associate it with visibility) is located in between BKGD and FTB. Clicking it updates the Switcher's T-Bar delegate and transition states so that the next TAKE or AUTO operation will remove all visible DSK or KEY layers from output. (On supporting control surfaces, press ALT & BKGD to trigger the QuickSelect feature.)

SECTION 9.7 COMPS AND MEMS

You'll also notice a button labeled *COMP* immediately above the *T-Bar*. Clicking it opens the *Comp Bin*, which provides powerful layer and effect control features.

We're going to discuss the *Comp Bin* in full soon (Section 14.8), but at this point we want to distinguish *Comps* from *Switcher* MEMs. Bumping the cursor at the left edge of the screen adjacent to the *Switcher* or an *M/E* reveals a *MEM bin* with features that are quite similar. The primary difference between *MEMs* and *Comps* is that the former retains (and apply) all settings in the *Switcher* – including source selection.

SECTION 9.8 EXPRESS MODE

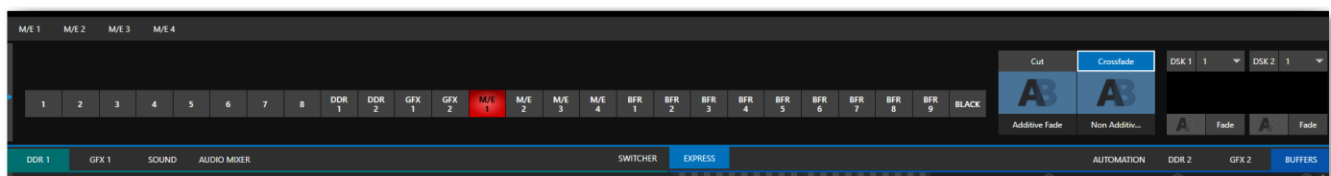


FIGURE 106

As mentioned earlier in this chapter, the *Switcher's* convenient *Express mode* simplifies the process for less complex productions.

To open the *Express mode* *Switcher* view, click or tap the *EXPRESS* tab right above the *Switcher*.

9.8.1 BACKGROUND TRANSITION

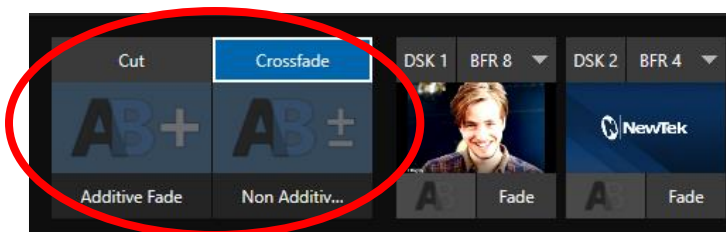


FIGURE 107

Note the quad-selector located between the single *Switcher* row, and the *DSK* controls at right. *Cut* and *Crossfade* mode selectors top this control group. Simply click or tap the *Cut* or *Crossfade* buttons to activate the corresponding *Background transition*.

Just below you will see two selectable transition controls. Tap or click these to activate the effect represented by the icon as the current *Background transition* instead. To open a transition selector to choose a different

transition for either of these slots, click the gear that appears at lower right when your mouse pointer is over the icon.

9.8.2 SWITCHING

Having selected the *Background transition*, simply click, or tap the button for the source you wish to send to *Program* output. There is no need to make a Preview row selection first, as you would need to do in the standard (2-row) *Switcher*. The *Background transition* you chose will be applied to display the new source.

9.8.3 DSKS

The two *DSK* control groups at right work just like their siblings in the standard *Switcher*, but it is particularly handy that you need simply click their viewport to show or hide the associated *DSK* layer.

Chapter 10 MONITORING YOUR VIDEO



The word “monitor” comes from the Latin “monēre”, meaning ‘to warn’, but has taken on additional meaning since Roman times. As a verb, these include such connotations as ‘keeping an eye’ on something, and ‘checking continually’. As a noun, we understand it to mean devices that permit one to do just that.

As you would expect, your NewTek system provides extensive and versatile monitoring – just what is needed to control your live productions. Monitoring features can also warn you of conditions that might affect output quality, as well as providing access to adjustments providing quality control and creative alternatives.

Among others you will find *Proc Amp*, *LiveMatte*, and *Edge* controls. (To give some of these the attention they deserve, they are treated individually in Section 7.5 and elsewhere.)

SECTION 10.1 INTERFACE AND MULTIVIEW

Really, the *Live Desktop* provides several multiview monitor displays – one on the *Live Desktop* (a.k.a., the “Interface”), normally comprising its upper third, and the others consisting of fully independent displays presented on secondary monitor outputs on the rear connector panel.

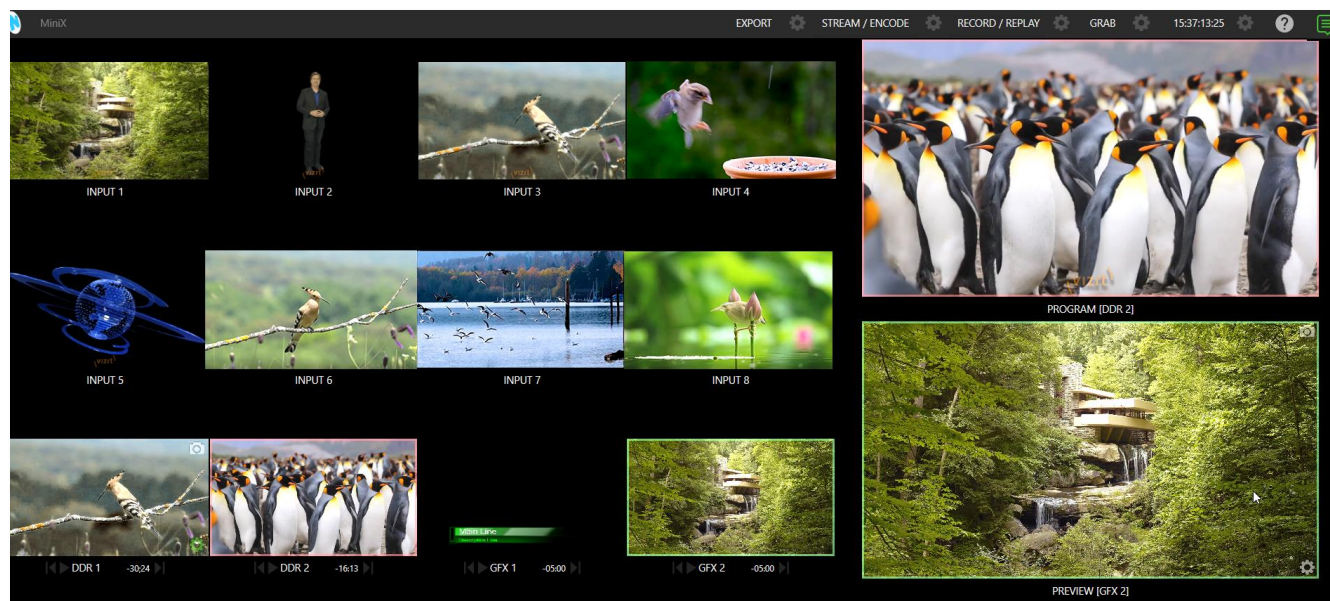


FIGURE 108

SECTION 10.2 LIVE DESKTOP MONITORS

Because the *Live Desktop* provides various control features along with a multiview pane, and controls can consume a greater or lesser amount of the screen, the multiview on the *Live Desktop* is adaptive.

To put this another way, the viewport layout of the *Live Desktop*'s multiview pane re-arranges itself as required to make optimal use of the space available.

For example, when tabbed modules (such as the *DDR*s) at the bottom of the *Live Desktop* are fully expanded, the *Program* and *Preview* viewports above are stacked one above the other. On the other hand, if the tabbed modules are minimized, these viewports are side by side.

Note: To learn how to use NDI KVM with multiviews, see Section 5.2 The Home Page

SECTION 10.3 WORKSPACE PRESETS

The basic layout of all multiview monitoring areas is established by assigning a *Workspace* preset to the screen. To access these presets, move the mouse pointer over the left-hand end of the *Dashboard* at the top of the *Live Desktop* to reveal the *Workspaces* menu.

Four presets, labeled *A-D*, are normally provided for each connected *Multiview* screen. Select a preset and assign a layout to it using the *Load Default* menu point. Continue to customize the display options for the viewports and recall the entire setup at a moment's notice by reselecting it.

To reset a preset, select it again and reload the default layout.

Each viewport in any layout offers diverse input or output source options, as discussed in Section 10.5 Viewport Options. The settings you select for individual monitors will be retained in the current *Workspace* preset.

Let's further consider an important *Workspace* layout option - *Scopes*.

Note: The Display Settings option in the Workspaces menu lets you choose the Resolution of connected monitors and apply Proc Amp settings to them. Select the native resolution of external display devices for best results. Changing Resolution can cause frames to be dropped, so modifications during live production are discouraged.

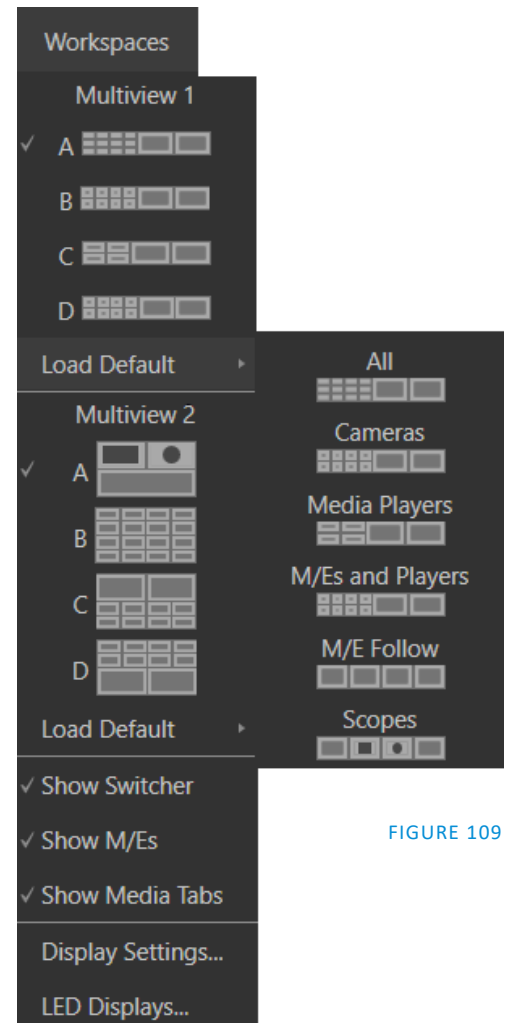


FIGURE 109

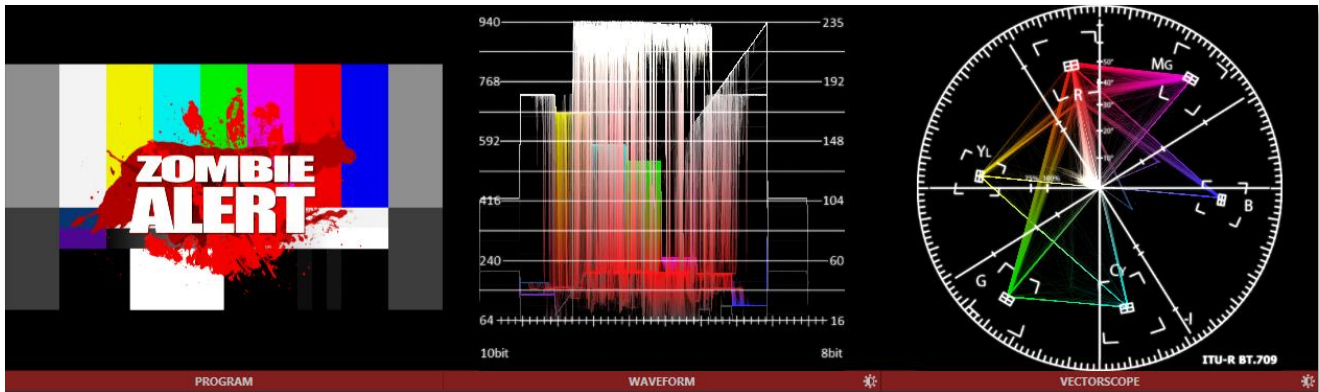


FIGURE 110

Scopes represent a very useful monitoring option, providing *Waveform* and *Vectorscope* displays.

Among other things, these help you to judge *Proc Amp* and *LiveMatte* settings (see Appendix C, Video Calibration). To display the *Scopes* view, select that entry from the *Load Default* options for a *Workspace*.

By default, scopes and the associated monitor refer to the *Look Ahead Preview* composition (including delegated *DSK* channels, or *FTB*), shown at left for reference. Right-click a scope to change the monitored source to the source you wish to view. Adjust the *Brightness* controls in the scope footers, and the trace overlay in full or solid colors using the context menu.

Hint: To use Scopes to calibrate a DSK source independently, first select it as the Preview row source.

Double-click a monitor in this *Workspace* to access *Proc Amp* and *LiveMatte™* controls for the current source.

10.4.1 LED DISPLAYS

The LED lights on your TriCaster Mini X can be configured in the *Lighting Panel*, (the last option under the *Workspace* options). The *Effect* pull down menu offers options such as color cycles, switcher operations, user input (clicks) and more. Once you have decided on an *Effect*, the *Options* menu provides an array of colors to choose from. The *Tool Tip* will describe what changes will incur.

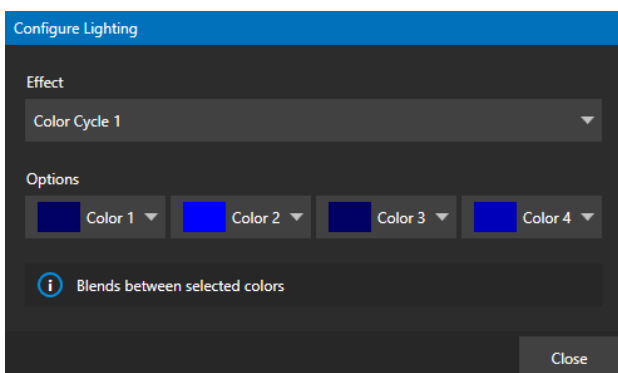


FIGURE 111

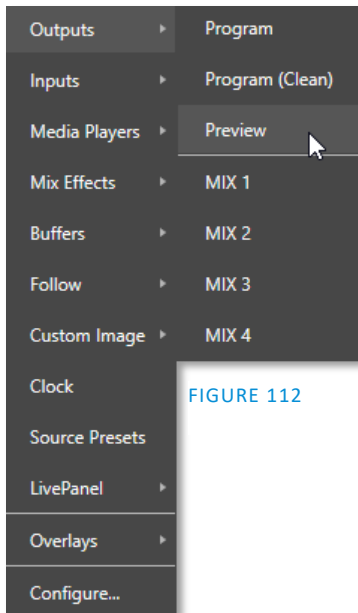


FIGURE 112

Right clicking an individual monitor viewport also opens a context menu. The first option group governs the monitor's source.

In addition to external video inputs (including *Network* sources), you can assign the output of *Media Players*; *M/Es*; main *Program* output (including *DSK* channels and effects); *Program (Clean)* without; the main *Switcher*'s look ahead *Preview*; another *Preview* showing *M/E source B* output (*Mix mode* only); or any *Output*.

Several special displays are also available. Selecting the *Clock* item replaces the video display with *Event Clocks* showing current time along with countdown style *Start* and *End* clocks. The *Custom Image* feature lets you display a network logo.

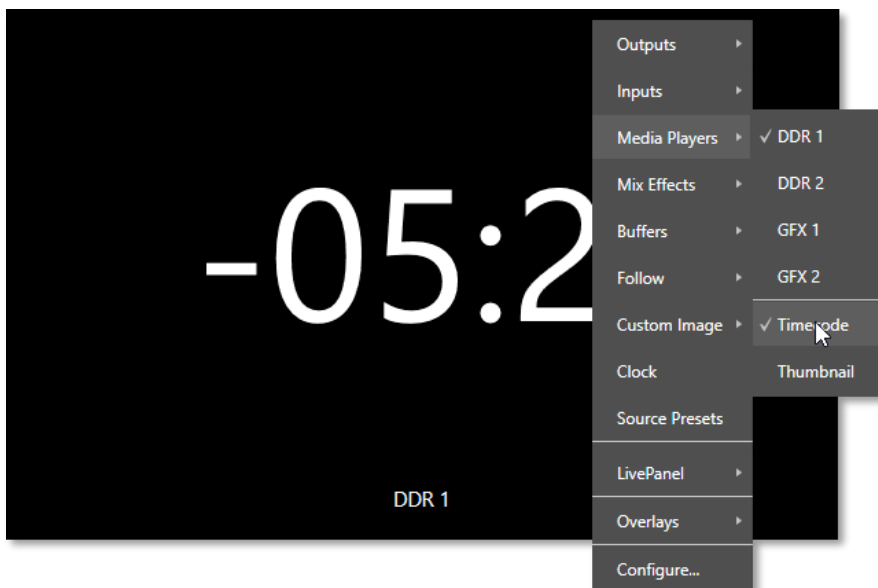


FIGURE 113

A further display you may find useful is the *Timecode Only* option that appears at the bottom of the *Media Players* sub-menu. This displays a time counter showing the current position of the playhead in the *Media Player* (respecting the specified *Player's Warning Colors* option, too).

Below the basic source selection options in the menu, you'll find another group that provides access to optional *Overlays* for the monitor port.

Available overlays include *Checkerboard* (shows a checked pattern wherever transparency exists in the source); *Title Safe* (note this is a 4:3 format safe margin); *4:3 Safe* (a true 4:3 frame edge boundary), *Center Cross* to help with alignment, and *Show Illegal* and *Show Alpha*.

In addition, you may choose to display *VU Meters* with the source.

Long-standing practice when using color bars to calibrate video signal color attributes requires special video monitors with 'blue only' (or 'blue-gun') displays. The *Blue Only* viewport option means you can now use any color monitor for this purpose.

The final *Overlay* option flips the images horizontally, useful for a variety of studio requirements such as talent orientation in greenscreen applications.

Note: Certain Overlay menu options are not suitable for all monitor sources hence are not listed when inappropriate or may not appear on all models.

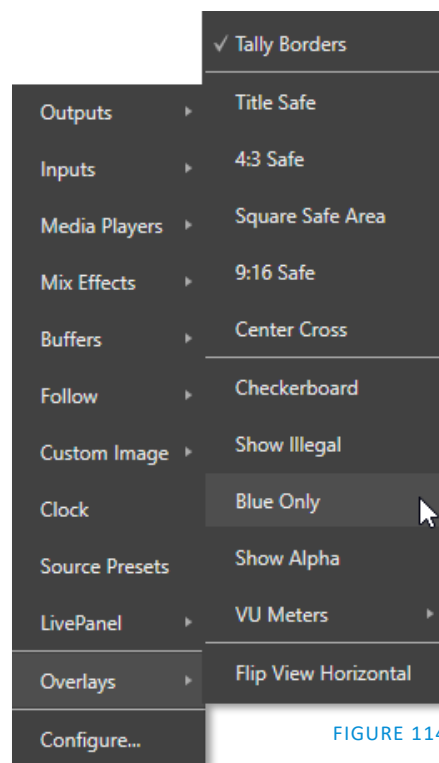


FIGURE 114

10.5.1 VIEWPORT PRESETS

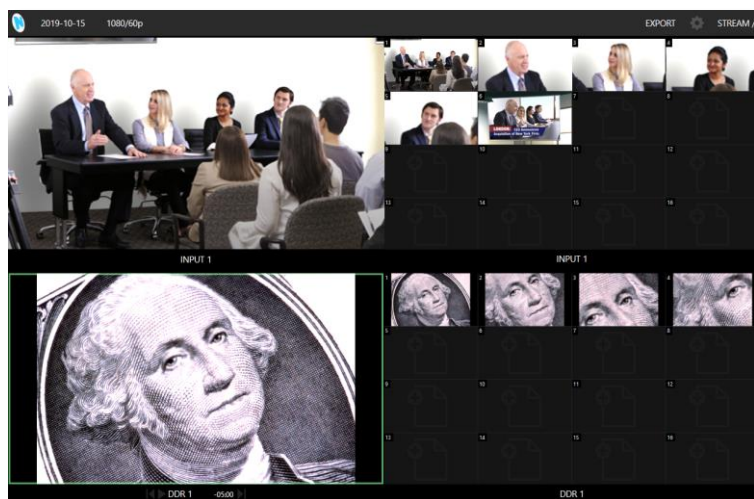


FIGURE 115

From corporate video to reality TV, PTZ cameras are playing an expanded role in modern production. TriCaster gives top billing to your PTZ presets right in its Live Desktop or any connected Multiview.

The viewport context menu option *Source Presets* lets you assign preset controls to any viewport. Large, colorful thumbnail icons represent PTZ presets for robotic cameras, *Pan and Scan* presets (a.k.a. 'virtual PTZ', requires Advanced Features), and *M/E Comps* (or zoom presets on systems lacking Comp support.)

With a click, or better yet a tap on a touchscreen, your source will gracefully move to its new position (*Pan and Scan* sources can also be set to Cut). Using TriCaster's Multiview workspace options you can create custom layouts providing one-click access to presets for multiple sources of different types.

Hint: Double-click a viewport to quickly open Input Configuration without using the gear or menu.

10.5.2 LIVEPANEL PRESET BUTTONS

In addition to *Source Presets* another option is to assign macros with *LivePanel*, which allows you to preset and edit macro buttons within any viewport with *LivePanel Buttons*.

The configure panel allows you to change the size and color of the preset buttons, the font style, macro selection and the option of initiating a *Two-State Toggle Button* (creating a turn on/off option). See the sub-heading *LivePanel Buttons* in Section 16.1.3 for more information.

One final item completes the monitor viewport options, *configure* opens the *Input Configuration* panel for the corresponding source.

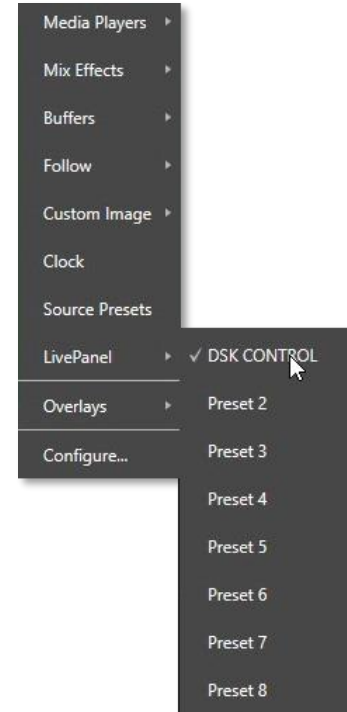


FIGURE 116

SECTION 10.6 VIEWPORT TOOLS

Viewports provide additional features when you move your mouse pointer over them.



FIGURE 117



FIGURE 118

As shown in Figure 117, when the *VU meter* overlay option is enabled, a volume knob (similar to those in the *Audio Mixer*) is shown. Drag this knob up or down to modify the audio source associated with this input.

Hint: When using a touchscreen with the Click Viewports to Show on PGM option enabled, swiping the monitor horizontally (rather than tapping it) will temporarily display viewport overlays.

Clicking the *Configuration (gear)* icon for a *Switcher* input, a *Preview* monitor, or *Program*, opens the corresponding *Configuration panel*. Notice, too, that a *snapshot* icon appears in the label below most viewports. Click this to quickly grab a still image (images grabbed in this manner will obey the *Add grabs to* option in the *Grab Configuration* panel; see Section 19.3).

The viewports for *Media Players* (Figure 118) show handy transport control in the label area – from left to right *Previous*, *Play/Stop*, and *Next*. In addition, a progress gauge is shown in the background to show the playhead position and warning colors as playback nears its end.

Hint: When LiveMatte, Proc Amp, or Crop settings are active for a source, the configuration gear gadget for a viewport is color coded and shown full-time.

SECTION 10.7 PROGRAM MONITOR

The *Program Output* monitor could hardly be more important, hence it's prominent default location at upper-right on the *Desktop*.

Normally, this viewport shows what the *Switcher* is sending to *Program* output at any moment. The display includes the *BKGD* video layer as well as any other video layers (such as one or more *DSKs*, or *FTB*) displayed above it.

Note: By default, Program output is sent to the MIX 1 output, subject to assignments made in the Output Configuration pane. When you move your mouse pointer over the Program monitor, a Configure button (gear) appears at lower right. Clicking this button opens the Configuration panel (see Section 8.2).

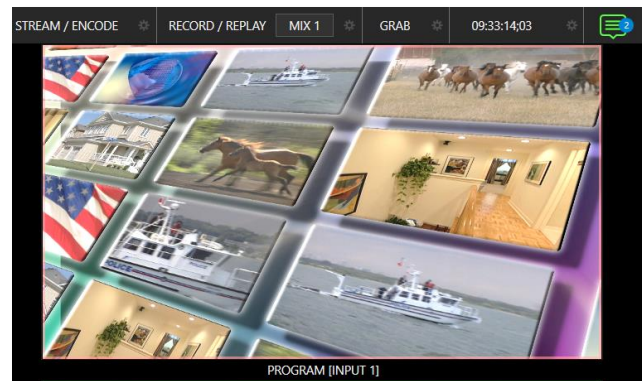


FIGURE 119

SECTION 10.8 LOOK AHEAD PREVIEW

Again, by default, the *Live Desktop* also prominently displays the *Look Ahead Preview* monitor (labeled simply *Preview*).

The *Look Ahead Preview* is versatile and powerful. Instead of showing just one video source (the *Preview* row selection), it displays what the outcome of a *Take* operation applied to all currently delegated video layers would be.

- Delegating the *BKGD* layer indicates you intend the next transition to swap the *Program* and *Preview* layers. The *Preview* monitor will show the *Preview* row selection as its background.
- When the *BKGD* is not delegated, that layer will not change during a transition. Consequently, in this case the *Preview* and *Program* monitors share identical backgrounds.
- Delegated *DSK* layers are shown above the background on *Preview* according to their current state. That is, if a delegated *DSK* layer is currently displayed on *Program* output it does not appear on *Preview* – since the next transition would remove it.



FIGURE 120

In other words, 'what you see is what you *will* get' – after performing the next *Take* or *Auto* operation as currently configured. This lets you set up the next shot, check its composition (including titles and overlay positioning), and switch to it with sublime confidence.

Chapter 11 MEDIA PLAYERS & BUFFERS



Media Players permit you to integrate video, stills and title pages into your live presentation. Media Players can play, stop, and advance automatically in response to Switcher operations. Animated buffer effects are perfect for many similar purposes, freeing up Media Players to be used for long form playback.

SECTION 11.1 MEDIA PLAYERS

NewTek live production systems includes a bevy of integrated *Media Players*, providing savings, convenience, and opportunities for automation. *Media Player* controls can be accessed in large, tabbed panes found in the bottom third of the *Live Desktop*, or by in tabs with similar features that appear in the *Input Configuration* panels of *Media Players*.

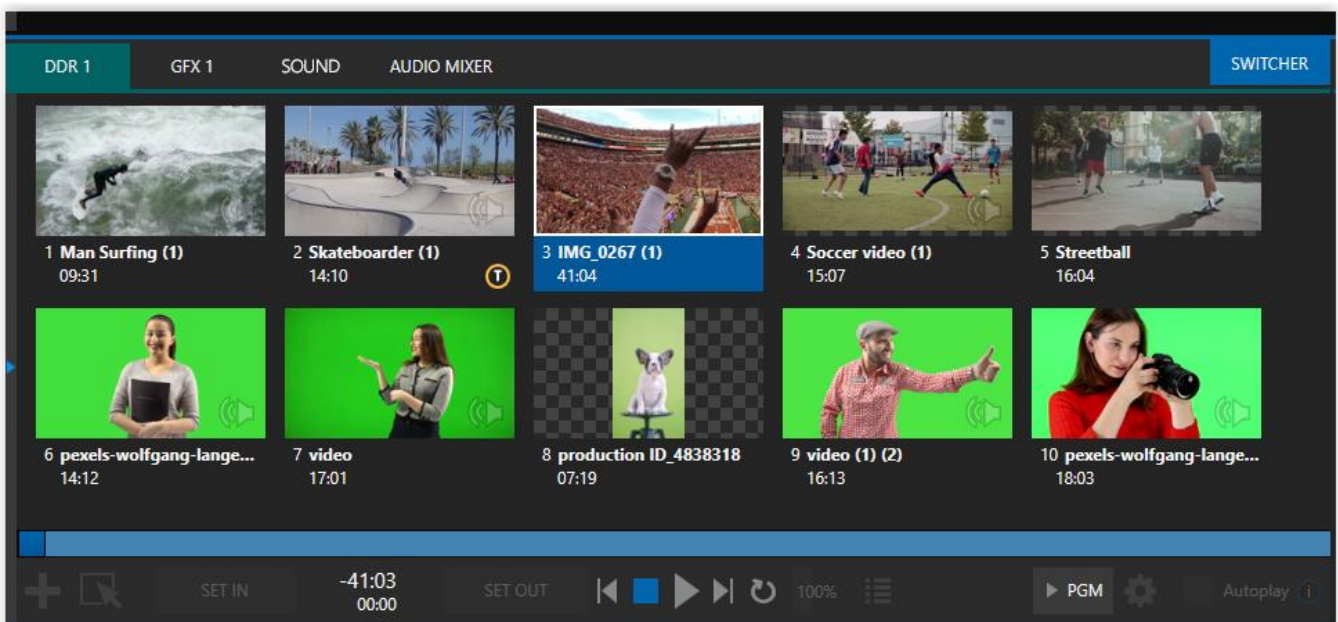


FIGURE 121

TriCaster Mini X provides two *DDR*s (clip players) and two *GFX* (graphics) players in addition to Sound and Audio Mixer.

Media Players can handle numerous media types:

- *DDR*s can display:
 - *Video clips* – including effects, motion titles such as scrolls, transitions, LiveGFX support and longer format movies.

- *Titles and still images* – editable title pages prepared for use in NewTek’s native titling system, CG pages prepared as image files, or other still imagery – photos, graphics, etc. (See also Chapter 20, Title Templates, for information on creating custom title pages.)
- Both *Sound* players and *DDR*s can play standard .wav or .mp3 format audio files.

Hint: DDR is short for “Digital Disk Recorder”, a legacy term that – while deeply enshrined as a term referring to a playback system – is technically incorrect since these DDRs don’t directly support recording.

11.1.1 PLAYLISTS

The dominant feature of all *Media Players* is a storyboard-style playlist, used to organize content for use during live productions.

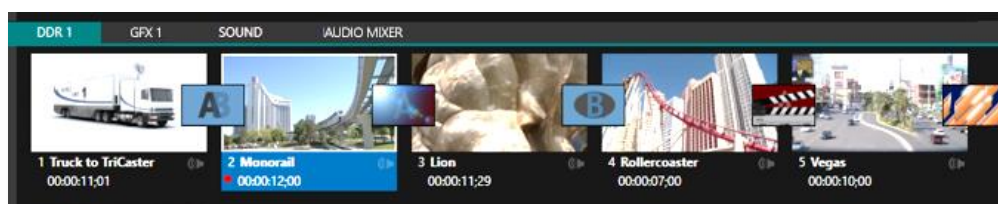


FIGURE 122

This arrangement offers easily visible thumbnail icons for each entry. A scrollbar at right accommodates long playlists when necessary. Icons in the playlist can be re-ordered quickly and easily using the familiar drag and drop workflow. The playlist can even be altered during playback (of course, if the currently playing item is removed, playback stops immediately).

An *Alias* (by default, the item’s filename) is displayed below the icon, along with the item’s duration.

Hint: The duration shown is the play time after trimming, and thus may be less than the file length on disk. In such cases, the In and Out markers on the Scrub Bar (spanning the width of the playlist immediately below it) show the effect of trimming operations, while the full width of the Scrub Bar depicts the total length of the file on disk.

Clicking an item selects it. The play position is automatically set to the *In Point* on selection. Standard *Shift + click* and *Ctrl + click* multi-selection operations are supported, and all selected items are denoted by a white border around their icon.

When a *Media Player* is stopped, its output to video monitors tracks selection and trimming operations. The current item is the *Media Player’s* output, and in consequence appears on monitors dedicated to the player, if any. Of course, only one item can be displayed on output at a time; the frame surrounding the thumbnail icon for the

Automatic Clip Trimming

Clips (but, for reasons that will become apparent, not still images or titles) in a playlist are automatically trimmed to accommodate the addition of transitions between clips when necessary.

(Otherwise, i.e., if there are no ‘unused’ frames to display during a transition, motion will appear ‘frozen’ during the transition, which is usually undesirable.)

To reset a clip to its full file-length boundaries, press the ‘g key (multi-selection is supported, too).

currently displayed item is illuminated. Double-clicking a thumbnail (or clicking the *Play* button) begins playback from the *In Point* of the current file.

Hint: Double-clicking elsewhere in the playlist pane opens the Media Browser (hold down shift when clicking to open a system file explorer instead of the custom Media Browser).

Note that even still images and static title pages added to the playlist are given a play duration. The default duration for these items is five seconds. Duration can be adjusted on an item-by-item basis (or as noted earlier, en masse for multi-selected stills or title pages).

During playback, the footer of playlist items illuminates as each is played in turn. When necessary, the playlist pane scrolls to display the icon for the currently playing item. A progress bar is displayed beneath the currently playing thumbnail, and the *Scrub-Bar* knob also tracks playback progress.

Hint: Selection status is independent of which clip is playing. Selected items have a white border in the Playlist pane.

11.1.2 FILE OPERATIONS

- Click the large + (*Add Media*) button at left beneath the playlist pane to open a custom *Media Browser* (see Section 11.1.8). Alternatively, double-click in an empty part of the *Playlist* pane.

Note: Add Media also supports compatible third-party asset management systems. Hold down the keyboard Ctrl key when clicking the + sign to access these tools.

- Newly added files become selected items in the *Playlist* pane.
- Drag (appropriate type) file(s) from one module's playlist to another module.
- Right-click in the playlist pane to show a menu with context-relevant items from the following list (operations affect selected playlist items):
 - *Cut*
 - *Copy*
 - *Paste*
 - *Remove*
 - *Split at Current Frame*
 -
 - *Audio Level* (clips with sound and audio icons) – note that the default 'per-clip' audio level applied to imported media files can be set in the Dashboard *Options* menu.
 - *Speed*
 - *Use Current Frame as Icon*
 - *Macros*
 - *Properties*
 -
 - *Transcode*
 - *Add to Export Media* – see Section 18.3.
 - *Send to > Buffer n* (Still image and title icons – see Section 11.3)
- Standard *Cut*, *Copy*, *Paste* and *Delete* keystrokes are supported for playlist entries.

- Un-playable (missing, corrupt, or unsupported) file icons are dimmed.

Hint: Multi-selection is supported for most operations, including Set Duration (applies to Still and Title only).

Most of the clip context menu items are self-explanatory, but let's talk about a couple of them in just a bit more detail.

11.1.3 SPEED

The playlist context menu item *Speed* permits you to give each clip a custom playback speed. It's important to realize that this *Speed* setting is separate from the Media Player's primary *Speed* control, located in the footer, below the *scrub bar*. The latter setting applies to all playlist items.

Both Speed settings are applied during playback. So, for example, if you play a clip with both Speed values set to 50%, the actual playback rate will be just 25%.

11.1.4 MACRO TRIGGERS

We've discussed elsewhere in this Guide how you can assign macros to any *Switcher* source, including *Media Player*. Beyond this, though, the playlist context menu item *Macro Triggers* allows every item in a playlist – every clip, still image, audio file or title page – to control its own unique macros.

- Any macro you can record or create can be triggered automatically on either playback or end of play for any and every individual playlist item.
- Multi-selection support in the playlist makes it a breeze to assign macros to multiple items.

Hint: You can use this capability, for example, to automatically show titles for certain types of clips and not others, give them different title page types, selectively adjust Proc Amps or apply LiveMatte keying automatically when needed for only certain items.

11.1.5 TRANSCODE

When clips are added to the *DDR* playlist, their suitability for real-time playback is evaluated. When appropriate, a small (T) icon is added to the entry's label, indicating that the item can be transcoded to a more suitable format.

To transcode the clip, select the *Transcode* option in the clip's context menu. The process will proceed in the background, and the *DDR*'s link to the original file will automatically be replaced when it is complete. Note that transcoding operations support multi-selection.

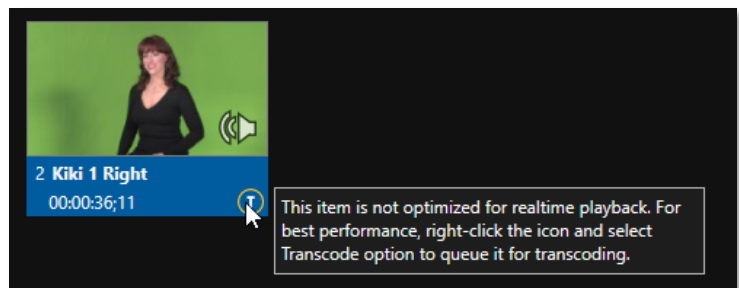


FIGURE 123

Note: Any time a clip is Transcoded, a new 'Transcoded' folder is added beside the original file. If you move a folder with the original clips and the Transcoded files to a new location, you will not need to transcode them again

11.1.6 PROPERTIES

DISPLAY NAME

The *Properties* panel allows you to edit the *Display Name* of a file in the playlist.

The *Display Name* fields defaults to the filename, but is a local alias, or 'nickname'.

Thus, editing the name does not change the name of the file on your hard drive. Roll the mouse over the icon *display name* to see the true filename and its path.

COMMENT

The *Comment* permits you to supply metadata with the file.

Hint: Both the Display Name and Comment box content for the selected clip are available as DataLink keys, which can in turn be served to live title pages or supplied along with the file to the Publish module.

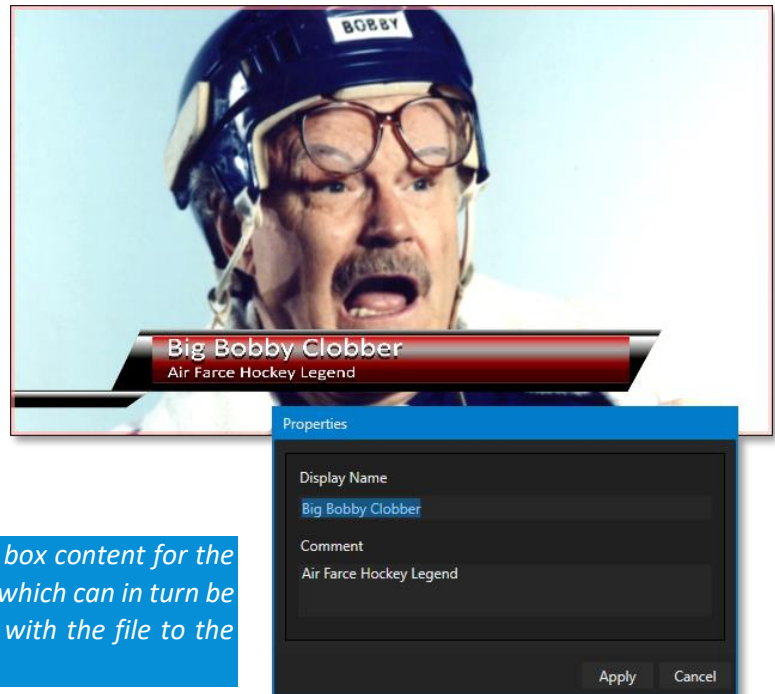


FIGURE 124

11.1.7 TRIMMING CLIPS AND THE SCRUB BAR

Just beneath the *playlist* pane is a full-width *Scrub-Bar*. The width of the *scrub-bar* represents the full run time of the current clip or other media file. Drag the knob to change time position.



FIGURE 125

In and *Out Points* for newly added Media are automatically adjusted when necessary to accommodate motion when transitions you add require it.

- To reset the file to its full length, press the 'g' key.
- To trim a clip manually, move the knob to the desired frame and press 'i' or 'o' on the keyboard (to set the *In Point* or *Out Point*, respectively).

During playback, the *Scrub-Bar* knob traverses the span between the *In* and *Out Points*. The duration (taking into account trimming operations) is displayed as a countdown in the upper of two timecode fields at left (Figure 125). The lower time display shows embedded clip timecode.

Drag the knob to move quickly backward or forward in the clip. Audio is normally muted during scrubbing; hold down CTRL to unmute it. Also note that holding SHIFT while dragging the scrub bar knob increases precision.



FIGURE 126

It's useful to note that (Quicktime only) clips that are still being captured continue to 'grow' even after addition to a DDR playlist (see Section 11.1.8). These growing clips, featuring the red 'recording' overlay (Figure 126) can be freely extended beyond the bounds of the 'in' and 'out' points they were given when initially added.

It can be very useful to recall that you can easily split clips using either the corresponding context menu item, or the "/" key.

ALPHA CHANNEL SUPPORT

For files with transparency such as 32bit image files, use non-premultiplied (or 'straight') alpha channels in *Media Players* (premultiplied files will not give correct results when overlaid on other imagery).

Hint: Monitors can optionally show a checkerboard pattern behind transparent content.

Time-Shifting

By the way, it's possible to use growing clips to 'time shift' your program:

- Initiate Recording.
- Add the captured clip to a DDR
- Press "g" on the keyboard to clear its *Out Point*.
- Trim the *In Point* to taste.
- Assign the *DDR* to an *output*.
- Wait some length of time and then begin playing the clip.
- (Of course, all of this could easily be performed by a macro, too.)

11.1.8 Media Browser

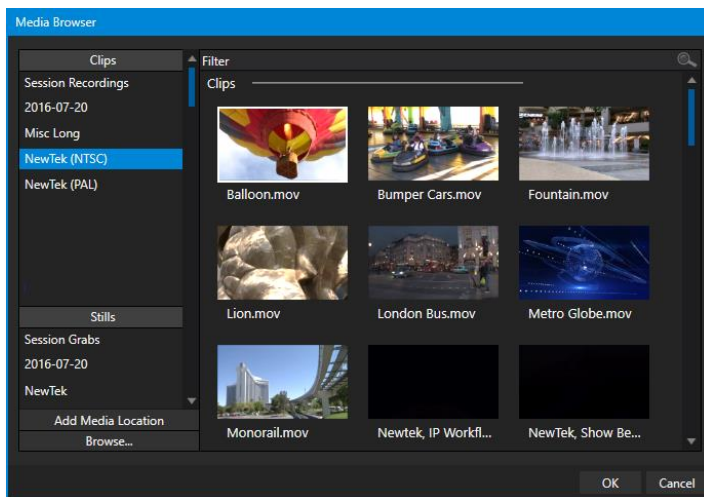


FIGURE 127

The custom *Media Browser* provides easy navigation and selection of content on your NewTek live production system or on the local network.

The *Media Browser* appears anywhere in the *Live Desktop* that you might wish to select content, transitions, or effects for use in your project (such as the *Media Players*, *LiveSet* and *Transitions* sections).

Its layout is principally comprised of two panes on the left and right that we'll refer to as the *Location List* and *File Pane*.

LOCATION LIST

The *Location List* is a column of favorite “locations”, grouped under headings such as LiveSets, Clips, Titles, Stills, and so on.

SESSION AND RECENT LOCATIONS

The *Media Browser* is context sensitive, so the headings shown are generally appropriate for the purpose for which they were opened.

A list of sub-headings appears under these main headings in the *Location List*. These may correspond to named sessions, or groups of content. When you select a sub-heading, the right-hand pane – the *File Pane* – is populated.

In addition to locations named for your stored sessions, the *Location List* includes two notable special entries. The *Recent* location provides quick access to newly captured or imported files, saving you time hunting through a hierarchy to find them. The *Session* location (named for the current session) shows you all files captured in the current session.

Hint: Clips that are being actively captured are marked with a red ‘record’ overlay. These clips continue to ‘grow’ and can be re-trimmed after their addition to DDR playlists or the Publish Queue.

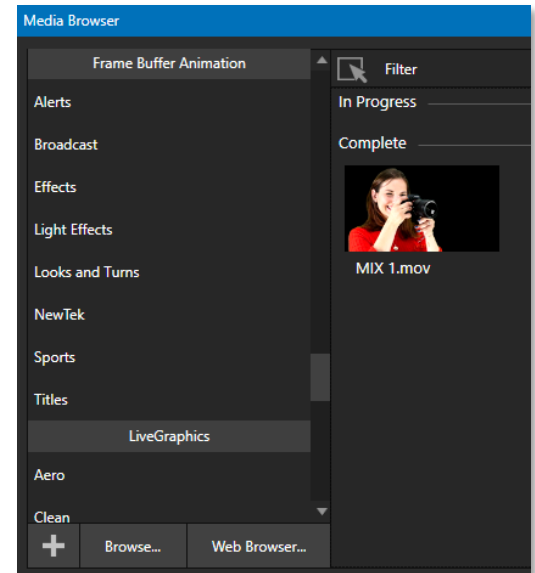


FIGURE 128

ADD MEDIA LOCATION & BROWSE

Clicking *Browse* opens a standard system file explorer, rather than the custom *Media Browser*.

Hint: To jump to the standard system file explorer (rather than the custom Media Browser) from a Media Player, hold keyboard Shift while clicking the Add button.

FILE PANE

Icons appearing in the *File Pane* represent content located inside the sub-heading selected at left in the *Locations List*. These are grouped under horizontal dividers named for sub-folders, which allows related content to be organized conveniently.

FILE FILTERS

The *File Pane* view is filtered to show only relevant content. For example, when selecting *LiveSets*, the browser only shows *LiveSet* files (.vsfx).

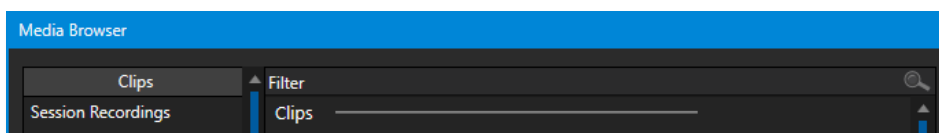


FIGURE 129

An additional filter appears above the *File Pane* (Figure 129). This filter quickly locates files matching criteria you enter, doing so even as you type. For example, if you enter “wav” into the filter field, the *File Pane* displays all content at the current location with that string as part of its filename. This would include any file with the extension “.wav” (WAVE audio file format), but also “wavingman.jpg” or “lightwave_render.avi”.

FILE CONTEXT MENU

Right-click on a file icon in the *right-hand* pane to show a menu providing *Rename* and *Delete* options. Be aware that *Delete* really does remove content from your hard drive. This menu is not shown if the item clicked is write-protected.

11.1.9 Player Controls

Transport controls and playback settings are located directly below the *Playlist* and *Scrub-Bar*.



FIGURE 130

TIME DISPLAY

During playback, the uppermost timecode field beside the transport controls displays the current countdown time for active playlist item or for the entire playlist when the *Playlist* mode button is enabled. (As mentioned earlier, the time show below is the embedded clip timecode.) Left click the field to type in a timecode, then press *Enter* to jump to that point in the file (*or* playlist).

Hint: The time display and scrub bar color provide visual indication that the playback is nearing its end. Ten seconds before the end of play for the current item, the digits in the time display and the scrub bar background turn amber. With only five seconds left, the color changes to red.

PLAYLIST MODE

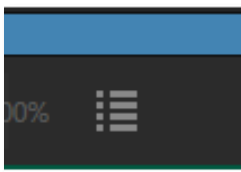


FIGURE 131

Normally, *Media Player* playback stops when the *Out Point* of the current playlist item is reached (unless *Loop* is also enabled, in which case playback of the current item repeats until manually interrupted).

Clicking the *Playlist* mode button (Figure 131) tells the *Media Player* to operate in continuous play mode, advancing through the playlist items until the last one has played.

TRANSPORT CONTROLS

A simple set of controls beneath the playlist pane provides all playback-related functions:

- **Previous* button – go to previous playlist entry
- *Stop* (clicking *Stop* when already stopped goes to the first frame)
- *Play*

- *Next button – go to next playlist entry
- (Not shown) – *Double-click* an entry in the playlist pane to begin playback at the start of that playlist entry

PREVIOUS, NEXT AND PRESENTATIONS

It's extremely useful to realize that, since transitions can be added between playlist items, pressing *Previous* or *Next* likewise employs transitions. Whether playback is underway or stopped, and whether the items are clips or stills, you can use *Previous* and *Next* to transition backward and forward between items.

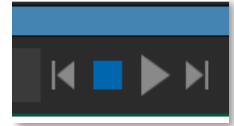


FIGURE 132

This provides a perfect way to manage presentations using *Media Players*, hence our inclination to refer to this as the “Presentation workflow”. With macros (and perhaps MIDI buttons) driving the presentation, this makes it easy to handle control graphics and video for seminars, to update weather graphics, and more.

Hint: In addition, since transitions support transparency, you can queue up a series of titles in a playlist, display it in a DSK, and transition backward and forward between titles interactively simply using Previous and Next.

SPEED

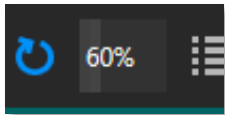


FIGURE 133

The *DDR* supports variable speed playback between 25% and 400% of the normal rate (100%). Speed can even be adjusted *during* playback.

Note: Certain highly compressed video file formats cannot successfully be played back at rates beyond 200%, even though Speed value is set to a higher value.

As is common for numeric input controls in the interface, drag left or right to adjust the *Speed* value, or click the slider to access a direct entry field allowing you to type a value using the keyboard.

Hint: Press Shift while double-clicking Speed to quickly reset to 100% default.

LOOP

Enabling *Loop* repeats playback continuously (respecting the *Single* switch state).

AUTOPLAY

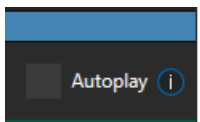


FIGURE 134

When enabled, the *Autoplay* switch has several important effects. First, it initiates playback automatically when the associated *Media Player* is placed on *Program Out* by a (Switcher) *Take* or *Transition* operation either directly (as a *Switcher* row selection) or *indirectly* – such as by being displayed via an *M/E* channel or *DSK* operation.

Second, if player output is displayed on *Program* the reverse *Switcher transition* occurs automatically as the end of play approaches, whether the *Out Point* of the current item for *Single* play or the end of the playlist. (This behavior is optional for *M/Es* – see Options in Section 14.2 and Section 7.1.)

Note: Unlike Switcher transitions, the DSK ‘out’ effect does not occur automatically as the end of play approaches.

In either case, when the *Media Player* is in standard operating mode (as opposed to *Playlist* mode), the current play position advances to the next item after playback is automatically stopped.

11.1.10 SHOW ON (...)

Another extremely powerful feature is nestled just to the left of *Autoplay* in the footer of each *Media Player*.

Labeled ► *PGM* by default this feature can target *M/Es* as well as the *Program* row of the main *Switcher*. It provides unique playback and display abilities that make instant replay a thing of joy, as well as serving other purposes. Click the configuration button (gear) beside the *SHOW ON* button to display the panel seen in Figure 135.

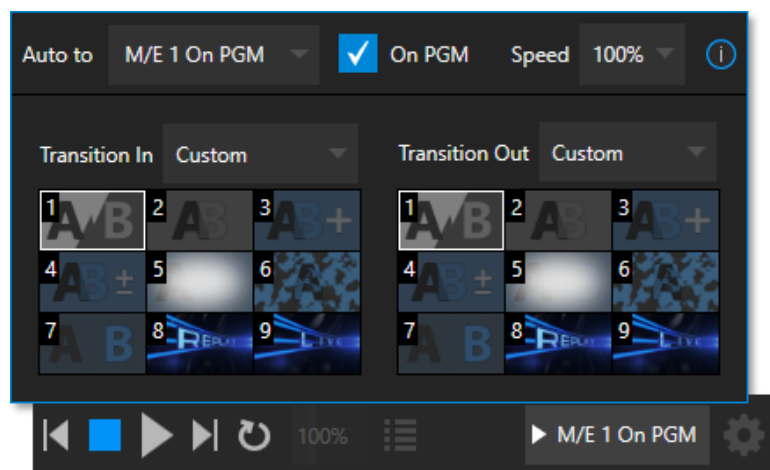


FIGURE 135

The largest part of this panel contains transition bins for custom *In* and *Out* transitions which will be used to display the current clip. Select *Custom* from the *Transition In/Out* menus to activate these bins. Otherwise, you can select *Cut* or *Current* options.

Note that the bin content displayed is synced with the *Background* transition bin for the target video bus, be it *PGM* or a designated *M/E*. Selecting *Current* uses the current selection in the *Background* transition bin of the target bus.

Hint: Unlike other transition bins, these icons do not show a “+” sign gadget on rolling the mouse over them. To replace the effect in slot, modify the content of the Background transition bin of the target you have selected for SHOW ON.

Once configured, clicking the *SHOW ON (...)* button will do the following:

- Swap the current *BG* transition for the main *Switcher* or a designated *M/E* for a custom *Animation Store* (such as "Replay!") or other specified transition.
- Transition the current *DDR* selection in on the main *Switcher*'s *PGM* row, or the *A* row of an *M/E*
- Play the clip (overriding the *DDR*'s own *Autoplay* setting, if necessary)
- Transition back to the original program using a custom transition (such as "Live!") when done
- Swap the background transition back to the original selection

Hint: Using Replay complementary features, SHOW ON functionality can be triggered by simply adding a clip (with custom length and speed defined in the Replay menu) from any source being captured to a DDR. Thus, a single button click or numberpad key press can trigger all of this.

M/E ON PGM

In a variation on this operation, the *On PGM* switch, which is available whenever the target is an *M/E*, provides a different mode of operation. In this case, the output of the Media Player is immediately selected as the top row source of the target *M/E*, and the *M/E* is transitioned in on the main Switcher's PGM (Program) row.

This unique feature allows you to use the main *Switcher* to display the *Media Player* output – including instant replays – with *KEY* layers (such as 'scorebugs') composed above it.

11.1.11 MEMS

Another very useful feature of *Media Players* is the *MEM Bin*, which provides quick and convenient access to stored playlists. *MEM slots* also store the state of various *Media Player* controls.

To display the bin, roll the mouse pointer to the (nearest) side of the screen in a tabbed *Media Player* or the *Audio Mixer* tab (for *Sound* and *Audio*, *MEM slots* are presented on the left side of the screen only).

When you change playlists by selecting another *MEM slot*, it's almost as though you are accessing another *Media Player*. Use *MEM slots* for quick access to different categories of content for use during a live presentation, playlists prepared for different clients, or for completely different programs you produce.

MEM slots can be named and can also display a small representative image (taken from the first entry in the playlist). As you move the mouse over a *MEM slot* icon it expands, providing an enlarged view of that playlist item's first frame. To populate a new *MEM slot*, simply click a blank icon.

To name a *MEM slot*, right-click it to open a context menu, and select *Rename*. Other menu options include *Delete*, *Export* and *Import* (the file will be saved with the extension '.pst').

PREVIEWING MEM SLOTS

At times you may wish to preview the contents of a different *Media Player MEM slot* without disrupting playback of the current playlist. When a clip is playing, a small white 'play indicator' is shown over the icon for the currently selected entry in the fly-out *MEM slot bin*.

If you select a different *MEM slot* while the current clip continues to play, the playlist bin view updates, but the original item continues to play. The transport controls (*Play*, *Stop*, etc.) at the bottom of the tab remain *dedicated to the *active* playlist – that is, the one playing, rather than the one currently being *previewed* in the tab.

To change to the *previewed MEM slot* immediately – ending playback and display of the current item – you can do so by any of the following means:

- Double-click an icon in the *previewed* playlist.

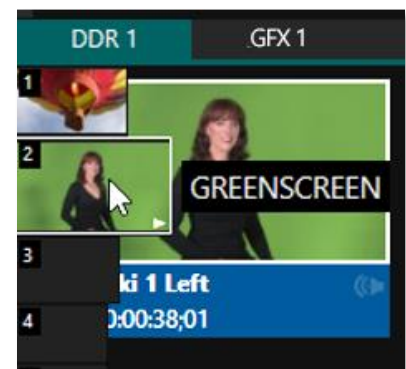


FIGURE 136

- Or press **Stop* twice, and then *Play* (playback of the active item ends with the first *Stop* command; the second sets the playhead to the start of the current item in the new playlist).
- * Normally, pressing *Stop* a second time when a clip is playing resets the playhead to the start frame in the *same clip*. The exception above occurs only when you are previewing the content of a different *MEM slot* during playback.

11.1.12 NETWORK SHARING

Default media file folders (*Clips, Stills, etc.*) are maintained for each session. This approach makes it easy to locate items using the *Custom File Browser* and is also convenient for other file management purposes.

The default media locations for the *active* session can be made accessible across your local network. *Share Media Folders and Buffers* in the *File menu* is on by default and allows network clients to update media in the active session, even while live. The following session media locations are dynamically shared:

- Audio
- Clips
- Stills
- Titles
- Buffers

Note: Several 'non-session dependent' locations in the main application folder are also shared. This allows applications such as NewTek's Virtual Set Editor, etc., running on external systems to export directly to the local system. These network shares are unaffected by the Share Media Folders and Buffers setting.

Transferring very large files across a network can time-consuming and impacts both disk access and network resources.

When these resources are already heavily taxed, dropped frames on streaming or video outputs, or sluggish response to controls may result.

We strongly urge you to gather media assets before starting your live productions whenever possible.

When suitable content is added to these shared folders, the *Media Browser* provides immediate access to it. Simply select the session name (under *Clips, Stills, etc.*) in the *Location List* at left to reveal the updated files in the *File Pane* at right.

This is wonderfully convenient but should be approached with a measure of common sense. Network bandwidth has fixed limits. Be judicious when transferring large files while 'live'. You may be using the network for many purposes – including streaming, or NDI video feeds from a 3Play®, as well as any unrelated traffic on the network.

SECTION 11.2 EDITING TITLE PAGES

On mouse-over, title page icons display a text edit gadget in their lower-left corner (Figure 137).

Click this button (or select *Edit Title* in the icon context menu) to open the pop-up *Title Page Editor*.

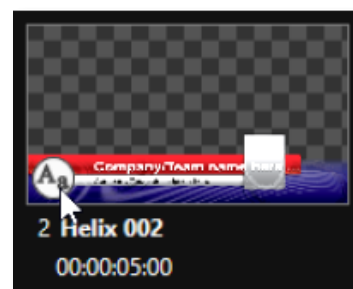


FIGURE 137

Opening the *Title Page Editor* during playback is permitted, so that *Title Page* content can be edited even during display (changes are detected and shown immediately).

Hint: The panel can be re-sized by dragging its lower-right corner and re-positioned by dragging its titlebar.

When you move the mouse over text in the *Title Page Editor's* preview pane, a white bounding-box appears. If you click once inside the box, it turns yellow (Figure 138), indicating the text object is selected, and a text edit field opens.

Hint: When the box is yellow the text can be nudged either a pixel at a time using ALT + the arrow keys or five pixels at a time using SHIFT+ALT+ the arrow keys.

Press *Enter* or click outside the box to complete editing operations or press *Tab* to advance to next entry field (press *Shift + Tab* instead to jump to the prior text field).

Note that the cursor keys – that is, the left/right and up/down arrows – allow you to navigate between text fields on the current title page. (If the text edit box is open, left/right arrows change the edit point as usual, but up/down closes the edit box, and subsequent cursor key actions move to the next object.)

Hint: A red line under a character or word indicates the spell-checker is questioning its spelling. Right click the word to open a menu suggesting alternatives. Click any entries shown if you wish to update the original.

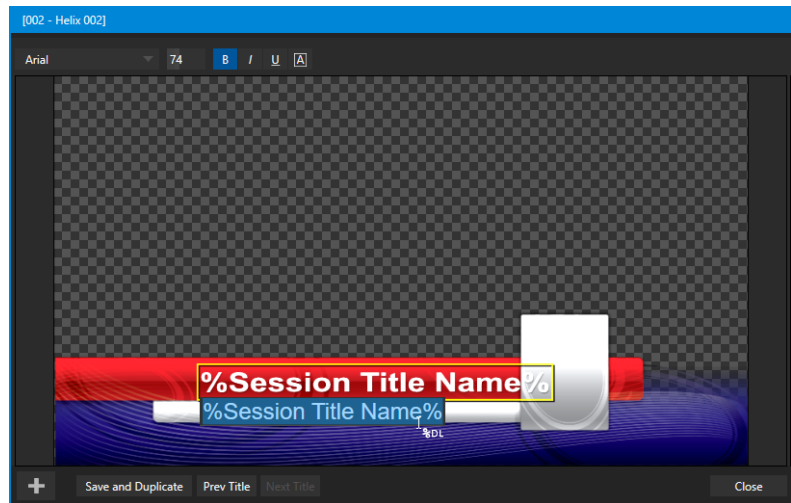


FIGURE 138

11.2.1 HEADER TOOLS

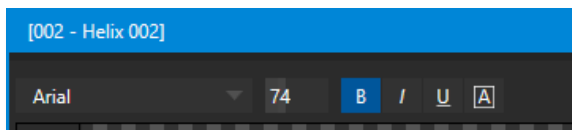


FIGURE 139

The header of the *Title Page Editor* holds an assortment of text attribute controls. These include a *Font* selector menu, numeric *Size* control, and *Bold*, *Italic*, *Underline* and 'ALLCAPS' switches.

11.2.2 FOOTER TOOLS



FIGURE 140

You can click *Close* when finished with the current edit, but often you'll find the *Save and Duplicate* feature handy. Click it to store your edits to the current title page, create a clone of item in the playlist, and load the new page for editing. This is a great way to quickly produce a number of matching pages.

Hint: Press Ctrl + s on the keyboard to perform this 'save and duplicate' operation without using the mouse. Similarly, PageUp and PageDown keys perform Previous and Next operations.

The *Prev* and *Next Title* buttons let you store your changes and move to another title page in the playlist without the tedium of closing the *Title Edit* pane.

STAND-IN IMAGES

Images embedded in *Title Pages* may be locked, or they may be editable stand-ins. When you roll the mouse over an embedded image and a white border is displayed around the image, the image is a *stand-in*. Click a stand-in to open the *Media Browser* and select a replacement image file (hold Shift while clicking to use the system file explorer instead).

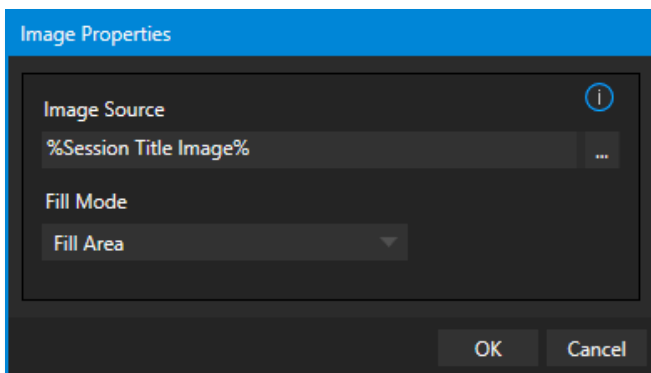


FIGURE 141

If, instead, you right-click a stand-in image, a context menu lets you select one of several optional methods of fitting the source image to its frame. Choosing *Stretch* causes the image to completely fill the frame. *Fill Area* retains the interested image's original aspect, cropping if necessary to fit inside the frame.

Show All Image also retains the original image aspect but fits the entire source image inside the frame (which may result in 'pillar-boxing' or 'letter-boxing'). This menu also allows you to open the *Image Properties* panel. It provides the same set of *Fill Mode* options but adds an *Image Source* box supporting direct entry. The *Image Source* box comes into play in connection with the next (extremely important) topic – *DataLink*.

11.2.3 DATALINK

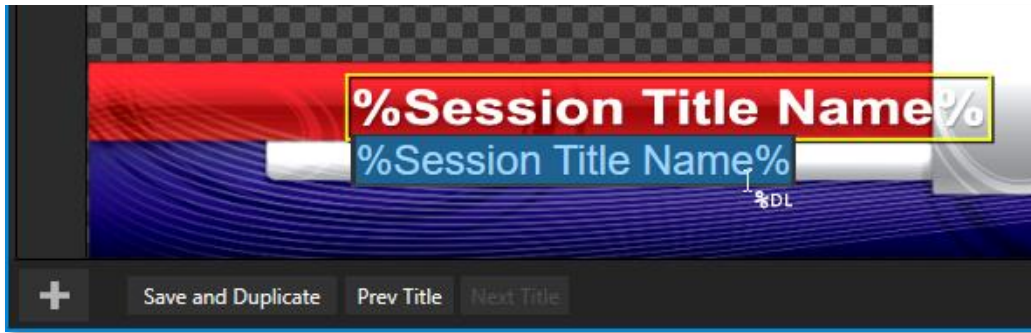


FIGURE 142

Since *DataLink* falls into the general realm of automation and has access to external sources, complete coverage of its features and capabilities is found in the companion *Automation and Integration Guide* included with this product. We'll mention here, though, that both text and images on title pages accept *DataLink* keys as input.

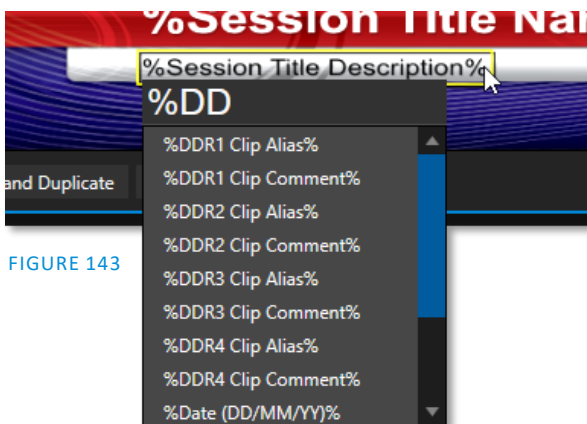


FIGURE 143

The Title Editor tells you which entry boxes accept *DataLink* keys by showing a custom mouse pointer, in the form %DL, as shown in Figure 142.

Making key entry even faster and easier to use, all available *DataLink* keys are shown in the drop-down menu as soon as you enter a % sign into an entry box (*key names* are in the format %*key name*%). If you continue typing, the list shown is filtered to show only relevant key names (Figure 143).

A line of text or image on a title page that has been set to a key name will be automatically replaced by the current value assigned to that key when the page is displayed.

There are endless uses for *DataLink*, and many ways to supply and update values assigned to *DataLink* keys.

For just one example, refer to the heading Configure *DataLink* Keys in Section 5.3.3. And do not fail to take advantage of the power and convenience of the *DataLink*™ web browser extension, which makes it easy to populate your title pages over the network from a web browser running on virtually any platform.

SECTION 11.3 BUFFERS

The system's powerful *Buffer* implementation provides many alternative graphics and animation sources for *M/Es* (including their associated *KEY* channels), the main *Switcher* and its *DSK* channels. *Buffers* are sometimes even more useful than similar imagery supplied from *Media Players*. The tabbed *Buffers* module (Figure 144) shows icons and controls for fifteen *Buffers*.

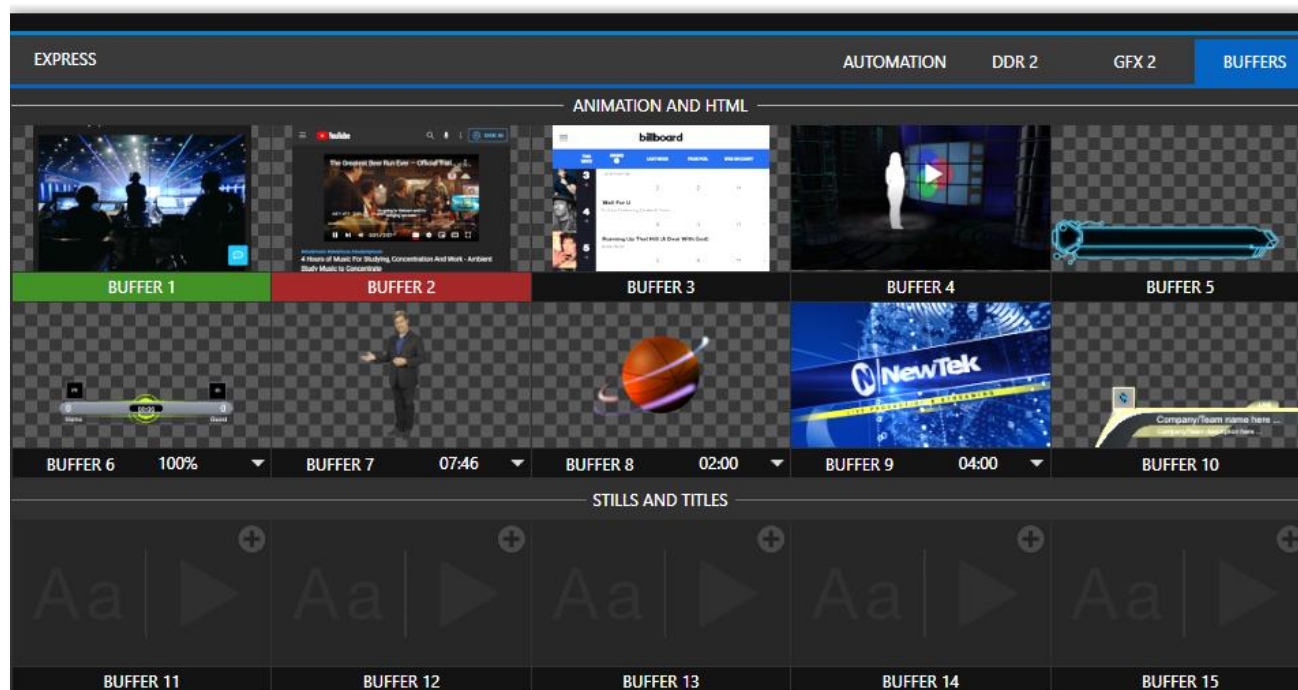


FIGURE 144

Buffers are retained in *Switcher Memory* for immediate recall and display. Using a buffer for a specific *DSK* or *M/E* channel (rather than a *Media Player*) eliminates the risk of accidentally displaying the wrong graphic, as might otherwise happen if the current *Media Player* item selection was not the one intended.

11.3.1 BUFFER TYPES

Buffers support the following media types:

ANIMATION EFFECTS

These are short duration full-motion video effects created in your favorite graphics or video applications and compiled using the add-on *Animation Store Creator* application.

- *Looping* effects – These animations play repeating endlessly, making them ideal for station ID ‘bugs’ and the like.
- *Auto-run* effects – These effects auto-run on display following a *Take* or *Auto*. Playback runs once and then holds the last frame.

GRAPHICS

- *Images* – 24 bit or 32 bit (with embedded alpha channel) images.
- *Title pages* – editable *title pages*, identical to those served up by the various *Media Players*.
- *LiveGraphics™* – animated graphics and titles.

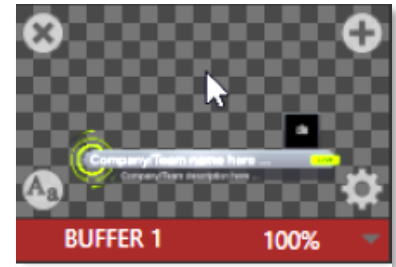


FIGURE 145

HTML BUFFERS

Display a live webpage in your TriCaster Mini X as a switcher source using HTML in a *Buffer* or bring in live graphics with an HTML outputting graphic renders-based website.

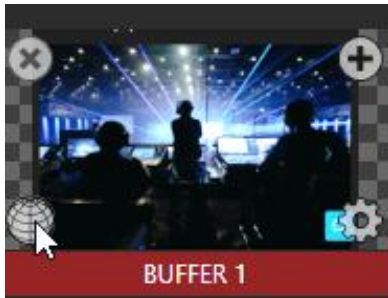


FIGURE 146

- Scaling and cropping are supported using the standard input and key layer configuration tools.
- Control within the webpage over mouse movement/clicks, back, forward, refresh, scrolling and keyboard input.
- Support for web output from CG providers.
- Supports audio, macros, and alpha transparency.

Note: For further information Live Link see Advanced Features: 11.3.5 Live Link

Selection and display of *Buffers* is controlled by buttons on main *Switcher* source rows, *M/E* rows, and similar options in *DSK* and *M/E KEY* source menus. The top ten buffer slots support all animated effect types, titles, graphics, and HTML web pages. The remaining buffer slots are restricted to graphics content, including popular image formats along with *title pages* (i.e., .cgxml files).

11.3.2 SELECTING CONTENT

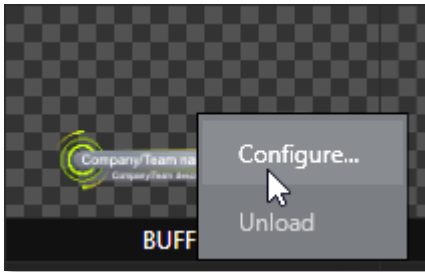
An *Add Media* [+] button appears at upper-right when you move the mouse over a *Buffer* icon. Click this button to display the *Media Browser*, and then pick a suitable file to populate a *Buffer*. Alternatively, drag an item from a *Media Player* onto a *Buffer* slot.

Or right-click a still image or title page icon in a *Media Player* and choose *Send to* from the context menu (select a specific buffer slot using the hierarchical menu). Note that, this *Media Player* menu item only appears for *still image* formats and *title pages*. *Animation effects* must be compiled using the *Animation Store Creator™* add-on application before they can be loaded into buffers.

Note: Generally, Buffers are linked to the original source file on disk. Having added a file to a buffer using the "Send to (Buffer x)" menu, the item can be removed from the Media Player. The buffer link will only fail if the original file is moved or deleted from the hard drive, with one exception: editable title page buffers are fully independent of their original source files.

11.3.3 BUFFERS MENU

Right-click a thumbnail icon in the *Buffers* pane to open a menu offering two items:



- *Configure* opens the standard configuration options, including cropping, color processing, and even *LiveMatte* keying.
- *The Unload* option clears the effect, freeing *Switcher Memory* reserved for effects.

FIGURE 147

When the mouse pointer is over a *Buffer* icon, the small [x] gadget shown in the upper-left corner provides another method of unloading an effect. Likewise, the *Configuration* (gear) icon that appears at lower right opens the *Input Configuration* panel for the *Buffer*.

11.3.4 ANIMATION FEATURES

The thumbnail icon for a buffer with an animated effect created in the *Animation Store Creator* application assigned to it provides several additional features. As mentioned earlier, *animation effects* can be of either *looping* or *auto-run* type (the type of determination is made in the *Animation Store Creator* utility when the effect is generated).

Loading either class of effect into a buffer result in a time control being displayed at right in the buffer label.



FIGURE 148

Hint: Select a Slow, Medium or Fast preset value from the drop-down menu, drag the mouse left or right over the time to change the current value, or click to enter a number directly using the keyboard.

For *auto-run* animation effects, a further control is added to the thumbnail icon when you move the cursor over it. The *Rehearse* button appears at the top-center (Figure 148). Clicking it plays the animation from its first frame through to the last and holds that image.

Hint: In addition to providing a way to preview the effect (by displaying it on the Preview monitor, for example), Rehearse offers a handy way to re-run animated alerts and similar overlays at will. Remember, too, that the Macros feature provides a great deal of useful functionality in connection with Buffers and their display.

Still image *Buffers* have another very valuable ability, discussed next.

11.3.5 LIVE LINK

The *Live Link* feature brings the power of the internet directly into a video switcher. Web page elements can be pulled into any production without a third party application.

HTML CEF WEB BUFFERS

The first 10 animation buffers in the TC Mini X support HTML/WEB Buffers supporting macros, alpha transparency for HTML5 and audio. To begin, click the Buffer's + (add) icon to add or replace a *Buffer* slot source (Figure 149).

This opens the *Media Browser*, click the *Web Browser* button as shown in Figure 150, to open an integrated browser. Enter the webpage (URL) to assign to the specified *Buffer* slot and click ok. It may take a few seconds for the thumbnail to update.



FIGURE 149

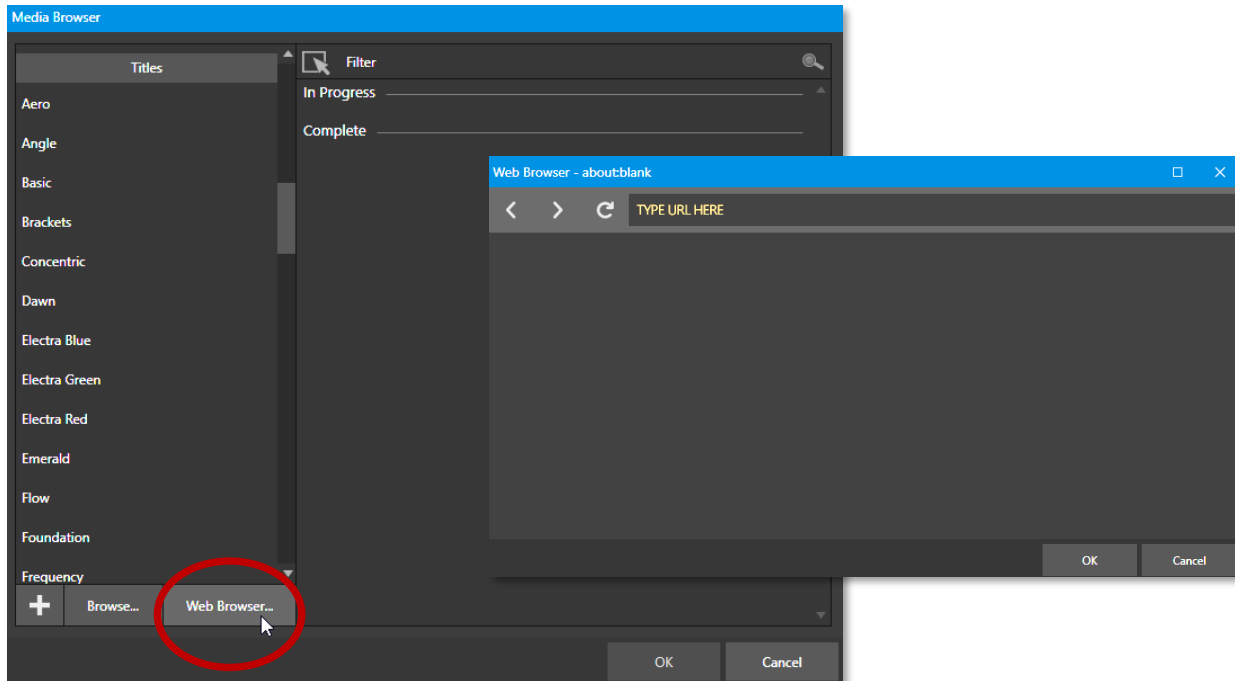


FIGURE 150

Note: You can simply type the name of the webpage and the browser will automatically enter the protocol of the URL.

Another option to assign a web page to a *Buffer* is to create an 'internet shortcut' url file by dragging a link from a web browser onto a file folder, much like a shortcut a browser creates. Open the *Media Browser*, select browse and find your internet shortcut file, click open.

When the web page is displaying in your buffer, hover over the lower left of the buffer's thumbnail and the *Web Icon* will appear (Figure 151).

The *Web Browser* can be used to preview the web page. Clicking the web icon will open the *Web Browser* window (Figure 152) allowing you to make changes to the web page. Interaction includes mouse movement/clicks, back/forward, scrolling and keyboard input. The URL being viewed can also be changed via the address field. Once changes are made, clicking OK will close the *Web Browser* window and apply the URL to the *Buffer*.



FIGURE 151

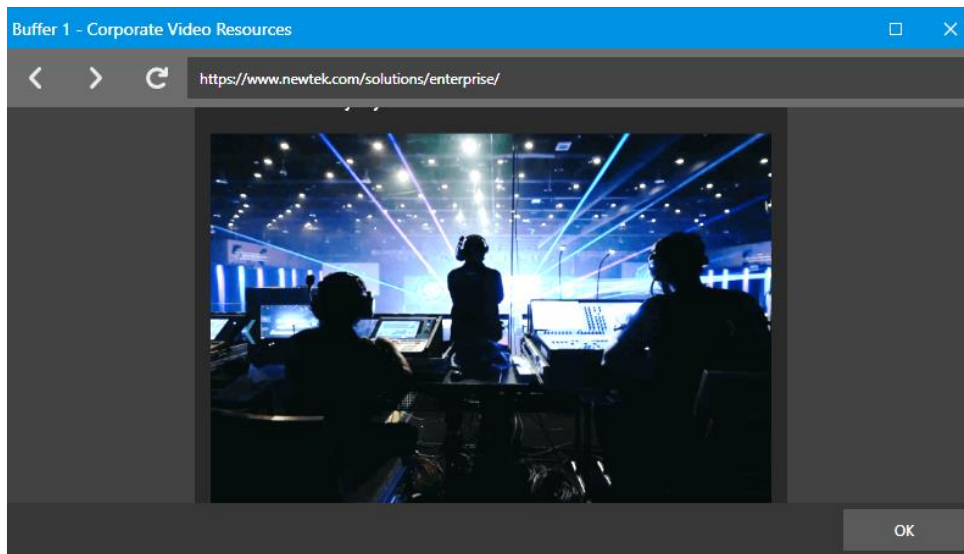


FIGURE 152

Once you have delegated the buffer to *Preview* or *Program*, you may want to adjust the appearance of the webpage. Many configuration options are available, simply hoover over the buffer thumbnail in the DSK channel and a gear will appear on the lower right of the buffer (Figure 153) click to open the *Buffer Configuration* panel and click on the DSK tab.

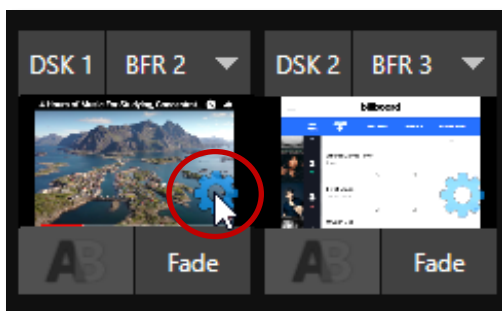


FIGURE 153

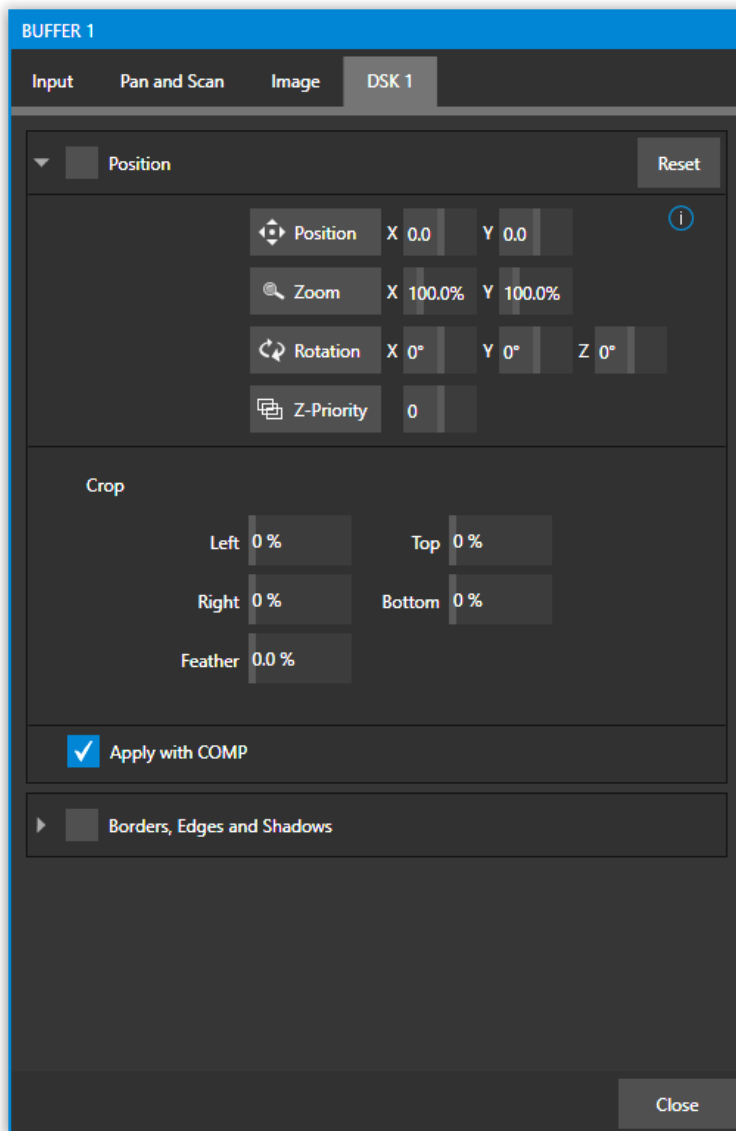


FIGURE 154

PERSISTENT THUMBNAIL

When a *Buffer* is assigned to a webpage a thumbnail is created, however it may not always be representative of the webpage. The *Update Icon* feature (found by right clicking on an HTML buffer thumbnail) will create a new 'persistent' thumbnail for the buffer. The thumbnail can be updated repeatedly using this feature. Clearing a *Buffer*, or changing it to another type of content will delete the persistent thumbnail file. If the *Update Icon* feature is not used, the existing post-page dynamic thumbnail will be displayed.

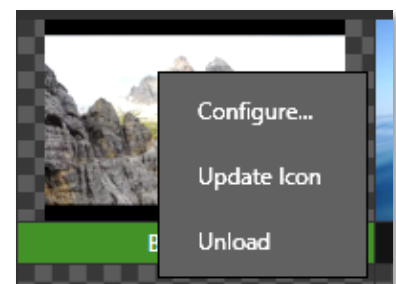


FIGURE 155

The *Position* control group includes *Position*, *Zoom*, *Rotation*, and *Priority*. *Position* settings can be toggled on and off together using the switch provided in the group header. The *Crop* controls are similar to those found in the *Input Tab*. For a deeper dive into all of the configuration options, see DSK Source Configuration in Section 9.6 Transitions and Effects.

Note: When scrolling in the Web Browser you may notice it doesn't move as smoothly as the output of the switcher, this is due to prioritizing the image on the switcher.

CG BUFFERS

Bring in professional live graphics into your production with an HTML outputting graphic renders-based platform from a CG website. By running the CG site on a separate system, simply copy the URL into an HTML buffer *Web Browser* URL. (Here we are using Singular.Live in this example, but you can find multiple CG websites online).

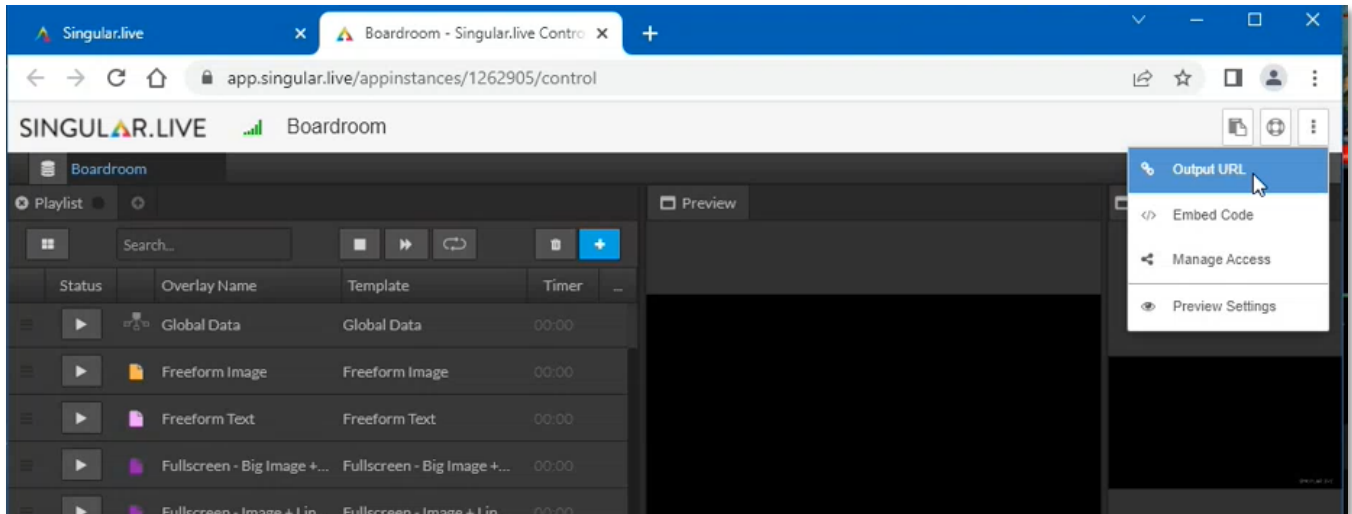


FIGURE 156

Changes made on the CG webpage (on a separate machine) will appear on the TriCaster in real time. Transparent files are easy to bring into the TriCaster as it sends over the alpha to the render-based platform.

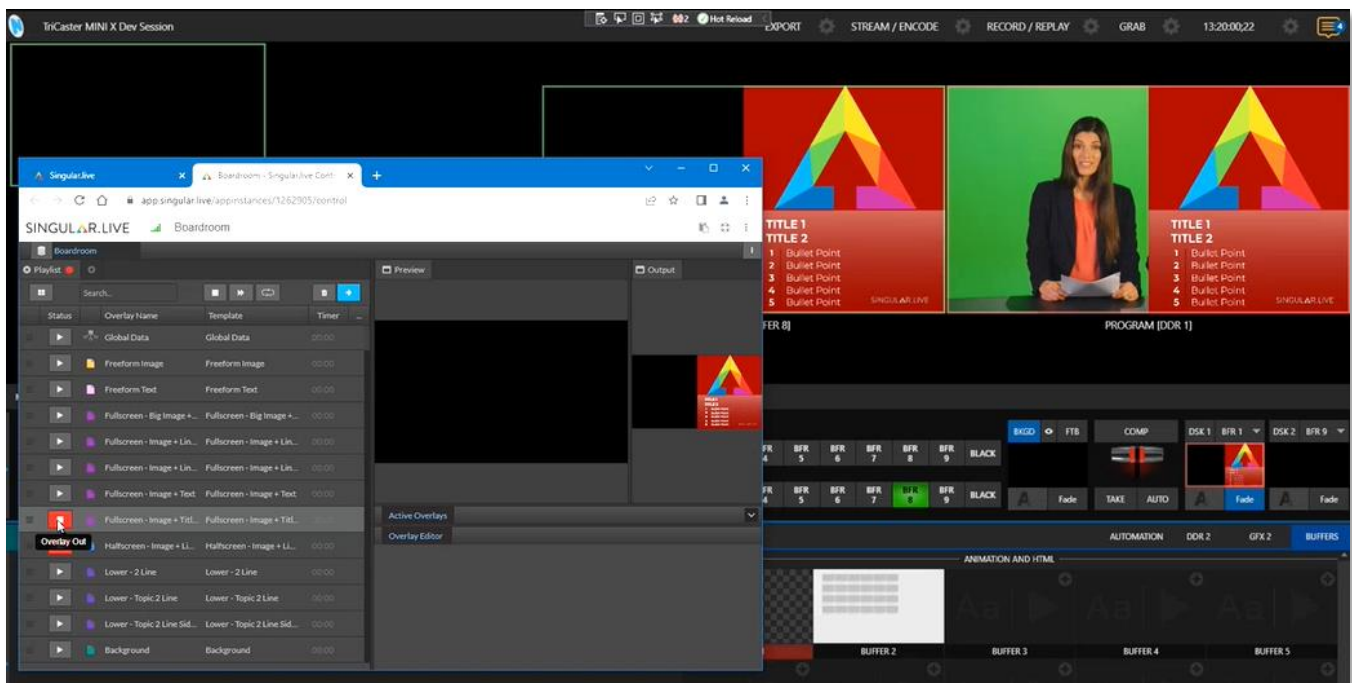


FIGURE 157

11.3.6 BUFFER WATCH FOLDERS

If you wish, you can share the *Frame Buffer* folder, which contains any *still images* used in *Buffers*, across a network. The *Share Media Folders and Buffers* is easily enabled even during a live production using the *File* menu at the left-hand end of the *Dashboard*.

The files which serve individual *still image* buffers are located in clearly named sub-folders of the *Frame Buffer* folder for each session. These folders are arranged as follows:

Sessiondrive:\Sessions\sessionname\Frame Buffer\Buffer (#)

e.g., *D:\Sessions\MySession\Frame Buffer\Buffer 3*

The *Frame Buffer* folder for the current session is accessible on the network when sharing is enabled. This allows *still image* buffers to be updated across a network using suitable graphics applications (such as Photoshop®).

Note: This special share name assignment is dynamic. Network sharing is activated, and the current Buffer path is automatically updated whenever you enter a session with the option enabled. This ensures that applications on networked systems can remotely access and update buffer graphics for the current production session.

The *Frame Buffer* folder is a “watch folder”. When you save an image for a given buffer, the buffer updates even if it is currently displayed, allowing you to refresh overlays instantly across the network.

- Still image *Buffers* are normally stored using the popular 32-bit PNG (Portable Network Graphics) format, but other formats are also supported.
- Specific filenames are not important in the buffer workflow. Multiple media files with different names can reside in buffer folders.
- The *Buffer* always shows the ‘newest’ file – that is, the one with the most recent modification date.

11.3.7 USING BUFFERS

Creative uses for buffers abound. You’ll notice that the sample media content supplied with your system includes several different styles of animated buffers – some full screen, suitable for looping backgrounds or similar applications, and others that are suited to overlay, bug, or alert tasks.

Buffers of all sorts can be displayed in the following locations:

- *Main Switcher*
 - *Program/Preview*) rows – all 15 buffers are directly accessible in the *Switcher*
 - *DSK* channels – choose a specific *buffer* using the *DSK*’s drop-down source menu.
- *M/E banks*
 - Select a *Buffer* on an *M/E* row just as you would for the main *Switcher*.

- *KEY* channels – choose a *buffer* for any *KEY* channel using its drop-down source menu.

Buffers allow abundant use of animation throughout your productions – as virtual set elements, layered graphics overlays, etc., all without tying up precious *Media Players*. (Consider that in addition to the *Buffer* cross-points in the main *Switcher*, *M/E*s also support *Buffers*). Given that *M/E*s are re-entrant, layering possibilities are nearly endless.

11.3.8 KEYING, PROC AMPS AND MORE

It's worth noting that you can apply *LiveMatte*, *Proc Amp* and *Edge* (cropping and feathering, when supported) settings to individual *Buffers* by opening its *Input Configuration* panel. Of course, you can also apply *Position* settings, including *Rotation* and *Scale*, in *DSK*, *M/E* and *KEY* channels.

11.3.9 EDITING TITLES

When a title page (.cgxml file) is loaded into a buffer, the *Edit Title* panel (see Section 11.2) can be opened by clicking the *Edit* gadget (AA) that appears at lower-left when you move the mouse over the icon. Changes made in the *Edit Title* pane take effect immediately.

11.3.10 BUFFER PRESETS

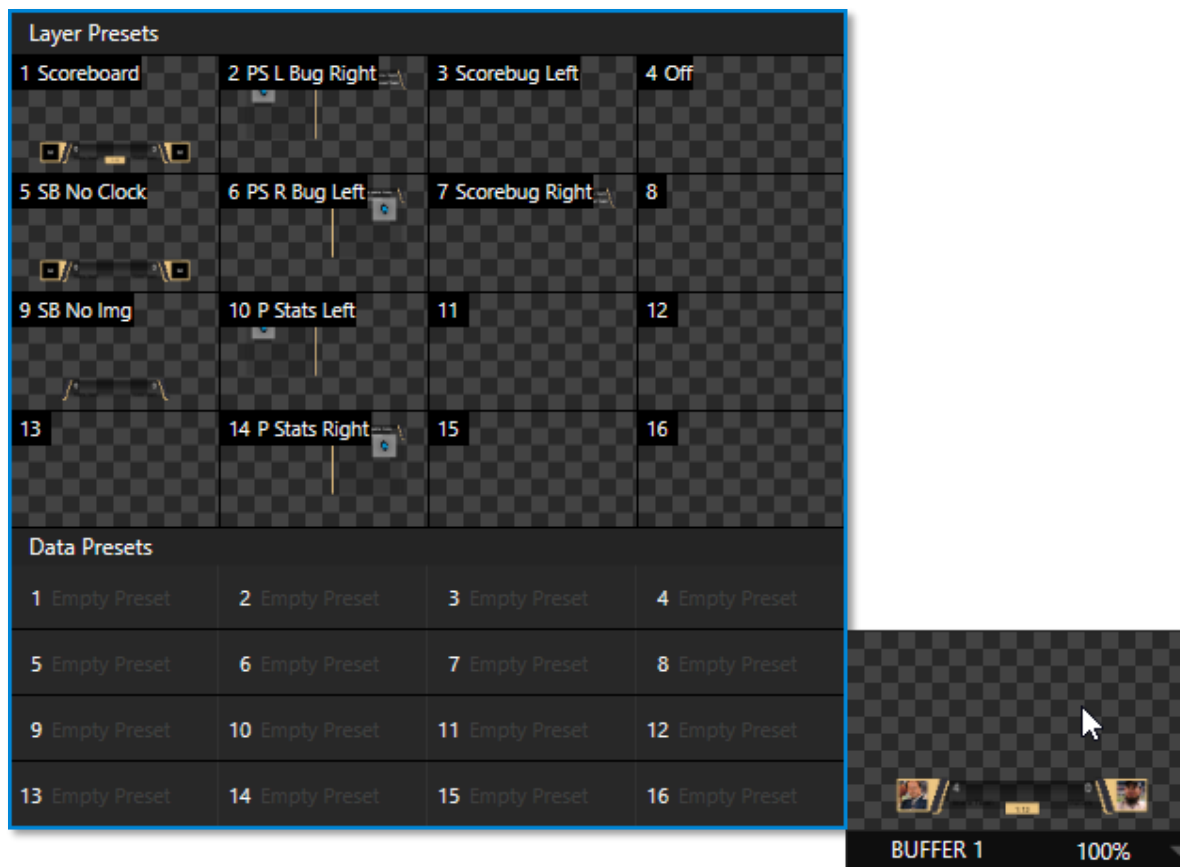


FIGURE 158

When a standard title page is loaded into a *Buffer*, clicking its thumbnail icon reveals a list of *Data Preset* slots. Each preset stores the text data and (replaceable) image content link for the title page. A single click recalls a complete data set and immediately updates the page.

This is incredibly useful, of course, because you can (for example) store the names, player numbers, images, and the like for the members of a sports team and recall these immediately with a click (or by executing a macro). And because the text values can include DataLink™ key names, it's even possible for statistics on a title page for a given player to be updated in real-time from external sources or text, xml, or CSV files in a watch folder.

Hint: If the Buffer contains a LiveGraphics effect, a set of thumbnail icons depicting Layer Presets is also shown. Selecting a preset allows you to dynamically change from the current animated state to another. More information about LiveGraphics usage is provided in Appendix A: Advanced Features

Chapter 12 PTZ CONTROL



Once primarily used in security applications, robotic or ‘PTZ’ (pan, tilt, zoom) cameras are appearing on the video production scene in ever-increasing numbers. The PTZ implementation in your TriCaster Mini X makes it easy to operate multiple (supported) cameras from the Live Desktop or a hardware control surface.

Any external *Switcher* input can be configured as a PTZ cam, controlled directly from the *Live Desktop*, or using a NewTek™ control surface. In addition to pan, tilt, and zoom control, you can adjust *White Balance* settings, *Focus*, *Iris* (or ‘brightness’), and *Animation Speed* (“travel speed”) settings for your PTZ cameras. A convenient visual *PTZ preset* system is included, too, allowing you to ‘lock in’ shots and access them quickly.

SECTION 12.1 INPUT CONFIGURATION, PTZ TAB

Connection and configuration settings for PTZ cameras are in the *Input Configuration* panel, specifically its first tab, labeled simply *Input* as shown below.

12.1.1 CONNECTING

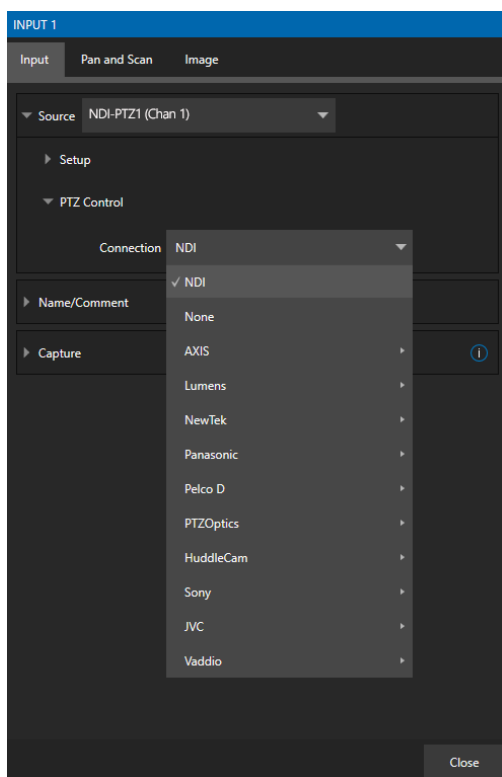


FIGURE 159

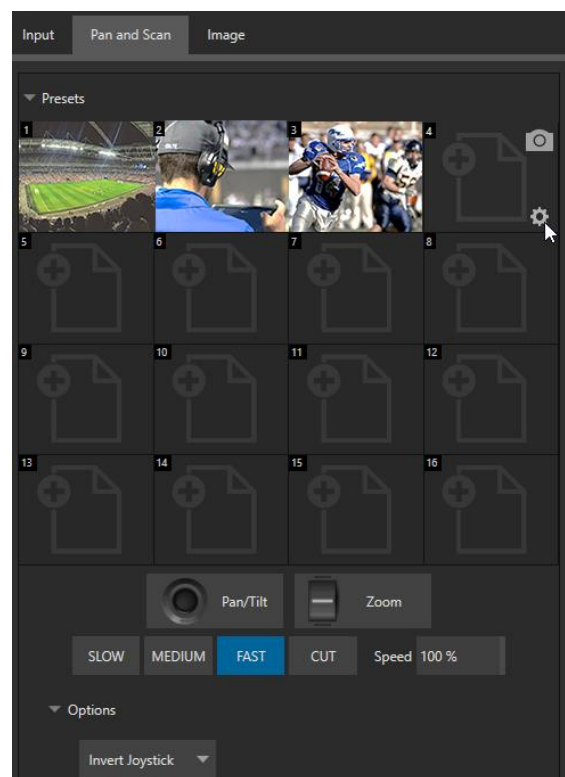


FIGURE 160

A few steps are required before you can control a PTZ camera.

1. First, connect the camera's output to a *Switcher* input – see Section 8.1.
2. Next you must configure the control connection to the camera by selecting its type.
 - Since the default *Connection* type in this group is *NDI*, you can skip this step for an NDI camera, which eliminates all the settings you would otherwise need to configure.
 - For non-*NDI* sources, use the *Connection* menu to choose the correct PTZ protocol for your device, and enter other data as required for that protocol.

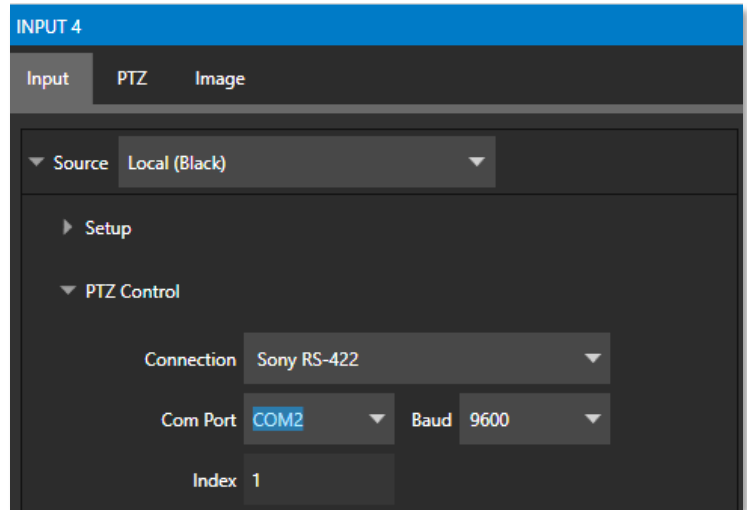


FIGURE 161

Having performed these steps, you are ready to expand and use the *PTZ Presets* control group in the next tab, labeled *PTZ*.

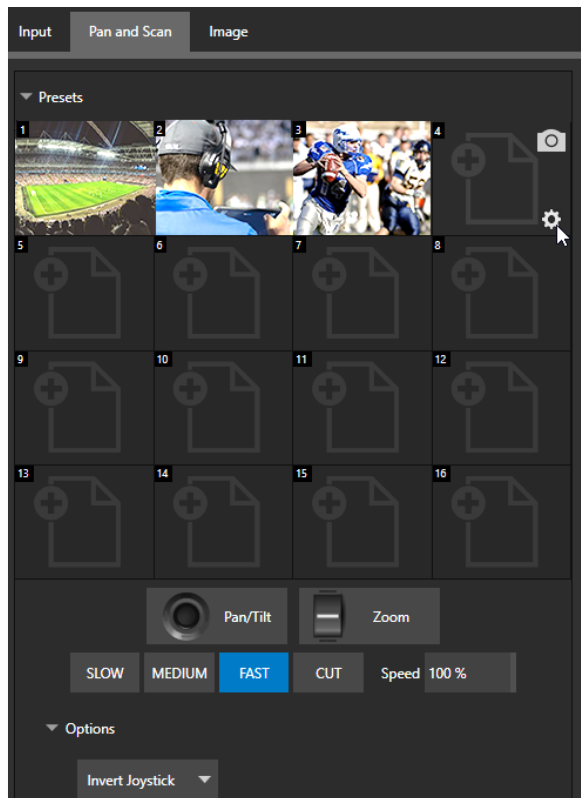


FIGURE 163

12.1.2 PTZ OPERATIONS

In the *PTZ* tab, a set of manual *Pan/Tilt*, *Zoom*, *Focus* and *Iris* controls sits just beneath a large *Presets* bin. Drag the mouse on these controls to make manual adjustments.

Speed controls, including *SLOW*, *MEDIUM*, and *FAST* presets along with a numeric speed control allow you to modify the speed with which presets are applied

OPTIONS

Expand the *Options* control below the PTZ speed group to access an Invert menu that lets you reverse the effect of joystick operations.

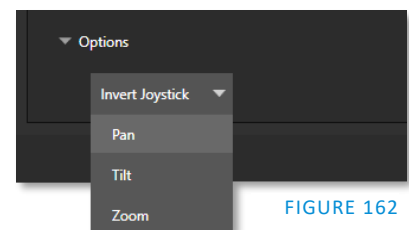


FIGURE 162

Note: Invert settings affect the related controls both in the Live Desktop and on hardware control panels.

12.1.3 PRESETS

The *Preset Bin* shows sixteen thumbnail icons for the current *PTZ camera*. Click a preset to select it, and the camera automatically begins moving to the new position.



FIGURE 164

To configure and store a preset:

1. Use the *Position*, *Zoom* and *Focus* controls mentioned in the previous section to navigate to the desired camera angle, etc.
2. Then move the mouse over the preset icon you wish to store (or update) and click the *Snapshot* (camera) gadget that pops up at upper-right corner of the icon.

Hint: Hold Shift when clicking the Snapshot icon to update the representative thumbnail image without modifying the preset already stored in that slot.

3. (Optional) Click the *Configuration gadget* (gear) to open a properties dialog that lets you assign a different *Alias* and *Comment* to each preset. (These entries also provide values for *DataLink* keys that can be used in the usual ways.)

Hint: See also Section 10.5.1, Viewport Presets, to learn how to display and use presets in your multiviews.

12.1.4 PTZ AND THE CONTROL PANEL

Pan, tilt, zoom control, and preset selection for connected robotic cameras are also directly supported on some NewTek hardware control panels.

Chapter 13 LIVEMATTE



Although its controls are deceptively simple, the LiveMatte™ feature employs powerful chromakeying technology capable of extremely high-quality results. Used alone or in conjunction with DSKs, M/E Overlays and LiveSet features, you will find LiveMatte can play a ‘key’ role in your live productions.

As we mentioned earlier, chromakeying – or simply “keying” – is used to combine images by eliminating a portion of a foreground image (effectively cutting a digital ‘keyhole’ in it) to reveal another background. This method is also used to insert talent seamlessly into virtual sets via *LiveSet™*.

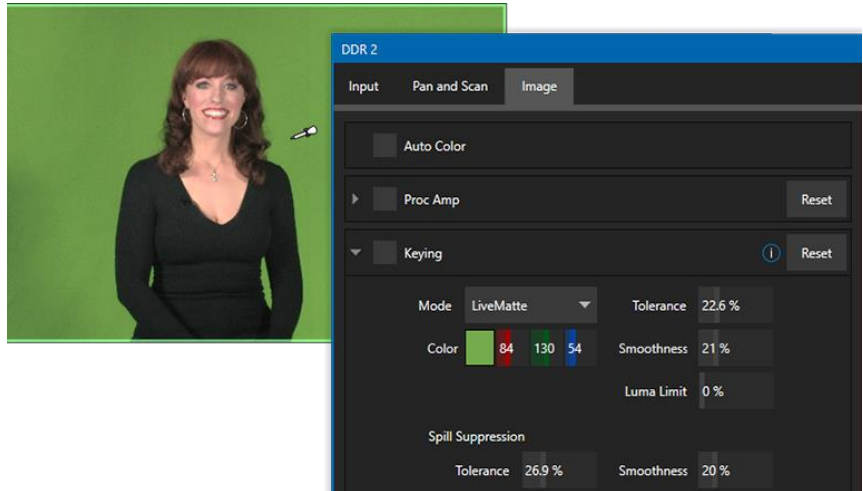


FIGURE 165

LiveMatte™ controls are in an expandable group in the *Image* tab of the *Input Configuration* panel for each *Switcher* source. To access them, click the *Configure* button (gear) that appears above an onscreen monitor in the *All Monitors* tab.

The method by which part of the image is defined as transparent is generically referred to as chromakeying, for its dependence on the color values (chrominance) of the video stream. (LiveMatte’s mature algorithms far exceed typical chromakeying techniques to provide real-time results, but for our discussion it’s not necessary to get into all the details; suffice to say it works very well and is easy to configure.)

Hint: Crop controls in the Input Settings tab can serve as a ‘garbage matte’ tool for chromakey sources.

SECTION 13.1 UNDERSTANDING KEYING

Chromakeying has become an essential tool in video and film production. Typically, foreground footage is shot in front of a blue or green screen, and then that background color – the *key* color – is treated as transparent, allowing another image to be inserted.

For example, when you see a tv meteorologist in front of a weather map, that person is almost certainly posed in front of a green screen. The background is ‘keyed out’, to be replaced by computer generated imagery.



FIGURE 166

Of course, if you choose poor settings, foreground areas may inadvertently be cut away as well. Or some areas that should be transparent may be only partially keyed. Good keying often requires judicious balance between ‘too much’ and ‘too little’. Let’s consider the tools provided to help you achieve a great result.

SECTION 13.2 MATTE

The term *Matte* refers to a black and white representation defining the transparent (background) and opaque (foreground) parts of an image during compositing.



FIGURE 167

Portions of the matte that are grey are treated as semi-transparent, which is very useful in progressively smoothing edges between foreground source material and inserted background imagery.

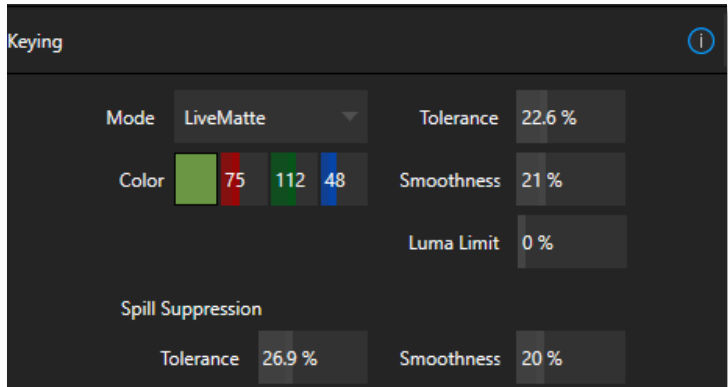


FIGURE 168

LiveMatte supplies a digital version of this traditional tool. As you would expect, controls in this group allow you to define and adjust the matte for the corresponding video input.

13.2.1 LIVEMATTE MODES

The *Mode* menu in the Keying control group lists several optional keying methods. Each has its advantages, and their controls differ as we will discuss next.

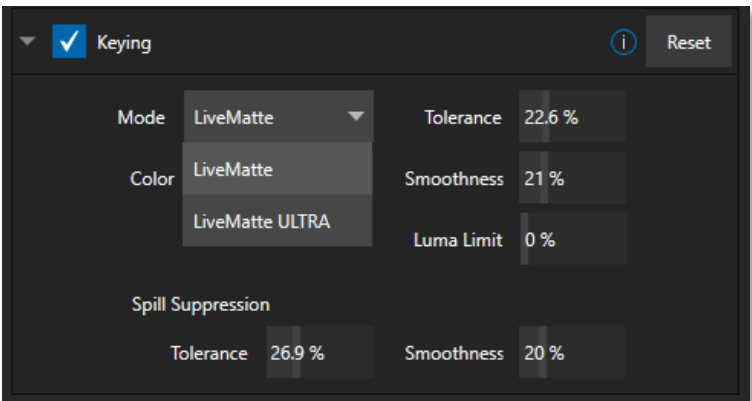


FIGURE 169

13.2.2 COLOR

All LiveMatte modes provide this control. Basically, you can think of *LiveMatte* as removing a specific color from the foreground image. The base color removed is chosen using the *Color* button. Click on the *Color*

sample box and keep the mouse button depressed. Then drag the *eyedropper* tool (mouse pointer) over one video monitor to choose the color you wish to *remove* and release it. The neighboring ‘color well’ is updated to show the color selected.

13.2.3 TOLERANCE

No physical greenscreen is *perfectly* comprised of one color. Wrinkles, folds, and shadows along with the seemingly inevitable uneven lighting result in difference. For this reason, when you choose the classic *LiveMatte* mode, a numeric slider labeled *Tolerance* is provided.

The *Tolerance* setting allows you to broaden the range considered as the key color, including more ‘near-neighbor’ colors to be included in the matte. A low tolerance removes only color values close to the primary or key color. As you raise the tolerance, you extend the range of values on either side of the primary color that will be treated as transparent. This allows you to deal with those imperfections we mentioned. On the other hand, it may be that there is (usually unplanned for) detail in the foreground that is somewhat similar to the key color. Reducing *Tolerance* may allow you to prevent unwanted holes appearing when the subject opts to wear his St. Patrick’s Day tie.

13.2.4 SMOOTHNESS

In the classic *LiveMatte* mode, *Smoothness* defines a further tolerance factor (*LiveMatte ULTRA* mode handles this automatically). We want our keyed foreground to blend smoothly into the background – rather than to stand out in hard relief like a postage stamp or decal. *Smoothness* serves this purpose, by defining a falloff zone of partial transparency. Don’t overdo it though, as aggressive settings can cause the foreground to become unnecessarily ‘muddy’.

13.2.5 LUMA LIMIT

When working with poorly lit backgrounds (or poor-quality footage), the color *Tolerance* range separating the foreground (talent) from the background can be extremely narrow. This problem can be aggravated by the subject’s choice of clothing, or when there are harsh shadows. There is often a strong chroma component (and associated chroma noise) in dark foreground areas. The ‘noise’ may be partially or completely transparent when tolerances are critical. Because the noise varies over time, ‘holes’ in the foreground can result, and even worse these may flicker on and off from one frame to another.

Classic *LiveMatte*’s *Luma Limit* control makes it possible to overcome this issue. In essence, it restricts the chromakey operation based on luminance (brightness) values. Dark foreground areas which typically cause the problems just described normally have quite different luminance values from the background color, which is usually brightly illuminated. In simplest terms, problem areas of this type can be decisively ‘pulled’ back into the foreground by pre-filtering the chromakey effect around a luminance threshold.

Generally, try to set up the best key you can *before* raising the *Luma Limit* from its default value of zero (no effect). Then gradually raise the limit until you are pleased with the result.

13.2.6 STRENGTH

LiveMatteULTRA uses a different method to define its matte, basically discriminating between background and foreground regions. Raising the *Strength* value can loosely be thought of as more aggressively defining

imagery as background. As with ‘tolerance’ in the classic LiveMatte, the lowest effective value is what you’re after with this setting.

13.2.7 OFFSET

The *Offset* setting is unique to LiveMatte Ultra and can be thought of as boosting the opacity level of partially transparent areas in your scene. Raise this value judiciously to solve problems with transparency in ‘borderline’ foreground regions.

SECTION 13.3 SPILL SUPPRESSION

The term ‘Spill’ refers to key color unintentionally reflected or ‘spilled’ onto the foreground subject. For example, a little green spill often appears on the shoulders of someone in a greenscreen shot.

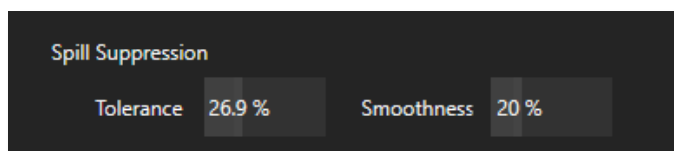


FIGURE 170

The *Spill Suppression* controls let you remove key color spill in your scene by reducing the amount of that color in the foreground, where it doesn’t belong. The net result is that the impression of spill color is eliminated, or at least reduced to the point where it is not objectionable.

Use the *Tolerance* and *Smoothness* controls (or, for LiveMatte Ultra, the *Strength* setting) in the *Spill Suppression* group in similar fashion to the controls by the same name discussed earlier. Endeavor to subdue spill without *overdoing* it, which can sometimes produce an unwanted gray fringe around foreground edges.

SECTION 13.4 COMPOSITING

When you enable (and configure) *LiveMatte* for an input, the onscreen monitor for that input shows the source keyed over a checkerboard pattern (when the Checkerboard option is selected in *Overlays* for the monitor.)

If you select the (*LiveMatte*-enabled) input as *Input A* in an *M/E* tab in *LiveMatte mode*, the keyed source is overlaid on sources in lower input rows. You will see the composite result when the output is assigned to either the *PGM* or *Preview* row.

SECTION 13.5 FINE TUNING

You’ll find *LiveMatte* easy to configure with a little experimentation – but a few handy workflow tips follow below. You may find it useful initially to turn *Smoothness* off or nearly so. Likewise begin with a low value for *Tolerance* – perhaps just 5-10, or so. Put the video source on *Preview* or *Program Output* before you do so, to provide a larger view to help you assess your settings.

Pick your primary *Color*, but – before releasing the mouse button – slide the eyedropper around to different parts the background. Watch the monitor as you do so to see how the area of transparency is affected by different Color choices. Release the mouse when you find the color that produces maximum results.

It's often preferable to pick an 'average' color from a location close to the boundary between the background and foreground regions. Now you can start to ramp up *Tolerance*. Bring it up slowly until most of the background color has been eliminated, cutting away most of the background to within a few pixels of the foreground/background boundary. Now raise *Smoothness* to fine tune that edge region, and you're nearly done.

Before considering your settings final, make sure to test the result using a moving source. This will sometimes reveal that overly aggressive settings cause small 'blocks' of pixels in the edge region to appear to snap on and off during motion - as they either qualify or disqualify for inclusion in the resulting matte. (A little reduction in *Tolerance* and increase in *Smoothness* will usually resolve this problem.)

Hint: It can be useful to zoom in using Position controls when fine tuning LiveMatte.

SECTION 13.6 LIGHTING FOR LIVEMATTE

We'd like to offer a few suggestions here to guide you in preparing your set. The single most important aspect of 'pulling a clean key' is lighting. The lighting should be even and diffuse. Bright 'hotspots' and shadows create different shades on the wall, and overexposed areas lack sufficient color for clean keying. (It is not how *much* light you have on the key wall, but how *evenly lit* that wall is.) Naturally, you want to keep your green (or blue) screen clean and free of wrinkles, ripples, folds, tears, or other blemishes, as well.

The distance from your talent to the wall behind them can make a big difference. When the subject stands too close to the key-colored background, the key color reflects onto the subject, creating a green or blue fringe that is difficult to remove. If you have available space, move your subject farther away from the wall.

When good distance is out of the question, you can improve things somewhat by placing lights above and behind the talent, lighting them from behind with a complimentary color filter over the light to 'cancel out' unwanted reflection (for green use a magenta filter; for blue, orange, or amber. Don't overdo back (or top) lighting, however. The limited dynamic range of the camera means there will be little useful color data in badly over-exposed highlights. This can make it next to impossible to separate fringe zones (such as hair detail) from the background (especially when this is also overexposed).

SECTION 13.7 CROP SOURCE

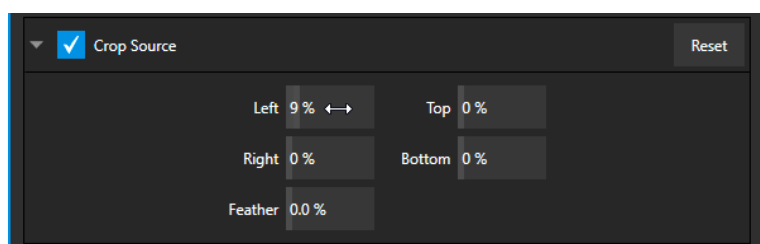


FIGURE 171

We discussed this feature earlier but wanted to bring it up again in our discussion of *LiveMatte*. It is common for unwanted items to remain after chromakeying. Common examples include microphones or lighting fixtures dangling from above, or perhaps a harsh crease, blemish, or tear in the background screen. *Crop Source*, located in the *Image* tab just below *LiveMatte*, can serve as a 'garbage matte' to remove this sort of intrusion.

Chapter 14 MIX/EFFECT (M/E) TOOLS



Mix/Effect (M/E) banks provide truly awesome production power and convenience. Pre-configure multi-layered compositions and switch them as easily as you would to any single input or use an M/E to control a special purpose sub-mix. This is also where NewTek's real-time virtual set technology, called LiveSet™, is located.

We introduced *M/Es* (Mix/Effect banks) way back in Section 2.2.6, but it's time for us to look at these extremely powerful tools in greater depth.

SECTION 14.1 OVERVIEW

The *Mix Effects*, labeled *M/E 1*, *2*, and so on, sit just above the main *Switcher* on the *Live Desktop*. Click an *M/E* tab (label) to expand or collapse the corresponding *M/E*. When expanded, *M/Es* occupy the area of the *Live Desktop* between the monitors and the main *Switcher*.

M/Es are presented individually in tabbed panes, each corresponding to a button in the main *Switcher*. Four *M/Es* are available to fulfill your diverse requirements.



FIGURE 172

As you would expect, selecting an *M/E* button on the (main *Switcher*) *Program* row displays the output from the corresponding *M/E* on *Program Output*. Likewise, punching an *M/E* button on the *Preview* row cues up that *M/E* bank's output for an upcoming *Take* or *Auto* operation. This makes it easy to *Take* or *Auto* directly to, from, or even between *M/Es* with a single click.

M/Es are re-entrant, meaning *M/E* buttons also appear on source selection rows and *KEY* channel source menus inside the *M/E* panels themselves. This is an exceptionally powerful feature, and one offering endless creative possibilities.

Note: Self-referential reentrancy (i.e., re-assigning a given M/E as one of its own sources) is not supported. However, you can often mimic effects of this sort by using multiple channels in another M/E .

SECTION 14.2 M/E MODES

At first glance, an *M/E* panel in its default operating mode ('mix', or transition) is scarcely distinguishable from the main *Switcher*. However, each *M/E* offers two *different* operating modes:

1. *Mix* (the default mode): A secondary switcher layout, with controls and options that are very similar to the main *Switcher*.
2. *Effect* mode: The *M/E* is configured for effects, including virtual set operations.

The *M/E* mode applied is automatically established based on the type of effect you load as its *Background* effect.

Effect selection is done in the usual way using the *Media Browser*.

Click the + sign gadget that appears at upper right when the mouse is over the thumbnail icon for the *Background* effect.

For the most part, you will initially see very little difference in the *M/E* panel when you change modes.

- In either mode, source button rows labeled *A* and *B* are arranged at left.

The number of layers shown varies according to the effect selection you make, from two effect layers to four.

- At right are two *KEY* control groups. In most respects these are identical in appearance and practice to the *DSK* controls located in the main *Switcher*.
- Even the central control group mimics the main *Switcher*, with its *Transition* controls and options, *T-bar* and so on.



FIGURE 173

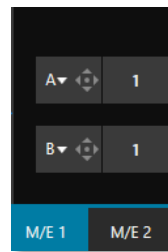


FIGURE 174

Autoplay and the M/E

One notable difference from an *M/E* in *Mix* mode and the main *Switcher* involves the *Autoplay* feature (see Section 11.1.9).

To avoid undesirable consequences for *Media Player* content aired on the all-important main *Program* output, *M/E*'s in *Mix* mode trigger *Autoplay* operations within the same *M/E* only.

As well, a switch in the *Dashboard Options menu* toggles *AutoPlay*'s triggering of 'Auto out' transitions.

14.2.1 MIX MODE

Really, an *M/E* in *Mix* mode is a switcher:

- Input rows labeled *A* and *B* behave just like the main *Switcher*'s *Program* and *Preview* rows
- *Transition* controls work in the usual manner
- *KEY* channels stand-in for *DSK*s and behave similarly
- *M/E* output can be routed to primary outputs and *Record* module

14.2.2 EFFECT MODE

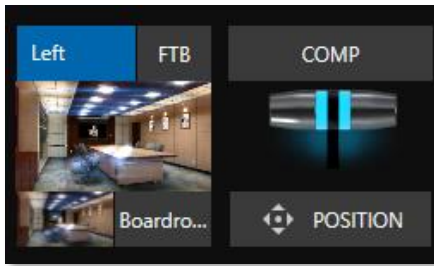


FIGURE 175

Even casual scrutiny reveals some (outwardly) subtle differences in the *M/E* control configuration when a *LiveSet* is loaded as the *Background* effect selection (Figure 175) as follows:

- The *Take/Auto* buttons located beneath the *T-bar* and *Transition icon* are replaced by a *Position* button.
- Two input rows (*A* & *B*) may be shown in *Effect* mode.
- *Positioner* controls appear for these input layers.

Let's consider the reasons behind these changes in the interface. An *M/E* in *Effect* mode allows preparation of a composite of two or more video sources, whether for direct output, or as a source for the main *Switcher*, or as an input for other downstream channels (including other *M/Es*).

Even in *Mix* mode, an *M/E* provides access to effects – including *Position* and transition effects (including *Animation Stores*) and animating *KEY* layers via the *Comps* feature. In *Effect* mode, however, an *M/E* asserts its real-time compositing and effects prowess more aggressively.

SECTION 14.3 THE T-BAR

In a mix effect, the *T-Bar* and associated controls match the main *Switcher*. It's different in an *M/E* displaying a *LiveSet* effect, however. In this mode, dragging the *T-bar* vertically adjusts the virtual 'camera distance' for the active *LiveSet* between 0% and 100%.

The *Duration* control works like similar numeric fields elsewhere in the *Live Desktop*: drag to adjust the value, click to type directly into the field using the keyboard, or press (keyboard) Shift and double-click to restore the default value. The drop-down *Duration* menu offers several convenient presets as well as a *Cut* option.

Hint: The maximum duration for an animated zoom is 30 seconds.

SECTION 14.4 INPUT POSITION CONTROLS



Position controls for all appear to the left of all input rows. Click this button to open the input's *Position* panel. These controls replicate those discussed previously, under the sub-heading DSK Controls in Section 9.6.1.

FIGURE 176

As mentioned, in *Effect* mode an M/E may support up to four input layers. The M/E doesn't *transition* between A and B in *Effect* mode. Instead, effects of various types are applied to the selected video inputs.

Let's consider the effects in the *Default* group as examples:

Click the + sign shown when you move the mouse to the effect icon located beside the effect duration. This opens the *Media Browser*. Select the effect group labeled *Default* under the *LiveSet* location on the left. These effects are straightforward compositing effects involving multiple video layers. *Effect* output (i.e., the *background* prior to *KEY* overlays) is the sum of all input layers.

If the source assigned to input A is fully opaque, any content in the layers below is hidden. When input A is partially transparent, the source assigned to input B is blended into the M/E background layer, and so on.

Hint: Don't overlook the fact that the input Position controls, including Borders, let you create very elaborate compositions even with these 'simple' compositing effects.

The output of active *KEY* layers is then added to the effect output before passing the combined result onward as the final M/E output.

UTILITIES

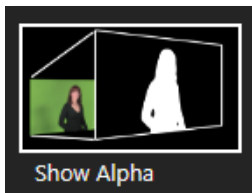


FIGURE 177

Effects can take other forms as well. We've looked at the *Default* effects group; now let's consider *Utility* effects.

If you were following along in the previous section, replace the current effect in M/E 1 with the *Show Alpha* effect from the *Utilities* group.

Applying this effect immediately causes the display on the *Program* monitor to update, showing the content of the alpha channel for input A (Input B is ignored).

The current input A source is keyed, so the effect displays black on *Program* out wherever transparency exists in the foreground, white for full opacity, and shades of gray for in-between blends.

With this in mind, it won't be hard to guess what *Show Inverse Alpha* does.

In contrast, the *Show Color* effect in this group passes the full color output of *LiveMatte* prior to being multiplied by the alpha channel. (It might seem as though this would be identical to the original source, but you may notice subtle differences. This is because of the *Spill Suppression* processing.)

Let's look at another of the effects in the *Utilities* group, *Color Correction*. Replace the current effect with the *Color Correction* effect. You'll notice that M/E 1's output on the *Program* monitor is now rendered in monochromatic grayscale. Drag the T-Bar down to affect the M/E's color saturation. Click the mouse on the *Position* button below the T-Bar and drag to modify *Hue*.



FIGURE 178

Finally, the *Make Legal* effect ensures the output of an M/E is within broadcast signal tolerances.

3D

The *Anaglyph (Red, Cyan)* effect found in the 3D category is a special purpose tool.

The anaglyph method of displaying 3D imagery depends on stereo video inputs that are filtered and composited into a single output stream.

In turn, this combined stream resolves into 3D when viewed through special glasses with red and cyan (blue-green) filters for left and right eyes respectively.

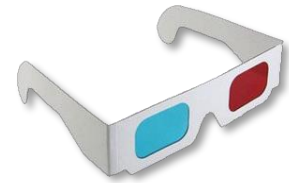


FIGURE 179

TriCaster provides easy access to anaglyphic technology by means of the *Anaglyph* effect. The effect combines 2 video inputs selected in an M/E. 3D output can then be switched easily like any other source. No complex configuration steps or tricky control panel operations are required.

PHOTOSHOP BLEND

The effects in the Photoshop Blend folder apply well-known blending modes to the A layer in the M/E as these are blended with the B layer. The resulting compositions can serve many purposes, such as adding animated 'bokeh' style overlays using DDR clips or adding interest to still overlays such as vignettes or titles.

SECTION 14.6 VIRTUAL SETS

We've considered the *Default* and *Utility* effects. Let's move on to a more glamorous species, the *virtual set*, presented as *LiveSet™* effects.



FIGURE 180



LiveSet is one of NewTek live production system's most powerful tools and can dramatically enhance a production. With it you can achieve the look of a large, sophisticated studio setting (Figure 180) within a very small studio space, all without the need for external equipment.

Broadly speaking, setup of a *LiveSet* effect is much like the *Default* effects previously considered (see Default Effects, Section 14.2.2). A typical *LiveSet* consists of a greenscreen shot (usually input A) with *LiveMatte* applied composited into a virtual set. *LiveSet* adds the foreground and background for the scene, and additional video inputs may add to the effect in various ways.

Hint: Most virtual sets require LiveMatte settings to be applied to Input A. Other inputs may also require keying, depending on the design of the virtual set.

To select a *LiveSet*, click the effect icon in the M/E's *BKGD* at lower-left control group to open the *Media Browser* (Section 11.1.8).

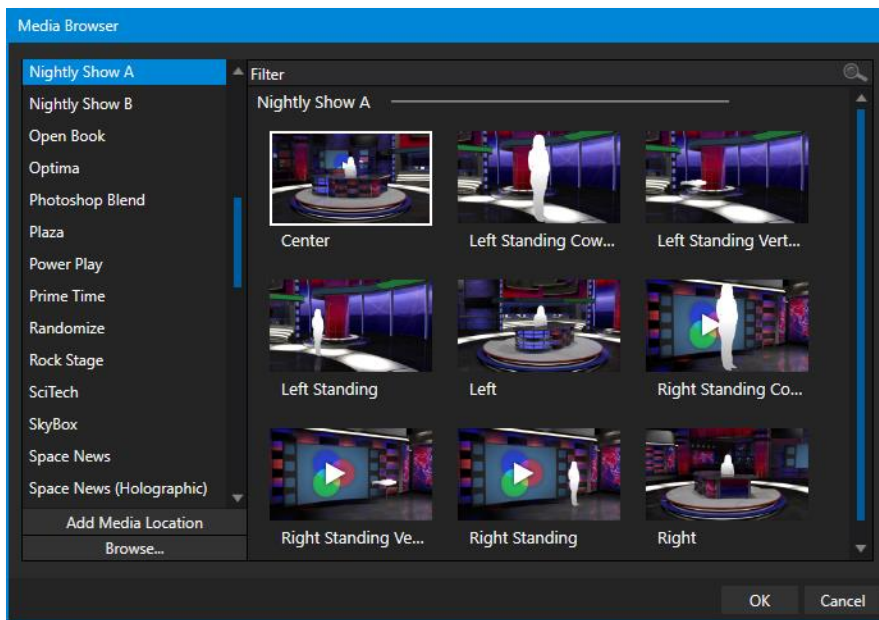


FIGURE 181

The *Location List* of the *Media Browser* lists any installed *LiveSet* groups under the heading “LiveSets”. Select an entry beneath to show thumbnail icons in the Browser’s *File Pane* (Figure 181 shows the content of the *NewTek* location under the *LiveSets* heading).



FIGURE 182

INPUT POSITION

We discussed *Positioning* controls for M/E video inputs previously (Section 14.4). It’s worth adding here that *Position* options do affect the scale, rotation, and position for *LiveSet* video inputs. You can often use *Positioning* controls to achieve a ‘good fit’ and natural appearance of talent or another source appearing in your virtual sets (greatly reducing the need to fuss with physical camera positions).

Hint: The Align group of LiveSet effects is specially provided to make it easier for you to adjust cameras and position talent on your physical set to suit the design of typical virtual sets.

14.6.1 HOLOGRAPHIC LIVESETS

A special variant of *LiveSet* effects are referred to as ‘holographic’. These effects can be amazingly lifelike and convincing, and are relatively easy to create.



FIGURE 183

Holographic LiveSet effects are loaded into an *M/E* in the same manner as any other effect. Simply move the cursor over the effect icon and click the + sign (*Add Media* button) that appears to open a *Media Browser*, then choose the effect you wish to load (several examples are included with your system). Adjust the current view for holographic *LiveSets* in similar fashion to standard effects using the *T-Bar (Zoom)* and associated *Positioner* controls.

The *Comp Bin*, discussed shortly, allows you to store and apply your favorite compositions.

Preset adjustments made using the mouse vary slightly for holographic effects. For a standard LiveSet effect, dragging the mouse left, right, up, or down changes the camera position in the frame. The right-mouse button zooms in or out. In a holographic LiveSet, right-mouse operations are the same. However, dragging the mouse on the canvas modifies camera *rotation*, rather than position.

Hint: New holographic effects, like standard LiveSets, can be created using the optional Virtual Set Editor application. A demo version of this utility is installed on your system, and its instruction manual can be located using the Help link on the Home page TriCaster's Launch screen.

SECTION 14.7 KEY CHANNELS

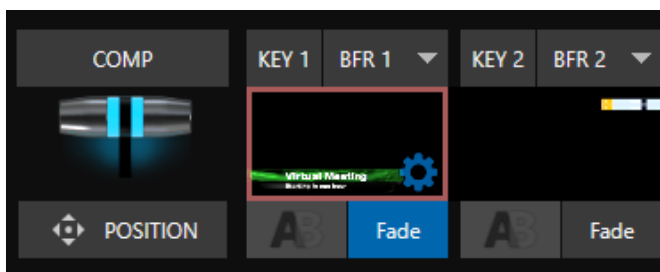


FIGURE 184

In *almost* all respects, the *KEY* channels in *M/E* panes match the *DSK* channels found in the *Switcher's* main *Transition* section. Unlike the *DSKs*, though, they constitute a 'pre-main *Switcher*' sub-layer. This means that *KEY* channels are applied *before* the composition is sent to the *Switcher* (or another *M/E*). Thus, content in a *KEY* channel appears beneath anything displayed via the (*Switcher*) *DSK* channels.

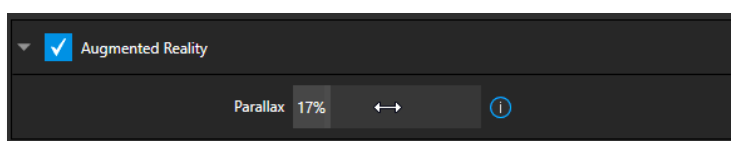
Likewise, *KEY* channel selection and *Position* controls largely work just like their *DSK* cousins, discussed earlier, with two notable exceptions. We'll consider these exceptions next.

14.7.1 KEY LAYERS AND AUTOPLAY

It's worth noting that the behavior of *Autoplay* for *Media Players* selected as sources for *KEY* channels conforms to the way it works for other *M/E* sources. That is, newly displaying a *Media Player* with *Autoplay* enabled in the *A* layer or a *KEY* layer for an *M/E* will trigger playback, but – by default – at the end of play the 'out' transition (and subsequent advance to the next playlist item) does not occur.

This default behavior can be overridden by checking the *Options* menu item *Enable Autoplay Out on M/Es*.

14.7.2 AUGMENTED REALITY



In one other unique and powerful departure from *DSK* features, the *Positioner* for *KEY* layers has an added feature labeled *Augmented Reality* (Figure 185).

FIGURE 185

Note: The Augmented Reality feature is for use in M/Es that are assigned to effects (such as LiveSets), and has no effect when a Mix effect (transition) is loaded as the Background effect for the M/E.



FIGURE 186

Any *KEY* layer with the *Augmented Reality* switch enabled is treated differently from a normal *KEY* layer in several respects:

- First, it does not merely appear *above* the main *M/E* layer composition you configure at left, as a typical *KEY* channel would. Instead, it is treated like another main layer added above the standard *M/E* layer rows.

- Thus, when you zoom or pan the *M/E*, the *KEY* layer zooms and pans right along with it, making its content appear as though it were embedded in the scene (Figure 186).

(This lets you effectively add one or more virtual layers to a *LiveSet* composition at any time, with complete control over positioning within the composition.)

- With the associated *Parallax* setting at 0%, the *KEY* source is locked to the background formed by the layers below it in a 1:1 relationship. When the ‘camera viewpoint’ changes, the *KEY* layer moves in the same amount and direction as the background.

Hint: You can use this ability to ‘pin’ a graphic element into a LiveSet.

- Raising the *Parallax* value modifies the motion of the *Augmented Reality KEY* layer during panning and zooming, making it appear closer to the camera than subject matter behind it. This enhances the impression of three-dimensional depth.

SECTION 14.8 COMPS

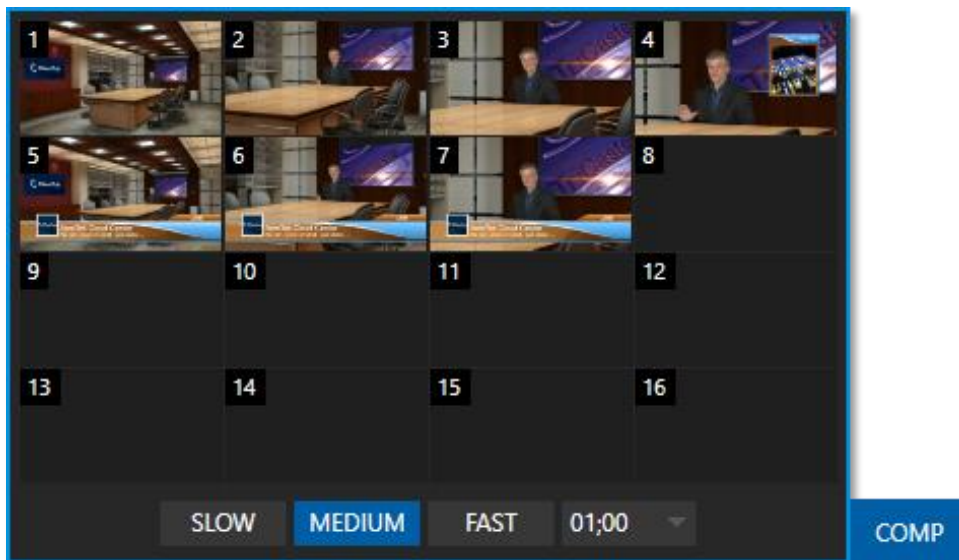


FIGURE 187

Different ‘virtual camera’ positions, along with other *M/E* attributes, can be stored in the *Comp Bin* associated with each *M/E*. In particular, *T-Bar* and most *Positioner* attributes for all layers in the module are stored in a *Comp* and are re-applied when you click the *Comp* icon later.

This includes such things as cropping and edge-feathering performed using *Positioner* settings (layer source selections are not stored, nor are *Border* or *Tracking* settings).

14.8.1 APPLY TO TRANSITIONS

Apply to transitions overrides the set for the individual Keys to match the *COMP* speed. For example,:

- *COMP* mem1 takes Key1 and Key2 on
- Key1 has manually set speed of 1:00
- Key2 has manually set speed of 2:00

If you have *COMP* set to Med (1:00) and switch to *COMP* mem1 - Key1 will take 1sec to transition and Key2 will take 2sec to transition on.

If you do the same thing but have *Apply to Transitions* enabled, both keys

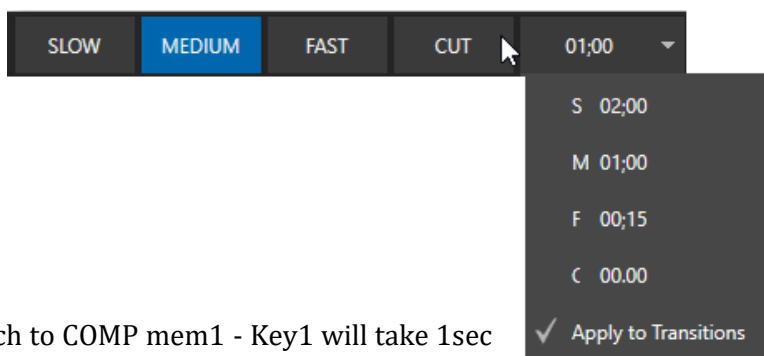


FIGURE 188

14.8.2 MANAGING COMPS

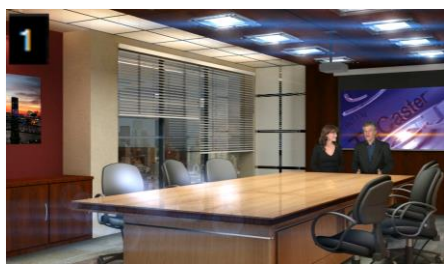


FIGURE 189

Having prepared a composition you wish to keep, storing a corresponding *Comp* is simply a matter of clicking the *COMP* button above the *T-Bar* to open the *Comp Bin*, and clicking an empty position. An image grabbed from output of the module is displayed to represent the *Comp*.

If you later wish to revise a *Comp*, you can do so either by rolling the mouse over it and clicking the ‘snapshot’ icon shown at upper-right (Figure 189) or by right-clicking the icon and selecting *Update* in the context menu that appears (Figure 190).

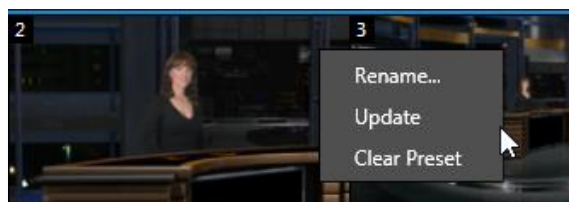


FIGURE 190

The context menu also allows you to *Rename* a *Comp*, update it, or clear it entirely.

14.8.3 ANIMATION

Selecting in a *Comp Bin* changes the current composition of layers displayed by the module to the new one. Changes can be immediate or animated over time. The timing controls in the footer of the *Comp Bin* determine the duration of the move from one position to another, while transition effects for layers are governed by their individual effect durations.

Hint: The main Switcher also hosts a Comp Bin. However, LiveSet effects are not supported in this case.

It's worth noting that, since *T-Bar* states are stored and applied by *Comps*, transition effects can be applied by selecting a *Comp*.

Let's discuss an example:

- Suppose *KEY 1* is a lower third type title overlay and has a 'fly on' type transition assigned to it.
- *Comp 1* was stored with *KEY 1* not visible.
- *Comp 2* was stored with *KEY 1* hidden.
- Click *Comp 1*, then – a moment later - *Comp 2*.



FIGURE 191

The result is that *KEY 1* will animate in and out according to the state stored in each *Comp*.

Comps can apply transitions to multiple layers at one time with a single click. Animation between the current layer states and settings is not limited to the effects provided by transitions, however. Let's consider a different example.

- Select *Comp 2*, ensuring that *KEY 1* is visible.
- Use the *Positioner* controls for *KEY 1* to slide the lower third title off the page, completely hiding it from view.
- Store a new *Comp*.

Switch back and forth between these two *Comps* and you'll notice that animating the layer's position of the layer has effectively allowed you to create a custom transition. Experiment with other settings, such as Z or Y axis rotation and you'll begin to see just how much power *Comps* give you.

Using these and other settings to animate the various video layers, very complex compositions can be introduced into your production with a single click in the *Comp Bin*.

Chapter 15 AUDIO



Less than 100 years ago, movies were silent. We've come a very long way since "The Jazz Singer" (1927, Warner Bros.), and great audio now plays a huge role in video production. NewTek live production systems provide an extensive set of professional quality audio tools, which we'll explore in this chapter.

External audio connections and basic configuration were touched on back in Sections 3.6.1 and 3.7.1. A tabbed pane labeled *Audio Mixer*, found just below the main *Switcher* presents more precise control over individual audio sources and outputs.



FIGURE 192

Compact sub-panels in this pane provide configuration and control for individual audio sources and audio outputs as follows, from left to right:

- *External* audio sources
 - Controls for audio *INPUTS 1-8*.
 - *TALKBACK* – a special source for Skype TX (and Live Call Connect on supporting models.)
- *Internal* sources – *Media Players*, and *Effects* (sound embedded in *Animation Store* transitions).
- *Aux 1* – control for *Aux* audio bus.
- *Phones* – volume control for the system's headphone output.
- *Master* – controls the primary audio bus.
- (Audio level controls for output recorders and streaming are in the *Output Configuration* pane.)

NOTE: Backplane analog outputs carry the first two channels assigned to *Master* and *Aux 1*.

SECTION 15.1 AUDIO SPECIFICATIONS

Analog audio conforms to SMPTE RP-155. The maximum input/output level is +24 dBu. Nominal input level is +4 dBu (-20dBFS), and the sample rate is 48 kHz. Levels above 0dBVU are shown in red in the *VU meters*, to caution you that overly high levels can result in clipping in recordings.

SECTION 15.2 HEADPHONES

By default, headphones connected to the ¼" stereo *Phones* jack carry the first two channels of the audio signal from the bus designated *Master 1* – that is, the same audio carried by analog output connectors marked as *Ch. 1* and *Ch.2* in the *OUT 1* group on the system's backplane.

The *Headphone* output is also affected by *Solo* switches located at the bottom of each control group in the *Audio Mixer*. When *Solo* is enabled for one or more sources – or outputs – *only* the soloed sources are sent to the connector. (See *Solo* in Section 15.7.2 for more information on this feature.)

SECTION 15.3 VU METER CALIBRATION

VU (Volume Unit) *meters* are located right above the *Volume* control sliders throughout the *Mixer*. The calibration of the *VU meters* can be changed to suit your preference. A menu at left beside the *Input* labels opens a small menu (Figure 193) offering three options as follows:

- *dBVU* – most familiar to users of typical analog audio mixers
- *dBFS* – dB 'Full Scale' – the digital standard; see the heading Audio Headroom in Section 3.11
- *dBu* – based on a voltage of 0.775 VRMS (a shy scale rarely seen in public, supplied for completeness, comparison, and the amusement of audiophiles)

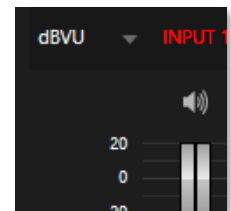


FIGURE 193

SECTION 15.4 EXTERNAL SOURCES

Columns in the first set of control groups are numbered as *Input 1-8*. Each of these can be assigned to any available audio source – whether analog (line level) sound supplied to rear panel connectors, HDMI embedded audio from one of the four inputs, available NDI source, or other system audio source (including network audio sources like Audinate's Dante™), with appropriate drivers.

**Note: Dante support requires an inexpensive user license from Audinate™.*

You may choose to think of these audio sources as being associated with the same-numbered *video* inputs in the nearby rows on the rear connector panel. This linkage, though the default, is not strictly enforced, however. You can connect directly to any external audio source in the *Advanced Configuration* panel (see Section 15.10).

Also, for reasons of your own, you may well prefer to treat various audio sources as independent of the nominally associated video input. The *Follow* (audio follows video) features play an important role in this context – see the sub-heading Follow Program Video in Section 15.10.2.

15.4.1 CONNECTION TYPE

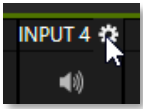


FIGURE 194

Naturally, only one connection can be active for a given audio input at any moment. To access the *Connection* menu, move the cursor into the label of an input, and click the *Configure* (gear) button that appears just at right. The *Advanced Audio Configuration* panel will be shown.

The drop-down *Connection* menu at the top of this pane allows you to assign any one of the supported sources connected to the system to this *Audio Mixer* control column.

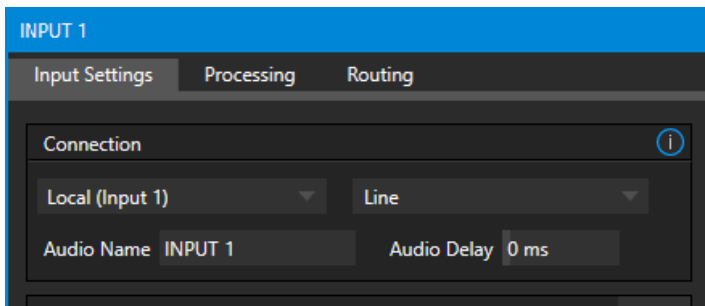


FIGURE 195

- *Local*
 - Physical inputs on the system’s backplane or faceplate listed, usually as *Input #*, but in some cases as “Mic” or “Line”.

Local sources support different connection types, as determined by a menu at right in the same group. This option typically defaults to *Auto-detect*, but depending on model may include *Line*, *Mic*, or *Embedded* (which is the correct setting for audio included with an HDMI or NDI video connection).
 - Network audio from supported protocols (such as Dante; may require third-party drivers).
 - *Skype TX Caller 1, 2* (sound from these two special input types, further discussed in the next sub-heading).
- *Follow (Video Source)* – this is the default selection and causes the control column to follow the same numbered video *Switcher* input.
- All available NDI audio sources.

Note: Channels 1 and 2 from Master and Aux 1 are always placed on Dante output channels 1-4, respectively.

15.4.2 NDI KVM AUDIO

NDI KVM sends audio through the default Windows Playback device at the time NDI KVM is launched. You can change what audio is being sent to NDI KVM by changing your default Windows Playback device and then rebooting. To learn more about NDI KVM, see Section 5.2 The Home Page.

Note: Available as a feature for all 7-3 or higher TriCasters that have Windows 10.

SECTION 15.5 LOCAL SKYPE TX CALLERS AND MIX MINUS

The default *Connection* option for external *Audio Mixer* inputs is *Follow Video Source*. When an audio input set this way is assigned to a Local>Skype TX Caller channel, the audio control group is automatically assigned to govern incoming audio from that remote Skype caller.

It is not necessary to use an *Aux* audio bus to configure a special audio ‘mix minus’ to return to the remote Skype caller in this case. The return audio for Skype TX is handled automatically, ensuring that (except when the *TalkBack* feature is engaged) the remote caller will receive a special mix comprised of the *Master* mix with the remote caller’s sound removed.

Note: These special a/v inputs are based on the professional Skype TX broadcast platform and require the use of its dedicated control application. Alternatively, you can connect directly to NDI a/v output from a dedicated NewTek TalkShow Skype TX unit, or the standard ‘consumer’ Skype client configured for NDI output using any Switcher input. See Appendix B: Skype and Skype TX for more information.

SECTION 15.6 TALKBACK

The Audio Mixer input labeled *TALKBACK* serves a special purpose, providing a way to converse with remote callers off-air (i.e., without intruding into your live program). You can choose what audio source to supply for *TalkBack* purposes from a variety of input types, as discussed in the previous sections.

As mentioned earlier, by default the *Connection* option for an external *Audio Mixer* input assigned to a Skype TX™ channel allows the corresponding control group to govern incoming audio from that remote Skype caller. In this case (that is, when an audio control is ‘listening’ to incoming audio from a Skype TX channel), a new button labeled *TALK* appears beside the group’s *SOLO* button.

When the *TALK* button is lit, audio you supply via the *TALKBACK* input (typically a microphone connected to one of the inputs on the system’s backplane, or perhaps the system *Stereo Mix* supplying audio with sound from a 1/8” motherboard audio input) is sent to the remote caller, temporarily replacing the normal mix-minus sound that remote caller would otherwise hear.

At the same time, the Mixer sends incoming sound from the remote Skype caller to its *Headphone* output, enabling you to have a two-way conversation off-air.

Hint: You will want to take steps (such as by enabling Mute, or perhaps Follow Program Video) to prevent sound from the Skype caller being sent to your live audio mix during TalkBack conversations.

SECTION 15.7 COMMON CONTROLS

Many important features and optional settings are common to most source types, and some even appear for outputs. Let’s review these before continuing.

15.7.1 MUTE

Mute switches for inputs and outputs appear as speaker icons located just above in the main *Mixer* panel. A single switch controls multiple channels, as appropriate for that source.

Enabling *Mute* removes the sound from that source from all downstream audio mixes and outputs.

Hint: Muted sources still show signal activity on the VU meters, but levels are drawn in gray rather than full color. This is also true for sources with Follow set that are not currently audible on output.

One important exception to *Mute* operation involves recording. The IsoCorder™ feature lets you capture any MIX a/v output - i.e., the primary output or outputs, or directly from hardware audio and video inputs with matching numbers. That is, the audio recorded with the video is taken from the audio input having the same number as the video input. In this latter case, the captured audio is routed directly to the recording module *prior* to most adjustments in the *Audio Mixer* – including *Mute*.

15.7.2 SOLO

Solo implementations and options vary widely in the audio industry, but broadly speaking, all variants provide very useful functions. The *Solo* feature offers remarkable flexibility without overly confusing complexity.

Consider a few basic design concepts:

- ❖ Enabling *Solo* for a source sends its (post-fader) sound to the *Headphones* output and removes all other sources from that output.
- ❖ *Solo* has no impact on audio mix (es) sent to any other output.
- ❖ This *Solo* feature is normally what it called ‘exclusive Solo’ (or ‘X-Or’ type). This means that enabling *Solo* for a given source disables all other *Solo* buttons.
- ❖ It is possible to use what is called ‘Solo latching’, ganging multiple sources for *Solo* output. Hold down the Ctrl key and click additional *Solo* buttons to add or remove their respective audio contributions from the latched *Solo* group.

AUDITIONING AN AUDIO SOURCE

It can be very useful to be able to preview one or more audio sources, doing so *without* allowing the test sound to be audible on *Program* output. Typically, this need arises in connection with testing microphones or other audio sources that will be used in the production.

To audition an audio source in this manner:

1. First *Mute* the source, removing it from primary outputs.
2. Then enable *Solo* to hear it on the *Headphones* output.

SECTION 15.8 INTERNAL SOURCES

Besides external audio sources, sounds played from internal storage volumes (including removable media such as external hard drives or ‘thumb’ drives) via the *Media Players* (DDR, etc.) can be added to the output mix.

15.8.1 MEDIA PLAYERS

Video and audio-only files in DDR playlists, along with audio files in the *Sounds* player, may contain one or more audio channels. At most, *Media Players* output the first four audio channels of multiple channels; additional embedded channels are ignored. Other options and controls in these sub-panels are similar to those provided for external audio sources.

15.8.2 EFFECTS (TRANSITIONS)

This control group governs the sound embedded in *Animation Store Transitions*. The remaining control groups in the *Audio Mixer* are dedicated to various audio outputs. We’ll come back to them in Section 15.9, but before we do so, let’s drill further down into more advanced audio options and tools.

SECTION 15.9 OUTPUT AND PRIMARY BUS CONTROLS

As has been discussed, the *Mixer* supports four primary audio busses – *MASTER* and *AUX 1*. Each of these is represented by its own control group in the *Audio Mixer* output section, and regulates sound sent to physical connectors or to ‘logical outputs’.

Hint: Shift + double click Volume knobs to restore their default values (0dB).

Settings in all the control groups in this section take effect downstream from all audio sources, further modulating and processing audio sent to outputs as the *AUX* and *MASTER* mixes, for recording, and for Internet streaming.

15.9.1 HEADROOM NOTES

In digital audio systems, signal levels that exceed maximum values are uniformly assigned the maximum value, a condition known as “clipping”. Clipping inevitably results in annoying audible issues.

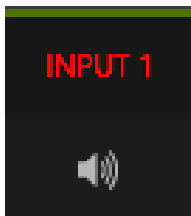


FIGURE 196

Worse, over-modulation that may not be apparent while listening during live production may nonetheless appear in

Secondary audio busses:

Beyond the primary busses mentioned here, the system maintains a large number of secondary internal busses.

For example, the *Solo* switch for each input (and output) is a ‘send’ that adds sound to a ‘Solo bus’.

Likewise, the *IsoCorder™* module permits discrete recording from the unmodified audio input associated with any single video source; this constitutes up to eight additional audio busses.

recorded files. This is often true even when levels *appear* to be below the ceiling level (0dBFS, the maximum allowable digital level).

Hint: When clipping has occurred, the label for the problem channel turns red briefly, as seen in Figure 196.

Due to this problem, digital audio system designs customarily allow substantial ‘headroom’ above the benchmark ‘alignment level’, making over-modulation much less likely. Often this allowance seems high to those familiar with analog audio systems; headroom levels between 18 and 24dB are not uncommon in professional digital audio realms.

You can apply level control to suit your own preference in this regard, using the separate *Record* (and *Stream*) level controls discussed shortly. For example, levels set at -20dBFS in the *Record Configuration* panel approximate typical professional practice. This has no impact on levels at the system’s audio outputs, and all but ensures clipping in recorded files will be avoided.

Advanced users can thus record files conforming to regional standards or personal preference, substantially reduce the possibility of audio clipping in recorded files, and even adjust the level on the fly if necessary.

The main point to remember from all of this is that for digital audio recording “less is often more”. When it comes to levels, go as high as necessary – but it’s equally practical to go no higher than necessary.

Hint: The Audio Mixer also provides Compressor/Limiters for each input and output. These can also be invaluable in defeating clipping due to over-modulation.

15.9.2 STREAM

Level controls for the stereo audio that accompanies *Streaming* outputs are provided in the *Output Configuration* panel (see Section 8.2).

SECTION 15.10 ADVANCED CONFIGURATION

The controls for all inputs (including internal audio sources) as well as the *Effects*, *Stream*, *Aux* and *Master* output groups include a configuration button shown when you move the mouse over the input label.

The familiar ‘gear’ icon opens the advanced *Audio Configuration* panel. We touched on this panel briefly when we discussed selecting and configuring *Connections* for external audio inputs. The *Audio Configuration* panel (Figure 197) offers many more important features and controls, however. Let’s explore these now.

15.10.1 INPUT TAB

AUDIO DELAY

Audio and video arriving at inputs in sync will *maintain* sync throughout the system to output or recording.

However, you should note that *upstream issues* can cause video to arrive at system later than the corresponding sound.

To mitigate this sort of external problem, the Audio Mixer provides an adjustable *Audio Delay* feature.

E.g., many cameras support simultaneous digital and analog audio output. In-camera processing can delay digital a/v output, resulting in analog audio output leading the digital output by a meaningful measure.

GAIN

For sources set to *Mic* input type, *Gain* knobs may appear in the *Input Settings* tab to allow you to compensate for microphone variances.

PAN

The *Audio Configuration* panel also provides *Pan* controls. *Pan* is a very useful feature; it adjusts *placement* of sound from source audio channels on the stereo channels comprising the audio mix(es). Using *Pan*, you can place all or part of *channel 1* onto *channel 2*, and vice versa.

- When *Pan* is set to the extreme left position for *channel 1*, its audio is sent exclusively to the first channel for the *Input 1* group.
- Centering the *Pan 1* knob splits the sound from *Input 1* equally onto *channels 1* and *2*.
- Sliding *Pan* for *channel 1* fully clockwise results in that source only being audible on *channel 2*.

Pan also modulates the sound levels on the left and right channels so that the overall volume neither rises nor drops as a result of adjustments.

Hint: “Pan” is not the same as “Balance”. The balance control in a stereo system varies the relative level of the left and right channels, but sound from the left channel will never come out of the right speaker, or vice versa (Pan does permit this).

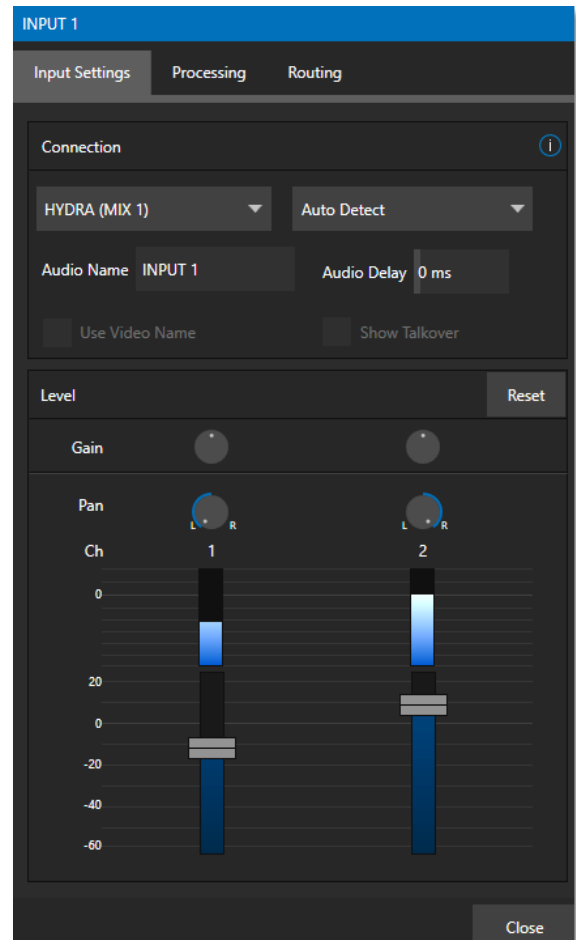


FIGURE 197

15.10.2 PROCESSING TAB

The second tab in *Audio Configuration* is named *Processing*, and likewise holds very valuable features.

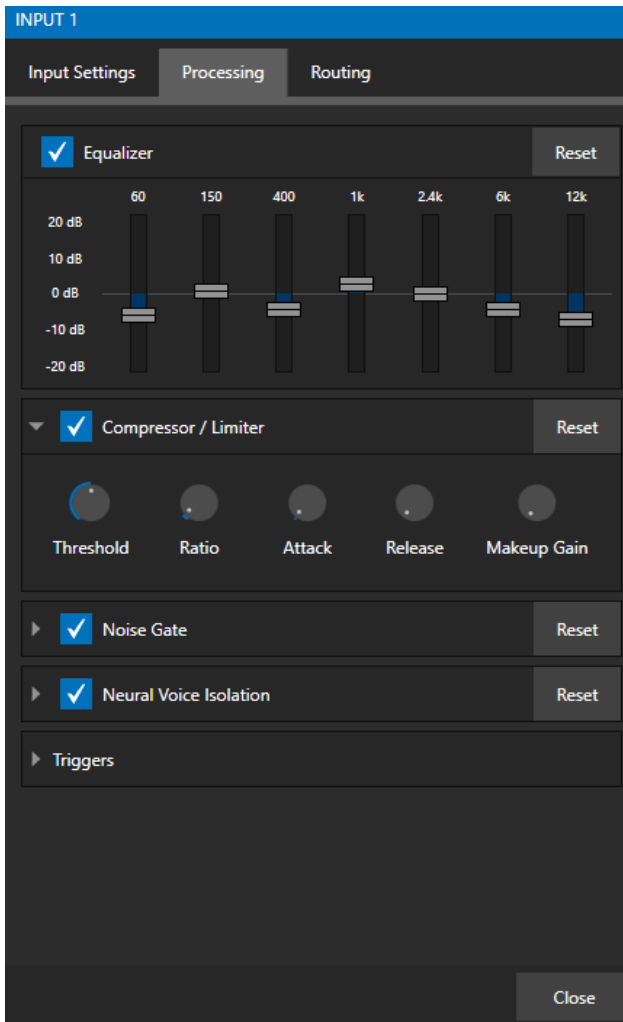


FIGURE 198

EQUALIZER

The seven-band equalizer allows you to ‘shape’ sound to taste, accommodate sources with different acoustic characteristics (say, mismatched mics), minimize feedback or roll off unwanted parts of the audio spectrum. Enable or disable the *Equalizer* using the switch beside the label above its control group.

The vertical sliders attenuate or boost the tonal range centered on the frequency shown at the top. The effect applied falls off gradually as sound draws closer to neighboring frequencies on either side. Click *Reset* to return all sliders to 0dB.

Hint: Naturally, reducing or increasing the level of one or more tonal bands affects the overall output level as well. This may call for you to trim the main level setting for the affected input or output.

COMPRESSOR LIMITER

The *Compressor/Limiter* can prevent clipping (see Section 15.9.1) from unexpected peaks or transients and making talent sound better than they do in real life, bringing voices, music, and other audio sources into an optimal dynamic range.

Being able to do this independently for each output too is icing on the cake, especially for Internet streaming, as it ensures correct levels at any time.

THRESHOLD

Sound above the set *Threshold* level will be compressed; the amount of compression and the way it is applied are both dictated by the other settings.

RATIO

A *Ratio* of 4:1 means that if input level is 4 dB over the threshold, the output signal level after compression will be just 1 dB over the threshold. The gain (level) is reduced by 3dB. Very high ratio settings are the reason the word “limiter” is part of the title for this feature.

The highest ratio setting will effectively reduce any signal that would rise above the threshold all the way down to the threshold level (except for a brief period during a sudden increase in source loudness, as dictated by the *Attack* setting).

ATTACK

Attack is also in milliseconds. The setting represents the amount of time it takes for the gain to change by a specified amount.

It would not be grossly incorrect to think of this setting as changing the slope of a graph depicting how aggressively the compressor pursues the target value (defined by applying the *Ratio* setting to the amount the signal surpasses the *Threshold*). Shorter values are more aggressive, while longer values are more subtle (and tend to be less noticeable to the audience).

RELEASE

Release is like *Attack* in many ways but refers instead to the speed with which the compression effect is removed as a source signal falls back on its own so that it no longer exceeds the *Threshold*.

GAIN

Naturally, compression impacts the overall output level of the source or output. The *Gain* control allows you to compensate, bringing the post-compressor/limiter signal back to a comfortable nominal range.

Hint: Different circumstances call for different Attack and Release strategies. For example, much less aggressive settings could work nicely for vocals, but fail badly when applied to a snare drum. Many websites provide suggestions on establishing the best compressor/limiter settings for different environments.

Compressor ... Limiter – what's the difference, anyway?

Compression and limiting are not really different processes, but rather a matter of degree and perceived effect.

Compression, ideally, takes the form of a subtle, almost imperceptible modulation of the sound level to bring it into a more pleasing and convenient range. A limiter is applied more for the purpose of managing, even 'crushing', unwanted spikes and transients.

That distinction aside, a limiter is essentially just a compressor set to a high ratio and, generally, a fast attack time. Audio engineers typically consider 'compression' with a ratio of 10:1 or more as 'limiting'.

NOISE GATE

The *Audio Mixer's* advanced options panel also include a configurable *Noise Gate* for each audio source, as well as all outputs. This lets you ensure that unwanted low-level sounds are prevented from inadvertently intruding into the mix.

NEURAL VOICE ISOLATION

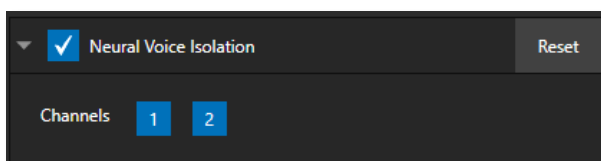


FIGURE 199

To enable/disable noise reduction, check the *Neural Voice Isolation* box and select your channel. AI audio can be individually selected for each channel of an input.

TRIGGERS

Several different per-input trigger (automation) features are found in the *Processing* tab.

FOLLOW PROGRAM VIDEO

Enabling *Follow Program Video* options for an audio source directs it to track switcher operations affecting the related video source.

Audio for sources with *Follow Program video* enabled in the *Audio Configuration* panel is automatically removed from mixed outputs until one or more specified video sources are displayed on *Program Output*.

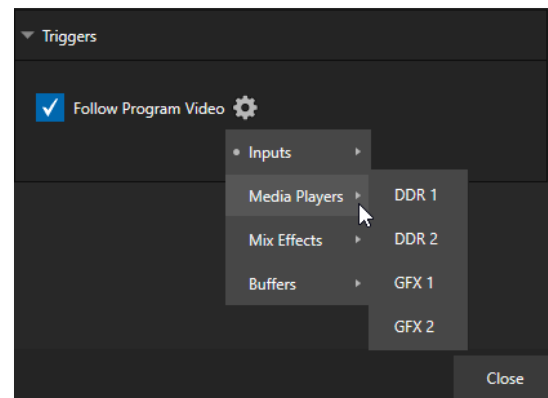


FIGURE 200

Hint: When the corresponding video source is not displayed on output, the audio source's VU meter level is displayed as a grayscale.

15.10.3 ROUTING TAB

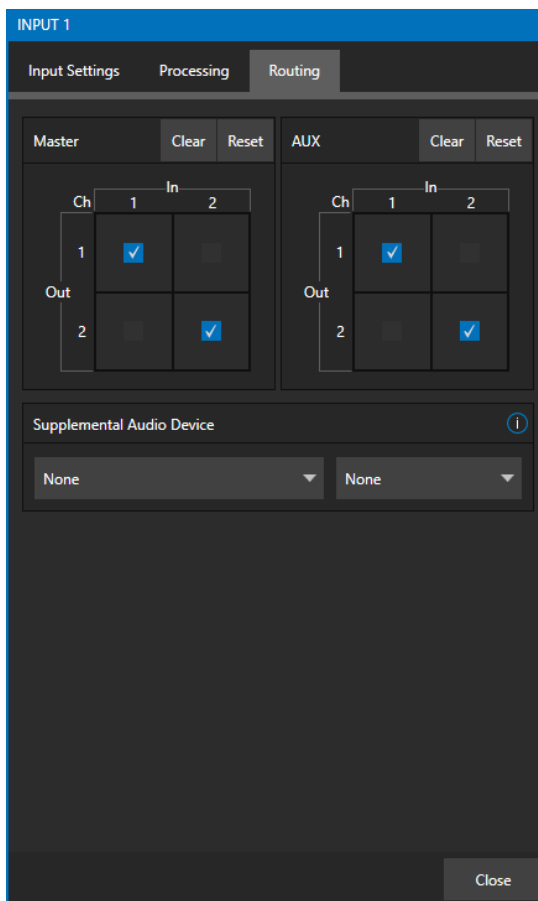


FIGURE 201

A main tab labeled *Routing* appears in the *Configuration* panel for all sources (Figure 201). The controls in this group determine output routing of the various channels supplied as inputs to the various output mix busses (*MASTER*, *AUX 1*, etc.) Let's discuss what an audio bus is and its uses before proceeding.

BUSSES AND OUTPUTS

Consider a very basic audio mixer. Its main audio signal path, from input to output, is properly called the 'master bus.' Sound supplied to one or more inputs is placed on this master bus (in the jargon of audio processing, this is called a 'send'), which ultimately flows to output connectors.

Slightly more advanced mixers often provide more than one 'send' for individual inputs. For example, the sound from all inputs may be sent to the master bus, comprising the 'master mix'. A different mix, sometimes called a 'sub-mix', might also be created by sending certain signals to a secondary ('auxiliary', or 'Aux') bus.

Hint: A secondary mix, prepared on an Aux bus, can serve many purposes. For example, you might wish to record a mix with all sound from talent microphones but that excludes any sound effects or music.

Let's summarize what we have learned so far: A 'send' pipes audio signals from an input to a discrete pathway called a 'bus'. Multiple sends can be used to place sound from a given source onto one or more internal busses.

What else should we know?

Each audio bus is discrete. Each can be directed along different output paths. And even when the blend of signals it carries is otherwise identical to another bus, it can be *processed* separately. Thus, its levels, equalization, and compressor/limiter settings can be unique.

The *Audio Mixer* provides four primary *audio busses*. These are identified in the *Audio Mixer* as:

- *MASTER*
- *AUX 1*

The *Audio Mixer* provides controls for each of these busses, allowing you to manage levels and signal processing. It is important to understand the distinction between *busses* and *outputs*. Now that we understand the former, let's consider the latter.

An *output* may be physical, or virtual – i.e., it may involve a connector on the rear panel, or not. For example, audio recorded internally does not require an output connector. Likewise, it may initially be analog or digital.

Note: Analog outputs 1 and 2 are permanently assigned to MASTER and AUX 1, respectively. In contrast, digital (or 'embedded') outputs are configurable in the Output Configuration panel.

SUB-MIXES AND 'MIX MINUS'

At times you may require specially configured audio mixes. For instance – some installations call for sending audio from one or more internal sources (such as a *DDR* or the *Sounds* player) to a secondary distribution system. Alternatively, you may want a 'clean' output from one or more sources for use apart from the main primary output mix.

Specialized sub-mixes of this latter sort are often referred to as 'mix-minus,' since one or more sources are deliberately subtracted from the main program. Mix-minus capabilities can be invaluable for productions like 'phone-in' shows. The remote caller needs to be able to hear the interviewer; but if you simply send the primary mix back to him, he is forced to endure a late-arriving echo of his own voice. This would be confusing and undesirable.

This approach eliminates annoying echoing, feedback and the like. Meanwhile, both participants can be heard on the main *Program* output. Also important, independent control and signal processing is provided for each part of the pipeline.

MIX MINUS FOR EXTERNAL TALKSHOW® DEVICES

The *Routing* tab provides four 2x2 matrix routing panels for each input, allowing for more sophisticated mixes than the example above. TriCaster Mini X supports 4 channel NDI output, so it's possible to route a unique mono mix-minus on each channel for a single Aux bus. A single NDI output can thus provide all the mix-minus needs for two Talkshow VS4000 systems.

Let's consider a simpler example – configuring unique mix minus audio for return to a single VS 4000.

The example assumes that you have assigned two *Switcher* inputs to receive the two individual TalkShow sources. We will use video *MIX 2* to supply *Program* video for return to each TalkShow caller. And we will assign *AUX 1* as the *Audio* source for *MIX 2*, assigning a unique (mono) mix minus to each of its three channels to supply return audio for each caller.

- In the *Output Configuration* panel, assign *Aux 1* as the *Audio* source for *MIX 2*.
- In the *Audio Mixer*, open the audio configuration pane for *Input 1*.
- Access the *Routing* tab and click the *Clear* button above the *Aux 1* routing matrix.

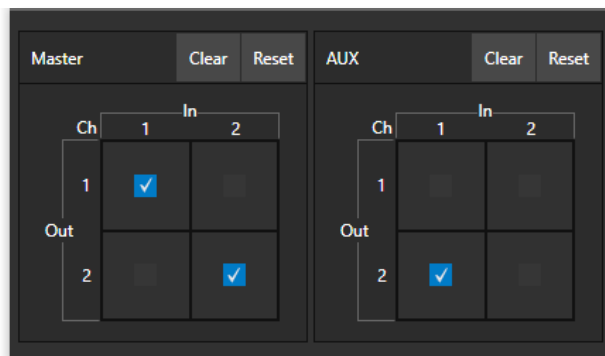


FIGURE 202

- Checkmark channel 2 in the column below *In 1*. This routes sound from the first incoming Skype caller to all *Aux 1* output channels *except* channel 1.
- Open the audio configuration pane *Input 2* and, in the *Routing* tab, clear the *Aux 1* matrix.

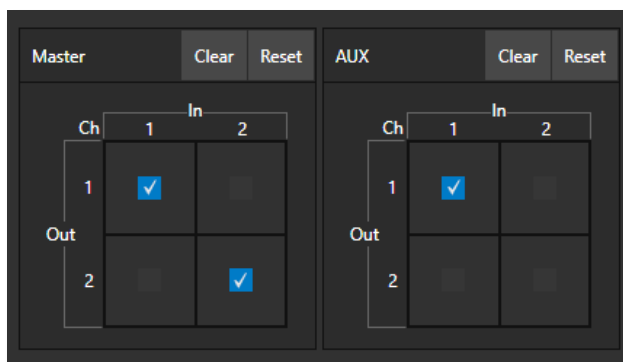


FIGURE 203

- This time, checkmark channel 1 in the column below *In 1*. This routes sound from the second incoming Skype caller to all *Aux 1* output channels *except* channel 2.

That's it – you've got the perfect mix minus configuration for return to your two Skype callers, using a single NDI output.

SECTION 15.11 SUPPLEMENTAL AUDIO DEVICES

TriCaster features two additional menus are provided at the bottom of the *Routing* tab, under the heading *Supplemental Audio Device*.

The left-most menu in this group lists any add-on output devices detected by the system, and a list of multichannel audio-only NDI outputs options.

For example, ‘transmit’ channels from Audinate’s Dante network audio protocol will be listed here if you have installed Dante Virtual Sound Card software. Or, in similar fashion, you may have installed AES67 drivers.

The audio connections on the motherboard are also listed here.

Making a selection result in the sound from source being sent to the corresponding audio channels of the designated output.

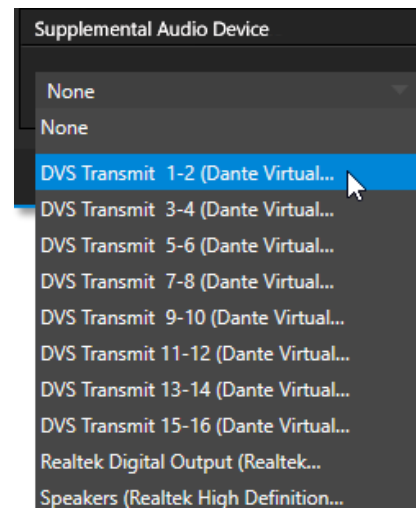


FIGURE 204

Note: Apart from the AUX and Master mixes, routing or mixing is not applied to supplemental audio. Each source channel is mapped to the corresponding output channel in 1:1 fashion, limited by the number of channels the output supports (e.g., a stereo output will always transmit channels 1 and 2 from the source).

SECTION 15.12 MEMS

Roll the mouse to the left edge of the screen in the *Audio Mixer* to show its *MEM* bin. Audio *MEMs* work just like their counterparts in the *Media Players* (see Section 11.1.11). *MEMs* are a convenient way to quickly store and recall audio steps and settings for different venues, productions, and users.

Chapter 16 MACROS AND AUTOMATION



Macros smooth out your workflow, reduce complex operations to a single button press, and make it easier to produce sophisticated programs. They provide many opportunities for both workflow streamlining and creative applications. (As well, macros can reduce or eliminate embarrassing operator errors.)

One of the hardest things about live switching is keeping up with the action. We're only human, limited as to how fast our fingers can move, recall, and perform important sequential steps, and so on. Macros are the answer to that dilemma. Record any sequence of events as a macro and play it back with one click. Alternatively, trigger it with a keystroke, or control panel operation.



FIGURE 205

Macros can do almost anything; Preload and play content, modify audio settings, automate complex switcher sequences, or perform synchronous operations. The compelling usefulness of macros justifies the prominent *Macros* menu placement in the *Dashboard* at the top of the *Live Desktop*.

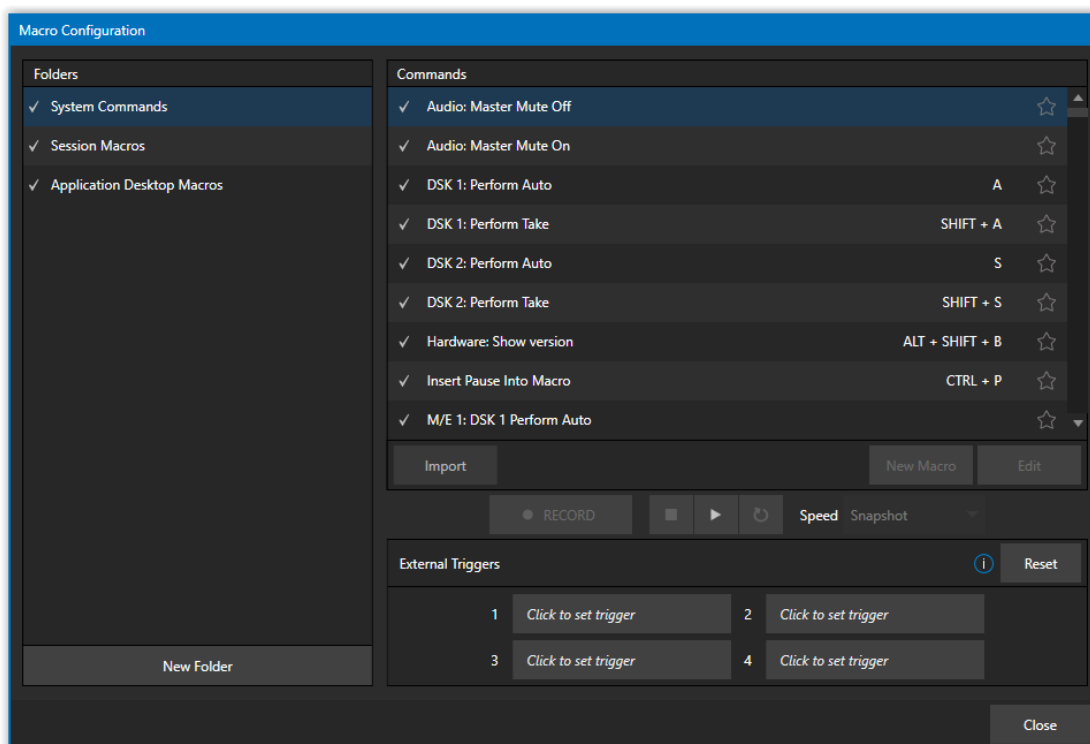


FIGURE 206

Click *Macros* to show a menu containing a *Configure Macros* item. This opens a large panel that allows to you create, organize, and even edit macros.

SECTION 16.1 CREATING MACROS

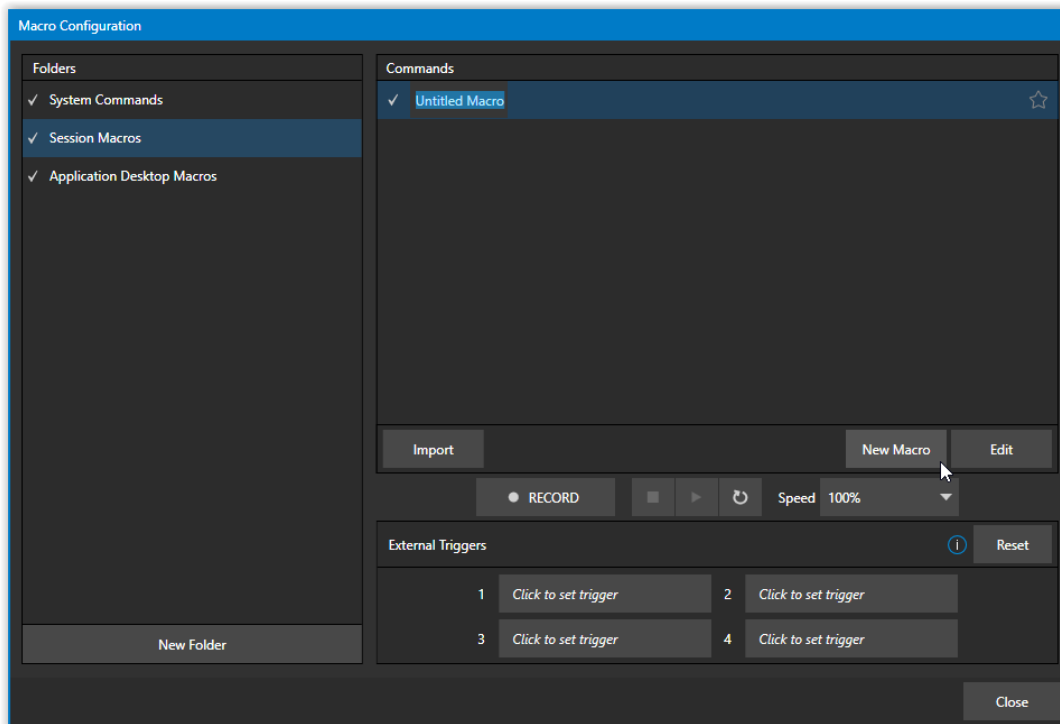


FIGURE 207

Creating a new macro is simple. Select a folder in the *Folders* column at left to contain the new entry (or add one, using the *New Folder* button below), then click the *New Macro* button below the *Macros Bin* at right.

Continue to define the macro by clicking the *Record* button at the bottom of the panel, and then just go ahead and perform the sequence of operations you wish to include in the macro. You can use mouse, keyboard, and *Control panel* operations when doing so.

When finished, click the *Stop* button to complete recording. Test the new macro by clicking the *Play* button. You'll notice that an animated bar in the background of the macro entry in the list tracks playback progress. Of course, you can modify the playback rate using the menu next to the *Record* button. You can even set macros to loop using the button at extreme right.

16.1.1 SPEED AND SNAPSHOT MODE

The *Speed* menu lets you modify the playback rate of your macro. One option in the *Speed* menu bears explanation: When you choose *Snapshot* as the macro's speed, you essentially force it to jump to its end result as fast as the system can get there. *Snapshot* mode is very useful for macros that configure the system to a particular state.

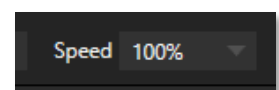


FIGURE 208

One example would be when you want to instantly reconfigure *M/Es* with different virtual sets for a scene change; or perhaps you want to quickly disable *LiveMatte* for all *Media Players* at once. The possibilities are endless.

Hint: You can record a macro that includes other macros. Depending on your order of operations, you may need to re-highlight the newly recorded macro in the list to show its Stop control (to end macro recording).

16.1.2 TRIGGERS

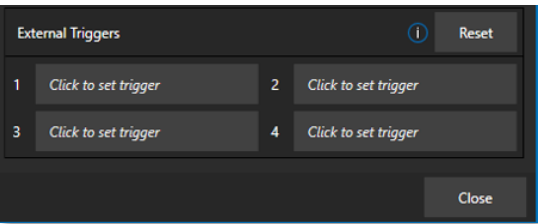


FIGURE 209

The External Triggers controls located near the bottom of the Macro Configuration panel allow you to configure one or more ways to launch macros. For example, select a macro, click in one of the Trigger boxes, and then press a suitable keyboard shortcut to assign it to that macro.

Note: Triggers can be a keyboard combination, Midi, X-Key, Control Surface or webpage button, or a GPI device signal. Click in a Trigger box and execute the desired trigger to register it.

16.1.3 LIVEPANEL BUTTONS



FIGURE 210

The *LivePanel Buttons* add the convenience of not only presetting macros, but the ease of editing them as well, all without ever leaving the user interface. Simply right click your mouse over any viewport to display the context menu, hover your mouse over *LivePanel* and eight *Preset* options will appear as shown in Figure 210.

To configure the buttons, right-mouse click over the viewport showing the preset buttons (Figure 211) and select *Configure* from the context menu. In the *Configure LivePanel Buttons* panel you can edit the size of the buttons, styling properties, the color of the buttons and/or text, and more (Figure 212).



FIGURE 211

CONFIGURE LIVEPANEL BUTTONS

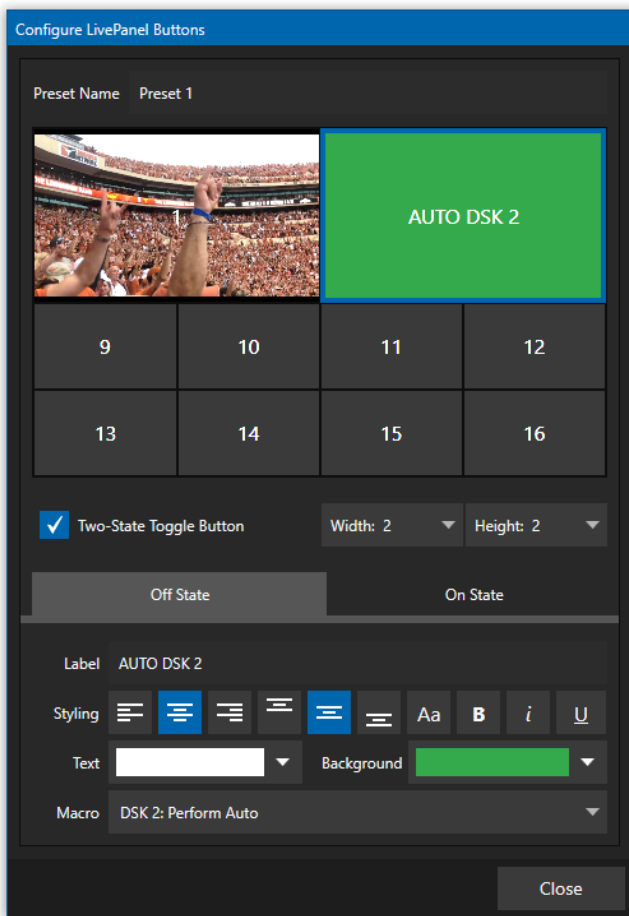


FIGURE 212

In the *Configure LivePanel Buttons* panel you can select the *Width* and *Height* of your buttons, (changing the size will 'absorb' other buttons, but they will not lose their properties). There is a *Two-State Toggle Button* checkbox to give you the ability to toggle on/off your macro with just a push of a button.

Just below the Two-State Button field, is the *Label* section and *Styling* options. Not only do you have many background color choices, but also an option to *Choose Image*, which can be helpful to remind yourself of what is on your viewport. In the *Macro* field, you have the option of using *System Commands* or *Application Desktop Macros* from the pull-down menu.

SECTION 16.2 MANAGING MACROS

The *Macro Configuration* panel has management features such as folders, rename, clone, and hotkey assignment, as well as *Import* and *Export* (use these to copy macros to other units, and for backup).

Hint: TriCaster includes additional and very useful import/export file format options for macros, including JSON, Excel (.XLSX), and even a working Javascript (.js) example.

16.2.1 SESSION MACROS

It's easy to keep macros designed for use with a specific production organized and accessible, thanks to the *Session Macros* folder in the *Macro Configuration* panel. Macros in this group are exposed within that session only (or new ones based on it, if the operator uses template sessions).

Chapter 17 STREAM/ENCODE



Live webcasting has dramatically altered the broadcast landscape. Information can now be shared among audiences around the world with very small transmission costs. The live streaming market provides many creative and profitable opportunities. When it comes to taking advantage of this new medium, your NewTek live production system places you in the forefront of all the excitement.

SECTION 17.1 INTRODUCTION

Encoding for a/v streaming and other purposes involves a wide variety of attributes. At times, too, account login details are required for your streaming service or CDN (Content Delivery Network). Further, you may maintain multiple streaming service accounts for different purposes, as well. In addition, you may capture encoded video files for other purposes.

TriCaster Mini's *Streaming Configuration* panel supports the creation and configuration of all the information and settings you are likely to need in this context.

SECTION 17.2 CONFIGURATION

To open the *Streaming/Record Configuration* panel click the *gear* at right of the *Dashboard's* large *STREAM/ENCODE* button.

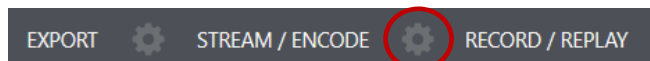


FIGURE 213

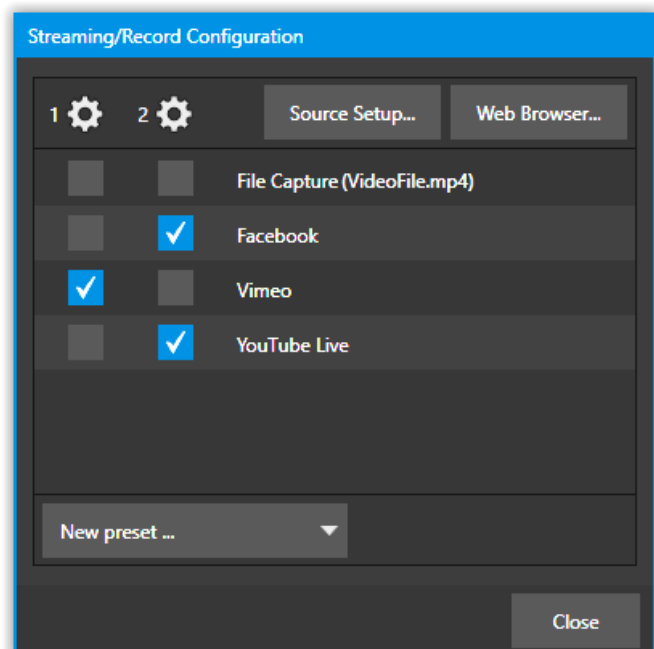


FIGURE 214

17.2.1 SOURCE SETUP

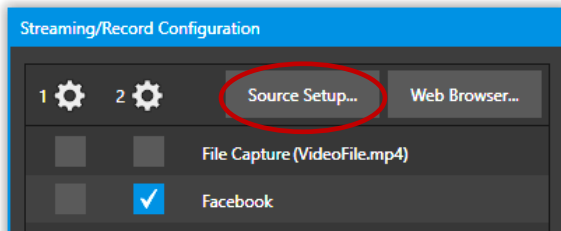


FIGURE 215

To assign video and audio sources to either streaming encoder, click the *SOURCE SETUP* button in the header of the *Streaming/Record Configuration* panel (Figure 215). Doing so opens the *Output Configuration* dialog, where you can continue to choose audio and video sources and adjust audio volume (Figure 216).

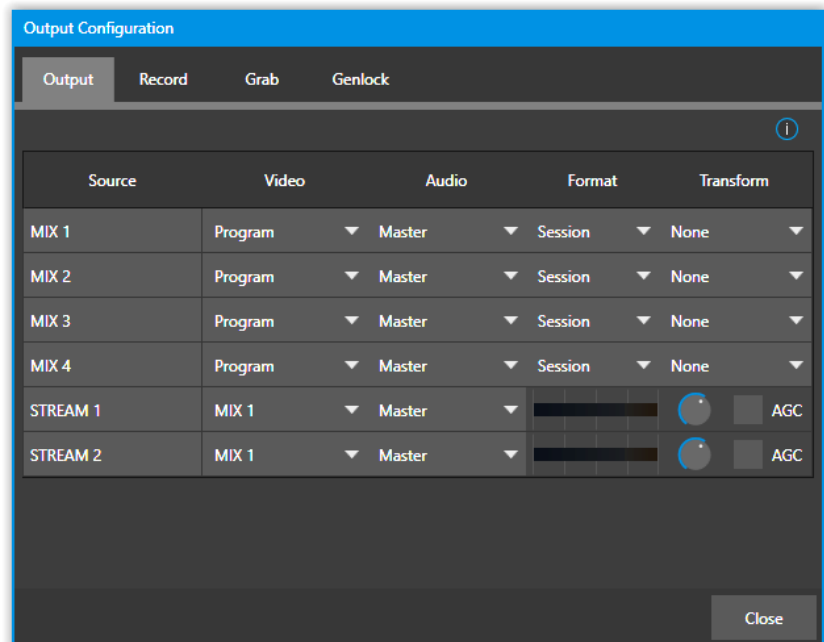


FIGURE 216

17.2.2 THE WEB BROWSER

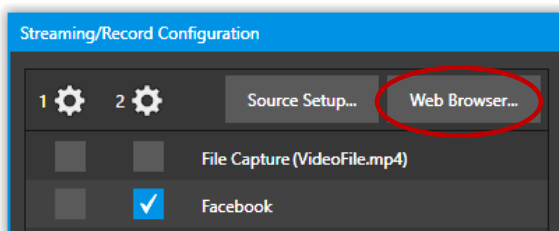


FIGURE 217

The *Web Browser* button in the header of the *Streaming Configuration* panel does what you would expect, opening a web browser. This feature is provided for your convenience when working with web streams or your content delivery network account. We do not recommend general purpose web surfing during live operations.

Hint: If the web browser is hidden from view because a Live Desktop operation took precedence, you can reuse this button to bring the browser window to the front again.

17.2.3 CONFIGURING ENCODERS

Settings for the two encoders are configured in the *Configure Encoder* dialog (Figure 218). Open this dialog by clicking the configuration (gear) icon for either encoder at left in the header of the *Streaming Configuration* panel.

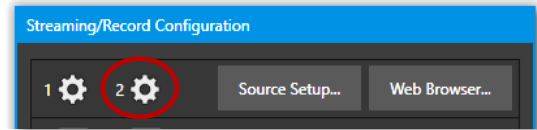


FIGURE 218

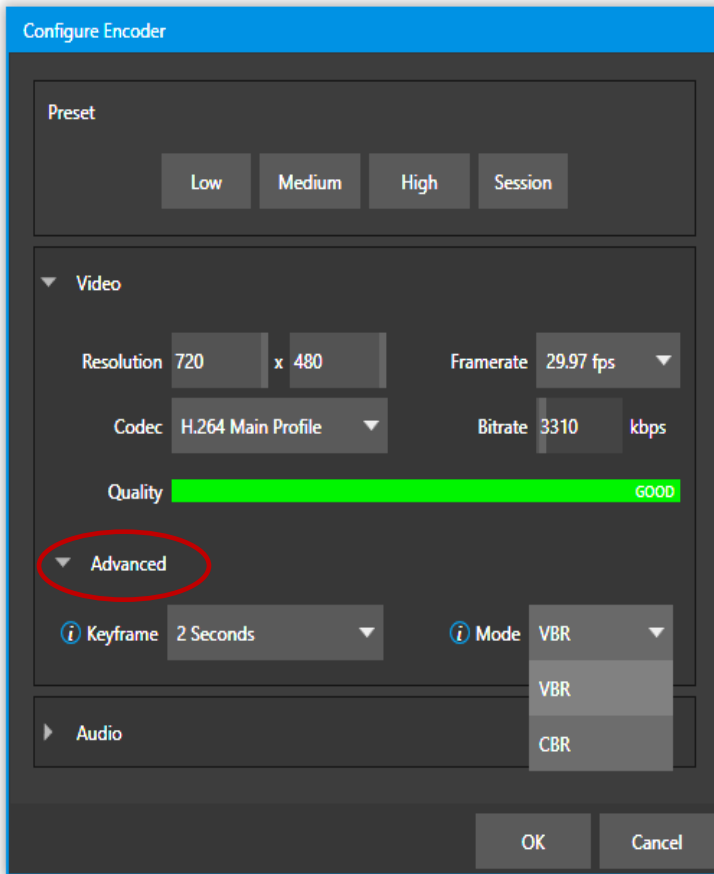


FIGURE 219

Select an encoder *Preset* (such as *Medium* or *High*), or manually configure custom settings using controls in the *Video* and *Audio* control groups. The *Codec* menu in this panel allows you to select from a number of H.264 profiles as well as HEVC.

Under the twirl-down *Advanced* menu you will see two options for *Keyframe* and *Mode*. *Keyframe* provides an encoder keyframe rate in (1 to 5) seconds and to the right an *Encoder Mode* option offers (constrained) variable bitrate (VBR) and constant bitrate (CBR).

Hint: With constant bit rate vs variable bit rate video, CBR is best for live streaming encoding, and VBR is best for on-demand videos.

If you choose to enable a preset whose target requires different encoder settings, TriCaster will identify potential conflicts and a yellow 'bang' will appear as shown in Figure 220. Hovering over the bang opens a pop up that will provide recommended encoder settings.

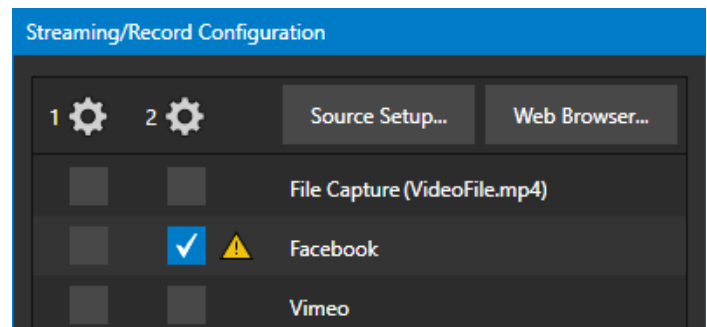


FIGURE 220

The information for a given CDN or other encoder target is also retained in a *preset* (e.g., Facebook, YouTube Live, etc.) you can add these presets as needed to the *Streaming Configuration* panel using the *New Preset menu* at left in the footer of the panel.

Hint: A default File Capture preset allows you to capture output from either encoder to a file.

At left in this panel are three columns of checkboxes; these allow you to send the output from either of the two *Encoders* to the presets at right.

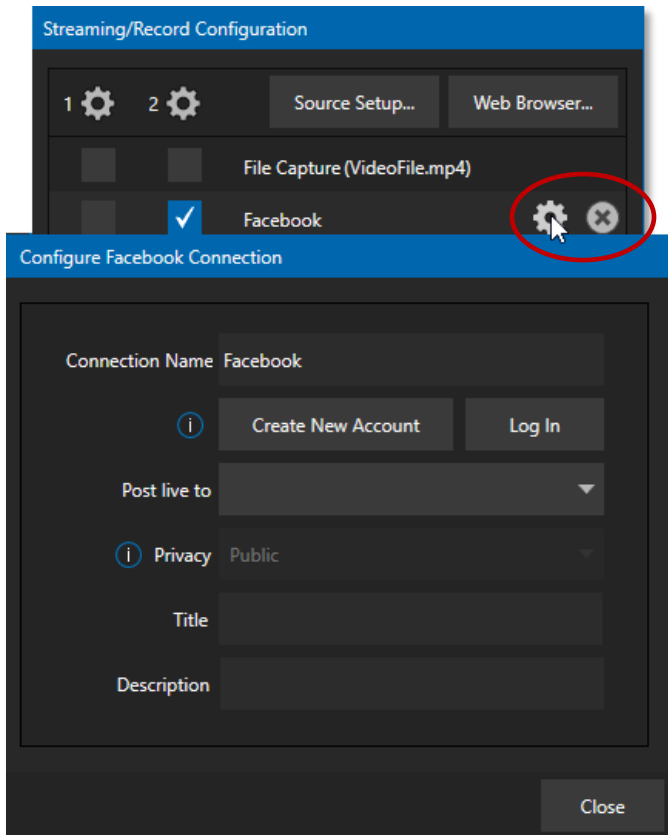


FIGURE 221

If you move the mouse pointer over a *Preset*, two new gadgets are shown at right (Figure 221). Click the familiar *Configure* gear to open a suitable dialog. It may invite you to enter your account credentials for a site and *Sign In* or provide suitable controls for other relevant settings that are required. Or, for a *File Capture* preset, you can choose the path for encoded file output.

Not surprisingly, clicking the little (x) gadget instead will delete the *Preset*.

17.3.1 CUSTOM PRESETS

In addition to the many recognizable names listed in the *New Preset* menu, you will observe a menu item labeled *Custom*. The dialog this selection opens allows you to enter the details typically required by generic streaming services.

In some cases, a CDN (Content Delivery Network) service may provide you with an XML file containing the recommended configuration settings related to your streaming account. (The *Import Settings* button in the Custom Connection dialog can be used to load this file, from which it will extract suitable values for streaming via the service automatically.)

SECTION 17.4 INITIATING THE STREAM

When you are ready, initiate streaming to the *Preset* targets you have check-marked in the *Streaming Configuration* panel by clicking the STREAM/ENCODE button in the *Dashboard* at the top of the *Live Desktop panel*. (The button will display elapsed time during streaming/encoding.) Once streaming has started, clicking the presets will start or stop that individual preset independently from the others.

Note: It should be noted that you cannot click a Preset streaming on one encoder and expect it to switch to another encoder. You must stop the first encoder, then select the other encoder.

SECTION 17.5 CAPTURING THE STREAM

To archive a live stream file as it is created by an encoder, simply configure and checkmark a *File Capture* preset for that encoder.

SECTION 17.6 STREAMING STRATEGIES

One of the best approaches when beginning (to stream your productions) is to establish a relationship with a commercial streaming media provider. A good provider can guide you past firewalls, provide public addresses for everyone to view your stream, and provide no end of valuable guidance.

And it may not be as expensive as you think (costs vary based on considerations such as how many viewers you expect, how much web bandwidth you use each month, and so-on). Some services based on an advertising model will even host your stream free.

17.6.1 ON DEMAND OR LIVE STREAMING?

Not all 'streaming' is 'live streaming.' The difference is similar to i) watching a television program you previously recorded at a time convenient for you, or ii) watching a live event.

On demand streams are stored on a server (often supplied by an external service provider), ready to be transmitted whenever a viewer wishes. Live streams are available at the time they are broadcast, such as during a live concert or event.

ON DEMAND HOSTING

The *Record* module permits you to capture your productions to a local hard drive. The resulting files can be hosted on a network later, so viewers can connect whenever they like. If you have the resources available, you can host the video yourself – but if many people will likely want to view your production, you will likely avail yourself of a service to stream it on your behalf.

Ideally, ‘on demand’ streaming video begins to play on request after a few moments. (Letting the stream get a bit ahead of the client playback device is called ‘buffering’ and helps ensure smooth playback). This stands in contrast to other types of online video distribution which requires the viewer to completely download the video file before he can begin play. Given a sufficiently high-speed connection between host and viewer, they may well be able to enjoy a seamless viewing experience without stuttering or other issues.

LIVE STREAMING

Live streaming is a growing international market, and one you may well wish to serve. This form of streaming is a somewhat more demanding implementation. Rather than record a file and deal with it later, live video is transmitted over the network (effectively in real-time, give or take a little ‘time in the pipe’ as it were.)

Delivering a good quality stream requires that you consider both your network connection capabilities and that of your viewers. Also, to ensure reliable delivery, you will ideally have some idea of the size of your audience.

Naturally, streaming video is highly compressed to reduce bandwidth demands and make it available to a wider group. The decision as to which encoding format to use for your live stream is up to you or – in some cases – your client. Here are some things to consider:

- Some corporate and institutional network administrators opt to support one or another format exclusively. (Check with your IT department to find out if this affects your decision).
- RTMP and RTSP combined have a very wide installed user base, and work well across multiple platforms (PCs, Macs, Linux, etc.).
- SRT is an open source protocol that is managed by the SRT Alliance. It can be used to send media over unpredictable networks, like the Internet. More information about SRT can be found here - <https://www.srtalliance.org/>.

RTSP STREAM DECODING

The processing demands from high-quality video applications and devices have increased in the last few years. As video content continues ever-expanding, technology evolves to handle the demand. Beginning with TriCaster version 7-4 dated January 2021 and later, TriCaster will take advantage of GPU hardware acceleration for all stream decoding.

Unfortunately, some streams are simply incompatible with the GPU decoder. We recommend that the originating stream vendors look to solve the compatibility and take advantage of modern GPU decoding. We also understand that users of TriCaster may not have that option and must wait for vendor development cycles.

As a workaround, if a stream is found to be incompatible, you can append the URL with a command that instructs TriCaster to not use hardware acceleration.

(optional components are enclosed in square brackets)

rtsp://\[username\[password\]@\]ip_address\[rtsp_port\]/server_URL\[?param1=val1\[¶m2=val2\]...\[&hw_accel=false\]\]

For example, the original URL of:

rtsp:// stream_IP_address.com:554/myStreamserver

Would change to:

rtsp://stream_IP_address.com:554/myStreamserver?hw_accel=false

BAND WIDTH CONSIDERATIONS

You'll often hear the term 'bitrate' in connection with streaming. This expression refers to data throughput per second (generally measured in Kilobits per second, or Kbps.) You could think of this as being like water flowing through a hose. You control the 'faucet', because you get to choose the *streaming Profile* setting in the system's *Configuration* panels. However, you don't own the 'hose' – or, at least, not the *entire* hose.

Once the stream leaves your immediate environment, even if you can supply good throughput locally, bandwidth may be constricted elsewhere along the transmission path. The level of Internet traffic can impose limits, but another major factor is the sort of connections your viewing audience may have.

Consider an example scenario: Even though you know that most of your audience is going to connect to your program using (relatively slow) wireless devices, you use a very high outgoing bitrate – thinking that this will surely be enough to fill the need. The fact is, though, a high bitrate ensures their experience will be poor. The client player tries to play at the specified bitrate, but (in this example) the wireless bottleneck impedes flow. It is as if you connected a fire hose on your end, giving them a suitable high-capacity nozzle for their end – but in the last stage of flow, the stream must pass through a small garden hose. Sadly, the stream will be quite insufficient, and output from the 'nozzle' (the client player) will falter badly.

For reliable performance, try to ensure the potential upload bandwidth from your system to the net is around twice the bitrate you choose. You *can* broadcast at a rate closer to your actual ceiling, but reliable performance cherishes headroom.

Also consider the expected download abilities of your viewers. Ideally, a safety margin 1.5 times the stream's bitrate is desirable. This may mean you need to consider using a lower resolution, or lower framerate for your stream – but doing so when required will generally deliver a smooth result and is the wise course. (Nothing inclines viewers to turn away quicker than a stuttering, start and stop stream. See "Speed Tests" in Section 17.8.1 for some useful resources.)

17.6.2 STREAMING MEDIA PROVIDERS

Using a commercial streaming media provider (sometimes referred to as a Content Delivery Network, or simply 'CDN') bypasses otherwise high-bandwidth requirements for the encoding computer. When you have made arrangements for a streaming media provider to distribute your stream, the encoder only needs enough bandwidth to get a single a/v stream to the provider. All end users connect to the *provider* to view the stream.

Most streaming providers have access to massive bandwidth (and often, with very little notice, they can scale up your allotment to meet a temporary need.) Since your local bandwidth is only used for uploading a single

stream, you can send a high-quality stream, secure in the knowledge that it will not degrade as soon as a second viewer attempts to see it.

Hint: A helpful way to find a good streaming service provider is to ask other members for recommendations in NewTek's online discussion forums.

SECTION 17.7 PRODUCTION AND CAPTURE CONSIDERATIONS

If you're not intent on live streaming, but wish to capture a live switching session, you would likely record at full resolution using the *Record* button (rather than *Stream*). The high-quality captured files can then be used later in a *DDR*, or perhaps be transferred to another computer (even on a different platform) for external processing or editing.

Hint: Use a portable hard drive to transfer the files between systems, or simply move them across a network.

You can always convert these files to a streaming file format if you later decide you'd like to supply them for 'on demand' Internet viewing. This lets you retain best quality right through to final output. When you eventually encode for streaming, you can choose settings that best suit the intended audience and streaming environment.

At the very least, if (perhaps to save conversion time) you capture video for web distribution using an encoder, it's best to capture it at least at the size that you intend for final output. This helps ensure satisfactory video quality for your viewers. When video is compressed (as it invariably is for web viewing) you can lose important detail; compressing a full-screen video down to a quarter or a sixteenth of its size is a lesson in humility!

OTHER FACTORS

Other variables to keep in mind when you're creating video for the web are contrast and motion. During video encoding for web distribution, a fair amount of video information and detail can be lost. For this reason, good lighting of your source video is essential.

Also, web streaming doesn't handle detail, transitions, and motion all that well -- so your best shots should be close up, and without a lot of movement. Too, audio from cameras and camcorders is rarely as good as that from external microphones. You should at least use a clip-on lavalier microphone, if not a directional or shotgun microphone to be sure you record only the audio you really want.

SECTION 17.8 DIAGNOSTICS AND TROUBLESHOOTING

Video streaming is becoming commonplace, but there are still a lot of things to consider. You have the necessary tools, but problems can occur. This section will point you in the right direction to overcome them.

17.8.1 TESTING YOUR STREAM

When it comes to using your system in a professional live production environment (i.e., your bread and butter depends on getting it right, and now - not tomorrow), failure to test beforehand is not merely unwise - it can be professional suicide.

You should already be aware of the need for redundancy in a professional environment (you didn't bring just one camera, did you?) As reliable as any device may be, Murphy's Law has not been repealed ... so you plan for this, bringing the appropriate equipment, such as uninterruptable power supplies, backup recording devices (there's no shame in having a VCR backing up your digital record – 'low tech' still has a place in the grand scheme.)

But you also need to perform onsite testing, to ensure your live stream is working well before 'zero hour.' No-one will thank you for excuses, no matter how brilliantly they point the finger at forces beyond your control.

1. Set up and enable a test stream.
2. You can use the integrated web browser to scrutinize the stream, but you should probably confirm using an external system, too.

Success at this point does not necessarily mean you're done. You may be able to see the stream locally, but can someone outside the local environment connect to it over the Internet? The best way to find out is to have someone at a remote location verify that your stream is streaming properly. If it is, great! Otherwise, keep reading...

TESTING WITH PING

Before your stream can be seen - whether on a local intranet or the Internet - client computers (or your service provider) need to be able to establish a network connection with your local system and its encoder.

Ping is a humble but effective tool to ensure the basic connection exists, thus it can help you with streaming, (and it works just fine in a multi-platform environment!)

Ping sends a small set of data packets to the target host (IP number), then 'listens' for an echo response in return. Ping estimates the round-trip time in milliseconds, records any data losses, and displays a summary when finished.

Bottom line, if you can't 'ping' your target, your connection has problems (the problem might be as simple as a bad cable connection). To issue a ping, you need to know the IP number of the target computer.

Finding the target IP number

For Windows XP®

1. Select *Run* from the Windows® *Start Menu* (look in the *Settings* sub-menu if it is not listed at the top level).
2. Type "cmd" (without the quotation marks) into the dialog, and press *Enter* on the keyboard.
3. In the command shell that opens, type "ipconfig" (without the quotation marks) and press *Enter* again.
4. The *IP Address* for the *system* will be reported in the window, along with other data.

For Windows Vista® (or later)

1. Type "run" (without the quotation marks) into the *Search* field, then press *Enter* on the keyboard.

2. Type “cmd” (without the quotation marks) into the dialog, and press *Enter* on the keyboard.
3. In the command shell that opens, type “ipconfig” (without the quotation marks) and press *Enter* again.
4. The *IP Address* for the system will be reported in the window (listed next to “IPv4 Address”), along with other data.

To find the IP Address for a system running OS X®

1. Click the Apple icon on the upper left on the Desktop and select *About This Mac*.
2. Click *More info ...* in the panel which opens.
3. Click *Network* in the *Contents* column at left.
4. The IP number for the system will be listed in the right-hand pane.

Issuing a Ping

Ping is a command line program and must be run from a command shell on the issuing computer. To open a command shell and send a ping, follow the procedure below that applies.

Windows®

1. Repeat the steps you performed above to re-open a command shell.

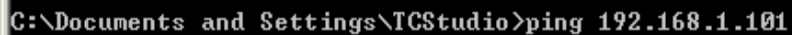


FIGURE 222

2. Type “ping” (without quotes) followed by a space and the target IP number, as in the image below – then press Enter.
3. Ping will go to work, and in a moment or two begin reporting results. A ping failure (indicating a network problem) will look like Figure 223. A success ping will display a report like Figure 224.

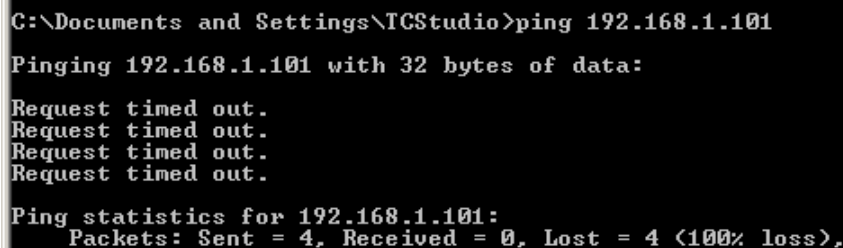


FIGURE 223

```
C:\Documents and Settings\TCStudio>ping 192.168.1.101
Pinging 192.168.1.201 with 32 bytes of data:
Reply from 192.168.1.201: bytes=32 time<1ms TTL=128
Reply from 192.168.1.201: bytes=32 time<1ms TTL=128
Reply from 192.168.1.201: bytes=32 time<1ms TTL=128
Reply from 192.168.1.201: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

FIGURE 224

Apple OS X®

For a system running Apple's OS X®:

1. Double-click Terminal in the Applications\Utilities folder.
2. Type the following command into the Terminal (without quotations) and then add the IP number, and press *Enter*:

"ping -c 4 *ipnumber*."

(So, for example, you might type: ping -c 4 192.168.1.101)

The response will be similar to the Windows® example described above. Again, a ping failure indicates a problem with the network connection.

17.8.2 SPEED TESTS

Are you sure your upload bandwidth is adequate to the bitrate you've set for your stream? Why not test and make sure? Again, several websites provide free speed testing. These will give you a basic idea of what your local bandwidth really is. One site which provides a list of online speed test resources is: <http://www.dslreports.com/speedtest?more=1>

17.8.3 WHERE IS THE PROBLEM?

With regard to streaming issues, don't overlook the fact that your NewTek live production system is just one of the pieces of the puzzle. There are many others that are just as important. One useful thing to try is to check the streaming file. Archive the streaming output file locally, and then examine it to see if it has any problems. This file corresponds exactly to what the encoder is sending to the downstream server. In cases of sync problems, low frame-rate problems, audio popping problems, etc., if it's an encoding issue, the problem will be seen in this file. On the other hand (if the file looks good), then the issue has to be after the encoder (perhaps a network problem or CDN configuration mismatch).

Chapter 18 EXPORT



The Export feature allows you to publish clips and stills from events like sports, entertainment, seminars, or news programming to social media sites, even while the live action and capture continues uninterrupted. Networks, corporate users and others with sophisticated website and distribution needs can deliver content moments after events occur for all manner of timely applications.

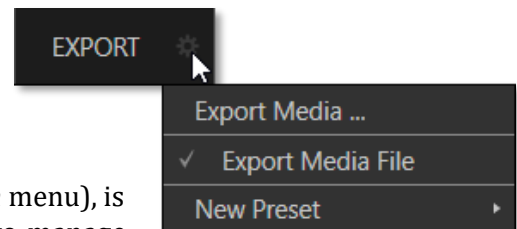
SECTION 18.1 OVERVIEW

In today's broadcasting world, a single program feed is often inadequate. Many viewers ingest media from multiple sources, even simultaneously. A live (and lively) online presence is critical for many productions with little or no traditional broadcast following. Publishing backstage feeds, locker room interviews, pre- and post-game chatter, etc.), to popular Internet sites can provide many opportunities for brand extension and monetization. In addition, the *Export* feature supports both transcoding and distribution to local or networked storage for archival or postproduction purposes.

Briefly, once you configure presets for your social media accounts and other publishing targets, the *Export* tools in the *Live Desktop* make it quick and easy to distribute selected content to multiple sites even while production and recording is still underway – just perfect for concerts, galas, and sport events.

SECTION 18.2 EXPORT MENU

The *Export* feature's user interface conforms in large measure to *Stream*, which we just discussed. Click the *Configure* gear next to the large *Export* button in the *Dashboard* at the top of the screen to open the menu (Figure 225).



At the top (where you would see *Web Browser* listed in the *Stream* menu), is an *Export Media* item. This opens the panel which allows you to manage media files you intend to export. We'll discuss this panel soon.

FIGURE 225

18.2.1 PRESET LIST

Below the *Export Media* menu item is a list of any presets you have configured as *Export* targets. (Before you create any presets for yourself, this list shows only the default *Export Media File* entry, a transcoding preset which cannot be deleted.)

Notice that each preset listed in the menu can be checked or un-checked. When you add media files to the *Export Bin* (by any method), the check-marked entries determine which *Export* targets are *automatically* assigned to them (you can also manually modify the presets for each item in the *Bin* at any time).

Hint: When the mouse pointer is over a preset in the Export menu, two gadgets are shown at right; the familiar gear allows you to modify the preset, and the (x) will remove it from the list.

18.2.2 NEW PRESET

Let's consider the *New Preset* menu item. Click this to display a sub-menu listing various supported *Export* targets (such as Twitter™, or Facebook™). Select an item here to open a dialog that lets you create a custom preset that will then appear in the list discussed just above.

For most external sites (i.e., social media sites or ftp sites), the dialog requires you to enter account credentials. Typically, you will need a *username* and password for your social media accounts, as well as FTP servers (credentials are tested when you press *OK*, and will report an error if a problem exists).

18.2.3 SOCIAL MEDIA SITES

Export can publish media to Facebook, Twitter, YouTube, and many more sites and services. Other connections may be supported as it becomes possible to do so.

18.2.4 TRANSCODE, SMTP, AND FTP

The *Transcode*, *FTP* and *SMTP* options let you handle various file conversions and publishing to local (or networked) storage volumes, or even as email. These are very useful output alternatives, effectively providing live export methods that do not force you to wait until production ends. This can be invaluable for collaboration, whether local or remote.

Transcode exports provide options to re-encode video in different formats before sharing. Select suitable options for your target device or application. For SMTP, replace the dummy email server entries with those of your own mail service.

18.2.5 WATERMARKING

To avoid unauthorized use of private or copyrighted media, you may want to add a watermark before exporting it to public sites. The *Watermark* feature in the footer of the various *Export* preset configuration panels allows you to select a suitable overlay image.

The image you select will be composited onto exported video or still image files. It should normally be a 32bit still image file (supported formats include popular formats such as PNG and Targa) that positions the watermark correctly in the frame considering the resolution and aspect of exported files.

SECTION 18.3 EXPORT MEDIA

Having discussed configuration of export accounts, let's move on to look at live operations. In the Dashboard's *File* menu, select the *Export Media* item to open this panel (Figure 226) which allows you to manage the list of clips and still images you wish to export.

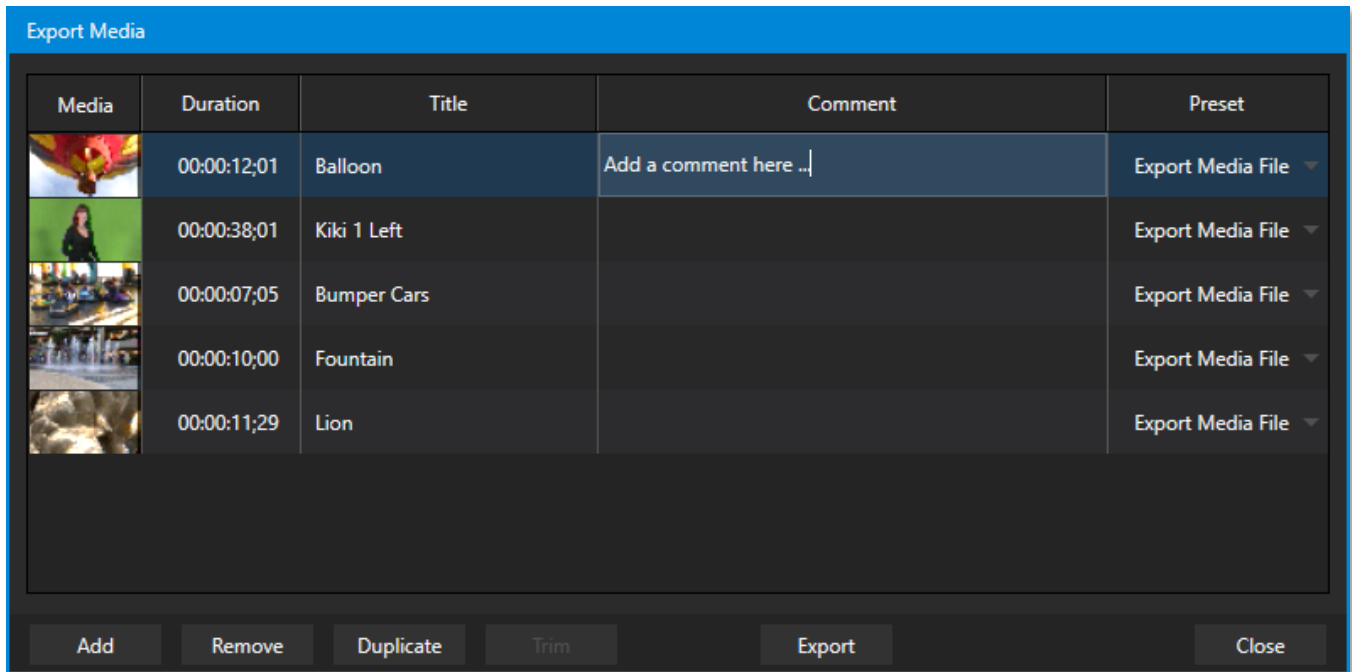


FIGURE 226

The features of the *Export Media bin* are powerful, but not hard to grasp. Click the *Add* button in the footer to open a *Media Browser*, which you can then use to select (and multi-select) content you wish to export. You can choose items from your current session or another location. These files will be added to the bin, but they will not be exported at this point.

Hint: You can Add and even Upload files that you are currently recording, even before pressing Stop. These files will be found in the virtual Session Recordings folder the Media Browser shows for the session.

18.3.1 METADATA

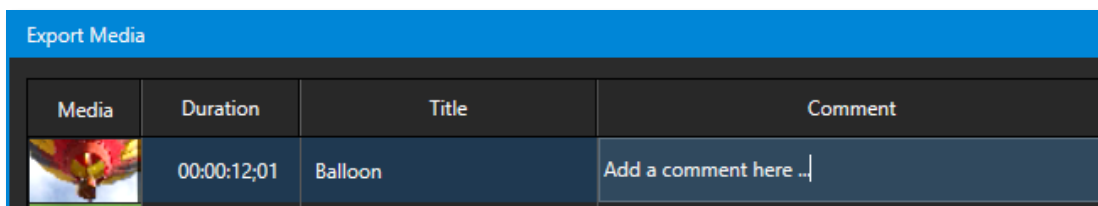


FIGURE 227

Information is displayed for each media file added, including the file name (or ‘alias’, in the case of files added from a *Media Player* playlist, as we will discuss later), *Duration*, and *Preset* (depending on settings, multiple entries pointed at different destinations may be added in one operation).

The *Comment* column allows you to add remarks that will be sent along with the upload to sites that support this. Click in this column to enable keyboard entry or navigate to it and simply start typing. Press *Enter* or click elsewhere to end editing. Similarly, type in the *Title* column entry to change the title supplied for the file to social media accounts but note that this has no effect on *FTP* or *Transcode* preset operations.

18.3.2 PRESETS

The *Preset* column provides a menu allowing you to change the export preset or even add multiple targets for each individual item in the bin. A checkmark beside a preset indicates the target is active.

Of course, *Preset* selections can vary from one item in the *Bin* to the next. Obviously, if no checkmarks appear here, the entry on that row is not fully configured, and will be ignored by the *Export* processor.

18.3.3 LIST MANAGEMENT

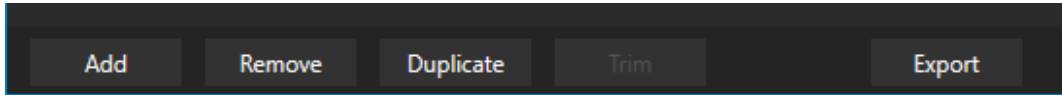


FIGURE 228

We touched on the *Add* button in the previous section. Three other nearby features help you manage your *Bin* entries.

- *Remove*, as you'd expect, deletes entries from the *Bin*, doing so without any effect on the source files.
- *Duplicate* clones selected entries. You may prefer this way of publishing an item to multiple targets (over adding multiple checkmarks to the *Preset* column for a single entry) at times.



FIGURE 229

For example, the encoding settings for one target may involve longer processing than you want to allow at the moment. Using a duplicate entry, you can defer processing that entry until a more convenient opportunity.

- Click the *Trim* button in the footer to open a trimmer pane (Figure 229) that allows you to set the *In* and *Out Points* for clips you plan to share (some file formats do not support trimming).

Clips that are added while still being captured can be re-trimmed to take advantage of file ‘growth’.

Hint: Checkmark the Still Frame switch to select a frame and convert it to a JPEG image file on upload.

18.3.4 THE EXPORT BUTTON



FIGURE 230

Also located in the footer of the *Export Bin*, the *Export button* is a toggle; that is, like the light switch in your foyer, it has two states – on and off.

When lit, the *Export processor* is activated, and is either watching for qualified entries *Bin* entries to appear for exporting, or actively processing those that are ‘ready to go’. (Conversely, of course, no light, no go ...)

Note: The footer Export button is linked to the Dashboard Export button, which does the exact same thing.

18.3.5 OTHER ‘ADD TO’ METHODS

Let’s handle the final details you need to become a social media maestro. It seems obvious that manually managing the minutiae of exporting could intrude into the already hectic life of a switcher operator. Live production already demands a lot of attention.

Using *Add* in the *Export Bin* panel, entering comments, etc., would be quite a distraction. For this reason, Export makes it as simple as possible to share your media, as follows

Once you have configured your *Export* presets, check-marked one or more in the *Export* menu, and decided whether to add checks to *Stills*, *Clips*, or both in the *Auto Queue* menu, you have eliminated a good deal of the fuss. But we can do better yet – in two ways:

- You can flexibly add both stills and clips to the *Export Bin* with a single click, keystroke, or button press. You can also add items from the playlist of a *Media Player*, or directly from the *Grab* and *Record* features (see the heading File Operations in 11.1.1).
- Second, you can even automate the matter of adding custom titles and comments. *Grab*, *Record* as well as the *Media Players* all support individual custom *Names* and *Comments*, which will supply the corresponding metadata for your *Export Media* additions.

Better still, the *Name* and *Comment* fields in *Grab* and *Record* support *DataLink* (compare Section 11.2.3). This lets you automatically supply unique and meaningful information to social media sites along with your *Export* uploads. You can even mix *DataLink* keys with literal text, to produce comments that embed things like the current time, the current score of a game, or the name of the person on camera at the moment into coherent sentences.

Chapter 19 RECORD, GRAB, AND REPLAY



You will often want to capture video clips from external sources, as well as to record your own live productions. Similarly, it can be quite useful to be able to grab stills from Program output for use in the current production. This chapter will provide everything you need to know about this topic.

SECTION 19.1 RECORD

NewTek™ live production systems provide a great deal of flexibility when it comes to capturing your program, selected elements of it, and a wide array of internal and external sources (with embedded timecode).

NewTek's IsoCorder™ technology delivers powerful recording capabilities. As TriCaster primary outputs are configurable, you can use them to capture *Program*, *Program (Clean)*, individual *Switcher* sources, *M/E*s, and more. You can record up to eight such “Mix” video sources simultaneously, and capture Switcher input sources (recording capabilities are affected by to your storage volume's bandwidth and capacity).

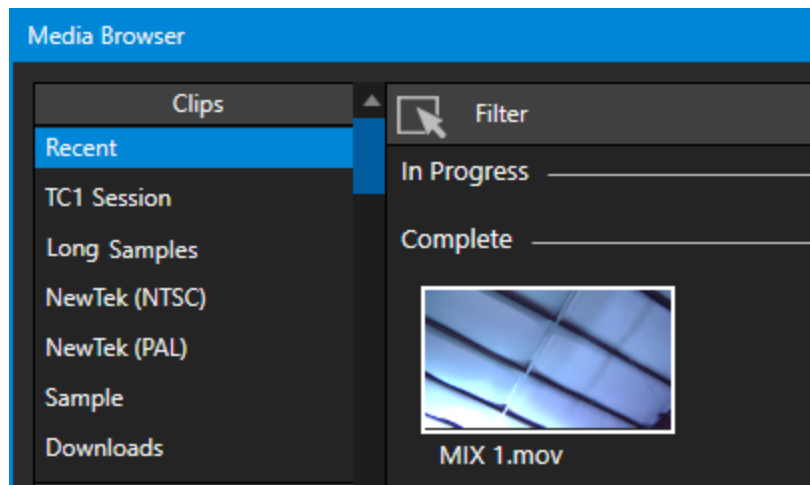


FIGURE 231

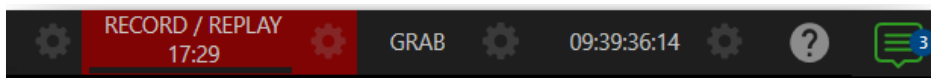


FIGURE 232

Recording is easily enabled and disabled by clicking the large *RECORD* button in the *Dashboard* (Figure 232). During capture, a nearby time counter tracks the length of clips recorded with the current base filename, and a horizontal VU meter beneath the button assures you that you are capturing audio.

Note: It is not necessary to interrupt recording to add a clip that is being captured to a DDR playlist or the Publish Bin. Clip icons show a red 'recording' indicator on clips currently being captured in playlists, the Publish Bin, and in the Media Browser.

19.1.1 RECORD CONFIGURATION

Of course, before you begin recording, you'll want to determine *what* to record, *where* to record it to, and so on. To support IsoCorder's flexible ability to capture almost anything, settings and controls for recording are provided in the individual *Input Configuration* panels as well as the *Output Configuration* pane. Click the *Configure* button (gear) for the control panel of the source you wish to configure to access these settings.

Let's review the features and settings found in these control groups.

19.1.2 CAPTURE CONTROLS

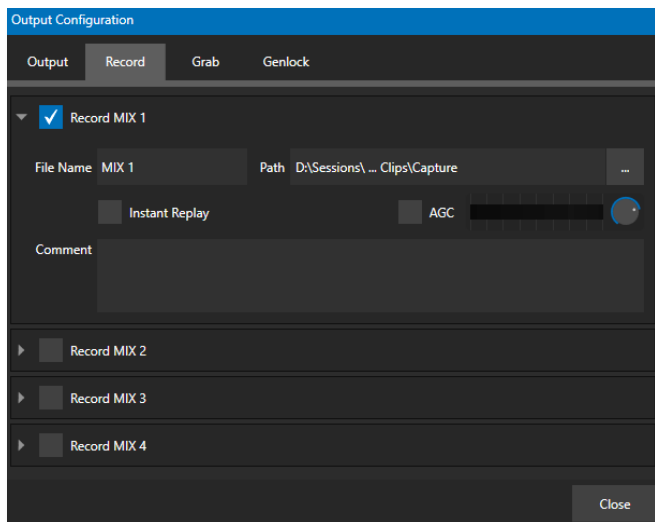


FIGURE 233

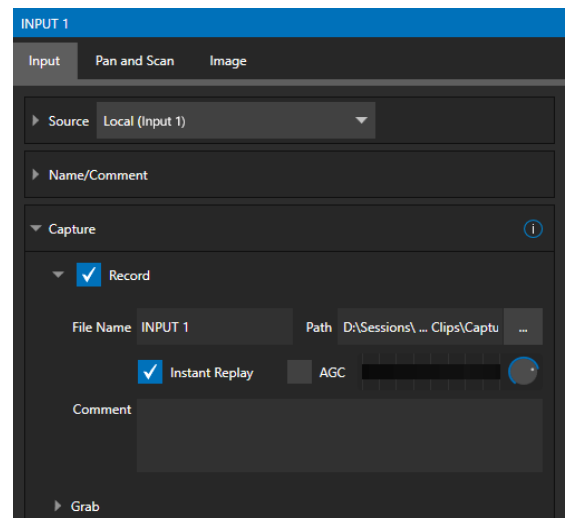


FIGURE 234

Whether you are setting up a recorder for a video input or output *Mix*, the *Capture* control group features are the same.

- A switch at the top is used to enable or disable the recorder
- Filename and path text boxes appear just below.
- A convenient *Drive Speed Test* button appears to the right of the *Path* field. Click it to open a utility that lets you evaluate the speed and capacity of your storage volumes.

Note: The same source cannot be selected for multiple recordings. For example: "IN 1" cannot be selected as the source for two recorders.

FILENAME AND COMMENT

IsoCorder uses the source name to supply the default filename for each recording (names are numerically incremented automatically as well), but these can be edited. Conveniently, NewTek's DataLink key system allows you to insert dynamic values from real-time production sources into the filename.

For example, every recorded or grabbed file can automatically incorporate a date stamp, identify the input selected on the *Program* row at the time of capture, and so on. This is useful not only in locating files from specific sources later, but also when posting media to social media services using the *Export* feature.

Note: IsoCorder captures a low resolution ‘preview’ file along with the high-quality recording. We do not recommend deleting or moving these supplemental files when using the recordings on a NewTek product, as doing so will result in higher resource usage during live production.

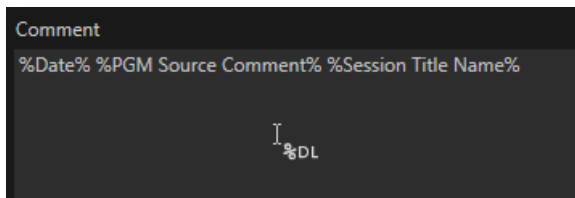


FIGURE 235

The *Comment* entry can likewise use *DataLink* values, and in turn can be used to provide metadata to title pages that refer to the captured clip’s comment.

PATH

Click the three dots at right beside the *Path* field to select from available storage volumes as targets for the recording.

Caution: It is recommended that any single drive be tasked to capture one or two video sources at most. A warning message is displayed if you exceed this number when assigning Destination settings. You are allowed to exceed this limit, however, when you are confident that very fast volumes can handle the load.

AUDIO LEVEL

The *Audio Level* control lets you set the volume independently for each source you capture and feature a convenient AGC option (Automatic Gain Control).

FILE FORMAT

IsoCorder records a high-quality Quicktime file. (You can use the *Transcode* function in *Export* to supply files in several different formats, when necessary.)

Hint: Download free NewTek codec packs for Windows® and Apple® computer platforms from the Downloads page on NewTek’s Support website. TriCaster record format is also supported by the NDI file plugin for Adobe Premiere included with the free NDI Tools bundle.

SECTION 19.2 REPLAY

Not too surprisingly, the *Replay* switch in each *Capture* control group enables instant replay features for individual recorders. Before considering how to perform a replay, let’s examine the *Dashboard* menu *Replay* options, which affect its playback settings.

- *Replay Duration* – determines the length of replay clips added to the *DDRs*

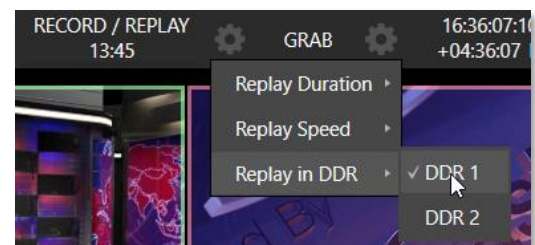


FIGURE 236

- *Replay Speed* – set the playback speed for clips added to the *DDR* using the replay workflow
- *Replay in DDR* – select a *DDR* as the target for replay clip playback

19.2.1 INSTANT AND DEFERRED REPLAYS

The method of triggering a replay varies depending on whether you want to replay a clip captured from a *Mix* (output) or directly from a *Switcher* input.

19.2.2 SWITCHER SOURCE

When the *Record* and *Replay* switches in the *Capture* control group for a *Switcher* source are both enabled, special replay controls are shown beneath its buttons on the *Program* and *Preview* rows of the *Switcher*

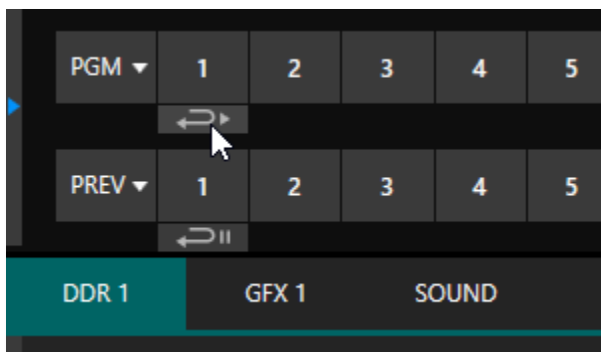


FIGURE 237

Click the replay button beneath the *Program* row to trigger an instant replay using the *DDR*'s *Show On* (PGM) feature and settings. (Or hold down CTRL and click the larger *Switcher* button itself to do the same thing.)

The similar button beneath the *Preview* row is even more useful; it adds the replay clip to the *DDR*, and leaves it selected, but does not play it on *Program Out* immediately. When you are ready to insert the replay clip into your program, press the keyboard *CTRL* key and the *Spacebar*. This will trigger the *Show On* (PGM) feature for the *DDR* currently assigned to replay duties. This workflow lets you defer replay playback until a more appropriate moment.

19.2.3 MIXED OUTPUT

Since outputs do not appear on the *Switcher* rows, the methods above do not apply. Instead, simply press a number pad key from 1-4 to trigger an instant replay from the corresponding output (Mix 1-4) recorder.

Hint: It's possible to assign an NDI output from the system to one of its own inputs if you prefer to use the same method as described earlier for your output replays.

SECTION 19.3 GRAB

At times, all you really want to capture is a still image from the current *Program Output* video stream (or perhaps, an input). This is the purpose of the *Grab* function.

Grab is represented by a large button in the *Dashboard* at the top of the screen, and smaller grab (camera) icons which appear at upper-right on *Switcher* source viewports. The former grabs stills from MIX outputs (when enabled individually), while the latter grabs an image from specific *Switcher* sources.

THE GRAB MENU

Click the *Configure* button (gear) next to *Grab* in the *Dashboard* at the top of the *Live Desktop* to choose a target *Media Player* for grabs. Otherwise, input and output grab configuration options are very similar to those described above for recording.

Note: De-Interlacing is automatically applied to avoid a 'comb' effect caused by inter-frame motion in fielded sessions.

Chapter 20 TITLE TEMPLATES



Your TriCaster Mini X ships with a huge number of stylish title template pages that can be edited even while live, as discussed in Section 11.2. Many include replaceable images embedded in them. This chapter explains how you can create your own custom title template pages.

Of course, you can create static titles and graphics in Adobe Photoshop® or other popular graphics and paint applications, and display these as images in (for example) a *DSK*.

However, you will often find it useful to create, instead, an *editable* title page in Photoshop that, once loaded and compiled by TriCaster as a native CGXML file, works just like those supplied with your NewTek system. The text remains editable, and embedded images can be marked replaceable too.

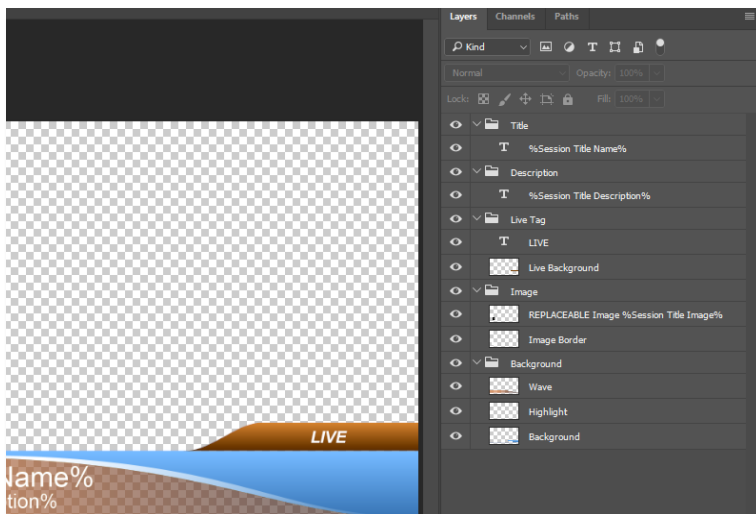


FIGURE 238

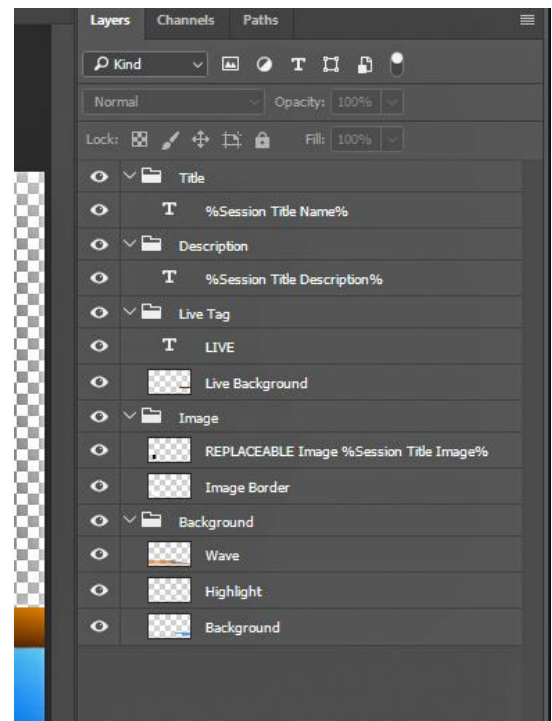


FIGURE 239

To make an image replaceable, simply add the string “REPLACEABLE” to the layer’s label in Photoshop. You can even assign a *DataLink* key to the layer in the same fashion – naming the layer in the format “REPLACEABLE %twitterpic%”, for example (without the quotation marks, of course).

As mentioned above, when you load the PSD into a *Media Player*, then open it in the *Title Editor*, the text is editable, and placeholder images can be replaced. And it gets even better if you load the file into a *Buffer*.

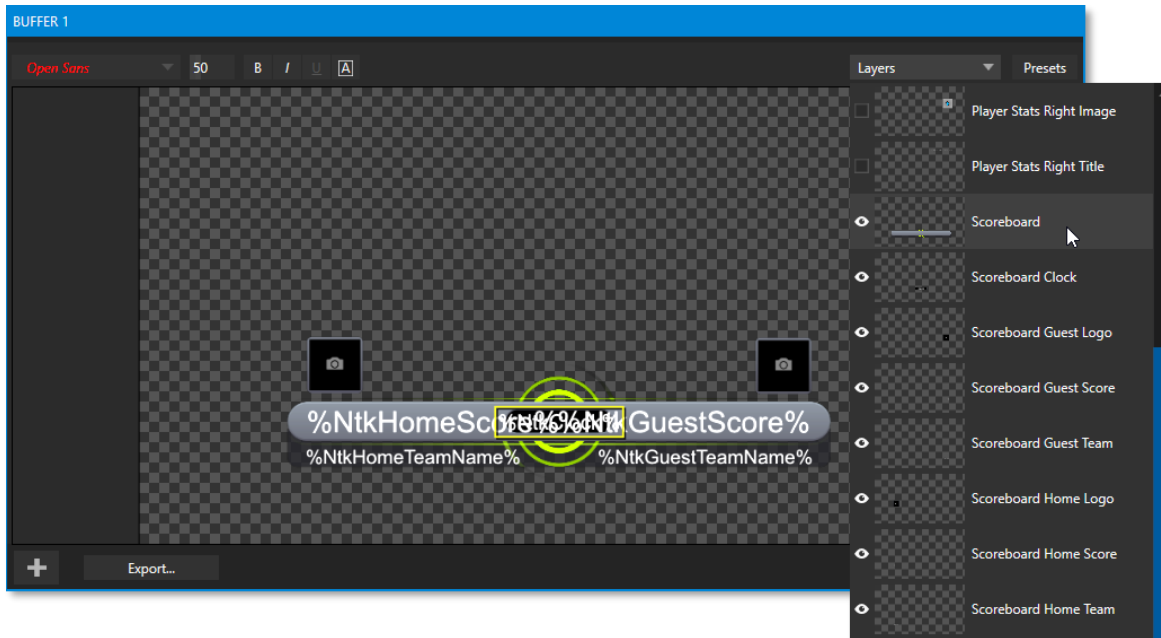


FIGURE 240

In that case, the *Title Editor* will show both *Layer* and *Data* presets, just like when you load a *LiveGraphic* title page, and store the result in a preset. Having done this, you can selectively hide or display different layers (or layer groups) of your Photoshop composition with a click or tap using the *Buffer* preset system.

In this manner (using *Layer presets*), a single title page can be used to display a whole theme pack of CG elements, and likewise (using *Data Presets*) a single title page can be updated to show individual player statistics for a whole team, and so on – all with a single click.

PART III (CONTROL PANELS)

Your Live Production system can be taken to new levels of convenience and functionality with the addition of a supported external hardware control panel, described in these Chapters.

Chapter 21 CONTROL PANEL: TRICASTER FLEX

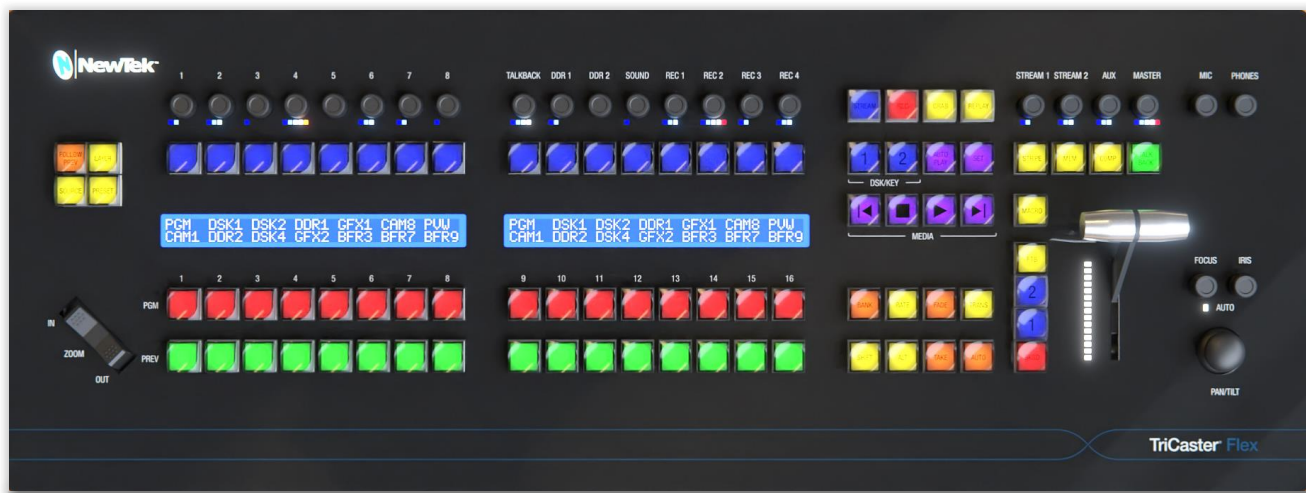


FIGURE 241

The Flex control panel is the ideal complement to supporting live production systems. It delivers new levels of control and usability while maintaining a convenient, compact footprint. In addition to Media Player and overlay controls, innovative PTZ and live switching features, Flex also provides extensive audio control, including integrated audio connectors. And it supports more extensive and user-friendly customization than any of our prior control panels.

In this chapter, we'll first explain how to connect and configure your new control panel, then continue to dig into its powerful controls and features.

SECTION 21.1 CONNECTION AND CONFIGURATION

Here are the basic steps needed to get you up and running with your new control panel.

- Connect the Flex control panel to your local network.
- Connect AC power to the control panel.
- Find the Flex control panel's IP address.
- Enter the Flex unit's IP address into the URL field of a web browser of another device on your network to access the configuration webpage.
- Check that the most recent firmware is installed.
- Confirm that your live production system software version includes Flex support.
- Choose a target system to control and connect to it.

For some people, the bullet points above will suffice. For the rest of us, though, this section will now continue with more elaborate details about each step.

Note: This equipment must be powered using a 3-prong connection.

21.1.1 SAFETY

Warning: Risk of Electrical shock. Disconnect all power sources before servicing.



This Protective Earth (ground) symbol identifies terminal which is intended for connection to an external conductor for protection against electric shock in case of a fault, or terminal of protective earth (ground) electrode.



This indicates that the equipment must be powered using 100 to 240 Volt Alternating Current.

Replacement fuse: 3A 250 V AC DC Fuse Cartridge, Glass Holder 5mm x 20mm, Slow Blow

21.1.2 CABLE CONNECTIONS

To begin, please connect your Flex control panel unit to your local network. In normal operation, this would be the network your live production system is connected to, but this isn't a requirement for initial setup (such as updating firmware or similar administrative tasks).

Hint: In normal use, the control panel and target live production system should be on the same subnet. Otherwise, for more sophisticated network environments, note that Flex also supports NDI Discovery Server.

Afterward, supply power to the Flex unit using the AC cord provided. After a few moments, Flex will complete its boot process, and briefly display the currently installed firmware version number.

Note: DHCP is enabled on Flex by default, so (assuming your network has a DHCP server) the unit will automatically connect to your LAN. If your installation requires static IP addresses, you can set this up later using the Flex configuration webpage. (For details, see the Network control group in Section 21.2.1).

21.1.3 FLEX CONFIGURATION WEBPAGE

Your Flex panel has a built-in webserver, which it uses to provide additional configuration settings and tools you can access in the web browser of another device (such as a laptop or tablet) on the same network.

Note: Together, the webpage described here and the panel's integrated control features host all necessary Flex configuration. (The control panel utility provided in the Add-Ons menu described in other sections of the User Guide does not support Flex.)



FIGURE 242

To access this webpage, first press the SHIFT, ALT and BANK buttons on the control panel simultaneously. This will display the unit's local IP address above the program (PGM) buttons. Enter the IP address into the URL field of a web browser on your LAN to open the local Flex configuration webpage (Figure 243).

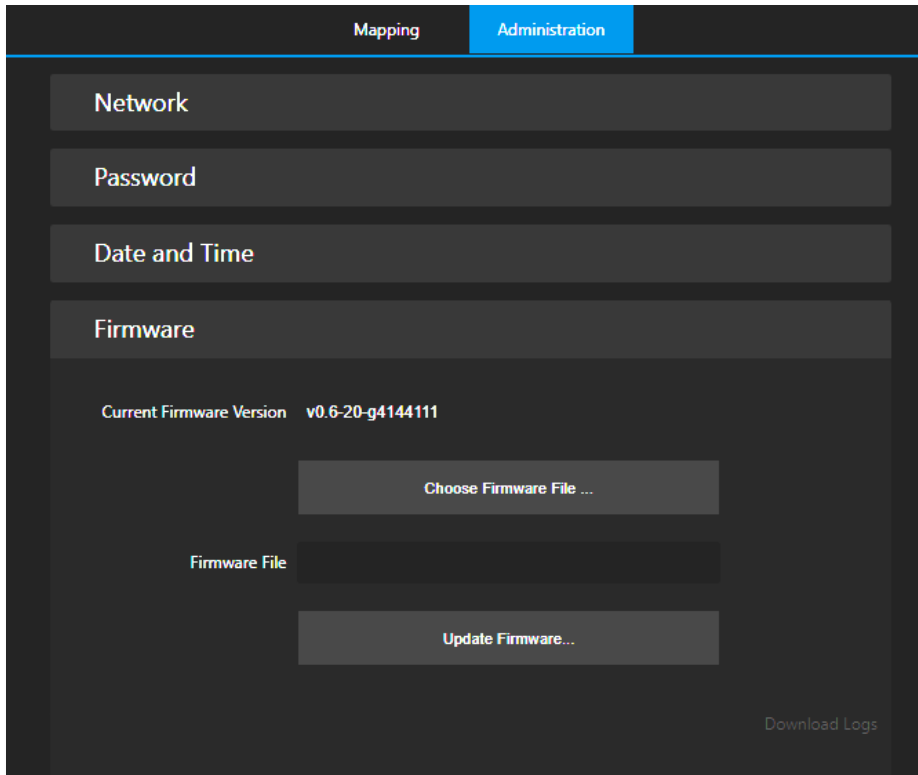


FIGURE 243

The first time you visit this webpage you'll be guided to create User and Password credentials to continue. Having done that, it's a good idea to check whether newer firmware has been released before continuing. On the Flex webpage's Administration tab, expand the Firmware control group to locate the Current Firmware Version number.

21.1.4 LATEST VERSIONS

Then, visit Newtek's Support>Downloads page to see whether a newer Flex firmware version is available. If so, download the update and then click Update Firmware on the Flex webpage to install it. (The webpage provides instructions and status messages to guide you during the process, which can take several minutes.)

With current firmware installed, the Flex panel is ready to connect – but there's one more consideration: Obviously, your live production system must be compatible with Flex, and needs to have a software version that includes Flex support installed (TriCaster version 8-0, or newer). As required, update your system in the usual manner (see Section 3.4 and the heading System Utilities in Section 5.2.4 in this User Guide).

21.1.5 CHOOSING A TARGET

At this point, you're almost ready to connect your Flex unit to your live production system.



FIGURE 244

With one or more supporting live production systems connected to the same network, open a live session; this notifies the Flex control panel that a suitable target is available. Open TriCaster Flex's *Utility* menu (SHIFT + ALT + SET) and observe the four options that appear in the LCD display on the left (Figure 244). Press the first lit button above the word *Connect* to choose an available target system to connect to. (Other options are *Restore* software, *Reboot* the control surface or *Done* to exit).

Notice that the Play (▶), Previous, and Next buttons (|◀ and ▶|) in the Media player controls light up (Figure 245). This is to indicate that you can use these buttons to navigate through a list of qualified systems when several are detected. To further help identify these systems, their individual IP addresses are also shown in the blue LCD panel to the right, above the firmware version (Figure 246).

Press the flashing green Play button (▶) to select a target from the list, telling Flex which live production system you wish to control. This will also close the Utility menu.



FIGURE 245



FIGURE 246

Note: Flex cannot connect to more than one live production system at a time.

You're all set: The Flex display will update to show the button names of any switcher inputs you have configured in your live production session. If you wish, though, you can continue use the tools provided on the Mapping tab of the Flex webpage to customize the order of source buttons in different banks on the Flex panel (see the heading labeled Mapping Tab in Section 21.2.2)

We briefly touched on the Flex configuration webpage when discussing initial setup and configuration. In this section, we'll take a more in-depth look at its features.

21.2.1 ADMINISTRATION TAB

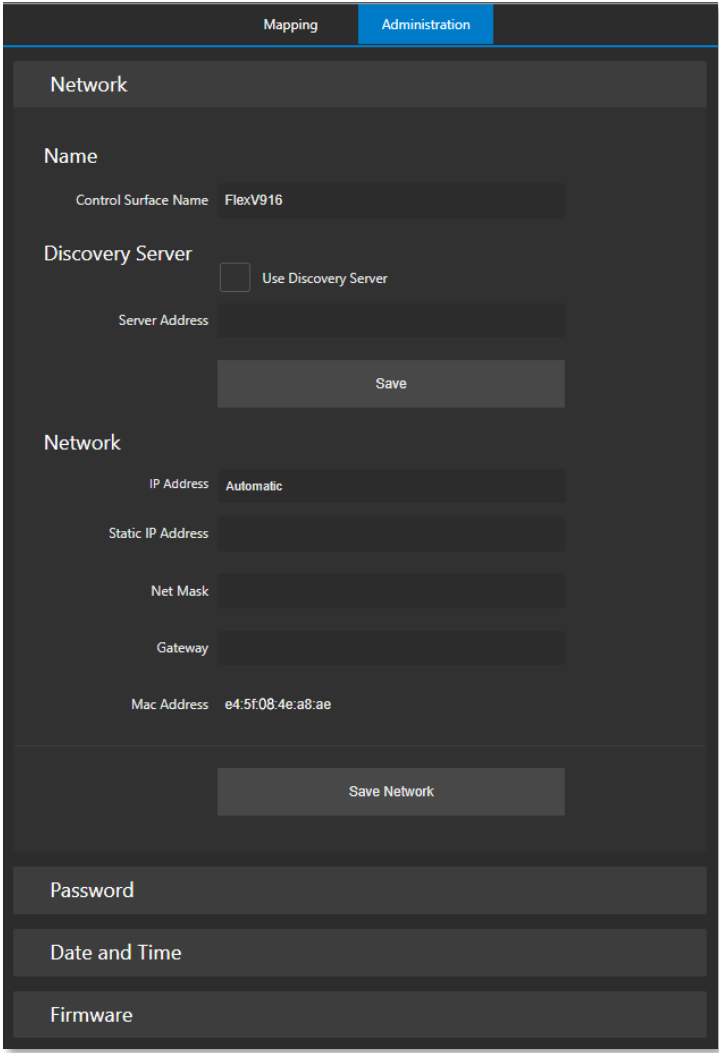
The screenshot shows the 'Administration' tab of the TriCaster Flex configuration interface. At the top, there are two tabs: 'Mapping' and 'Administration', with 'Administration' being the active tab. Below the tabs, there are several sections. The first section is titled 'Network' and contains a 'Name' field with the value 'FlexV916'. Below this is a 'Discovery Server' section with a checkbox labeled 'Use Discovery Server' which is currently unchecked. Underneath is a 'Server Address' field. A 'Save' button is located below these fields. The second section is also titled 'Network' and contains several fields: 'IP Address' with the value 'Automatic', 'Static IP Address', 'Net Mask', 'Gateway', and 'Mac Address' with the value 'e4:5f:08:4e:a8:ae'. A 'Save Network' button is located below these fields. At the bottom of the form, there are three more sections: 'Password', 'Date and Time', and 'Firmware', each with a corresponding input field.

FIGURE 247

The *Administration* tab contains all necessary network settings, divided into groups nested under collapsible ‘accordion’ widgets. The *Control Panel Name* field initially shows your unit’s Serial Number, but you can replace this a name of your choosing (the serial number remains visible at the top of the TriCaster Flex webpage).

To connect via an NDI Discovery server, checkmark the *Use Discover Server* simply box, and add your server’s IP address in the provided field. Additional *Network* settings, such as *IP Address*, *Static IP Address*, *Net Mask*, *Gateway*, and *Mac Address* are provided in this control group, too.

NOTE: If no DHCP server is detected when FLEX is connected, it automatically fails over to a default static IP address. After a restart, Flex will attempt to search for DHCP again.

Change the IP Address Mode in the Network setting group to supply a 'permanent' static IP address if this is needed. Click the button below to *Save Network*. A pop-up message will appear to confirm network changes.

PASSWORD

Expand the *Password* control group to enter the password of your choice (twice). Click *Update Password* to confirm your choice.

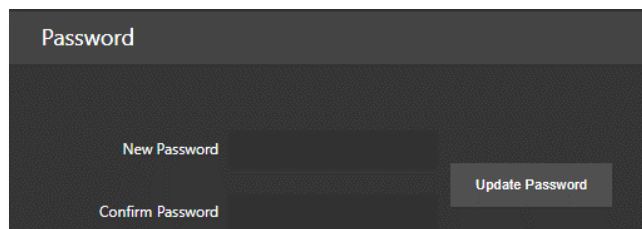
The screenshot shows a dark-themed interface for the 'Password' control group. It features two input fields: 'New Password' and 'Confirm Password'. To the right of the 'Confirm Password' field is a button labeled 'Update Password'.

FIGURE 248

DATE AND TIME

The *Date and Time* controls allow you to chose from several different methods for setting the date and time, using the combo box widget. Click on the *Set Date* drop-down menu to choose between *Automatically*, *Using NIST Time Server*, or *Manually*:

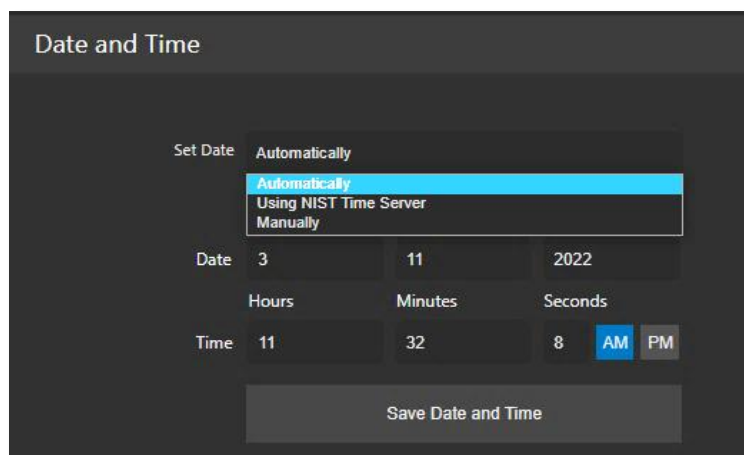
The screenshot shows the 'Date and Time' control group. The 'Set Date' dropdown menu is open, displaying three options: 'Automatically' (highlighted in blue), 'Using NIST Time Server', and 'Manually'. Below the dropdown, there are input fields for 'Date' (3, 11, 2022) and 'Time' (11, 32, 8, AM, PM). A 'Save Date and Time' button is at the bottom.

FIGURE 249

The default option is *Automatically*, will allow the system to set the time. Be sure to click *Save Date and Time*.

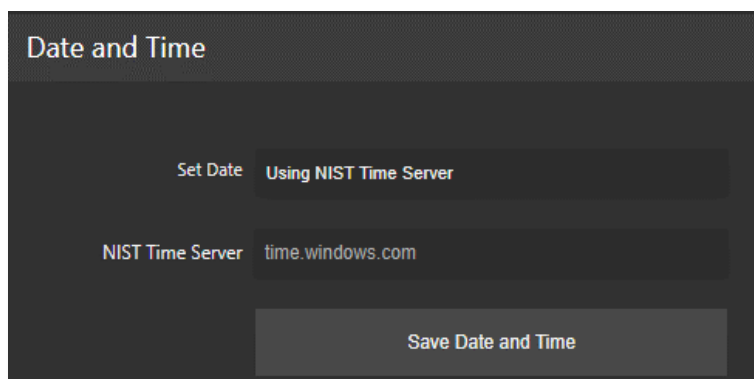
The screenshot shows the 'Date and Time' control group with the 'Set Date' dropdown menu set to 'Using NIST Time Server'. Below this, there is an input field for the 'NIST Time Server' with the value 'time.windows.com'. A 'Save Date and Time' button is at the bottom.

FIGURE 250

Using NIST Time Server uses the complex suite of algorithms that is defined in the NTP (Network Time Protocol) specifications to ensure that clocks on computers throughout a network are as accurate as possible. Once complete, click *Save Date and Time*.

Date and Time

Set Date **Manually**

Day Month Year

Date 3 11 2022

Hours Minutes Seconds

Time 11 32 27 **AM** PM

Save Date and Time

FIGURE 251

A *Manual* option is offered for those who need it. Once completing your entry, click *Save Date and Time*.

FIRMWARE

Firmware

Current Firmware Version v0.6-20-g4144111

Choose Firmware File ...

Firmware File

Update Firmware...

FIGURE 252

In the *Firmware* panel, the *Current Firmware Version* is displayed with options to *Choose Firmware File* and *Update Firmware* as discussed in Section 21.1.4, under Latest Versions.

21.2.2 MAPPING TAB

The *Button Mapping* configuration panel displays a graphic of the Control Panel you are editing. The *Bank* buttons let you to determine which sources are in which button bank. Choose the button map appropriate for the Product you wish to control using the menu at the top of the page.

The number of banks shown on the webpage reflects the product you select. For example, Figure 253 illustrates the web interface when you chose “TC Mini X/ Mini 4K/ TC 410+”; this option provides controls for mapping *Bank 1* and *Bank 2*. Figure 254 displays the drop-down menu offering more options such as TC1, TC1 Pro and TC2Elite (see Section 21.4 Switcher for more information assigning *Banks*).

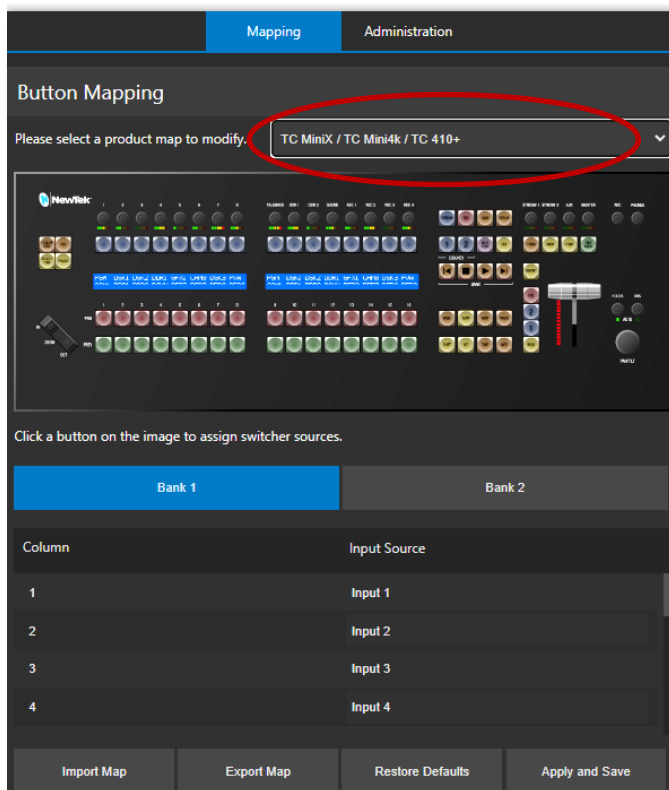


FIGURE 253

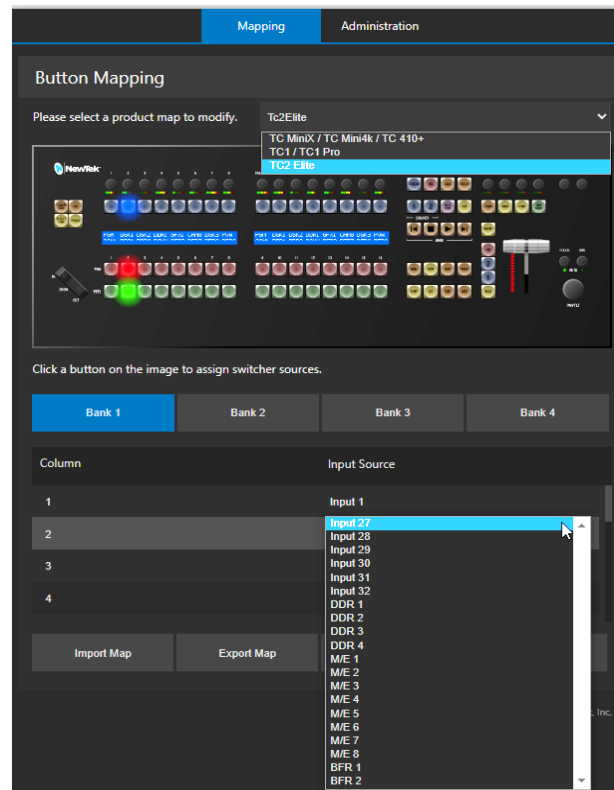


FIGURE 254

Clicking on the Input Source column will open all the available inputs to choose from to map your control surface as you wish (Figure 254). As you make your selections, the buttons light up on the graphic of the control panel in the UI. When done, click *Apply and Save*; a pop-up will appear to confirm success (Figure 255).

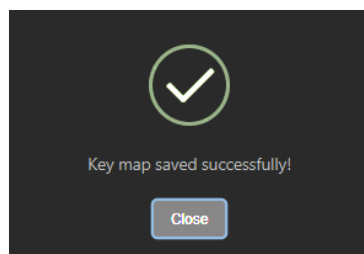


FIGURE 255

Note: If the Control panel is connected to a TriCaster when Apply and Save is clicked, the control panel must reconnect. Afterward, the control panel will automatically load the Bank profile corresponding to the model connected.

Import/Export Map and *Restore Defaults* buttons perform just as you would expect by importing/exporting map configurations and resetting to default. In this manner, you can easily switch between mapping configurations you have prepared in advance and stored for different productions or other purposes.

If you prefer to *Restore Defaults*, you will receive a confirmation message (Figure 256) stating the restore was successful. After clicking *Apply and Save* your TriCaster Flex will restart with the new mappings applied.

Hint: Clicking the Mapping Tab's Restore Defaults button will restore the working configuration for the key map currently displayed on the Webpage - only.

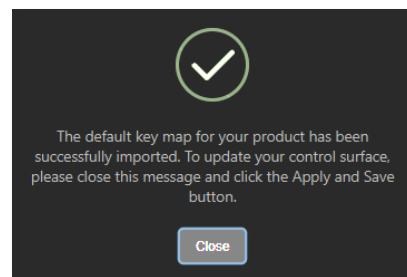


FIGURE 256

SECTION 21.3 CONTROL LAYOUT



FIGURE 257

The various control groups are organized into groups as the following:

- | | |
|------------------------------|-----------------------------------|
| 1 – Switcher | 6 – Media Group |
| 2 – Display | 7 – T-Bar |
| 3 – DSK/KEY Row | 8 – Mini Joystick, Focus and Iris |
| 4 – Audio Connections | 9 – Zoom/Joystick Delegates Group |
| 5 – Qualifier & Action Group | 10 – Zoom rocker |

- Yellow buttons typically work as qualifiers and require another selection to do anything.
- Amber buttons are action buttons, and immediately perform an operation, setting or selection.
- Button illumination brightens when the button is pressed (or are in an On/Selected state).

21.3.1 DISPLAY

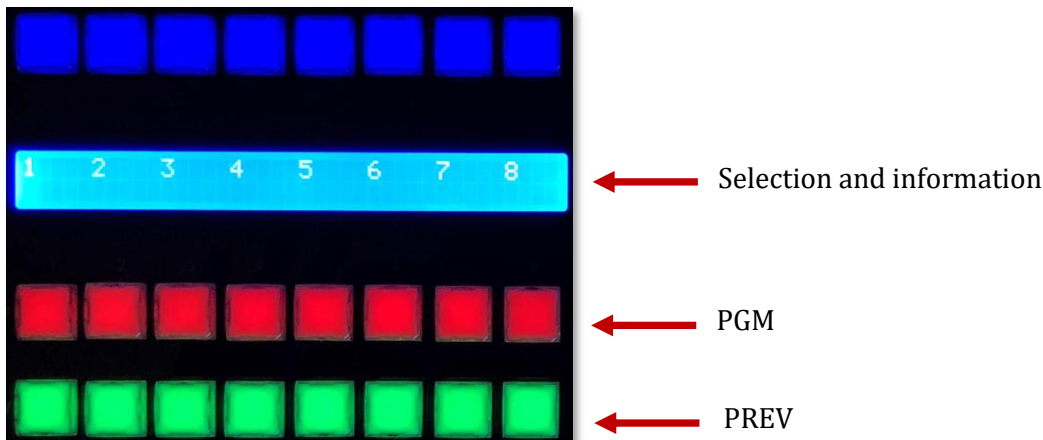


FIGURE 258

TC Flex control panel features helpful indicators and system feedback by means of illuminated displays.

- The blue LCD panel just below the *KEY* row shows labels identifying the source selection that would result from pressing a button in the same column.
- PGM/PREV – A/B row lighting color follows the UI colors for the delegated bus (Switcher or M/E).

SECTION 21.4 SWITCHER

21.4.1 BANK

The *Switcher* crosspoints of TriCaster systems are organized in banks comprising between 8 and 32 columns (varies by model).

- Holding down BANK shows the opposite BANK in the LCD panel temporarily (on release, Flex reverts to the original Bank).
- Add SHIFT to cycle backwards.
- Double punch BANK to latches to the new Bank.

Hint: For models with 3 or 4 BANKS, double punching repeatedly to advance can be tedious. As an alternative, press ALT + BANK to pick a Bank directly using the Program row.



FIGURE 259

The default Switcher mapping by model is shown below:

TriCaster Model	Bank 1	Bank 2	Bank 3	Bank 4
TC Mini X TC Mini 4K TC 410 Plus	Input 1-8, DDR 1-2, GFX 1-2, M/E 1-4	Buffer 1-15 BLACK		
TC1/ TC1 Pro	Input 1-8, DDR 1-2, GFX 1-2, M/E 1-4	Input 9-16, DDR 1-2, GFX 1-2, M/E 1-4	Buffer 1-15, BLACK	
TC2 Elite	Input 1-8, DDR 1-4, M/E 1-4	Input 9-16, DDR 1-4, M/E 5-8	Input 17-32	Buffer 1-15, BLACK

21.4.2 DSK/KEY

Pressing the DSK/KEY buttons, labeled 1 and 2, delegates or assigns the buttons in the 16-button *KEY* selection row to the left to the corresponding DSK/KEY layer(s) for the current bus (MAIN or M/E (1-n)).

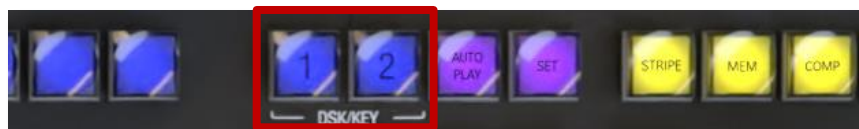


FIGURE 260

Row delegates for DSK/KEY 3 & 4:

- Press SHIFT + 1 to toggle DSK3/Key3 (the DSK 1 button pulses slowly when DSK 3 is delegated. If *both* DSK/KEY 1 and DSK/KEY 3 are delegated, the button pulses faster).
- Press SHIFT + 2 to toggle DSK4/Key4 (the DSK 2 button pulses slowly when DSK 4 is delegated. If *both* DSK/KEY 2 and DSK/KEY 4 are delegated, the button pulses faster).

21.4.3 STRIPE



Holding down the STRIPE button allows the Switcher and Transition control groups to be delegated to MAIN, M/E 1-n, (varies by model) or *Pre-Viz* using the LCD display and PGM row buttons.

FIGURE 261

21.4.4 MEM AND COMP



When the *MEM* button is held down, the first 9 columns in the displays update to list the names of *MEMs* for the bus(es) delegated to the stripe. Punching a button in the selection row below a *MEM* name recalls the corresponding preset for the bus assigned to the stripe.

FIGURE 262

When the *COMP* button is held down, the first 16 columns in the display update to the list the names of *COMPs* for the delegated buss(es).

- To clear a MEM/COMP, press a button (1-16) in the DSK/KEY row.
- To recall a MEM/COMP, press a button (1-16) in the PGM/A row.
- To store (or update) a MEM/COMP, press a button (1-16) in the PREV/B row.

SECTION 21.5 PTZ CONTROLS

Depending on the TriCaster model and feature set, PTZ (pan-tilt-zoom) style joystick operations are not limited to 'real' PTZ cameras. Static cameras, *Media Players*, and *Buffers* are among the different source types that may benefit from 'virtual PTZ' functionality.

The current Zoom/Joystick delegate state is fully independent of the Stripe Delegate (and is always based on either FOLLOW PREV or SOURCE).

21.5.1 CONTROL BUTTON GROUP

FOLLOW PREV

Follow Preview is probably the most useful joystick delegate mode. You will likely not be surprised to learn that, when enabled (as is the default in new sessions), the Zoom/Joystick delegate state tracks your current PREV row selection.

SOURCE

This button is an alternative to *Follow Preview*.

- While SOURCE is pressed, the PGM row button for the current selection is lit (only sources in the current Bank are shown; change banks if necessary to access other sources).
- Naturally, pressing a different button delegates the Zoom/Joystick controls to control the new source.

LAYER

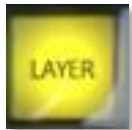


FIGURE 264

The LAYER button delegate modes let you further refine the target for your Zoom/Joystick operations.

Depending on the controlled source type, the LAYER display shows a list of controllable layer options. The layer list shown at any time varies both by selected source type and by product, but will be comprised of appropriate entries from those listed below:

- BKGD
- DSK 1
- DSK 2
- DSK 3
- DSK 4
- LAYER A
- LAYER B
- LAYER C
- LAYER D
- KEY 1
- KEY 2
- KEY 3
- KEY 4

NOTE: The Main DSK layers are not available when the Zoom/Joystick controls are delegated to a specific Switcher source by either the FOLLOW PREV or SOURCE buttons. To reveal these layer options for selection, first double-punch the Source button, then press the LAYER button.

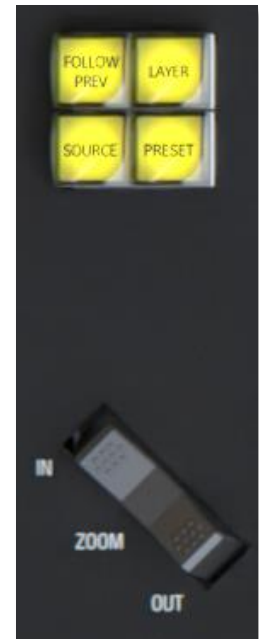


FIGURE 263

PRESET



FIGURE 265

In addition to manually controlling the PTZ camera with the *Joystick*, of course, you will often want to store and recall *PTZ presets*.

- Hold down to list presets on the LCD at right.
 - To clear a preset, press (1-16) in the KEY row.
 - To recall a preset for the delegated source, press a button (1-16) in the PGM row.
 - To store (or update) a preset, press (1-16) in the PREV row.

ZOOM ROCKER

This control works as you'd expect: Zoom in and out by rocking forward and backward.

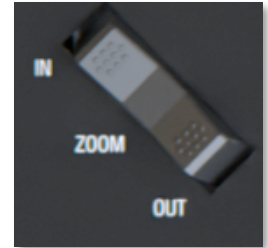


FIGURE 266

SECTION 21.6 PAN/TILT



FIGURE 267

The Mini-joystick pans/tilts delegated PTZ Cameras

FOCUS/IRIS KNOBS

These knobs sit to the right of the joystick, push to toggle auto-focus or auto-iris (exposure). The LED is lit when AUTO is enabled for FOCUS or IRIS.

SECTION 21.7 TRANSITIONS

The Delegate buttons determine which players are governed by operations in this group. Multi-selection is supported.

- Delegate buttons (BKGD, (DSK/KEY) 1, (DSK/KEY) 2. Press SHIFT with button 1 or 2 to delegate DSK/KEY 3 and 4, when supported.

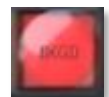


FIGURE 268

Hint: Double-punch any delegate button to remove perform an AUTO on the corresponding layer. (To perform a TAKE instead, assign CUT as the layer's transition effect.)

21.7.1 FTB



FIGURE 269

To perform a *Fade to Black* operation, press *SHIFT* + the *FTB* button (the *SHIFT* button is required as a safety measure, since *FTB* is a somewhat dangerous operation). Revert to normal output by pressing *FTB* alone.

- *SHIFT* flashes if *FTB* is pressed without it.
- *FTB* pulses slowly while on.

21.7.2 FADE & TRANS



FIGURE 270

These two buttons provide a quick way to control the *Transition Bin* selection for the delegated switcher layer(s).

- Pushing *FADE* assigns Crossfade as the effect for the currently delegated layers.
- Push the *TRANS* button to activate the last-used (non-Fade) Transition effect to the currently delegated layers.

21.7.3 RATE



FIGURE 271

Press the *RATE* button and the LCD display will show SLOW - MEDIUM - FAST options for the effect assigned to the selected delegate(s); use PGM row buttons to select.

SECTION 21.8 T-BAR

The T-Bar is perhaps the most recognizable component of a professional video control panel, and arguably one of the most important. The T-Bar can manually be pulled to modify the progress of a transition between delegated video layers. An LED in the nearby vertical row displays the percent of completion of the current effect.

- LEDs light progressively from top-to-bottom or bottom-to-top depending on the direction the T-Bar must travel to complete the currently delegated effect.



FIGURE 272

21.8.1 TAKE & AUTO

The *TAKE* and *AUTO* buttons perform a cut or transition respectively, affecting only the currently delegated video layers.

Hint: To quickly TAKE/ AUTO any individual layer, double punch its Layer Delegate button.

- Double-punch BKGD-> auto Background
- Double-punch 1 -> auto DSK 1
- Double-punch 2 -> auto DSK 2
- Shift + double-punch 1 -> auto DSK 3
- Shift + double-punch 2 -> auto DSK 4
- For an individual Take, set the layer's effect to CUT (effect bin slot 1)

21.9.1 BACKPLANE CONNECTIONS



FIGURE 273

From left to right you will find ports for DISPLAY, 2 USB, and the ETHERNET port. These are followed by line level inputs for TALKBACK, MIC, and PHONES (headphones). Lastly, two pairs of line level AUDIO IN and two pairs of AUDIO OUT connectors are provided.

21.9.2 VOLUME KNOBS



FIGURE 274

TriCaster Flex has taken audio features to the next level. In most control panels, operating the audio mixer requires you to access the UI in your live production system. A helpful new implementation of *Volume Knobs* on TriCaster Flex (at the very top of the control panel) give you fingertip access to control audio levels for *Audio Mixer* inputs and output busses as follows:

- IN 1-8
- TALKBACK
- DDR 1, 2
- SOUND
- REC 1-4

Hint: The LCD list other Switcher sources with recording enabled when you hold down REPLAY.

- STREAM 1, 2
- AUX
- MASTER
- MIC and PHONES
- MUTE/UNMUTE
- VU Meters (LED colors follow the UI meters, with the last (RED) LED reserved to indicate clipping).
- SOLO (For sources supporting Solo, press ALT and the associated Volume knob to toggle Solo mode).

In addition to adjusting the volume levels, push a *Volume Knob* to toggle mute/ unmute the channel. Record options have enable/ disable AGC (audio gain control) without having to go into the Output Configuration panel to make those changes.

21.9.3 TALK BACK

The button input labeled TALK BACK serves a special purpose, providing a way to converse with remote callers off-air (i.e., without intruding into your live program).

- *TALK BACK* is a PTT button (Push-to-Talk)
 - Hold it down to activate the *TALKBACK* feature, sending the audio source designated in the UI to all Mixer connections with *TALKBACK* capability.
 - Double-punch *TALK BACK* to lock it ON (the button will pulse light in this state).

SECTION 21.10 STREAM, CAPTURE AND REPLAY

STREAM & RECORD

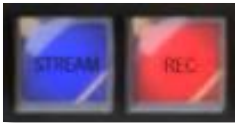


FIGURE 275

- *STREAM* - Push to enable or disable TriCaster's live streaming feature.
- *RECORD* - Pressing this button enables TriCaster's *Record* feature.

Note: As a safety measure, pressing the REC button when recording is underway does not stop recording. Instead, the SHIFT button flashes to remind you that you must hold it down at the same time as pushing REC to end recording.

GRAB



FIGURE 276

GRAB is a qualifier button:

- Hold down GRAB and punch a PGM row button to grab the associated source.
- To grab from Mix 1-(4 or 8, varies by model), punch the corresponding numbered button in the PREV row.

REPLAY

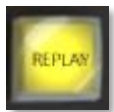


FIGURE 277

Hold down REPLAY to list Instant Replay enabled sources by name on the LCD display.

- The LCD(s) will list the enabled *Switcher* sources starting from the left, followed by any recorders enabled in the *Record* tab in Output Configuration panel.
- Punch the PGM row button for the desired recorder source you wish to show on as an instant replay.

Note: Instant Replay uses the Show On feature of the DDR designated in the Replay Configuration menu. As such, the replay clip can be shown on PGM or an M/E, or even an M/E on PGM.

- Or defer playback of the instant replay as follows:
 - Add the replay clip to the DDR playlist without playing it by clicking the source's PREV B/D row button (with REPLAY held down).
 - When you're ready to trigger the instant replay, press SHIFT + AUTO to initiate the replay DDR's 'Show On' operation.
- In either case above (instant or deferred replay), double the length of the replay by double-punching the recorder's button.

21.10.1 MACRO

- On TriCaster's *Live Desktop*, open the *Macro Configuration Editor* (See Chapter 16 Macros and Automation).
- Select the macro you wish to assign in the onscreen lister.
- Click the mouse in the *Listen* field at bottom left.
- Hold down the *MACRO* button and press a suitable button on the control panel.

Hint: To clear an assigned macro from a button, while holding down MACRO, double-punch the button you wish to clear.



FIGURE 278

21.10.2 MEDIA PLAYER GROUP

This control group provides convenient control over TriCaster's most important Media Player functions and configuration options.



FIGURE 279

SET



FIGURE 280

Press to show (from right to left):

- MEM names for slots 1-12, LOOP, DDR 1-2, and Sound in the displays at left. (PGM row buttons light to show the current selections when SET is pressed).
- Punch a button in the PGM row at left to:
 - Delegate the MEDIA group controls to the media player specified.
 - Or to apply a MEM (1-10) to the currently delegated media player
 - Or to Toggle Loop mode

AUTO PLAY



FIGURE 281

Click button to toggle *Autoplay* mode for the delegated players (button is lit when Autoplay mode is on).



FIGURE 282

- **PLAY** - Push to initiate playback for delegated *Media Players*.
- **STOP** - Push once to end playback for delegated *Media Players*; push a second time to return to the start position. (This operation respects the *Single* setting for individual *Media Players*)
- **PREV** - Press this button to go to the previous playlist entry in delegated *Media Players*. (The selection cycles to the last playlist entry when necessary.)
- **NEXT** - Push this button to go to the next playlist entry in delegated *Media Players*. (The selection cycles to the first playlist entry when necessary.)

CONTROL SURFACE: TRICASTER MINI



This chapter introduces the control surface offered by NewTek for your TriCaster Mini helping you to see how it can complement your system and add ability to your production setup. We will also discuss connecting to the control panel from your live production system.

The TriCaster Mini CS, the ideal traveling partner, provides studio-style control and a small footprint to deliver professional result from the office, an event, or almost anywhere you like, delivering portable, professional control for live productions of any size.

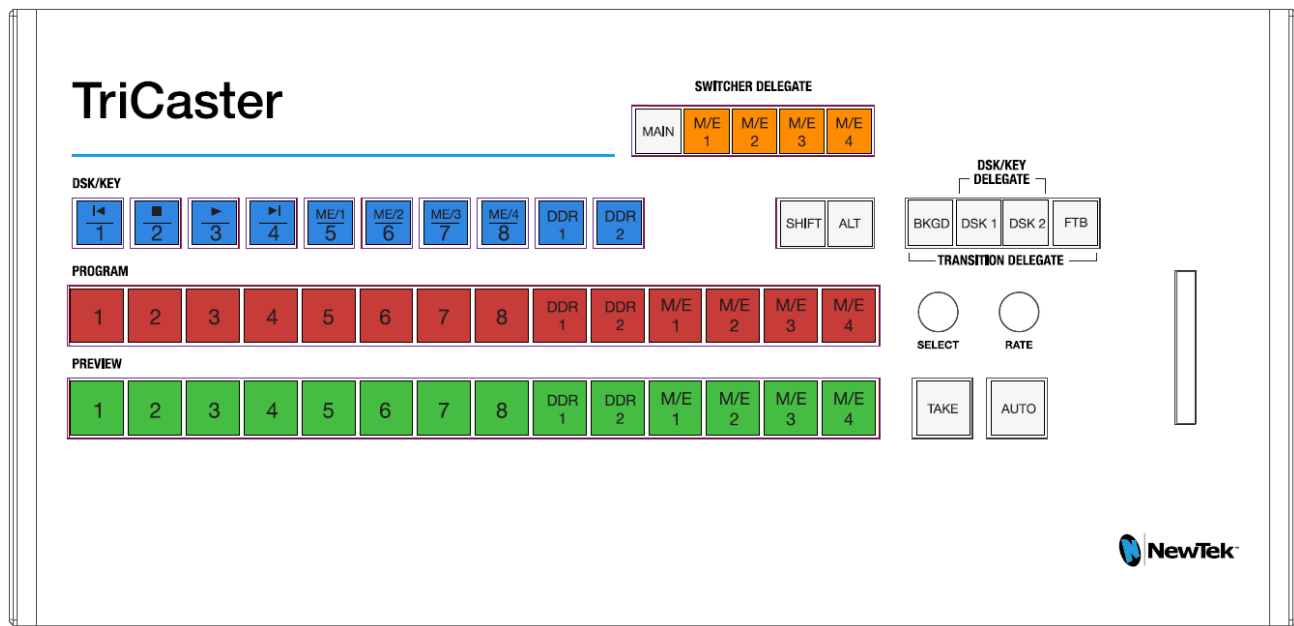


FIGURE 283

- When SHIFT is pressed, DSK/KEY row buttons execute their secondary function (shown above on the button label), as follows:
 - The first four DSK/KEY row buttons, when 'shifted', provide Media Player transport control.
 - To delegate these operations to a specific Media Player:
 - Hold SHIFT + ALT together, and punch DDR1 or DDR2 in the same row
 - The next four DSK/KEY row buttons, when 'shifted', select M/Es 1-4.
- Holding down ALT while making a PROGRAM, PREVIEW or DSK/KEY row selection addresses sources from the second *bank* of Switcher sources, in the order listed below:

BFR 1	BFR 2	BFR 3	BFR 4	BFR 5	BFR 6	BFR 7	BFR 8	BFR 9	M/E 1	M/3 2	M/E 3	M/E 4	BLACK
-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------	-------

PART IV (APPENDICES)

Certain topics which benefit from in-depth coverage are in this section, along with cross-references and concluding with a keyword index; Appendix F: lists all shortcut keys.

Appendix A: [ADVANCED FEATURES](#)



TriCaster Advanced Features is your 'backstage pass' to the latest power user tools and features from NewTek's industry-leading developers. The benefits of Advanced features are already substantial and, best of all, will continue to accrue as further innovations are added to super-charge your workflow and take your productions to new heights of visual excellence.

A.1 INTRODUCTION

TriCaster Advanced Features gives you unlimited access to a growing list of cutting-edge tools and technology. Let's consider them one by one beginning with one of the most exciting developments of all.

A.2 LIVE STORY CREATOR

This powerful advanced feature delivers numerous and powerful benefits. At the lowest level, you might use it simply as a very convenient and useful teleprompter for your show's talent. Coupled with the NDI Studio Monitor included in the free NDI Tools bundle, this is a great feature. There is a great deal more to Live Story Creator, however: Live Story is first and foremost a powerful natural language automation system.

OVERVIEW

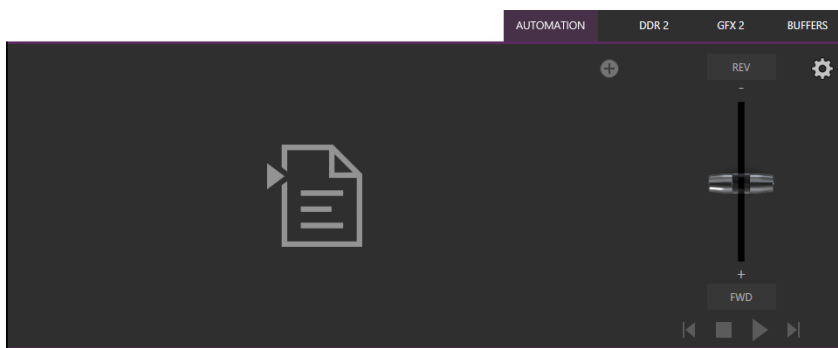


FIGURE 284

When you first open the new *Automation* tab, located at right below the main *Switcher* (near *Buffers*), you'll see the empty script pane and some (rather lonely looking) controls at right (Figure 284).

Click or tap the (+) plus sign gadget below and to the right of the tab to open *File Explorer*. Here you can navigate to a location where you have previously stored a .docx file such as those created in Microsoft Word (or compatible software with similar features). Figure 285 illustrates the result. Let's discuss a few of the Live Story interface controls and features before continuing.

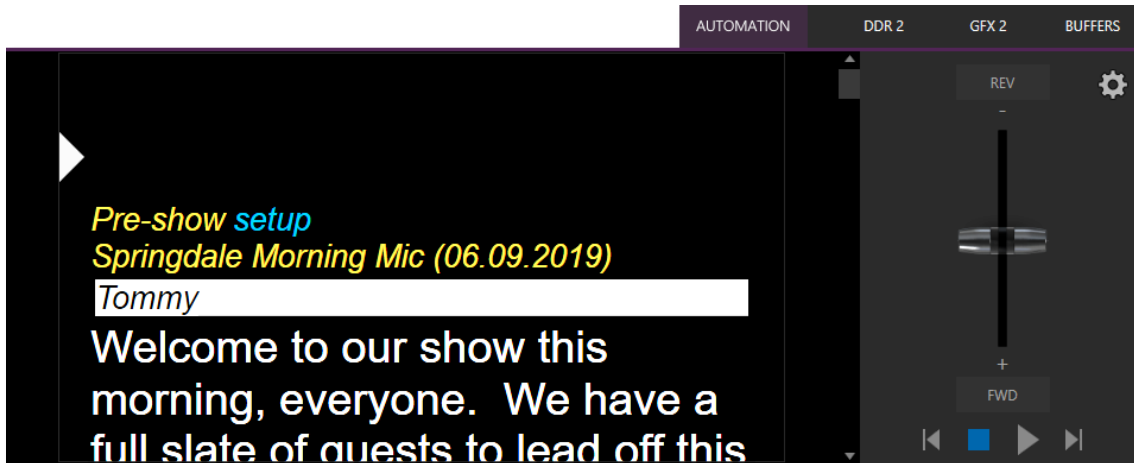


FIGURE 285

Examining the tabbed Live Story panel from left to right, you'll see:

- An (x) gadget at upper left. Click this to unload the current script (.docx file).
- The eyeline control is a white caret (triangle) control that can be dragged up and down to adjust the position of the similar eyeline indicator on a teleprompter display.
- The script pane, or canvas, displays the content of the .docx document, as interpreted by Live Story.
 - Colors and font styles have special meaning:
 - Generally, text shown as yellow and italicized is not meant to be read aloud.

For example, in the image above, the first line (“Pre-show setup”) is used to issue a command to TriCaster to place Black on output before the show begins. The line has Word’s *Subtitle* style applied to it, indicating it is not intended to be read aloud. Live Story shows this in yellow italics.

You’ll notice, though, that the word “setup” is blue in the script pane, despite also being in *Subtitle* style in the document. This is because the command we inserted to cause black to be shown on output is in a comment applied to this word in the document.

Blue identifies commented text in the document, and comments can be used to issue commands (we’ll discuss all of this in more depth shortly).

Hint: Move the mouse pointer over blue text in the script pane to see a tooltip showing how Live Story Creator has interpreted your comment.

Notice that the show’s title (“Springdale Morning Mic”) is displayed in yellow, too. Live Story recognized this as the SHOW TITLE because the *Title* style is applied to this text in the document.

- Document styles can trigger corresponding actions, too. For example, the *Title* style can automatically display a specially prepared LiveGraphic™ animation to introduce your show – all without any manual intervention.

Note: The SHOW TITLE animations require the latest LiveGraphics content to be installed. The Broadcast templates in the updated content include a special Show Open preset specially prepared for this purpose.

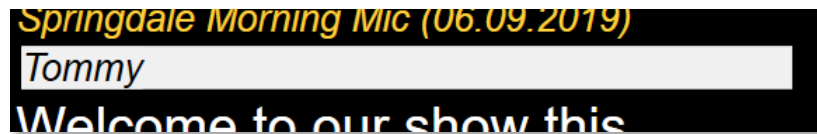


FIGURE 286

- In similar fashion, the document style *Heading 3* denotes a CUE, used to introduce someone who will be speaking next. CUEs are shown in inverted color, black text on a white background (Figure 286).

Like the *Title Style*, CUEs can also automatically trigger TriCaster operations. For example, if a *Switcher* input name matches a name in the CUE, that source will be placed on output, and a title identifying the newly introduced talent will be shown. (After a few moments, the title is automatically removed.)

- Text meant to be read aloud is shown in a large white typeface.
- Live Story's operating controls are located in a small panel just right of the script pane (Figure 287).

This *Control Panel* is dominated by a vertical *Speed* control. You can modify the scroll rate during playback or traverse the script backward or forward by dragging the *Speed* bar up or down.



FIGURE 287

Transport controls below the *Speed* bar include *Stop* and *Play*, as well as *Previous* and *Next Segment* buttons.

Hint: Divide your show into different segments identified by applying the Heading 2 style to lines in your .docx file.

- Click the familiar gear gadget in the upper-right corner of the *Control Panel* to open Live Story's configuration panel (Figure 288).

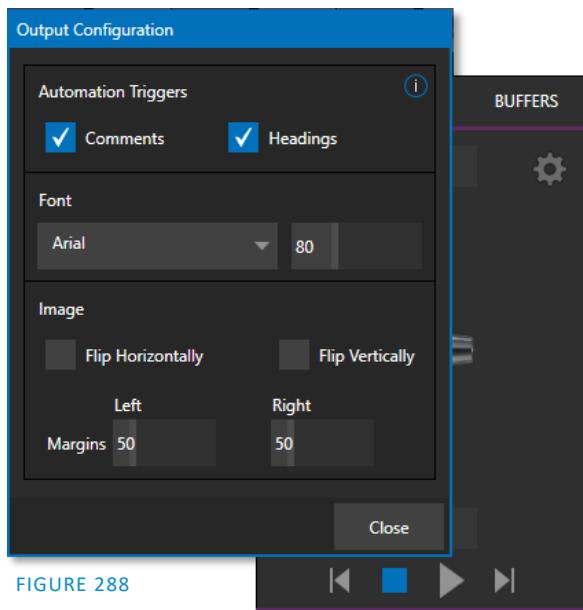


FIGURE 288

- The two switches in the *Automation Triggers* toggle automation in response to *Comments* or *Headings* (styles) applied in the .docx document.

When both switches are off, Live Story does not process automation commands, and operates as a simple teleprompter.

- Font controls let you adjust text display for legibility when you are using an external teleprompter device.
- The *Flip* switches in the *Image* group likewise allow you to conform the teleprompter output to match your local devices, while the margin sliders let you move text confine teleprompter output to best suit your preference.

STYLE-BASED OPERATIONS

As discussed earlier, various styles applied in the .docx document serve different purposes in the Live Story Creator context, and some styles can even trigger automatic operation.

Document Style	Applies to	Displayed on output	Description	Sets value for DataLink key	DataLink Key Description
Title	Show Title	Yes	e.g., "State of the Union Address"	%SCRIPT_ShowTitle%	Main show title
Heading 1	Show Descriptor	No	Sets value for a DataLink key	%SCRIPT_ShowDescription%	Main show descriptor
Heading 2	Segment	No	e.g., "Intro Package", "Welcome Monologue"	%SCRIPT_ShowSegment%	Current show segment
Heading 3	Cue	Yes (inverse)	e.g., talent name: "Billy Bob"	%SCRIPT_CueName%	Name of speaker
Subtitle	Information text	Yes (yellow, italics)	Displayed but not meant to be read (e.g., "walk behind desk")	%SCRIPT_Info%	Current/Last information text
Normal	Body text	Yes (white)	Text to be read by on-air talent.	n/a	n/a

The following styles automatically trigger TriCaster operations when the *Headings* switch is enabled:

- **Title** – triggers automatic display of a *LiveGraphics* show opening title animation. This title uses the text from the *Title-style* line to populate the first line of the animated title and takes (hidden) text from a line entered using the *Heading 1* style for its second line (when used).
- **Heading 3** – identifies a *CUE*. When the *Headings* switch is enabled, a lower-third title is automatically displayed for a short duration when the eyeline reaches a CUE for the first time in a *segment*. The first line of the title shows the full *Switcher* input name set in *Input Configuration* for that source. The second line of the title is drawn from the (Input Configuration) *Comment* entry for the source.

Hint: Titles as discussed above default to a pre-selected LiveGraphics title, and transition in and out automatically. To display a different title page, preload the desired title into Buffer 1 beforehand. This can be done using a command entered as a Comment in the docx file. You can also use Comments to change the title's transition effect, display duration, and so on.)

COMMENT-BASED COMMANDS

While heading styles provide some useful automation possibilities, the use of *Comments* in the .docx file provide much more powerful and detailed options.

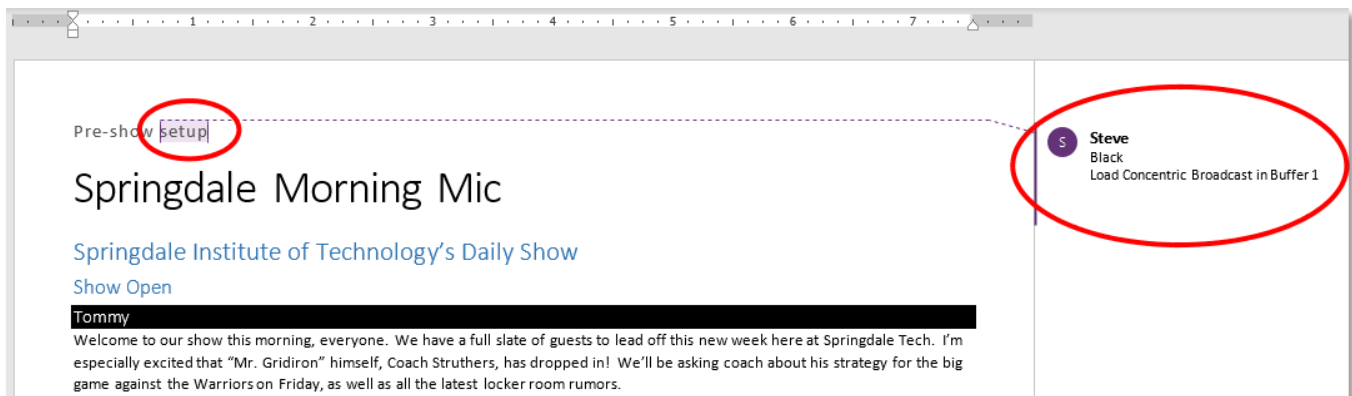


FIGURE 289

Note the *Comment* inserted into the text in Figure 289. You'll see that it has two lines. The first is simply "Black", which tells TriCaster to select the input named "Black" on its *Program* row.

FUZZY LOGIC

Interestingly, the very same thing would occur if the command entered in the *Comment* had instead said "Show black", "Put black on program", "Send Black to Program", or "Output black". Live Story uses "fuzzy logic" to try to find the best match to your entry.

Note: Extraneous text is typically ignored so, for example, you could instead write "Put black on program while I have lunch" and still be successful (assuming your lunch was satisfactory).

We mentioned earlier that you can move the mouse pointer over blue text in the script pane to see a tooltip showing how Live Story interprets your comment. It's helpful to know that Live Story determines the probability that the interpretation is correct. If the wording of a *Comment* entry is ambiguous, and could be interpreted differently, the text color in the script pane is red rather than blue.

Hint: If a comment has multiple lines in it, Live Story adds an asterisk to any ambiguous lines in the tooltip. The entry "Output black" would trigger this behavior, even though it is correctly interpreted, with the result that black is shown on output.

LIVE STORY CREATOR VS. MACROS

Let's pause at this point to consider how Live Story commands differ from the somewhat similar shortcuts used by macros.

Like Live Story commands, a macro shortcut is entered on a single line, and may support arguments, or parameters. Macro shortcuts, however, require you to use very specific syntax. By contrast, though, Live Story commands are entered using 'natural language' – the way you might normally speak or write. A macro shortcut will generally fail with an error if your syntax isn't perfect, while, as mentioned previously, Live Story uses fuzzy logic to try to find the best match for your command, and only fails 'reluctantly'.

Hint: Live Story documents are evaluated on loading. When Live Story determines its evaluation of a specific command in a Comment could be incorrect, it posts a message to TriCaster's Notification panel to note the ambiguity, in addition to marking the commented text in red in the script pane.

Additionally, a single *Comment command* can do things that would otherwise require multiple commands entered on separate lines in a macro. Consider for example, when Live Story encounters the simple *Comment* entry "fade tom slow", it results in the following steps:

1. Select an input named "Tom" (or "tom", or with a similar string, such as "Tommy" in the input Comment box) on the Switcher's *Preview* row.
2. Set the Switcher's *Background* transition effect to Fade
3. Set the effect speed to *Slow*
4. Perform an *Auto*.

To do something similar using a macro, you'd need to enter four separate lines, getting the syntax correct for each macro shortcut – and since macro commands require an input index parameter rather than a name, you'd need to know in advance which camera would be pointed at Tom. If Tom was moved to a different chair on your set at the last minute, the macro would need to be modified. Using Live Story Creator, you could simply rename the camera.

Better yet, continuing with the example above, if – rather than naming an input "Tom", you prepared a PTZ preset (or Pan and Scan preset, for a non-PTZ source) and named it "Tom", TriCaster would automatically select that preset at step 1 above.

Hint: Imagine what this means if you are using a few PTZ cameras to cover a city council meeting, with PTZ presets prepared for different participants. If two people change seats at the last minute, all you need to do is update the PTZ presets and Live Story Creator will take care of everything else for you!

So, not only are Live Story Comment commands easier to write, often performing multiple operations with a single entry, but they are ‘smarter’, and easier to maintain and update. Which brings us to another key feature.

LIVE UPDATES

As clever as Live Story is, at times you will want to experiment, or need to tweak your wording to bring about the result you intend. As we mentioned, you don’t even need to press *Play* to test modifications, you can simply reload the .docx file and hover the mouse over commented text to see the way the embedded commands have been evaluated.

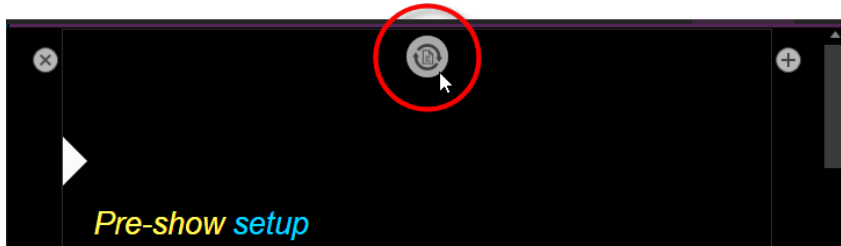


FIGURE 290

Making this exercise even better, Live Story Creator has a built-in *Update* feature, and monitors changes to the script file you loaded. Suppose you are using a laptop to edit your script in Microsoft Word. You might opt to save the file across a network connection to (for example) the shared Public\Documents folder on your TriCaster.

Then, when you modify the document and save it (which typically requires just a quick keyboard operation to perform - CTRL + s) Live Story will immediately show an *Update* icon at the top of the script pane (Figure 290). Click this to reload the .docx file with the changes you made.

Note: This will reset Live Story’s eyeline indicator to the top of the script, so don’t update when your talent is reading from the teleprompter you are quite confident that they will miss you with whatever they throw!

DEFAULT BEHAVIORS

We mentioned that Live Story’s *Comment commands* can perform compound operations and are simpler to use than macros. They are designed to provide a useful result even when you don’t supply complete information. Let’s look at an example.

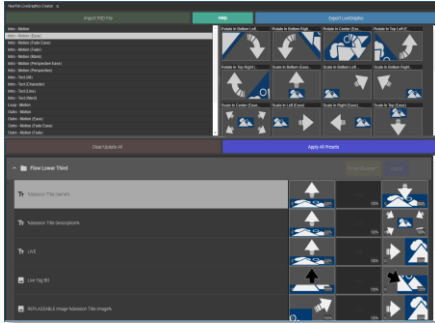


FIGURE 291



FIGURE 292

Figure 291 above illustrates the result of the *Comment command* “Fade Football Star on in PiP” (the simpler entry “PiP football star” would have produced the same result). We didn’t supply much detail about our intentions, so Live Story produced a ‘default’ picture in picture effect, as shown.

But perhaps you want to make some changes. In Figure 292, we have edited the Comment command to read “PiP Football Star large top right”. Since we defined the position of the overlay, and specified a size, we achieved a more pleasing result.

The *Comment command* “title” provides another example of default operation. Add the simple entry “title brackets lower third” to a comment. Live Story evaluates the comment when the *eyeline* indicator reaches it, and the predetermined defaults are used so you may not even need to add any more detail. The designated title page (Brackets Lower Third) is automatically i) loaded into a Buffer, ii) assigned to a DSK, iii) faded on, iv) held briefly, and then v) faded off).

In this example, you might kick things up just a bit by changing the effect used to introduce the title page. The *Comment command* “Show Title brackets lower third Edge LtoR(H)” works quite nicely for this purpose.

Hint: You might want a title to remain in view longer than the default time. To do this, add the word “hold” to the command you use to display the title. Then insert a new comment into your script at the point where you want to remove the title from view and enter the command “title off”.

Try different things – you can change effect speed (Fast, Medium, Slow), specify the DSK to use, and so on.

MORE COMMENT COMMANDS

We can’t provide an exhaustive listing of every possible command and combination of options, but we’ll identify the main TriCaster Advanced features Live Story currently supports and give some examples and a few alternatives that work just as well. The best way to learn is to just try typing in what you want to happen!

AUDIO MIXER

- “Set input 2 Volume to -5”; “volume input 2 5dB”; “input 2 9dB”; “Set volume for input 2 to -20”
- “volume master 5dB”; “Set Saster Volume to 5”; “set volume to 5”
- “mute”; “mute on” – mute master output
- “unmute”; “mute off” – unmute master output
- “mute out 2”; “mute out 2”; “mute aux 2”; “mute aux 2 on”; “aux 2 mute on” – mute named output
- “aux 2 mute on”; (etc.)
- “mute input 3”; “mute input 3 on” – mute the named input

- “mute input 3 off”
- “solo input 6”; “solo input 6 on” – solo the named input
- “solo input 6 off”
- “unsolo input 6”; “solo input 6 off” – disable solo for the named input
- “follow input 6”; “follow input 6 on” – solo the named input
- “follow input 6 off”
- “input 4 eq”; “input 4 eq on”; “input 4 enable equalizer”
- “input 4 eq off”; “input 4 disable equalizer”; (etc.)
- “input 5 compressor on”; “input 5 enable compressor”
- “input 5 compressor off”; “input 5 disable compressor”; (etc.)
- “AudioMemName”; “select AudioMemName”; “recall AudioMemName”; “mixer AudioMemName”; “audio AudioMemName”; “audio select AudioMemName”; “audio recall AudioMemName”
- “audio select MEM 3”; “audio select 3”; “audio recall 3”; “audio recall mem 3”; “mixer select 3”; “mixer select mem 3”; “mixer recall 3”; “mixer recall mem 3”

COMPS

- “load MyComp”; “select MyComp”; “myComp” – loads the named main Switcher *Comp*.
- “comp 3”; “load Comp 3”; “select comp 3”; “go to comp 3”
- “m/e 1 comp 2”; “me1 comp 2”; “select comp 2”; “load main comp 2”; “go to myMENAME myCompName”; (etc.) – loads the designated *Comp* into the main *Switcher* (default) or named *M/E*.
- “myComp medium”; “myComp normal”; “myComp 1.0”; “go to myComp slow”; (etc.) – loads a *Comp* using the designated speed.

MEDIA PLAYERS

- “play lion”; “play lion on main”; “take lion on me1”; “cut to lion on m/e 1”; (etc.) – locates, loads and plays a clip named “lion” on the main *Switcher* (default) or named *M/E*; performs an auto or take when playback ends. The scrolling of the script pauses until playback is complete.
- “circle(h) lion” works as above but uses a Circle wipe to display and remove the clip.
- “play lion ddr2” forces the operation to use DDR 2 (rather than selecting a default DDR)
- “add lion to ddr 2; “load lion ddr 2” – adds the clip without displaying it
- “play lion at 9 db”;
- “play lion voiceover”; “play lion vo” – the script continues to scroll during playback
- “play lion at 5 db”; “play lion 5db”; “lion 5 db”
- “DDR 1 stop”
- “DDR 1 Next”; “DDR 1 Previous”

DSKs

- “Toggle DSK 1”
- “toggle DSK 1 slow”
- “auto on dsk 2 slow”; “auto on M/E 2 key 2 slow”; “Fade M/E 2 keyer 1 slow”;
- “fast auto off dsk3”
- “Set main dsk 1 as input 1”; “Set input 1 on main dsk 1”; “set DSK 1 to GFX 1”; “set M/E 1 Keyer 1 to GFX 1”; “assign GFX 1 to Me2 key 2”; (etc.)
- “Play lion in dsk2”; “Play lion on dsk2”; “Show lion in dsk2” – plays a clip named lion in DSK2, fading on and holding the last frame.

- “Play lion on key 2 in M/E 1 with slow circle(h)” – as above but uses a slow Circle wipe in M/E 1.
- “Show lion on dsk 3 with fast auto”; “Show lion on dsk 3 with take”
- “set main DSK 3 transition to circle(h)”; set circle(h) Me2 key 1”; “change transition on me4 for key 2 to circle(h)”; (etc.)
- (etc.)

See also dedicated PiP (Picture in Picture) commands)

BUFFERS

- “Add globe to buffer 6”; “buffer 6 load globe” – loads the *Globe Frame Buffer Animation* effect to the target buffer
- “Add Hexagon lower third to buffer 6” – loads the named static title page into the target buffer
- “buffer 3 load Hexagon Broadcast” - loads the named LiveGraphic into the target buffer
- “Buffer 1 set speed medium”
- “Buffer 1 speed 2”; “Buffer 1 set speed 2x”; “Buffer 6 globe 200%” – set speed to 2x the default (Medium).
- “set buffer 1 to layer preset 2”; “set buffer 1 to index 2”; “set buffer 1 to preset 2”; “buffer 1 select preset 2”; “buffer 1 preset 2”; select buffer 1 index 2 (etc.) – select the designated *Layer Preset* for a LiveGraphics title page loaded in the buffer slot specified.
- “set buffer 1 to data preset 2”; etc.– select the designated *Data Preset* for a LiveGraphics title page loaded in the buffer slot specified.

MACROS

Of course, at times you may want to run a more complex macro from within a script, using a *Comment command*. The commands below will let you do that.

- “macro_name”; “macro macro_name”; “run macro_name”; “play macro macro_name”; etc.
- “stop macro_name”; “stop macro macro_name”; macro macro_name halt”; macro_name end”; etc.
- “stop all macros”; “stop macros”; “halt macros”; etc.
- “Continue Paused Macro”

PIp (PICTURE IN PICTURE)

- “Input 5 pip”; “pip input 5”; “ots input 5” – assign input 5 to a default DSK and display it using default options
- “pip globe” – show the matching file (in the example, a Framebuffer Animation) using default PiP settings
- “pip globe center large”; “pip globe middle large”
- “pip me1 key 2 globe small bottom right”
- “pip input 5 large center” – assign input 5 to an unused DSK and display it large size at center screen
- “pip main dsk 3 input 5” – assign input 5 to DSK3 and display it using default options
- “pip me1 dsk 1 input 2” – assign input 2 to DSK1 in M/E 1, and display it using default options
- “pip input 5 fly in L” – assign input 5 to an unused DSK and fly it in from left
- “fade pip off”; “ots off” – remove the default PiP from output using a fade effect
- “pip main dsk 2 off” – remove the PiP in DSK 2 from output
- “pip myPtzPreset” – recall the named PTZ preset and show that source in a PiP
- “pip dsk1 myPtzPreset”; “pip me1 key1 myPtzPreset”; “take pip myPtzPreset”; (etc.)

PTZ

- "myPtzPreset "; " myPtzPreset on program"; "put myPtzPreset on program"; "take myPtzPreset on program"; "select myPtzPreset on program"; "take to myPtzPreset "; (etc.) – locate a PTZ preset named "myPtzPreset" and display it on Program output.
- "put myPtzPreset on preview"; "select myPtzPreset on preview"; etc.
- " myPtzPreset on M/E 1 b row"; "me1 put myPtzPreset on b row"; "me1 select myPtzPreset on b row"; "set me1 b row to myPtzPreset"
- See also PiP-related PTZ preset commands.

RECORD, GRAB

- "record"; "record on"; "recording on"; "start record"; "begin record"; "start recording"; etc.
- "record off"; "recording off"; "stop recording"; "end recording"; etc.
- "Grab still"

STREAM

- "stream"; "stream on"; "streaming on"; "start stream"; "begin streaming"; etc.
- "stream off"; "streaming off"; "stop streaming"; "end stream"; etc.

SWITCHER

- "auto"; "transition" – perform a *Background* transition on the main *Switcher*
- "take"; "cut"
- "fade"; "crossfade"; fade slow; etc.
- "input 4"; "input 4 on program"; "put input 4 on program"
- "bob"; "take to Bob"; "take Bob on program"; "fade to Bob" – perform a *Background* (Take or Fade, respectively) transition to show a source (or PTZ preset) named "Bob" on the main *Switcher*.
- "input 4 on preview"; "put input 4 on preview"; "set preview to input4"; "input 4 on b row"; "set layer b to input 4"; etc.
- "me1 put input 4 on preview"; "set M/E 1 preview to input 4"; etc.
- "set M/E 1 b row to input 6"; "set me1 layer b to input 6"; "input 6 on me1 b row"; etc.
- "auto to input 4"; "transition to input 4" etc.
- "fast auto to input 4"; etc.
- "auto to input 4"; "transition to input 4" etc.
- "M/E 3 bob"; me3 auto to input 4"; "transition me3 to input 4"; etc.
- "ftb"; "fade to black"; "ftb off"; etc.

DSK

- "DSK1 Auto"; "dsk 1 perform auto"; etc.
- "dsk2 take"; etc.

DATALINK

- "set datalink key myVar to I'm a Yankee Doodle Dandy"
- "datalink key myVar equals 12"; "set datalink key myVar to value 12"; "set datalink myVar to 12"; "set myVar to 12"; "myVar equals 12"

TITLES

- "Aero Broadcast"; "Title Aero Broadcast" – for a LiveGraphic title, sets the default layer preset (10), then transitions the designated title in a default buffer assigned to a default DSK, removing it after 8 seconds;
- "Aero Broadcast hold"; "title Aero Broadcast hold" – as above but does not automatically remove the title
- "title Aero Broadcast Fly In L medium"; etc. – as above using the designated transition and speed
- See also title-related commands listed in the Buffers subheading.

ALIASES AND COMMENTS

You can use these *comment commands* to eliminate the need to pre-configure Switcher source names. Instead, you can set them right in the .docx script. This also allows you to change the metadata associated with a Switcher inputs at various points in your show as you wish.

- "Sam Smith is on input 4"; "assign input 4 to Sam Smith"; "set input 4 to Sam Smith" - set an alias for the designated input's name value

Hint: You'll likely want to use the full name here, for use in connection with titling.

- "input 4's description is CEO of Megadyne Inc."; "Input 4 memo is CEO of Megadyne Inc."; etc. – sets the descriptor for the input, which is used as the value for the second line of default lower third titles

TELEPROMPTER OUTPUT

Even apart from all the foregoing (for example, if you disabled both the Comments and Headings support), Live Story Creator would still be very useful as a native teleprompter. Coupled with file watcher functionality to easily update your script right up to 'air-time', this feature saves setup time, expense, and provides exceptional output flexibility.

CONTROL

We discussed Live Story's onscreen teleprompter controls earlier in this section, but these primarily serve for use during setup.

For live use, most prefer a dedicated physical controller device, whether for use by the talent or another teleprompter operator in the studio. Live Story natively supports the ShuttleXpress USB controller by Contour Design Inc. (Figure 293).

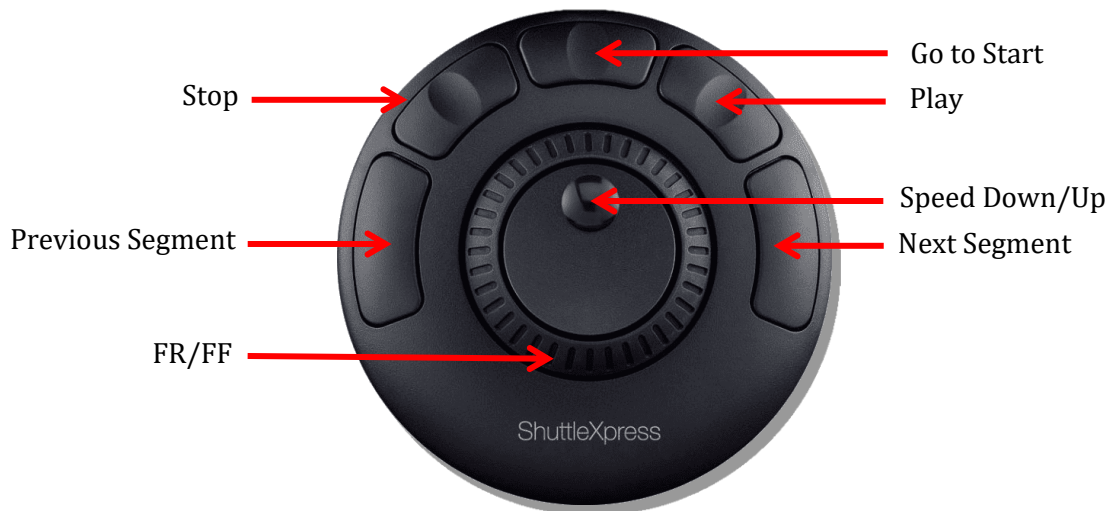


FIGURE 293

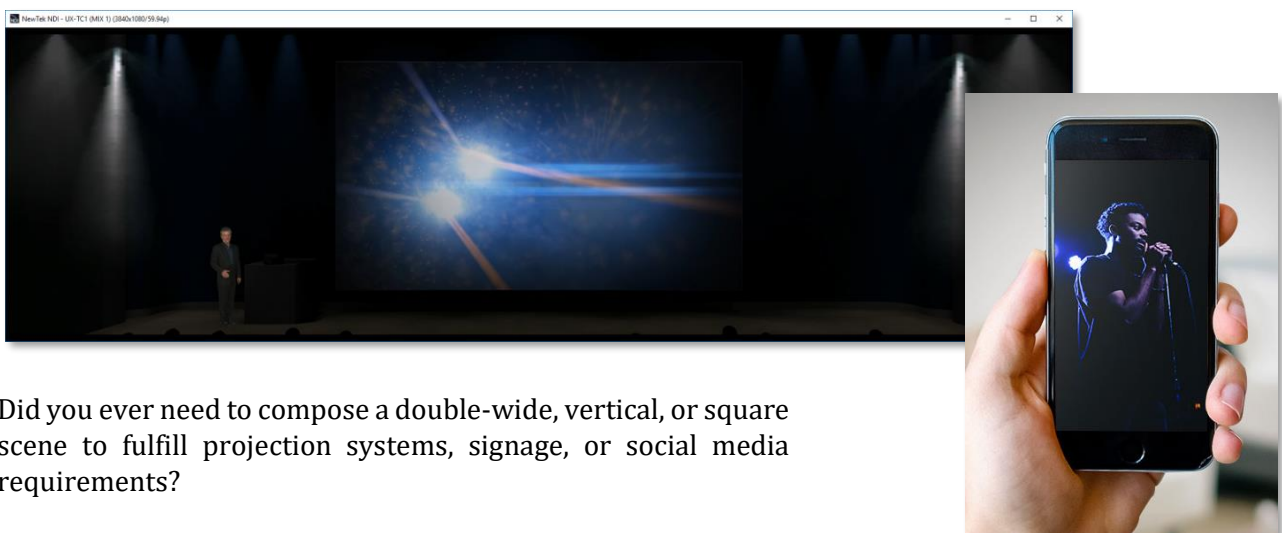
You will find this device provides common teleprompter controls in an easy to use and affordable manner.

OUTPUT

To support a wide array of external teleprompter hardware, Live Story assigns the scrolling text output to the highest numbered Switcher output by default. You can then assign this signal to a video Mix output supplying a physical video output connector to feed an external device.

Of course, in an NDI® environment, you have other powerful options. TriCaster automatically supplies the teleprompter output to your network as an NDI stream, offering many display options. For example, the Studio Monitor application included with the free NDI Tools bundle can be used to display your teleprompter output on Windows® and OS X® platforms. Another option, (in a new session) Live Story assigns the highest numbered external Switcher source to the Teleprompter's NDI output, making the Teleprompter output available to Multiviews.

A.3 FORMAT INDEPENDENT PRODUCTION



Did you ever need to compose a double-wide, vertical, or square scene to fulfill projection systems, signage, or social media requirements?

Advanced Features and resolution-independent NDI make this all possible, providing precision aspect control at every stage of the process.

- Select from numerous non-traditional frame formats to work in when creating a session, including double-wide and many more
- Flip, rotate, scale, and crop individual inputs
- Flip, rotate, and choose the resolution for individual outputs
You can even send left/right or quarter screen segments, and optionally enable automatically blended Overlap

SESSION OPTIONS

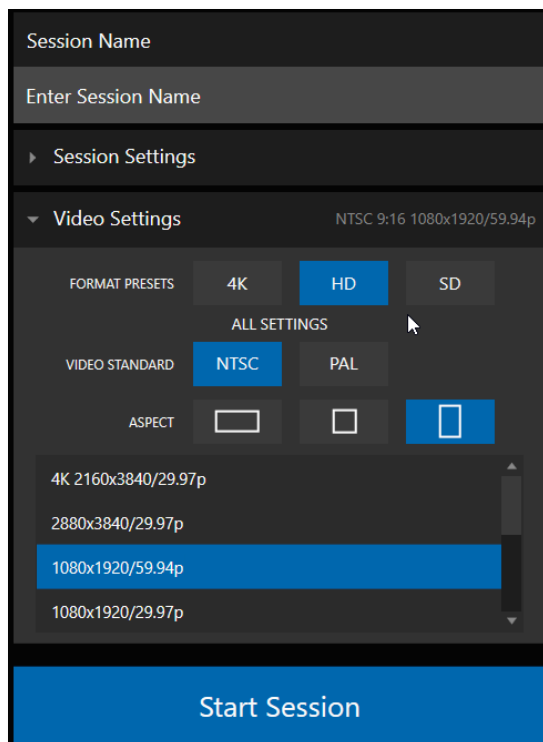


FIGURE 294

You'll first encounter these powerful features on the *Home* page of the *Launch screen*. Here you will notice a new group of buttons labeled *Aspect*, which allow you to choose between landscape (horizontal), square, or portrait (vertical) aspect modes.

In the Format selector, you will find that traditional video formats have been complemented by appropriate non-standard options including, for example, portrait aspect 1080x1920/59.94p and 1080x1920/29.97p options, and so on.

INPUT ROTATION

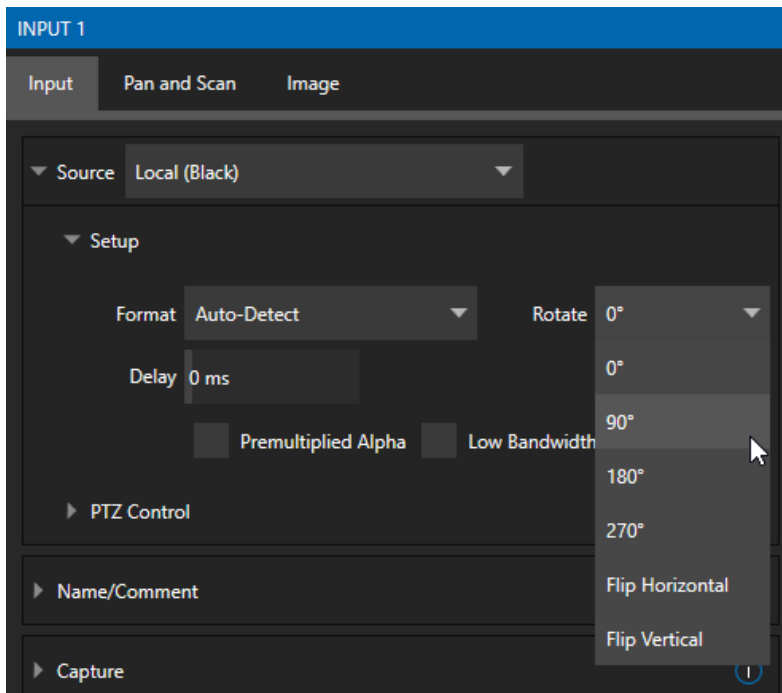


FIGURE 295

To complement non-traditional sessions and handle unusual *Switcher* sources (e.g., non-landscape mobile device output) each input now features a new *Rotate* menu, which also includes *Flip* options.

OUTPUT ROTATION

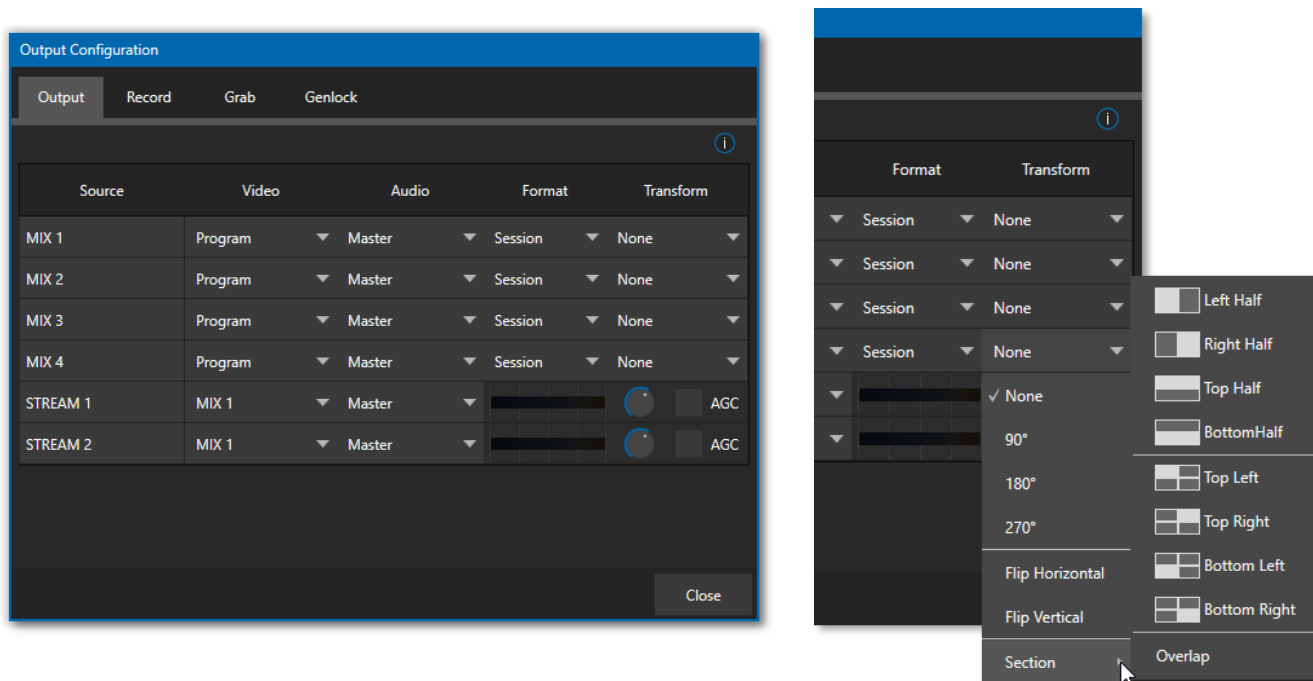


FIGURE 296

Extending these capabilities even further, all *Mix* outputs now support independent format selection, rotation, and flip control, as well as *Section* options.

A.4 LIVEGRAPHICS

LiveGraphics® represents a revolutionary approach to title and CG page creation, combining the power of the most complete production systems on the planet and the world's leading creative platform—Adobe® Creative Cloud®. Of all the Advanced Features benefits, LiveGraphics is the one of those most likely to dramatically elevate your productions in ways that grab the attention of your viewers.



FIGURE 297

LiveGraphics comprises several key elements: It includes an extensive set of professionally prepared animated title templates and the interface to control and update these in your NewTek live production system, but this is just the beginning.

Imagine designing animated titles, motion graphics, and looping effects using popular Adobe® Creative Cloud® tools—then importing them directly into your NewTek live production system, which will then allow you to both animate and update them dynamically.

With the LiveGraphics™ Creator plugin to simplify authoring in After Effects® CC for users of all experience levels, enhanced real-time system functionality allowing powerful customization, playback, and even automation options, LiveGraphics lets you produce and present spectacular live graphics faster and easier than ever before.

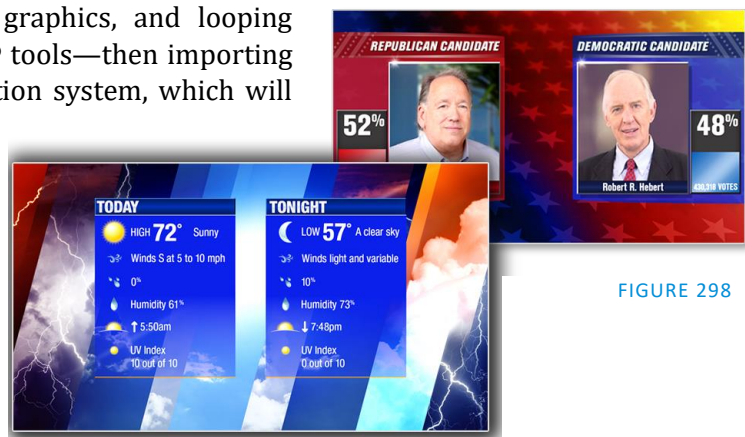


FIGURE 298

Note: Please download and install the After Effects plugin "LiveGFX Creator Plugin Installer" located on the NewTek Download page. (<https://www.newtek.com/downloads/>)

USE LAYERED PSD FILES

Create individual graphics or an entire package in Adobe® Photoshop® CC, then bring the PSD files directly into your NewTek live production system.

LiveGraphics recognizes the layers automatically, allowing you to manage elements independently and store up to 16 macro-ready variations per system buffer.

ANIMATE IN AFTER EFFECTS

Animate graphics faster and easier than ever before with the included LiveGraphics Creator plugin for Adobe® After Effects® CC.

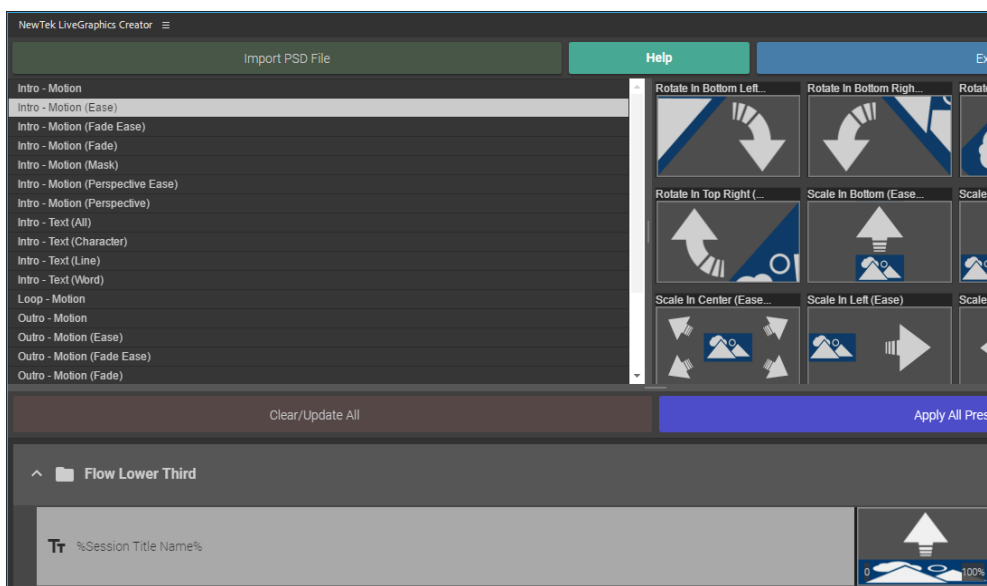


FIGURE 299

With LiveGraphics Creator, AE novices can simply import a layered Photoshop® PSD file, drag and drop presets to apply motion to its different layers, then export the result ready for live use.

Or use the included SmartBuilder™ to have LiveGraphics analyze your content and add animations for you with a single click! Of course, more advanced users can refine their creations using standard methods right in After Effects.

Hint: The use of NewTek LiveGraphics Creator is documented in its own manual, installed with the plugin.

USING LIVEGRAPHICS

You can also simply load any of the 100+ production ready LiveGraphics templates supplied with Advanced Features into your system's Buffer slots and get started right away.

The *Title Editor* lets you change text and images on the title pages and hide or show layers using the *Layers* menu (Figure 300). Layers animate into place as you do so. Then, store the results as presets – display the *Layer* preset bin by clicking the *Preset* button at upper right (Figure 301).

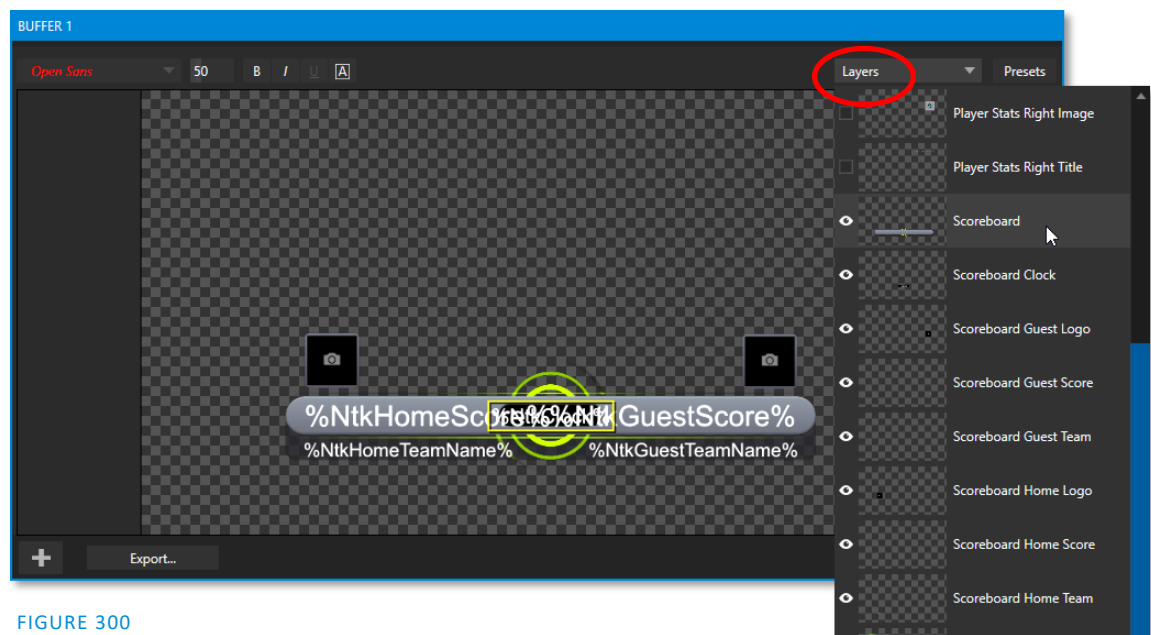


FIGURE 300

To preserve the *Layer* and *Data* presets you generate for your LiveGraphics pages outside of the current live session, you can use the *Export* button at left in the *Title Editor's* footer. Afterward, load your creation into a *Buffer* slot, and use its preset bin to animate from one state to another at will – see Section 11.3.10, Buffer Presets for more detail.

LIVEGRAPHICS AND DATALINK

Naturally, LiveGraphics supports NewTek's DataLink system for dynamically updating title page details even during live display (information about DataLink appears in several places in this manual, but you might start your exploration at Section 11.2.3).

DataLink™ technology lets you use external hardware scoreboards, spreadsheets, and other remote data sources to populate and update your graphics automatically.

LiveGraphics templates even support text and images direct from the Web when combined with the *DataLink Web Browser extension*.

In addition, the next Advanced Feature we will discuss (LivePanel™) provides several powerful ways to update *DataLink* key value (such as its integrated *Scoreboard* control pages for

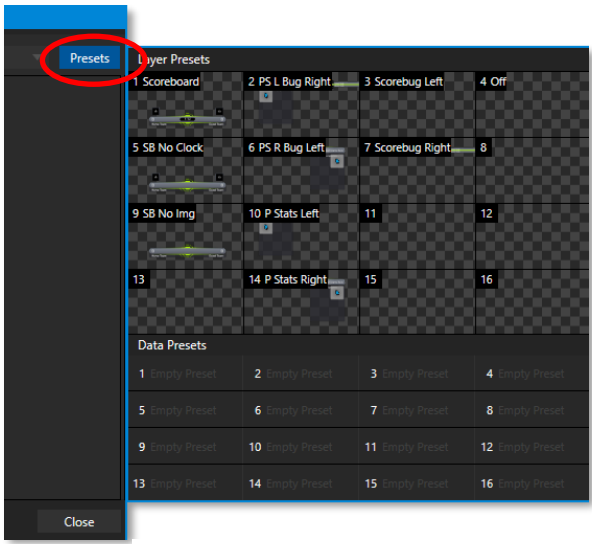


FIGURE 301

various sports). And thanks to the included *Builder* web app, you can even build custom control pages capable of driving your motion graphics from any device on your network with a web browser.

A.5 LIVEPANEL

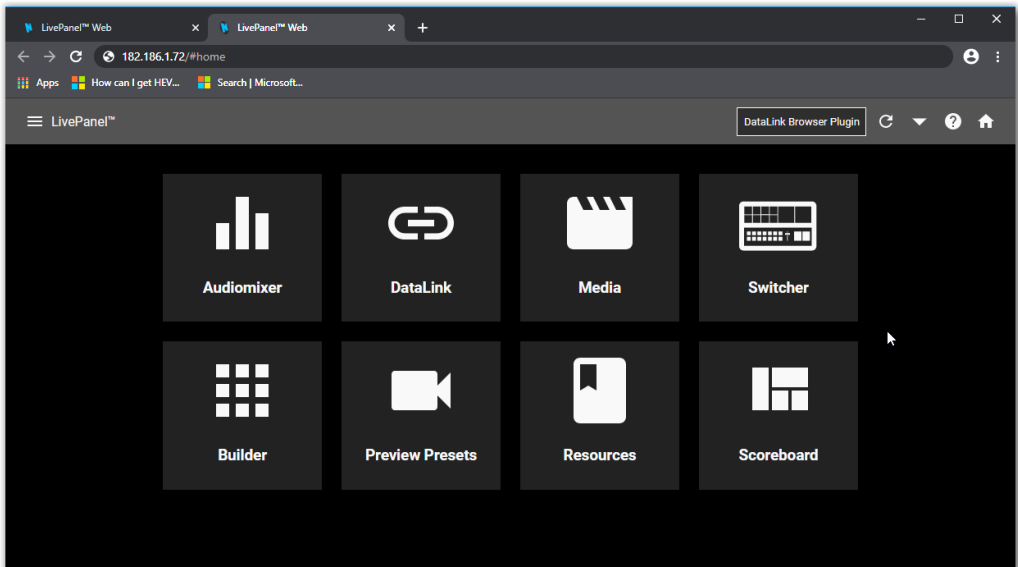


FIGURE 302

LivePanel, included with your TriCaster Advanced Features, provides potent tools to extend your production workflow, and is also available for separate purchase.

Among other features, LivePanel lets you create custom control panels for your NewTek live production system accessible in web browser of a device on your network. LivePanel also provides several production-ready tools for remote video mixing, media playback, audio mixing, and macro automation.

A.6 NETWORK ACCESS TO LIVEPANEL

To access LivePanel, click the *Notification icon* (see Section 7.7) at upper-right on your NewTek live production system's *Live Desktop*, to open the Notification Panel.

Note the URL shown beside the *Web Browser* button at left in the footer of the *Notification Panel*. Enter this into the address bar of a web browser on another device on the same network.

Note: You may be asked for a username and password to access LivePanel – see Section 4.1.

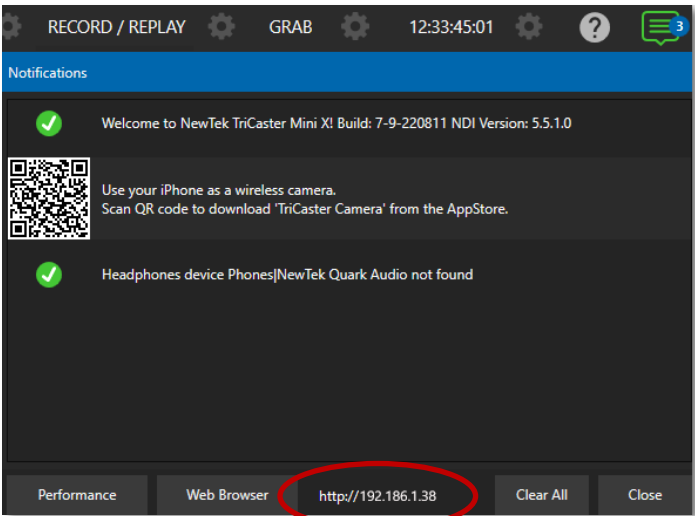


FIGURE 303

A.7 HOME PAGE

After entering security credentials (if necessary), Live Panel's *Home* page is displayed in your web browser.

Note: We recommend using a modern web browser (such as Chrome or Firefox) available for your platform for best results when using LivePanel.

All the existing LivePanel tools can be accessed from the Home page, either by using the 'hamburger' menu at left in the titlebar, or by clicking the large icons on the page.

Hint: Click the (?) icon at right in the titlebar to view information about LivePanel tools on the current page.

A.8 AUDIO MIXER

LivePanel's prepared web applets are easy to use. In many cases the controls mimic similar items in the main interface for your system. This can be seen in Figure 304, which shows LivePanel's *Audio Mixer* applet.

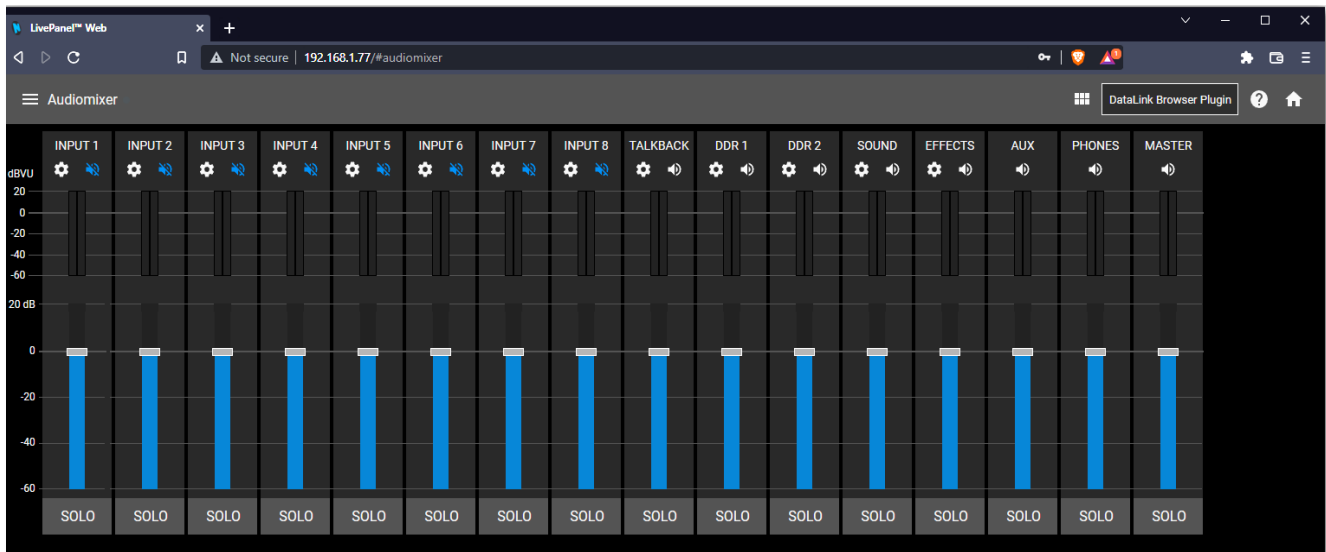


FIGURE 304

A.9 DATALINK

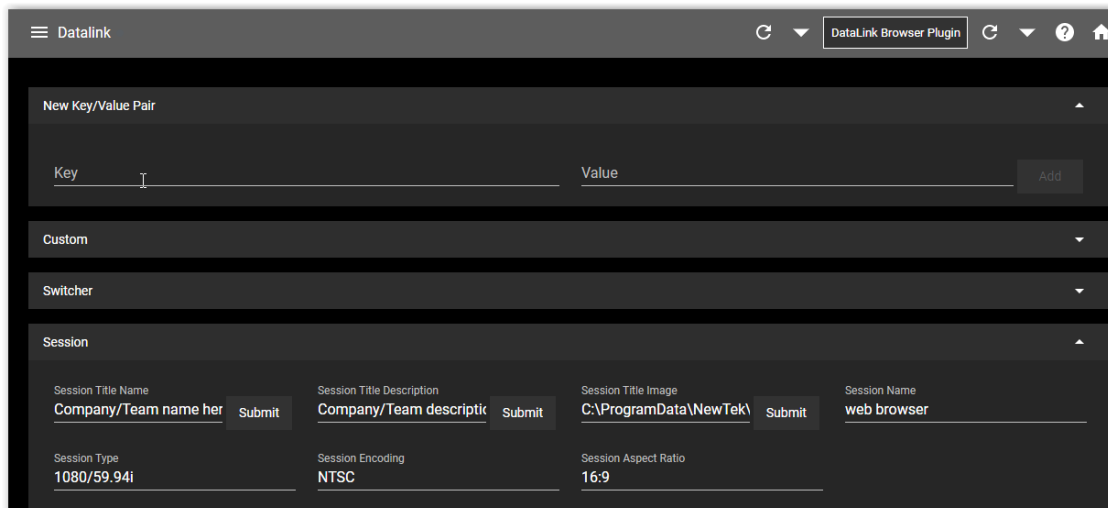


FIGURE 305

The *DataLink* page lets you selectively review the values for all current DataLink keys, edit these, and even generate entirely new key-value pairs.

A.10 MEDIA

Likewise, LivePanel's *Media* page provides you with remote access to the system's *Media Players*, allowing you play, trim, add and remove media files in the playlist.

Toggle the Media Player's *Autoplay* and *List/Single* playback modes, and trigger the *Show On* feature to push *Media Player* output to display on *Program* output or in a designated *M/E*.

You can even upload images from a local host device (such as a tablet or smart phone) to the playlist on your NewTek live production system.

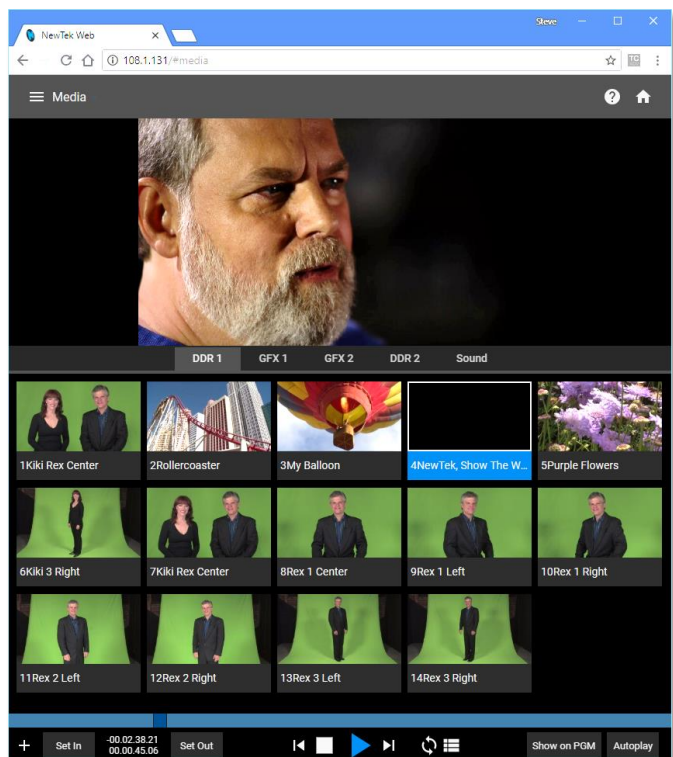


FIGURE 306

A.11 SWITCHER

The LivePanel Switch applet is designed to make it easy to switch less demanding productions without a complex interface. Simply tap a viewport for a video input to *Take* it to Program output or swipe it horizontally to do so with a transition.

Select transition effects for the *Switcher*, an *M/E*, *DSK*, or *KEY* layer, and – most powerful of all – select a *Comp* to change your composition entirely with a single tap.

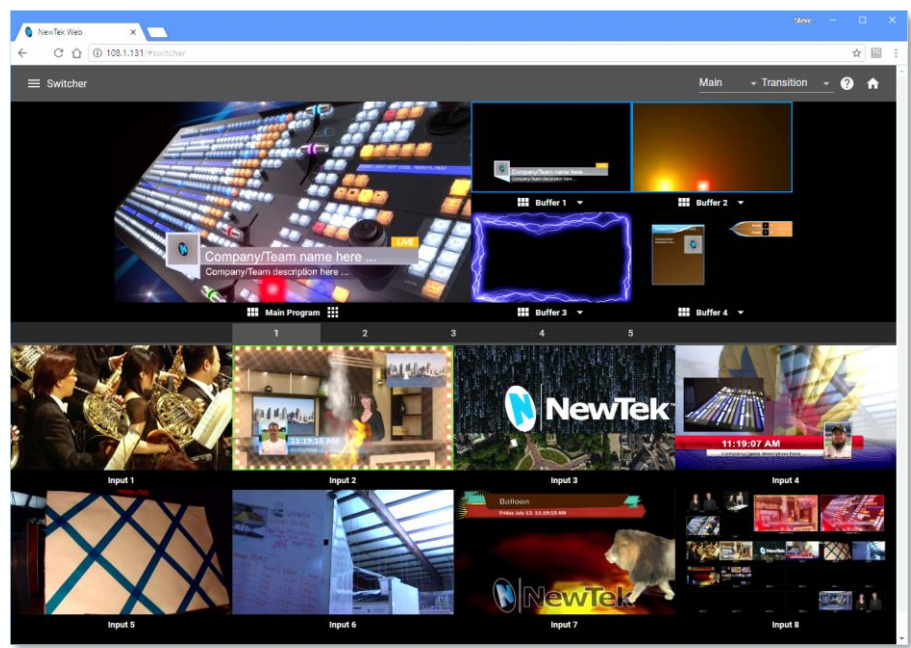


FIGURE 307

BUILDER

Unquestionably the icing on the LivePanel cake, the Builder Applet allows you to easily create custom control panels to meet a virtually limitless number of needs.

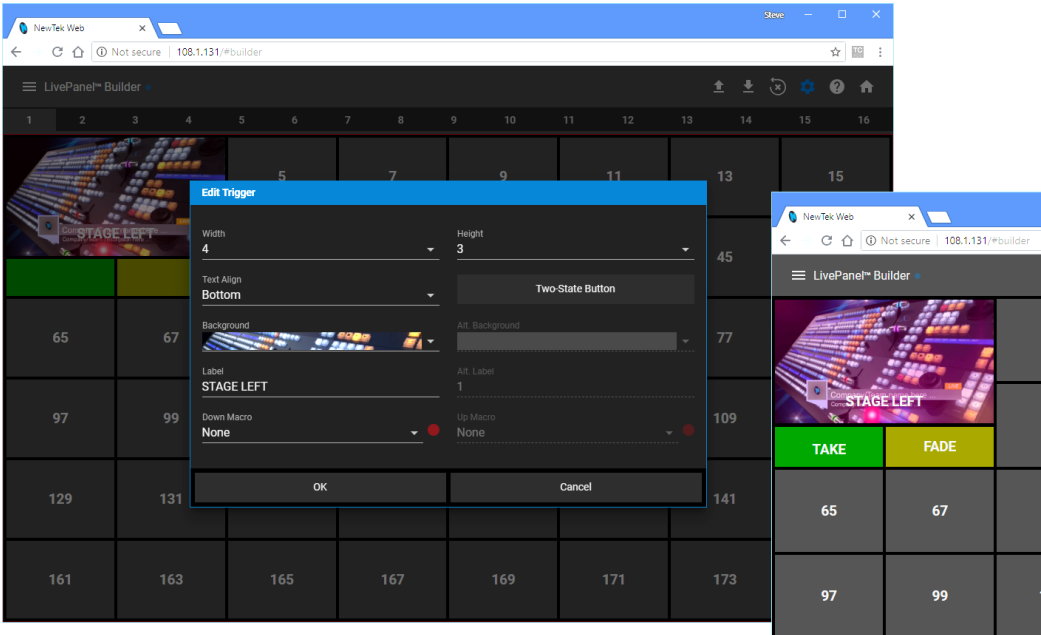


FIGURE 308

Flexibly assign a macro to a button, or perhaps two – one to operate when the button is pressed, and another to be applied when you click it a second time. Give the button a color or fill it with a still image of your choice.

The Builder applet lets you gang buttons together to give them more prominence, and you can even assign moving video from an external *Switcher* input, *Media Player*, or output to a button (effectively turning it into a monitor) and creating your own custom *Switcher* interface.

Hint: Each individual live production system hosts its own LivePanel instance, so Builder pages can be exported for use on additional systems.

A.12 SCOREBOARD

As discussed elsewhere, DataLink can be directly connected to several popular scoreboard systems.

With the new *Scoreboard* applet included in LivePanel, however, you can be your own scorekeeper using the web browser in virtually any laptop or tablet. Better yet, Advanced Features includes full motion *LiveGraphics* scoreboard title pages already configured to work with the *Scoreboard* applet.

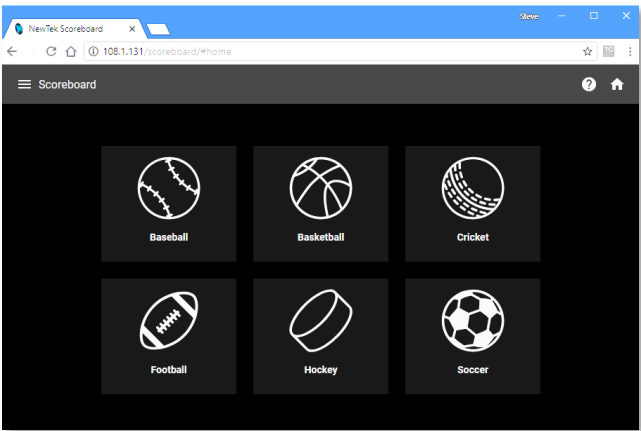


FIGURE 309

Having chosen the *Scoreboard* for your sport, and a matching scoreboard title page, the applet lets you set the team names, supply individual team logos, and track The *Channel* menu controls just one of two related channel settings – this one (on the local host), and another channel used by the control surface itself. These combine to let you connect to and control alternate live production systems.

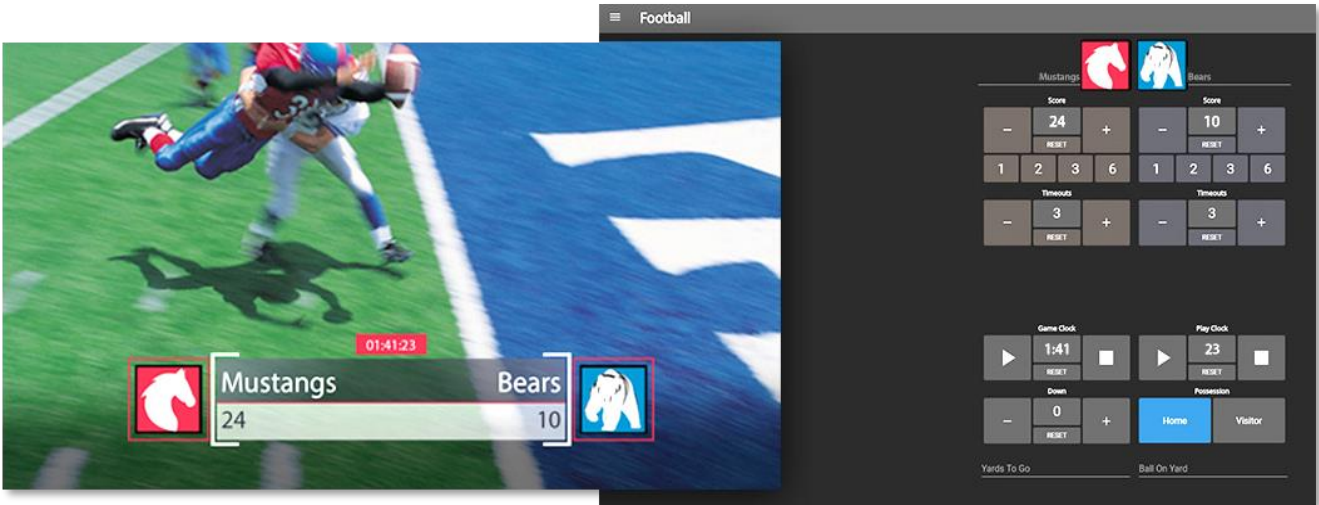
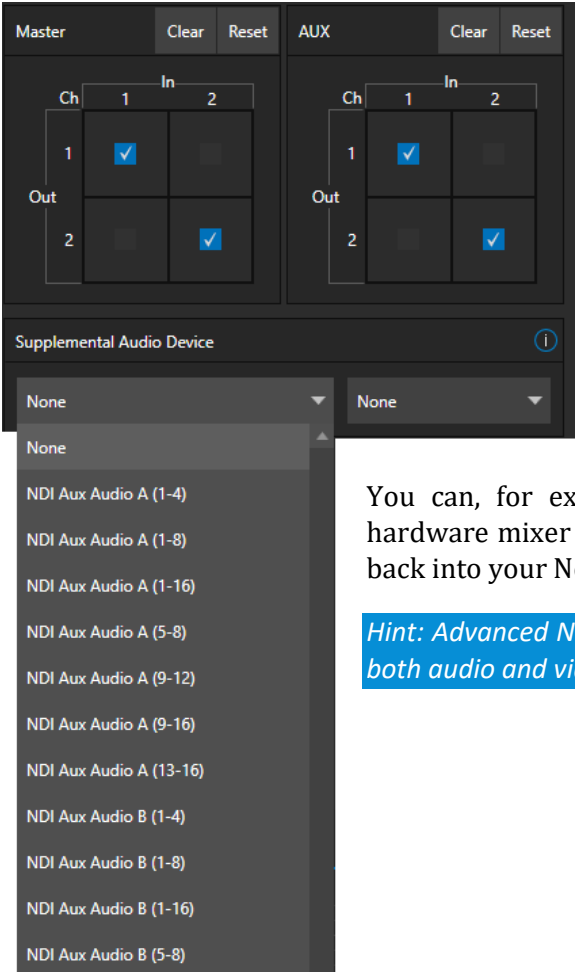


FIGURE 310

Hint: You can open the Switcher applet or a custom Builder page in another browser tab (or on another device) to display and removed the various elements of your animated LiveGraphics scoreboard, too.

A.13 ADVANCED AUDIO I/O



Advanced Features software provides our most complete support for audio solutions such as Audinate’s Dante and AES-67 implementations (with required third-party drivers).

NewTek’s *Advanced Audio* support for ASIO drivers allows you to easily transmit or receive audio using these popular ‘audio over IP’ protocols.

Send selected channels from any *Mixer* input, *Media Player* or output bus and channels you wish to send and match these to the transmit channels provided by your third-party driver.

You can, for example, send your *Media Player* output to an external hardware mixer supporting the same protocol, and route that mixer’s out back into your NewTek system for output with your video.

Hint: Advanced NewTek systems also provide independent per input delays for both audio and video, for a perfect match every time.

FIGURE 311

A.14 VIRTUAL PTZ

Virtual PTZ, also referred to as “pan and scan”, is a unique *Switcher* input feature applied to ‘standard’ video sources, i.e., those which are not actually connected to physical PTZ (Pan, Tilt, Zoom) cameras.

Naturally, the latter (real PTZ cameras), when connected and configured, show *Pan*, *Tilt*, and *Zoom* controls in their *Input* tab of the corresponding *Input Configuration* panels.

The *Virtual PTZ* feature adds similar controls to all *Switcher* sources (except M/Es, which already have similar capabilities thanks to *Comps*). This even includes *Media Players* and *Buffers*.

Since the *Virtual PTZ* implementation includes *Presets* and *Speed* controls, this means is that you can zoom in/out and pan around within the frame supplied by any source.

Of course, this is a particularly attractive feature in the context of UHD video sources but can also be used to mimic the so-called “Ken Burns effect” (a.k.a., ‘pan and scan’) when applied to imagery in *Media Players* (Figure 312).

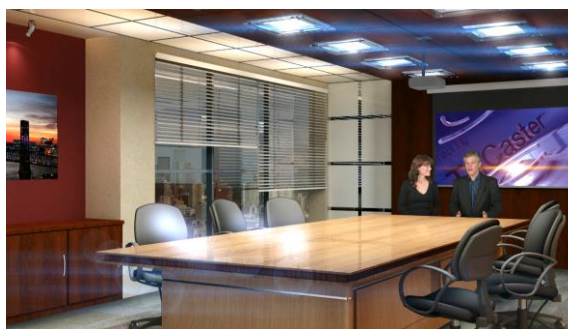


FIGURE 312

A.15 VIRTUAL SET LIBRARY

Advanced Features also provide you with the use of the entire NewTek collection of optional, LiveSet™ virtual sets, covering an array of useful broadcast, business, and industrial applications. A sampling from this collection is shown below.





At the time of writing, the collection includes thirteen professionally designed multi-camera virtual set environments ready-to-use for your productions.

Note: You can also purchase this collection outright in NewTek's online store, which has added benefit for Virtual Set Editor owners who will receive the corresponding customizable VSE projects with their purchase.

A.16 SOUND AND MUSIC

This 100-track bundle of songs includes a variety of license-free, full-length audio tracks to engage your audience and give you plenty of soundtrack options. The multiple musical genres in this bundle allow you to develop a theme or evoke specific emotions with background music for live content or edit these unlimited-use music clips into pre- and post-production content.

Appendix B: SKYPE AND SKYPE TX



Skype™ and SkypeTX™ may sound alike, but they are not really the same thing. Each delivers specific capabilities, and their connection and configuration details differ as well. This chapter will help you understand these distinctions, so you can take full advantage of these important connection options.

Literally hundreds of millions of people use Microsoft Skype™, so it's very likely that you are already acquainted with its use on mobile and desktop platforms. It's hard to imagine a better technology to extend your productions to remote participants, whether around the corner or in far-flung corners of the world. NewTek products provide several approaches to working with Skype.

B.1 SKYPE TX

Until recently, however, there were 'missing links' in the i/o chain between Skype and the video broadcast realm. Microsoft introduced its Skype TX platform, comprising purposed-designed hardware, software, and control room tools, to link the two worlds. NewTek was honored to be selected as the sole North American manufacturer of authorized Skype TX hardware, the NewTek TalkShow® product line.

B.2 TALKSHOW®



FIGURE 313

Each Skype TX-based TalkShow™ VS4000 unit provides the features needed to connect four high quality a/v streams between Skype callers and standard studio systems. NewTek live production systems make connection particularly convenient, by leveraging NDI® (Network Device Interface) IP protocol. Thanks to NDI, a single gigabit network connection easily handles all a/v traffic between the two units.

AUDIO AND VIDEO CONNECTIONS

The basic IP connection setup between the NewTek production system and TalkShow is quite simple. Individual a/v outputs for each remote Skype caller from TalkShow can be assigned to the *Switcher* and *Audio Mixer* like any other NDI a/v input source.

Similarly, the video output from your program can be assigned as the return-to-caller source on the TalkShow system. You will likely want to prepare a unique mix minus audio feed for the audio return to each individual caller. See the sub-heading Sub-Mixes and 'Mix Minus' in Section 15.10.3 of this manual for more detail on configuring these.

B.3 SKYPE TX CONTROLLER

A vital part of the Skype TX™ platform is the call-center application referred to as Skype TX Controller (available [here](#) for free). You might think of Skype TX Controller as the 'switchboard' for your calls. The operator can initiate outgoing calls, answer incoming calls, and more.



FIGURE 314

The Skype TX Controller console provides numerous professional conveniences, including contact and call management tools, auto-answer, detailed call quality metrics, and fallback images to be displayed automatically should network conditions for a remote caller fall below designated minimums.

B.4 SKYPE TX CALLER

TriCaster Mini X also features native Skype TX capabilities independent of a TalkShow system.

This unique support lets you directly connect Skype™ calls from anywhere in the world to either of two special *Switcher* inputs by leveraging the Skype TX Controller.

All you need to use this feature is a network connection between your NewTek system and the computer running the (free) Skype TX Controller application that provides call management.

AUDIO AND VIDEO CONNECTIONS

Section 8.1.1 explains how to assign a *Skype TX Caller* to a selected video *Switcher* input.

In this implementation, by contrast with the use of an external TalkShow system discussed earlier, it is not necessary to manually create a special mix minus audio return feed to send to the remote caller(s). You will find details on the special ‘mix minus’ that is automatically supplied in Section 15.5.

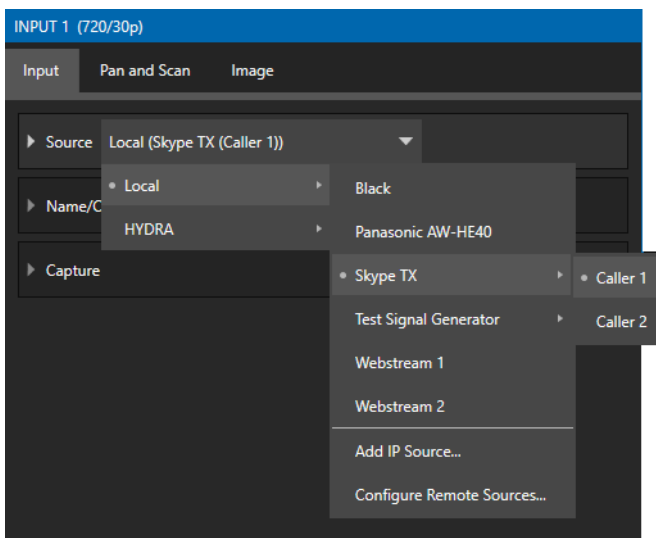


FIGURE 315

In addition, you will find it worth reading Section 15.6, which explains the unique *TalkBack* feature provided in the *Audio Mixer* to let the production system operator talk ‘offline’ with selected callers, without fear of the conversation being overheard on program audio output.

Hint: The return video sent to remote Skype callers can show a tally overlay. The overlay image is named SkypeTally.png, and is located in the file folder at one of the locations below, by model:

C:\ProgramData\NewTek\TriCaster\Configuration\Skype\ or ...

This image can be deleted to disable the feature, or replaced to modify it.

B.5 SKYPE FOR CONTENT CREATORS

Microsoft offers several variants of Skype. The Desktop version of Skype provides new possibilities for Skype users including broadcasters, streamers, and vloggers. including output using NDI® protocol. This support for NDI provides yet another way to bring Skype calls into NDI-enabled software and systems.

Input setup for Skype a/v output is like connecting any other NDI source. The Skype client does not directly support NDI input from you to remote callers, but you can use the NDI Virtual Input application (included with the free NDI Tools pack) to connect program video output and a mix minus audio return feed prepared as discussed in the sub-heading Sub-Mixes and ‘Mix Minus’ in Section 15.10.3 of this manual.

Appendix C: PERFORMANCE CONSIDERATIONS



In this section, we'll consider the most common questions you may have (and of course we'll provide the answer, too). Answers are intentionally brief – perhaps just a reminder of one or two steps required to perform some operation. For this reason, we'll also point you to explanatory information elsewhere in this manual whenever that would be useful.

C.1 TESTING, ONE TWO ...

Professionals simply do not leap into new environments blindly. They prepare, plan, plan some more, and then – most importantly – they test. This allows them to tackle the tough jobs with confidence.

C.2 IMAG AND LATENCY

What's IMAG? It's a compression of the expression "Image MAGnification." Typically, in modern IMAG applications, video cameras supply live imagery to projection systems, magnifying speakers and performers so that audience members further back in large venues can still see what's going on.

IMAG is a very tricky task at the best of times, one that calls for excellent planning, and where possible, testing. Those designing an IMAG installation have, not just one, but *two* inter-related broadcasts to consider – in the form of the audio and video streams.

C.3 RELATIVITY AND THE SPEED OF LIGHT

Wouldn't it be nice if audio and video travelled from their respective broadcast devices at the same speed? Then, wherever you were seated in the audience, the sound from hypothetically perfect speakers and the video image from huge video displays co-located at the front of the auditorium would reach your ears and your retinas at precisely the same moment!

This is not the case, however. Sound travels quite slowly – so slow, in fact, that even in relatively small venues it reaches those in the rear of the audience noticeably later than those in the front.

In loose terms, for a mid-size auditorium 600 feet long, it takes around a half-second for the audio to reach those in the back. For this reason, audio engineers often position speakers throughout the 'house', then introduce carefully considered delays by electronic means – to ensure 'late sound' from front speakers does not arrive after sound from the nearest speaker to those further back.

Light, on the other hand, travels so much faster that for all intents transmission can be considered instantaneous in the same setting. So, a person in the rear will see the image on a screen at the front long before sound from a *co-located* speaker arrives.

If transmission of the video signal from the camera lens right through to the projection screen were instantaneous (it's not, mind you), we'd likely need to *find* a way to delay it. Viewed in this light, a certain amount of latency is actually "A Good Thing!"

C.4 LATENCY AND YOUR AUDIENCE

"Latency" – what's that you ask? In this context, latency refers to the time it takes for the video signal to transit one part of the IMAG path, from camera lens to final display. Latency is usually expressed either in milliseconds or in video "frames" (typically either 25 or 29.97 to the second.)

As we said above, really a little video latency is not a bad thing as viewed from a surprisingly short distance back into the audience. This is just as well, since for all practical purposes a little latency is also unavoidable.

Even so, as long as audio and video are in sync *at your seat*, only a rather 'significant' degree of latency will be objectionable – unless you happen to be very near the stage. For those in the front rows, a few extra frames of latency *may* be rather disconcerting. (It's true that IMAG was conceived primarily for the benefit of those further back, but if the latency is too obvious for those nearest the front it can be disconcerting distraction.)

For this reason, it's desirable to keep video latency to an agreeable minimum – but put away any notion of 'zero latency.' Not only would this require bending the laws of physics, but it would also be a bad idea.

Even before considering minimizing latency in the device chain, acknowledging that there is always going to be some latency calls for some creative thinking with regard to practical staging. For example, if you design your IMAG layout in such a fashion that those in the front row are unlikely to be able to see the screen(s) without lifting their eyes from the onstage talent, they are extremely unlikely to notice a small amount of latency.

C.5 LATENCY AND YOUR NEWTEK SYSTEM

NewTek systems excel at IMAG, but inevitably constitute just one (albeit critical) link in a chain of devices. It is common for each device to contribute a little to the combined total latency for the system.

Your NewTek system's portion of the total latency is well within acceptable standards for devices of this sort – roughly between 1 and 2.5 frames (the actual amount can vary slightly within this range, depending on several factors).

For example, suppose a 'video frame' supplied by a camera arrives at an input one millisecond after a different frame has been sent to the output. Obviously, the new arrival must wait its turn; it can't be transmitted until the correct duration for its predecessor has elapsed. Thus, the newly arrived frame must 'wait' almost one entire frame, until its turn comes.

How can you achieve the lowest possible latency for the switcher's section of the IMAG chain? One trick is to 'Genlock' your cameras to the switcher output. This bypasses input time-base correctors, ensuring latency is consistent during switching operations. For genlocked sources, consider disabling *Frame Sync* – see Section 8.1.1.

Hint: You can assess the latency of your pipeline easily. Run timecode directly to 1) a monitor and at the same time 2) through the switcher to a second (identical) monitor. Take a photograph that encompasses both monitors, and compare the timecode shown.

C.6 OTHER SOURCES OF LATENCY

Often, significant latency is added by other devices in the IMAG chain that come *after* the switcher. Projectors are a common contributor, but at times the cameras themselves are a factor.

Here are some helpful points to consider when designing and connecting your system:

- If you use Multiview output from the switcher to the projector – if at all possible, match the format sent from the switcher to the native format of the projector. (On some projectors this allows the unit to avoid using its own internal scaling, which is often a significant factor in unwanted latency.)
- When possible, try supplying the projector with analog video. This can eliminate a lot of the complexity from the process (of course, this is not always possible.)
- Certain projectors provide a ‘low latency mode’ to disable features of the unit that carry a heavy toll in latency. Enabling this mode can make positioning the projector slightly more challenging (as you may sacrifice some ability to position and scale the image using projector menu functions) – but the latency reduction can be very worthwhile.
- Some cameras include features that add more latency than you would expect. For instance, image stabilization (by definition) adds one field of latency and sometimes more. Disable anything of that sort that you can.
- Latency may be slightly lower for progressive sessions, so for lowest latency, (genlocked) 720p cameras and session are theoretically ideal (again, for genlocked sources, consider disabling *Frame Sync* – see Section 8.2.3).

Appendix D: VIDEO CALIBRATION



An oft-heard expression is “Don’t worry – we’ll fix it in post”. Post-production techniques are invaluable, but they do not invalidate another often repeated axiom – “Garbage in, garbage out.” And, in the case of live production, the results are immediately obvious to your audience ... for good or for bad.

For these reasons and many more, it’s important to control the visual quality of your work all the way through the production pipeline. Adequate lighting, good cameras, and high-quality connections are critical, but there are other things to consider.

Human vision is remarkably adaptable. In one study, participants equipped with inverting lenses initially saw everything upside-down. After a few days, people reported that things appeared right-side-up again.

Even when things are dramatically wrong our brains compensate to a remarkable degree. Have you ever looked at a television in a store without noticing anything untoward, and then realized that the colors on another unit nearby looked much better?

This built-in tolerance makes it difficult for us to judge whether the blacks in our video productions are black –or just dark gray; whether reds are purple or tinged with a slight greenish caste, and so on. What we really need is a reference for comparison – just as we find it easier to pick the best-looking display from a row of television sets on a store shelf.

D.1 WHAT (AND WHERE) TO CALIBRATE?

Tip one – forget ‘point and shoot’. Lighting, set design, and countless other factors have an impact on the video attributes our cameras ‘see’.

As a first step, cameras need to be properly calibrated. *Ideally, this is done right at the camera.* Corrective measures taken downstream never fully compensate for problems at the first link in the chain. Most cameras, even inexpensive consumer models, provide adequate controls for this purpose. Some other video sources do not offer much if anything in the way of adjustment, however. You may need to adjust the *Proc Amp* settings to improve previously recorded video played from a DVD player, or a networked ‘video chat’ system.

In both cases above, the *Waveform* and *Vectorscope* monitors will help. We also need to consider *downstream* devices, though.

Computer displays are inherently different than television sets. Thus, the video you see on your computer interface can vary quite a bit from what your viewers see on their televisions. Too, you will want to be able to correctly judge colors when preparing titles and graphics. It will help you a great deal if your downstream video monitors are properly set up to allow you to make these judgments.

D.2 CALIBRATING VIDEO SOURCES

The obvious disparity between neighboring televisions on display in a store amply demonstrates that even identical (and brand-new) video devices can differ dramatically. When mixing multiple cameras, we need to ensure that their output matches. Switching to a camera with obviously different color characteristics will be seen as a glaring error by your viewers.

Even when we restrict matters to a single unit, color characteristics can change as the system warms up, and as it ages. For this reason, it's important to allow a device to warm up before attempting to calibrate it.

D.3 SETTING BLACK AND WHITE

Naturally, the color range available for transmission and recording is bounded at the upper level by white and at the lower level by black. Anything else falls somewhere in-between.

Consider what happens if you gradually raise the brightness control on your television. Beyond a certain point (and unlike claims made for laundry detergent) your whites do not become whiter. They can't – the upper limit (white) is firmly fixed. Instead, parts of the image that are nearly white are boosted, eventually also becoming white. Meanwhile, black portions of the picture are tending towards gray. Since white cannot become whiter, and black has become gray, we could say that the dynamic range of the image has been narrowed. The net result is a less vibrant image.

The same thing is true for video from your cameras. If the black and white levels from the camera are incorrect, you are effectively losing either shadow or highlight detail. For this reason, the first thing many do is calibrate their camera for correct levels.

WAVEFORM MONITOR

From days of old, for video engineering purposes the scale between black and white was defined in IRE units (IRE being an acronym for "Institute of Radio Engineers"). White was pegged at 100 IRE. For PAL (and NTSC-J) countries, black was defined as 0 IRE. For NTSC lands, black properly sat at 7.5 IRE.

Thankfully, the day of strict adherence to these analog video concepts has virtually disappeared. Today, confirming that the black and white levels your camera is sending are correct is as simple as sending first black, and then white (or the brightest part of your scene, and reading values from the scale alongside the *Waveform* scope.

Hint: Your system automatically compensates for the traditional 7.5 IRE black (also known as "setup" and "pedestal") in SD format NTSC sessions.

Connect your camera to the correct input, block the lens so it receives no illumination, and check the level shown in the *Waveform* monitor. Generally, it should be 16 on the 8bit scale.

To check white, use either the brightest part of your scene or a white card, ensuring that it is evenly illuminated with the same lighting your main subject will receive. Move in or zoom to fill the viewfinder with this, and confirm that the *Waveform* monitor is showing 235 on the same scale. Otherwise, you might try using your camera's Auto White Balance feature with the white card – your camera manual will provide instructions. Afterwards, check the black level again.

Some more professional cameras offer full manual controls for white balance and/or black level. Use these as instructed to ensure your camera is providing the correct white and black levels.

If you cannot make source adjustments, or can't get it quite right by these means alone, you can use the Brightness and Contrast controls in the *Proc Amp* for that input to tweak black and white levels. (Of course it is always best to perform adjustments at the source if possible.)

D.4 ADJUSTING COLOR

We're going to move into color calibration next, but first we can use our black and white signals for some further tests.

VECTORSCOPE

While we're still working with black and white levels, we can introduce the *Vectorscope*, and perform an initial test of the camera's color balance.

A vectorscope (Figure 316) can be likened to the familiar 'color wheel' (Figure 317) which sweeps radially through the colors of the spectrum – yellow, red, magenta, and so-on, around the arc of a circle. Colors are more progressively intense (saturated) towards the outside of the circle, while color saturation is zero at its center.

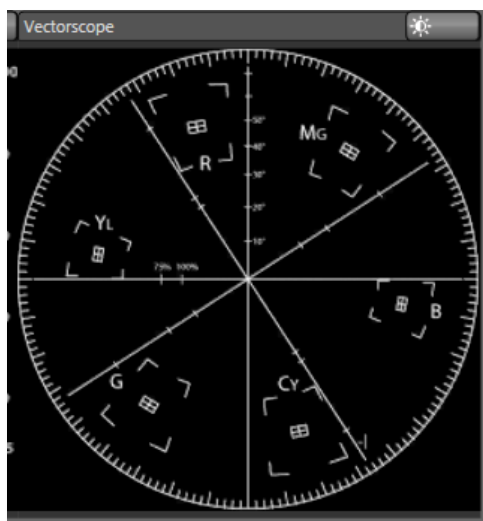


FIGURE 316

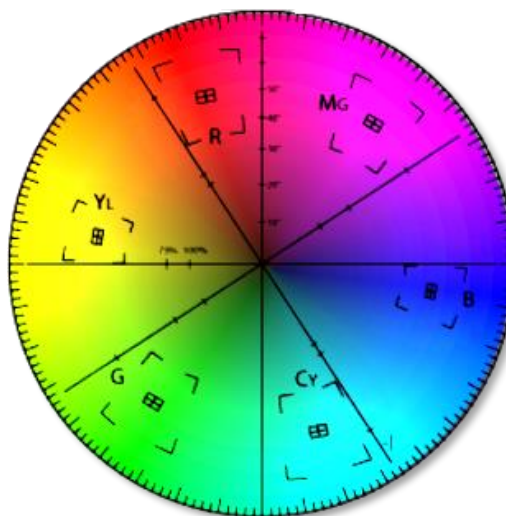


FIGURE 317

As it happens, from the vectorscope point of view, neither black nor white properly have any color saturation. Thus, with the lens cap on (or with a white card filling the viewfinder), the vectorscope should show only a small fuzzy trace at its center. If the fuzzy dot is off-center horizontally or vertically, this would indicate that the camera is incorrectly calibrated, tinting gray areas.

When the trace is off center, the direction and distance of the offset tells us what sort of tint (and how much) is represented by the deviation. You may be able to use the color controls at your camera to correct for this

offset, or you can use the Proc Amp's *U Offset* and *V Offset* controls to do so (as always, source controls are best). Adjustments to *U Offset* move the trace left or right, while *V Offset* changes adjust its vertical position.

Let's move on to a slightly more rigorous testing.

D.5 COLOR METRICS

At this point, we've assured ourselves that the signal from the camera is neither too bright nor too dark, that its output falls within broadcast legal luminance limits, and that the black & white part of the signal does not have an unwanted color cast.

We haven't done anything yet, though, to assure our reds are red, not slightly brown, or that our blues are not slightly green or magenta, etc. The *Vectorscope* can provide much more specific information about your camera's color signal. Let's see how it can assist you to ensure your colors are accurate.

USING COLOR BARS

You'll no doubt have seen the familiar color bars used as a standard reference for video signal calibration. Two examples are shown here. Figure 318 (NTSC) is an example of the color bars used in NTSC countries, while Figure 319 (PAL) is a PAL example, common throughout European nations.



FIGURE 318 (NTSC)

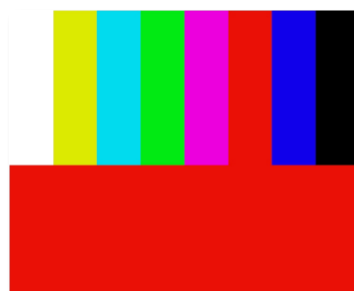


FIGURE 319 (PAL)

You can use color bars in conjunction with *Waveform* and *Vectorscopes* to make sure the video supplied to is consistent, accurate and broadcast legal. Most video cameras are capable of displaying color bars – check your camera manual to see how to display these (given a choice, use 75% bars). Then look at the *Vectorscope* to see how it traces the individual colors comprising the image.

The *Vectorscope* graticule has six distinct rectangular *targets*, one each for Yellow, Red, Magenta, Cyan, Blue, and Green. The targets are small rectangles with a crosshair superimposed on them.

When a source is properly calibrated, the trace from the different colored segments of the color bars displayed will fall right inside their individual targets (see Figure 320).

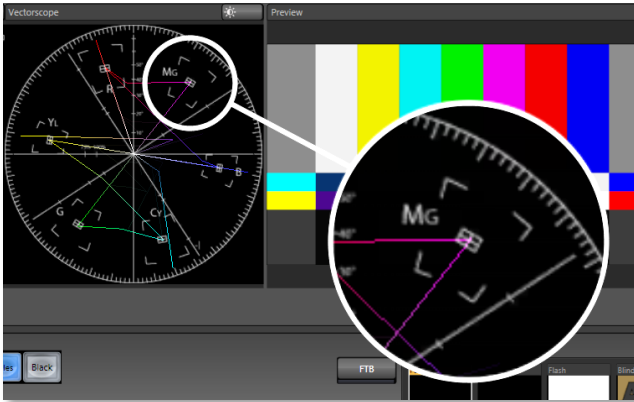


FIGURE 320

If the trace vectors do not line up as they should even after performing a white balance at the camera, you can use *Proc Amp* controls to tweak the signal.

Adjust the *Hue* control to rotate the vectors around the center point to line them up correctly on their respective targets. Increasing *Saturation* will move the trace further out towards the edge of the scope. Decreasing *Saturation* lowers color intensity, bringing the trace back closer to the center.

Hint: Naturally, you should repeat the steps above for each connected source, to ensure a perfect match when switching from camera to camera during your live productions.

At this point, your video signal should be reasonably accurate, and broadcast legal. Naturally, there are other devices between that signal, you, and your viewers. Let's discuss ways to calibrate downstream video monitors to ensure that you see your video at its best.

D.6 CALIBRATING YOUR MONITORS

In most cases, you will use a computer monitor of one type or another to display your system's controls – the user interface. Even though the television monitors you likely use for final program output may look somewhat similar to your computer monitor, they do differ in a number of important respects. For this reason, we will approach their calibration separately.

D.7 COMPUTER MONITOR

This is a subject that could (and does) easily fill multiple volumes, but which we will sadly give short shrift. It may not be *quite* as important to achieve 'spot-on' color calibration for the monitor used for the user interface. Not surprisingly, video output color is best judged on larger downstream video monitors.

Nevertheless, you may find a trip to one of any number of websites providing test imagery and calibration hints useful. A search for "computer monitor calibration" will yield an endless list, along with some commercial utilities for those who feel the need. Here are links to several such sites:

<http://epaperpress.com/monitorcal/>
<http://www.lagom.nl/lcd-test/>

<http://www.jasc.com/support/kb/articles/monitor.asp>
<http://www.photofriday.com/calibrate.php>
http://www.onecomputerguy.com/windows7/windows7_calibrate_color.htm

D.8 PROGRAM OUTPUT MONITOR

Again, a search of the Internet for “video monitor calibration” will yield many resources, both free and commercial. If you have television monitors connected to your outputs, though, you can make further use of the color bars. The approach is somewhat similar to what we did for our video sources, in that the first adjustments are to ensure that black and white levels are correct. Color calibration is performed afterward.

First, warm up your monitor for at least 10 minutes (some sources recommend a half-hour), to stabilize its output. Dim room lights and window light, to eliminate light sources that might skew your color perception. Pass the color bars from your camera through to program output (or use a color bars image in one of the *Media Players* for this purpose.

D.9 COLOR ADJUSTMENTS

Having set the black and white levels earlier, let’s continue...

NTSC

1. If your monitor has a “Blue-gun only” setting, enable it. Depending on the brand of your monitor, the alternating bars will appear either as light gray and black, or blue and black (Figure 321).

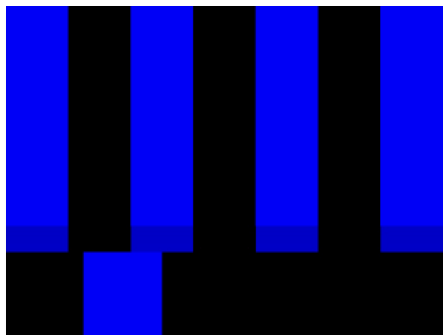


FIGURE 321

2. If the monitor lacks a “Blue-gun only” feature, you can obtain a colored filter such as a Kodak Wratten #47B gel and hold that between your eyes and the monitor for a similar result.
3. Tweak the *Color* (or *Saturation*) knob until the small rectangular patches at very bottom of the tall upper bars on the extreme right and left of your screen merge with the tall bars above.
4. Tweak the monitor’s *Hue* control until the small rectangles beneath the third and fifth tall upper bars likewise blend into bar above.

5. You may need to redo the last two steps several times, balancing the adjustments of these two controls until all four of the lighter bars evenly match the smaller rectangles just beneath them. (Note that a perfect match simply may not be possible on some monitors.)
6. Check your results with some familiar imagery and make any further tweaks you feel are required.

PAL

1. If your monitor has a “Blue-gun only” setting, enable it. Raise the *Color* or *Saturation* setting until the three right hand bars all appear to be the same shade.
2. If the monitor lacks a “Blue-gun only” feature, you can obtain a colored filter such as a Kodak Wratten #47B gel and hold that between your eyes and the monitor for a similar result.
3. Turn off the “Blue-gun only” feature (or dispense with the filter if you used one) and examine the red bar. It should appear not to be so vivid that it bleeds onto the nearby bars.
4. Check your results with some known familiar imagery and make any further tweaks you feel are required.

Appendix E: KEYSTROKE SHORTCUTS

E.1 SWITCHER...

Program Row, Inputs 1 to 24	F1 – F12 plus [Shift + F1-12]
Preview Row, Inputs 1 to 24	1 – 0, -, =, and [Shift + 1-0, -, =,]
Transition Delegate – select BKGD	[key (left square bracket)
Transition Delegate – multi-select BKGD	Shift + [
Transition Delegate – select DSK 1	] key (right square bracket)
Transition Delegate – multi-select DSK 1	Shift +]
Transition Delegate – select DSK 2	\ key (backslash)
Transition Delegate – multi-select DSK 2	Shift + \
Transition Delegate – select DSK 3	Backspace
Transition Delegate – multi-select DSK 3	Shift + Backspace
Transition Delegate – select DSK 4	Insert
Transition Delegate – multi-select DSK 4	Shift + Insert
Transition Delegate – select FTB	b
Transition Delegate – multi-select FTB	Shift + b
Select FTB Delegate and perform Fade	Ctrl + b
Auto DSK 1 On/Off	a
Auto DSK 2 On/Off	s
Auto DSK 3 On/Off	d
Auto DSK 4 On/Off	f
Take DSK 1 On/Off	Shift + a
Take DSK 2 On/Off	Shift + s
Take DSK 3 On/Off	Shift + d
Take DSK 4 On/Off	Shift + f
Select Transition (prev/next)	, and . (comma and period)
Select Fade Transition	Ctrl + f
Restore Last Transition	Shift + Ctrl + f
Transition – Slow ... Medium ... Fast	z...x... c
Transition – Increase Speed	Shift + c
Transition – Decrease Speed	Shift + z
Toggle Reverse Transition On/Off	e
Ping Pong Transition (off/on)	Alt + e

E.2 T-BAR

Take	Return Key
Take T-bar Back to Top	Shift + Return
Transition ... Go/Pause	Spacebar

Auto Reverse	Shift + Spacebar
Adjust T-bar... Down/Up	t / Shift + t (faster add Ctrl)
Halfway T-bar... Take	h

E.3 RECORD, GRAB, AND STREAM

Record on/off	r/ Shift + r
Grab (still image)	p
Stream Start	Ctrl + /
Stream Stop	Ctrl + Alt + /

E.4 TABS

Show (DDR, PTZ, Mixer, etc.) tab	Ctrl + (F1 to Fn)
Select M/E 1- 8	Ctrl + Shift + F1-F8
Show M/E pane	Ctrl + m
Hide M/E pane	Alt + m
Toggle M/E pane	m

E.5 WORKSPACE

Show Interface A	Alt + Numpad 0
Show Interface B	Alt + Numpad 1
Show Interface C	Alt + Numpad 2
Show Interface D	Alt + Numpad 3
Show Multiview A	Ctrl + Numpad 0
Show Multiview B	Ctrl + Numpad 1
Show Multiview C	Ctrl + Numpad 2
Show Multiview D	Ctrl + Numpad 3

E.6 MEDIA PLAYERS

Stop	k
Play	l (lower case L)
Go to previous playlist item	j
Go to next playlist item	; (semi colon)
Autoplay On/Off	u
Loop On/Off	' (apostrophe)
Single On/Off	q
Navigate through Playlist items	Up/Down/Left/Right Arrows
Set In Point for current Playlist Item (Clip or Audio file)	i

Set Out Point for current Playlist Item (Clip or Audio file)	o
Split clip at current frame	/
Reset In and Out Point to the file's first and last frame	g

E.7 GENERAL

E.8 EDIT TITLE PANE

Save the current title page	Ctrl + Shift+ s
Save the current title page, and duplicate it in the playlist	Ctrl + Shift+ s
Save the page and move to the previous playlist item	Page Up
Save the page and move to the next playlist item	Page Down

E.9 SELECTION AND NAVIGATION

Select All	Ctrl + a
First/Last Item	Home/End
Navigate through Playlist items	Up/Down/Left/Right Arrows
Select All Previous/Following Items	Shift + Home/End
Playlist - select to Top/Bottom	Shift + Home/End
Playlist – add to selection (Up/Down)	Shift + Up/Down arrow

E.10 MISC.

Cut	Ctrl + x
Copy	Ctrl + c
Paste	Ctrl + v
Delete	Delete Key
Show version	Alt + b

Appendix F: CUSTOMER SUPPORT



It is frequently said that “All good things must come to an end” ... but your NewTek live production system is designed to work as well on any day as it did on the day you bought it. NewTek’s Customer Support features and options are meant to help you achieve that goal. This chapter identifies the many assets and services available to you.

F.1 PRODUCT SUPPORT

Product support is available for as long as you own your NewTek system. Find complete details on our main Support page at <https://www.newtek.com/support/get-support/> or request a ticket with customer support <https://support.newtek.com/hc/en-us/requests/new>.

- **Training:**
 - **NewTek University:** Our new online training will help you take advantage of the countless creative possibilities your unit offers (you can also become a Certified Operator once you pass the exam) – see NewTek U
 - **Discover:** Watch our online videos to learn the benefits and features of NewTek professional video production systems NewTek's YouTube channel
- **Software Updates:** visit Downloads and Documentation
- **Knowledge Base:** Our extensive online knowledge base will help you troubleshoot issues and clear up any misunderstandings – see NewTek Knowledge Base
- **NewTek User Forums:** Join the conversation, get answers, learn techniques, and share ideas with users like yourself at NewTek Forums

F.2 THE PROTEKSM ADVANTAGE

The option ProTek plan goes far beyond basic service, offering you NewTek-provided priority service for however long you own your product. ProTek enhances your support experience—every step of the way.

- With ProTek coverage, your part of a select group of customers whose needs are our first priority.
- Every ProTek Ultra customer gets the white-glove treatment by their local Ultra Service Provider.
- ProTek Ultra customer calls and emails are answered first.
- ProTek gives you priority access to the people who can answer your questions, help you with configuration, and walk you through solutions, step by step.
- And if disaster should strike and your NewTek product needs repair, we’re standing by ready to provide the best access. You get free overnight shipping.
- If you opt for a repair rather than a replacement, you get a guarantee that within three to five days of receiving your product it will be on its way back to you.
- If your product needs to be back in service immediately ProTek Ultra customers can usually get a replacement shipped right to your doorstep the very next day. Find full details of the ProTek program at <http://new.tk/ProTek>.

Appendix G: RESTORE FACTORY DEFAULTS

G.1 CREATE A 'USER BACKUP' DRIVE

The *Backup and Restore* system permits you to create a bootable clone of TriCaster's system drive on another (same size) hard drive inserted into one of the removable drive bays or connected externally for this purpose.

The clone operation includes:

- The hidden, factory prepared *Restore partition*
- **It is critical that you back up before restoring and move your folders to D: drive. This procedure restores your system drive ("C:") to its 'as-shipped' state. The "D:" drive, which holds content and sessions won't be modified.**

Afterward, the backup drive can be removed and stored, to be used later, if necessary, to i) restore the internal drive should, or ii) in the case of a catastrophic drive failure, to be connected internally to completely replace the original drive.

Note: Since opening TriCaster's case is a breach of warranty, please call Customer Service to obtain direction before proceeding with an internal drive replacement.

The actual creation process for the backup is discussed in the next section.

G.2 RESTORE TRICASTER SOFTWARE

We firmly expect you will never need it, but isn't it good to know that comprehensive *TriCaster Restore* features are available in the event of an unforeseen problem? You can use one of two methods to access TriCaster's *Restore* features, depending on your situation.

To restore your TriCaster system software *after the system launches normally*:

1. Click the Shut Down icon on the Home Page of the Startup Screen.
2. Click the Administrator Mode link at right to exit the TriCaster environment.
3. In the Administrator Mode screen, select the link labeled Backup or Restore System.
4. Select the Restore Factory Defaults bootup option.

If TriCaster should ever fail to boot up properly, you may need to try a different approach to restore your TriCaster system software, as follows:

1. Select the menu item labeled *Restore Factory Defaults* from the black boot screen that appears shortly after powering TriCaster up. (If this screen does not automatically appear, reboot and press F8 a few times in quick succession, say once per second, immediately after powering up.)

Either method described above will ultimately present you with powerful system backup and restore tools. The management screen initially presents you with 3 options as follows:

- *Restore System Partition to Factory Defaults*: overwrites the C partition (only) on the existing system drive from the disk image in its local Restore partition.

Note: Following a Reset to Factory Defaults operation, the Windows® system software must be re-activated. To do this, you will need the operating system serial number, which you will find on a sticker affixed to the exterior of your TriCaster.

This procedure restores your system drive (“C:”) to its ‘as-shipped’ state. The “D:” drive, which holds content and sessions won’t be modified. However, any TriCaster software updates (possibly including optional TriCaster expansion packs) will be overwritten – so use this function only when necessary. When you do restore, remember to update the TriCaster software afterward.

- *Create User Backup Drive*: create a bootable clone of the entire system drive (as it exists at the time) on either an external HDD or a drive mounted in a removable hard drive bay.

The clone operation includes:

- The existing (factory prepared) Restore partition
- The complete C partition

In cases of catastrophic drive failure requiring drive replacement, a service technician can simply connect the *User Backup* ‘clone’ drive in place of the original internal system drive, and you’ll be back in production (prudence would call for creating a new *User Backup* drive as the first order of business.

- *Restore System Partition from User Backup Drive*: overwrites the C partition (only) on the existing system drive with the C partition on the external user backup drive. This allows you to insert a *User Backup* drive into a removable drive bay (or otherwise connect it) and regain a functional system partition as stored on the clone drive.

Note: This feature is intended to allow a restore operation of the system drive that the clone was prepared on. It is not intended to allow restoration from a ‘foreign’ TriCaster. Using the ‘Restore System Partition from User Backup Drive’ in the latter manner will almost certainly cause multiple problems. In a dire emergency, however (such as if a system drive fails when no User Backup from the same unit has been prepared) a User Backup created on a similar model TriCaster can be helpful, as follows:

- *Install the ‘foreign’ User Backup drive to replace the defunct system drive.*
- *Power up, and – at the boot selection screen – select ‘Manage TriCaster’.*
- *Choose ‘Restore System Partition to Factory Defaults’, and follow prompts.*

It will be necessary to re-activate Windows, re-enter the registration unlock code, and possibly update the TriCaster software after the restoration process in this case. Be aware that the approach above is not encouraged since minor hardware differences between the two TriCasters involved may well result in issues in the resulting Windows installation. Also, the new system drive will no longer have a factory restore partition. Still, it might be better than nothing in a crisis.

INDEX

3

3D, 135

A

Add-Ons, 44
Administrator Mode, 27, 40, 41, 262
Alpha Channel, 85
Anaglyph, 135
Animation Store. *See* Live Desktop: Transitions: Animation Store
Asset Management, 99
Audio Mixer. *See* Live Desktop:Audio Mixer , *See* Live Desktop:Audio Mixer

B

Buffers, 111, 118

C

Calibration, 249
Chroma Keying. *See* LiveMatte
Clocks. *See* Multiview Output and Sessions
Color Correction, 134
Comps, 139
Connections
 Cameras, 16
 Output, 17

D

Default Effects, 134, 135

E

Eject, 38
External Audio. *See* Live Desktop:Audio Mixer

F

Feature Key, 41
Files
 Export, 107
 Import, 107

G

Genlock
 Configure. *See* I/O Configuration

H

Hard Drive
 Eject, 38
Home Page. *See* Startup Screen
Home Page, 18

I

I/O Configuration
 Audio, 25
 Genlock, 75
 Proc Amp, 7, 70, 91
 Video Input, 23
 Video Output, 21, 72
IMAG, 18, 245, 246, 247
Internal Audio. *See* Live Desktop:Audio Mixer

K

Keyboard, 14
Keying. *See* LiveMatte

L

Latency, 245, 246, 247
Launch Screen, 3, 18, 35
 Home Page, 4
 Open/Manage Session, 38
 Shutdown, 40
 Exit to Windows, 40
Live Desktop, 4, 20, 49, 50
 Audio Mixer, 11, 143
 External Audio tab, 144
 Internal Audio tab, 147
 Mute, 146
 Presets, 156
 Solo, 147
 VU Meters, 144
Grab, 182
Media Browser, 102
 Filter, 103
Media Player, 12, 147
 Asset Management, 99
 Autoplay, 105
 Presets, 107
 Title Pages, 108
 Editing, 109
 Stand-In Images, 110
Options Menu, 56
 Autoplay Out, 57

- Default Media File Level, 57
- Lock Mouse, 57
- Tabs Follow, 56
- Record, 67, 75, 168, 179
- Switcher, 57, 78
- Take, 86
- T-Bar, 86
- Transitions, 5, 7, 57, 77, 79
 - Animation Store, 8, 44, 81, 133, 143, 148
 - Waveform/Vectorscope, 249, 250, 251, 252
 - Workspace Presets, 6, 23, 52, 90
- LiveMatte, 9, 125, 126, 127
 - Luma Limit, 127
 - Spill Suppression, 128
- LiveSet, 10
 - Zoom T-Bar, 133
- Luma Keying. *See* LiveMatte

M

- M/E
 - Position, 133
- Macros, 157
 - Session, 160
 - Triggers, 159
- Monitors
 - Computer, 14
- Multiview Output
 - Broadcast Clocks, 59

N

- Network
 - Connecting, 168
 - Ping, 169
 - Sharing Buffers, 108
 - Sharing Media Folders, 108
 - Testing, 168
- Network Sharing, 108
- Network Sharing, 118

O

- Open/Manage Session. *See* Launch Screen

P

- Password
 - Default, Windows, 15
 - Export, 174
 - LivePanel, 29, 43
 - Streaming, 164
 - Web, 29, 43
- Pause, 257
- Ping, 169
- Play, Pause, & Stop, 257

R

- Rename Input, 51
- Restart. *See* Startup Screen:Shutdown System
- Restore, 42
- Restore TriCaster, 262

S

- Session, 18, 35
 - Auto-Launch, 39
 - Backup/Restore, 40, 47
 - Session Page, 19
 - Volume (Storage), 39
- Share Media, 173
- Show Alpha, 134
- Show Inverse Alpha, 134
- Shutdown. *See* Launch Screen:Shutdown
- Social Media
 - Password, 174
- Spell Check, 109
- Spill Suppresion. *See* LiveMatte
- Startup Screen
 - System Utilities
 - Restore TriCaster, 262
- Stop, 257
- Streaming, 161, 165, 166
 - Bitrate, 167
 - Capture, 165
 - Configuration, 162, 163
 - Encoders, 163
 - Password, 164
 - Presets, 164
 - Production Tips, 168
 - Profile, 167
 - Providers, 165, 167
- Switcher. *See* Live Desktop

T

- Touchscreen
 - Configuring Inputs, 40, 52
 - Switching, 7, 57

V

- Video
 - Calibration. *See* Calibration
- Virtual Set. *See* LiveSet

W

- Windows Login, 15
- Windows™
 - Exit to. *See* Launch Screen:Shutdown

CREDITS

Product Development: Alvaro Suarez, Artem Skitenko, Brad McFarland, Brian Brice, Bruno Deo Vergilio, Cary Tetrick, Charles Steinkuehler, Dan Fletcher, David Campbell, David Forstenlechner, Erica Perkins, Gabriel Felipe Santos da Silva, George Castillo, Gregory Marco, Heidi Kyle, Ivan Perez, James Cassell, James Killian, James Willmott, Jamie Finch, Jarno Van Der Linden, Jeremy Wiseman, Jhonathan Nicolas Moriera Silva, Josh Helpert, Karen Zipper, Kenneth Nign, Kyle Burgess, Leonardo Amorim de Araújo, Livio de Campos Alves, Matthew Gerner, Menghua Wang, Michael Gonzales, Mike Murphy, Monica Luevano Mares, Naveen Jayakumar, Ryan Cooper, Ryan Hansberger, Sergio Guidi Tabosa Pessoa, Shawn Wisniewski, Stephen Kolmeier, Steve Bowie, Steve Taylor, Troy Stevenson, Utkarsha Washimkar

Special thanks to: Andrew Cross, Tim Jenison

Third Party Licenses:

This product uses a number of third-party software libraries under license. Related license requirements are defined in documentation installed on the product. To view these licenses, please click the Additional Licenses link provided in the Help menu on the Startup>Home page shown upon launching the product.

Trademarks:

TriCaster, 3Play, TalkShow, Video Toaster, LightWave 3D, and Broadcast Minds are registered trademarks of NewTek, Inc. MediaDS, Connect Spark, LightWave, and ProTek are trademarks and/or service marks of NewTek, Inc. All other products or brand names mentioned are trademarks or registered trademarks of their respective holders.

